

1

CHAPTER

Real Numbers



Previous Year Questions

Multiple Choice Questions

1. If two positive integers p and q can be expressed as $p = 18a^2b^4$ and $q = 20a^3b^2$, where a and b are prime numbers, then LCM (p, q) is: [CBSE 2024]

(a) $2a^2b^2$ (b) $180a^2b^2$
(c) $12a^2b^2$ (d) $180a^3b^4$

2. If the HCF ($2520, 6600$) = 40 and LCM ($2520, 6600$) = $252 \times k$, then the value of k is [CBSE 2024]

(a) 1650 (b) 1600
(c) 165 (d) 1625

3. If $a = 2^2 \times 3^x$, $b = 2^2 \times 3 \times 5$, $c = 2^2 \times 3 \times 7$ and LCM (a, b, c) = 3780, then x is equal to [CBSE 2024]

(a) 1 (b) 2
(c) 3 (d) 0

4. A pair of irrational number whose product is a rational number is: [CBSE 2024]

(a) $(\sqrt{16}, \sqrt{4})$ (b) $(\sqrt{5}, \sqrt{2})$
(c) $(\sqrt{3}, \sqrt{27})$ (d) $(\sqrt{36}, \sqrt{2})$

5. The HCF of two numbers 65 and 104 is 13. LCM of 65 and 104 is $40x$, then the value of x is: [CBSE 2024]

(a) 5 (b) 13
(c) 40 (d) 8

6. The greatest number which divides 281 and 1249, leaving remainder 5 and 7 respectively, is:

(a) 23 (b) 276
(c) 138 (d) 69

7. The LCM of three numbers 28, 44, 132 is: [CBSE 2024]

(a) 258 (b) 231
(c) 462 (d) 924

8. (HCF $\times$ LCM) for the numbers 30 and 70 is [CBSE 2023]

(a) 2100 (b) 21
(c) 210 (d) 70

9. The number $(5 - 3\sqrt{5} + \sqrt{5})$ is [CBSE 2023]

(a) an integer
(b) a rational number
(c) an irrational number
(d) a whole number

10. The ratio of HCF to LCM of the least composite number and the least prime number is [CBSE 2023]

(a) 1 : 2 (b) 2 : 1
(c) 1 : 1 (d) 1 : 3

11. HCF of 92 and 152 is [CBSE 2021]

(a) 4 (b) 19
(c) 23 (d) 57

12. HCF of two consecutive even numbers is [CBSE 2021]

(a) 0 (b) 1
(c) 2 (d) 4

13. The (HCF $\times$ LCM) for the numbers 50 and 20 is

[CBSE 2021]

- (a) 1000 (b) 50
(c) 100 (d) 500

14. For which natural number n , 6^n ends with digit zero? [CBSE 2021]

- (a) 6 (b) 5
(c) 0 (d) None

15. The exponent of 5 in the prime factorisation of 3750 is [CBSE 2021]

- (a) 3 (b) 4
(c) 5 (d) 6

16. What is the greatest possible speed at which a girl can walk 95 m and 171 m in an exact number of minutes? [CBSE 2021]

- (a) 17 m/min (b) 19 m/min
(c) 23 m/min (d) 13 m/min

17. Three alarm clocks ring their alarms at regular intervals of 20 min, 25 min and 30 min respectively. If they first beep together at 12 noon, at what time will they beep again for the first time? [CBSE 2021]

- (a) 4:00 pm (b) 4:30 pm
(c) 5:00 pm (d) 5:30 pm

18. The greatest number which when divides 1251, 9377 and 15628 leaves remainder 1, 2, and 3 respectively is [CBSE 2021]

- (a) 575 (b) 450
(c) 750 (d) 625

19. If a and b are two coprime numbers, then a^3 and b^3 are [CBSE 2021]

- (a) Coprime (b) Not coprime
(c) Even (d) Odd

20. If n is a natural number, then $2(5^n + 6^n)$ always ends with [CBSE 2021]

- (a) 1 (b) 4
(c) 3 (d) 2

21. The LCM of two numbers is 2400. Which of the following CANNOT be their HCF? [CBSE 2021]

- (a) 300 (b) 400
(c) 500 (d) 600

22. HCF of 144 and 198 is [CBSE 2020]

- (a) 9 (b) 18
(c) 6 (d) 12

23. 225 can be expressed as [CBSE 2020]

- (a) 5×3^2 (b) $5^2 \times 3$
(c) $5^2 \times 3^2$ (d) $5^3 \times 3$

24. The total number of factors of a prime number is [CBSE 2020]

- (a) 1 (b) 0
(c) 2 (d) 3

25. The HCF and the LCM of 12, 21, 15 respectively are [CBSE 2020]

- (a) 3, 140 (b) 12, 420
(c) 3, 420 (d) 420, 3

Short Answer Type Questions

26. In a teachers' workshop, the number of teachers teaching French, Hindi and English are 48, 80 and 144 respectively. Find the minimum number of rooms required if in each room the same number of teachers are seated and all of them are of the same subject. [CBSE 2024]

27. Can the number $(15)^n$, n being a natural number, end with the digit 0? Given reason. [CBSE 2024]

28. Two numbers are in the ratio 2 : 3 and their LCM is 180. What is the HCF of these numbers? [CBSE 2023]

29. What is the HCF of smallest prime number and the smallest composite number? [CBSE 2018]

30. Given that $\sqrt{2}$ is irrational, prove that $(5 + 3\sqrt{2})$ is an irrational number. [CBSE 2018]

Case Based Type Questions

31. Teaching mathematics through activities is a powerful approach that enhances student's understanding and engagement. Keeping this in mind, Ms. Mukta planned a prime number game for class 5 students. She announces the number 2 in her class and asked the first student to multiply it by a prime number and then pass it to second student. Second student also multiplied it by a prime number and passed it to third student. In this way by multiplying to a prime number, the last student got 173250.

Now, Mukta asked some questions as given below to the students: [CBSE 2024]

- (i) What is the least prime number used by students?

(ii) (a) How many students are in the class?

OR

(b) What is the highest prime number used by students?

(iii) Which prime number has been used maximum times.

32. Khushi wants to organize her birthday party. Being health conscious, she decided to serve only fruits in her birthday party. She bought 36 apples and 60 bananas and decided to distribute fruits equally among all.



Based on the above information, answer the following questions :

- (i) How many guests Khushi can invite at the most? [CBSE 2023]
- (ii) How many apples and bananas will each guest get? [CBSE 2023]
- (iii) (a) If Khushi decides to add 42 mangoes, how many guests Khushi can invite at the most? [CBSE 2023]

OR

(b) If the cost of 1 dozen of bananas is , the cost of 1 apple is and cost of 1 mango is ₹20, find the total amount spent on 60 bananas, 36 apples and 42 mangoes. [CBSE 2023]

Long Answer Type Questions

33. Prove that $\sqrt{5}$ is an irrational number. [CBSE 2024]
34. Prove that $5 - 2\sqrt{3}$ is an irrational number. It is given that is an irrational number. [CBSE 2024]
35. Show that the number $5 \times 11 \times 17 + 3 \times 11$ is a composite number. [CBSE 2024]
36. Three bells toll at intervals of 9, 12 and 15 minutes respectively. If they start tolling together, after what time will they next toll together? [CBSE 2024]
37. Prove that $\frac{2-\sqrt{3}}{5}$ is an irrational number, given that is an irrational number. [CBSE 2024]
38. Prove that $\sqrt{3}$ is an irrational number. [CBSE 2023]
39. Given that $\sqrt{3}$ is an irrational number, show that $(5 + 2\sqrt{3})$ is an irrational number. [CBSE 2020]

OR

An army contingent of 612 members is to march behind an army band of 48 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march? [CBSE 2020]

40. Prove that $\sqrt{2}$ is an irrational number. [CBSE 2019]
41. Find HCF and LCM of 404 and 96 and verify that $\text{HCF} \times \text{LCM} = \text{Product of the two given numbers}$. [CBSE 2018]
42. Prove that $3 + \sqrt{2}$ is an irrational number. [CBSE 2009]
43. Prove that $2 - 3\sqrt{5}$ is an irrational number. [CBSE 2010]



Explanations

Previous Year Questions

Multiple Choice Questions

- (d) Factor of $p = 2 \times 3 \times 3 \times a \times a \times b \times b \times b \times b$
Factor of $q = 2 \times 2 \times 5 \times a \times a \times a \times b \times b$
Therefore L.C.M. of p and q is $= 2 \times 2 \times 3 \times 3 \times 5 \times a \times a \times a \times b \times b \times b \times b = 180a^3b^4$.
- We know that $\text{HCF} \times \text{LCM} = \text{Product of two number}$
Given $\text{HCF} = 40$ and $\text{LCM} = 252 \times k$
Then $40 \times 252 \times k = 2520 \times 6600$
On solving this equation we get $k = 1650$
- $a = 2^2 \times 3^x$
 $b = 2^2 \times 3 \times 5$
 $c = 2^2 \times 3 \times 7$
 $\text{LCM}(a, b, c) = 3780$
 $= 2^2 \times 3^3 \times 5 \times 7$
 $\therefore x = 3$
- (c) In the given options.
for option (a) $\sqrt{16}$ & $\sqrt{4}$ are not irrational number i.e. their real values are 4 & 2 respectively. If cannot be correct answer.
Now multiplying other option, we get
(b) $\sqrt{5} \times \sqrt{2} = \sqrt{10}$
(c) $\sqrt{27} \times \sqrt{3} = \sqrt{81} = 9$
(d) $\sqrt{36} \times \sqrt{2} = \sqrt{72}$
So, correct answer is option (c).
- (b) The product of the two numbers using the formula: $\text{HCF} \times \text{LCM} = \text{Number 1} \times \text{number 2}$
 $\Rightarrow 13 \times 40x = 65 \times 104$
 $\Rightarrow 520x = 6760$
 $\Rightarrow x = 13$
- (c) $281 - 5 = 276$ and $1249 - 7 = 1242$
The required number is the HCF of 276 and 1242.
 $276 = 2 \times 2 \times 3 \times 23$
 $1242 = 2 \times 3 \times 3 \times 3 \times 23$

$$\text{HCF}(276, 1242) = 2 \times 3 \times 23 = 138$$

Hence, the greatest number that divides 281 and 1249 leaving remainders 5 and 7 is 138.

- (d) Prime factorization of the numbers:
 $28 = 2 \times 2 \times 7$
 $44 = 2 \times 2 \times 11$
 $132 = 2 \times 2 \times 3 \times 11$
 $\text{LCM}(28, 44, 132) = 2 \times 2 \times 3 \times 7 \times 11 = 924$
- (a) $\text{HCF} \times \text{LCM} = 30 \times 70 = 2100$.
- (c) $5 - 2\sqrt{5}$ (An irrational number)
- (a) Least composite number is 4 and least prime number is 2
 $\Rightarrow \frac{\text{HCF}(2,4)}{\text{LCM}(2,4)} = \frac{2}{4} = \frac{1}{2}$
- (a) $92 = 2 \times 2 \times 23$
 $152 = 2 \times 2 \times 2 \times 19$
 $\text{H.C.F.}(92, 152) = 2 \times 2 = 4$
- (c) Let numbers be $2x$ and $2x + 2$
 $2x = 2 \times x$
 $2x + 2 = 2(x + 1)$
- (a) $\text{HCF} \times \text{LCM} = \text{Product of numbers}$
 $= 50 \times 20 = 1000$
- (d) For 6^n , where n belongs to natural number, the given number never ends with zero.
- (b) $3750 = 2 \times 3 \times 5 \times 5 \times 5 \times 5$
 $= 2^1 \times 3^1 \times 5^4$
- (b) $95 \times 5 \times 19$ and $171 = 9 \times 19$
 $\Rightarrow \text{HCF}(95, 171) = 19$
- (c) $\text{LCM}(20, 25, 30) = 300$ minutes
 $= 5$ hours
- (d) Greatest number $= \text{HCF}[(1251 - 1), (9377 - 2) \text{ and } (15628 - 3)]$
 $= \text{H.C.F.}[1250, 9375, 15625] = 625$
- (a) a^3 and b^3 will be co-prime, if a, b are co-prime.
- (d) Unit digit of 5^n and 6^n are 5 and 6 respectively.
[$\because n$ is a natural number]
 $\therefore$ Unit's digit of $2(5^n + 6^n) = 2 \times (5 + 6)$

$$= 2 \times 11 = 22 \text{ (i.e. 2)}$$

21. (c) 2400 is not divisible by 500.

22. (b) $144 = 24 \times 32$

$$198 = 2 \times 32 \times 11$$

$$\text{HCF} = 2 \times 3^2 = 18$$

23. (c) Prime factorisation of 225

$$\begin{array}{r|l} 3 & 225 \\ 3 & 75 \\ 5 & 25 \\ 5 & 5 \\ & 1 \end{array}$$

$$225 = 3^2 \times 5^2$$

$$198 = 2 \times 32 \times 11$$

24. (c) Total number of factors of a prime number is 2.

Hence, option (c) is correct.

25. (c) $12 = 2 \times 2 \times 3$

$$21 = 3 \times 7$$

$$15 = 5 \times 3$$

$$\therefore \text{HCF} = 3$$

$$\text{LCM} = 2 \times 2 \times 3 \times 5 \times 7 = 420$$

Short Answer Type Questions

26. To find a minimum number of room we need to find maximum number of teacher seated in a room.

So the Required number of teacher in a room = HCF of (48,80,144)

$$\text{HCF of } 48, 80, 144 = 16.$$

$$\text{No. of rooms} = \frac{48+80+144}{16} = 17$$

27. If 15^n end with digit zero, then the number should be divisible by 2 and 5.

$$\text{As } 2 \times 5 = 10$$

This means the prime factorization of 15 should contain prime factors 2 and 5.

$$\Rightarrow 15 = (3 \times 5)^n$$

It does not have the prime factor 2 but have 3 and 5, Since 2 is not present in the prime factorization, there is no natural number n which 15^n ends with digit zero.

So, 15^n can not end with digit zero.

28. Let the numbers be $2x$ and $3x$

Given, $\text{LCM} = 180$

Clearly, $\text{HCF} = x$

$$\text{LCM}(a, b) \times \text{HCF}(a, b) = a \times b$$

$$\Rightarrow 180 \times x = 2x \times 3x$$

$$\Rightarrow x^2 - 30x = 0$$

$$\Rightarrow x(x - 30) = 0$$

$$x = 0 \text{ or } x = 30$$

$\therefore x = 0$ is not possible as HCF can't be 0

$$\text{HCF} = x = 30.$$

29. **Smallest** prime number is 2

Smallest composite number is 4.

Therefore, HCF is 2.

30. Let us assume that $(5 + 3\sqrt{2})$ is rational. Then there exist co-prime positive integers a and b such that

$$5 + 3\sqrt{2} = \frac{a}{b}$$

$$3\sqrt{2} = \frac{a}{b} - 5$$

$$\sqrt{2} = \frac{a - 5b}{3b}$$

$$\Rightarrow \sqrt{2} \text{ is irrational.}$$

$$[\because a, b \text{ are integer, } \therefore \frac{a - 5b}{3b} \text{ is rational}]$$

This contradicts the fact that $\sqrt{2}$ is irrational.

So, our assumption is incorrect.

Hence, $(5 + 3\sqrt{2})$ is an irrational number.

Case Base Type Questions

31. First, we need to find the prime factors of 173250

$$\text{So, } 173250 = 2 \times 3 \times 3 \times 5 \times 5 \times 5 \times 7 \times 11$$

(i) Hence 3 is the least prime number used by students.

(ii) (a) There are a total of 7 student

OR

(b) 11

(iii) The prime Number 5 used in maximum time.

32. **Number** of apples = 36

$$\text{Number of bananas} = 60$$

$$\text{Number of mangoes} = 42$$

$$\therefore 36 = 2^2 \times 3^2$$

$$60 = 2^2 \times 3 \times 5$$

$$42 = 2 \times 3 \times 7$$

$$(i) \text{ HCF}(36, 60) = 2^2 \times 3 = 12$$

$\therefore$ Khushi can invite atmost 12 quests.

(ii) Each guest will get 3 apples and 5 bananas

$$(iii) (a) \text{ HCF}(36, 60, 42) = 2 \times 3 = 6$$

$\therefore$ Khushi can invite atmost 12 quests.

OR

(b) Cost of 1 dozen of bananas = ₹60

Cost of 1 apple = ₹15

Cost of 1 mango = ₹20

$\therefore$ Total amount spend on 60 bananas, 36 apples and 42 mangoes

$$= 5 \times 60 + 15 \times 36 + 20 \times 42 = ₹1680$$

Long Answer Type Questions

33. We need to prove that $\sqrt{5}$ is irrational

Let us assume that $\sqrt{5}$ is a rational number

So it can be expressed in the form p/q where p, q are co-prime integer and $q \neq 0$

$$\Rightarrow \sqrt{5} = \frac{p}{q}$$

On squaring both the sides we get,

$$\Rightarrow 5 = \frac{p^2}{q^2}$$

$$\Rightarrow 5q^2 = p^2 \quad \dots(i)$$

$$\frac{p^2}{5} = q^2$$

So 5 divides p^2 , p is a multiple of 5

$$\Rightarrow p = 5m$$

$$\Rightarrow p^2 = 25m^2 \quad \dots(ii)$$

From equations (i) and (ii), we get

$$5q^2 = 25m^2 \quad \dots(ii)$$

$$\Rightarrow q^2 = 5m^2 \quad \dots(ii)$$

q^2 is a multiple of 5

$\Rightarrow q$ is a multiple of 5

Thus, p, q have a common factor 5. This contradicts our assumption that they are coprime. Therefore, $\frac{p}{q}$ is not a rational number.

Hence, $\sqrt{5}$ is an irrational number.

34. Let $5 - 2\sqrt{3}$ be a rational number.

$$\therefore 5 - 2\sqrt{3} = \frac{p}{q} \text{ [where } p \text{ and } q \text{ are integer, } q \neq 0$$

and q and p are integer]

$$\Rightarrow -2\sqrt{3} = \frac{p}{q} - 5$$

$$\Rightarrow -2\sqrt{3} = \frac{p-5q}{q}$$

$$\Rightarrow \sqrt{3} = \frac{p-5q}{-2q}$$

We know that p and q are integers, so $\frac{p-5q}{-2q}$ is a rational number.

So, if RHS is a rational number, LHS will also be a rational number, which means $\sqrt{3}$ is a rational number.

But $\sqrt{3}$ is an irrational number.

This contradicts our assumption

Hence, $5 - 2\sqrt{3}$ is an irrational number.

$$35. 5 \times 11 \times 17 + 3 \times 11 = 11(5 \times 17 + 3) = 11(88) \\ = 11 \times 2 \times 2 \times 2 \times 11$$

So, the number mentioned above has more than two prime factors. But any prime number only has 1 and the number itself as its prime factor.

So, it is a composite number.

36. Three bells toll together

$$= \text{LCM}(9, 12, 15)$$

$$= 180 \text{ they next toll together after } 180 \text{ min}$$

37. Let us assume that $\frac{2-\sqrt{3}}{5}$ is a rational number.

Thus $\frac{2-\sqrt{3}}{5}$ can be represented in the form of $\frac{p}{q}$,

where p and q are integers $q \neq 0$, p and q are co-prime numbers.

$$\frac{2-\sqrt{3}}{5} = \frac{p}{q} \Rightarrow 2-\sqrt{3} = \frac{5p}{q} \Rightarrow 2 - \frac{5p}{q} = \sqrt{3} \\ \Rightarrow \frac{2q-5p}{q} = \sqrt{3}$$

Since, $\frac{2q-5p}{q}$ is rational $\Rightarrow \sqrt{3}$ is rational.

But, it is given that $\sqrt{3}$ is an irrational number. Therefore, our assumption is wrong.

Hence, $\frac{2-\sqrt{3}}{5}$ is an irrational number.

38. Let us assume $\sqrt{3}$ is rational

So there exist co-prime integers p and q , $q \neq 0$

such that $\sqrt{3} = \frac{p}{q}$

Squaring both side

$$3 = \frac{p^2}{q^2}$$

$$p^2 = 3q^2 \quad \dots(i)$$

$\Rightarrow 3$ is a factor of p^2

$\Rightarrow 3$ is a factor of p

$\Rightarrow p = 3m$, where m is an integer

$\Rightarrow p^2 = 9m^2 \quad \dots(ii)$ [Squaring both side]

From equation (i) and (ii), we get

$$\Rightarrow 3q^2 = 9m^2 \Rightarrow q^2 = 3m^2$$

$\Rightarrow 3$ is a factor of q^2

$\Rightarrow 3$ is a factor of q also

So both p and q have 3 as their common factor which contradicts the fact that p and q are co-prime

So are assumption is wrong

Hence $\sqrt{3}$ is a irrational.

39. Let $5 + 2\sqrt{3} = \frac{p}{q}$, be a rational number.

$5 + 2\sqrt{3} = \frac{p}{q}$, where a and b are co-prime integers.

$$\Rightarrow 2\sqrt{3} = \frac{p}{q} - 5 = \frac{p-5q}{q} \Rightarrow \sqrt{3} = \frac{p-5q}{2q}$$

Hence, $\frac{p-5q}{2q}$ is rational as p and q are integers.

But it is given that $\sqrt{3}$ is irrational.

$\Rightarrow$ LHS is irrational and RHS is rational.

which condtradicts our assumption that $5 + 2\sqrt{3}$ is a rational number.

$\therefore 5 + 2\sqrt{3}$ is an irrational number.

OR

For maximum number of columns, we need to find highest common factor i.e. HCF of 612 and 48.

$$\text{Now, } 612 = 48 \times 12 + 36$$

$$48 = 36 \times 1 + 12$$

$$36 = 12 \times 3 + 0$$

$\therefore$ HCF of 612 and 48 is 12.

$\therefore$ Maximum number of columns in wich they can match is 12.

40. Let $\sqrt{2}$ be rational. Then, there exist positive integer a and b such that $\sqrt{2} = \frac{a}{b}$,

[Where a and b are co-prime, $b \neq 0$]

$$\Rightarrow (\sqrt{2})^2 = \left(\frac{a}{b}\right)^2 \Rightarrow 2 = \frac{a^2}{b^2} \Rightarrow 2b^2 = a^2$$

$\therefore 2$ divides $a^2 \Rightarrow 2$ divides $a \quad \dots(i)$

Let $a = 2c$ for some integer c .

$$a^2 = 4c^2 \Rightarrow 2b^2 = 4c^2 \Rightarrow b^2 = 2c^2$$

$\therefore 2$ divides $b^2 \Rightarrow 2$ divides $b \quad \dots(ii)$

From (i) and (ii), we get

2 is common factor of both a and b

But this contradicts the fact that a and b have no common factor other than 1.

$\therefore$ Out supposition is wrong.

Hence, $\sqrt{2}$ is an irrational number.

41. Using the factor tree for the prime factorization of 404 and 96, we have

$$404 = 2^2 \times 101 \text{ and } 96 = 2^5 \times 3$$

To find the HCF, we list common prime factors and their smallest exponent in 404 and 96 as under:

Common prime factor = 2,

Least exponent = 2

$$\therefore \text{HCF} = 2^2 = 4$$

To find the LCM, we list all prime factors of 404 and 96 and their greatest exponent as follows:

Prime factors of	Greatest Exponent
404 and 96	
2	5
3	1
101	1

$$\therefore \text{LCM} = 2^5 \times 3^1 \times 101^1 = 2^5 \times 31 \times 101^1 = 9696$$

$$\text{Now, HCF} \times \text{LCM} = 9696 \times 4 = 38784$$

Product of two numbers = $404 \times 96 = 38784$

Therefore, $\text{HCF} \times \text{LCM} = \text{Product of two numbers.}$

42. Let assume $3 + \sqrt{2}$ is a rational number.

$$\therefore 3 + \sqrt{2} = \frac{p}{q}$$

$\{p, q \text{ are co-prime integers and } q \neq 0\}$

$$\Rightarrow \sqrt{2} = \frac{p}{q} - 3$$

$$\Rightarrow \sqrt{2} = \frac{p-3q}{q}$$

Since, $\frac{p-3q}{q}$ is a rational number but we know

$\sqrt{2}$ is an irrational.

$\therefore 3 + \sqrt{2}$ is not a rational number.

43. Let assume $2 - 3\sqrt{5}$ is a rational number.

$$\Rightarrow 2 - 3\sqrt{5} = \frac{p}{q},$$

$\{\text{where } p, q \text{ are co-prime integers and } q \neq 0\}$

$$\Rightarrow 2 - \frac{p}{q} = 3\sqrt{5} \Rightarrow \frac{2q-p}{3q} = \sqrt{5}$$

Since, $\frac{2q-p}{3q}$ is a rational number but we also

know $\sqrt{5}$ is an irrational.

$\therefore$ **Rational** $\neq$ irrational

$\Rightarrow$ **Our** assumption is wrong.

$\therefore 2 - 3\sqrt{5}$ is an irrational number.

2

CHAPTER

Polynomials



Previous Year Questions

Multiple Choice Questions

1. If the sum of zeroes of the polynomial $p(x) = 2x^2 - k\sqrt{2}x + 1$ is $\sqrt{2}$, then value of k is:

[CBSE 2024]

- (a) $\sqrt{2}$ (b) 2
(c) $2\sqrt{2}$ (d) $\frac{1}{2}$

2. The zeroes of a polynomial $x^2 + px + q$ are twice the zeroes of the polynomial $4x^2 - 5x - 6$. The value of p is:

[CBSE 2024]

- (a) $-\frac{5}{2}$ (b) $\frac{5}{2}$
(c) -5 (d) 10

3. **Assertion (A):** If the graph of a polynomial touches x -axis at only one point, then the polynomial cannot be a quadratic polynomial.

Reason (R): A polynomial of degree $n(n > 1)$ can have at most n zeroes.

[CBSE 2024]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A)
(c) Assertion (A) is true but Reason (R) is false
(d) Assertion (A) is false but Reason (R) is true
4. If the sum and the product of zeroes of a quadratic polynomial are $2\sqrt{3}$ and 3 respectively, then a quadratic polynomial is:

[CBSE 2024]

- (a) $x^2 + 2\sqrt{3}x - 3$ (b) $(x - \sqrt{3})^2$
(c) $x^2 - 2\sqrt{3}x + 3$ (d) $x^2 + 2\sqrt{3}x + 3$

5. If α, β are the zeroes of the polynomial $6x^2 - 5x - 4$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is

[CBSE 2024]

- (a) $\frac{5}{4}$ (b) $-\frac{5}{4}$
(c) $\frac{4}{5}$ (d) $\frac{5}{24}$

6. If α and β are the zeroes of the polynomial

$5x^2 + 3x - 7$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is

[CBSE 2024]

- (a) $-\frac{3}{7}$ (b) $\frac{3}{5}$
(c) $\frac{3}{7}$ (d) $-\frac{5}{7}$

7. If α and β are the zeroes of the polynomial

$2x^2 - 9x + 5$, then $\alpha^2 + \beta^2$ is

[CBSE 2024]

- (a) $\frac{1}{4}$ (b) $\frac{61}{4}$
(c) 1 (d) $\frac{71}{4}$

8. If α and $\beta(\alpha > \beta)$ are the zeroes of the polynomial $-x^2 + 8x + 9$, then $(\alpha - \beta)$ is

[CBSE 2024]

- (a) -10 (b) 10
(c) ± 10 (d) 8

9. What should be added to the polynomial $x^2 - 5x + 4$, so that 3 is the zero of the resulting polynomial?

[CBSE 2024]

- (a) 1 (b) 2
(c) 4 (d) 5

10. A quadratic polynomial, one of whose zeroes is $2 + \sqrt{5}$ and the sum of whose zeroes is 4, is:

[CBSE 2024]

- (a) $x^2 + 4x - 1$ (b) $x^2 - 4x - 1$
(c) $x^2 - 4x + 1$ (d) $x^2 + 4x + 1$

11. If a polynomial $p(x)$ is given by $p(x) = x^2 - 5x + 6$, then the value of $p(1) + p(4)$ is:

[CBSE 2024]

- (a) 0 (b) 4
(c) 2 (d) -4

12. If α and β are the zeroes of the polynomial $p(x) = kx^2 - 30x + 45k$ and $\alpha + \beta = \alpha\beta$, then the value of k is:

[CBSE 2024]

- (a) $-\frac{2}{3}$ (b) $-\frac{3}{2}$
(c) $\frac{3}{2}$ (d) $\frac{2}{3}$

13. **Assertion (A):** Degree of a zero polynomial is not defined.

Reason (R): Degree of a non-zero constant polynomial is 0.

[CBSE 2024]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A)
(c) Assertion (A) is true but Reason (R) is false
(d) Assertion (A) is false but Reason (R) is true

14. If $p(x) = x^2 + 5x + 6$, then $p(-2)$ is

[CBSE 2023]

- (a) 20 (b) 0
(c) -8 (d) 8

15. A quadratic polynomial whose sum and product of zeroes are 2 and -1 respectively is

[CBSE 2023]

- (a) $x^2 + 2x + 1$ (b) $x^2 - 2x - 1$
(c) $x^2 + 2x - 1$ (d) $x^2 - 2x + 1$

16. If α, β are zeroes of the polynomial $x^2 - 1$, then the value of $(\alpha + \beta)$ is

[CBSE 2023]

- (a) 2 (b) 1
(c) -1 (d) 0

17. If α, β are the zeroes of the polynomial $p(x) = 4x^2 - 3x - 7$, then $\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$ is equal to:

[CBSE 2023]

- (a) $\frac{7}{3}$ (b) $\frac{3}{7}$
(c) $\frac{3}{7}$ (d) $-\frac{3}{7}$

18. A quadratic polynomial having sum and product of its zeroes as 5 and 0 respectively, is

[CBSE 2021]

- (a) $x^2 + 5x$ (b) $2x(x - 5)$
(c) $5x^2 - 1$ (d) $x^2 - 5x + 5$

19. Zeroes of a quadratic polynomial $x^2 - 5x + 6$ are

[CBSE 2021]

- (a) -5, 1 (b) 5, 1
(c) 2, 3 (d) -2, -3

20. The zeroes of quadratic polynomial $x^2 + 99x + 127$ are

[CBSE 2021]

- (a) Both negative
(b) Both positive
(c) One positive and one negative
(d) Reciprocal of each other

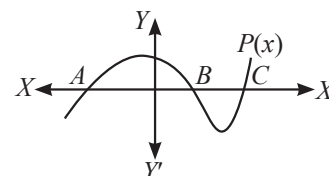
21. The graph of a polynomial $P(x)$ cuts the x -axis at 3 point and touches it at 2 other points. The number of zeroes of $P(x)$ is

[CBSE 2021]

- (a) 4 (b) -2, 4
(c) 2, 4 (d) 0, 4

22. In figure, the graph of a polynomial $P(x)$ is shown. The number of zeroes of $P(x)$ is

[CBSE 2021]



- (a) 1 (b) 2
(c) 3 (d) 4

23. A quadratic polynomial, the product and sum of whose zeroes are 5 and 8 respectively is

[CBSE 2021]

- (a) $k[x^2 - 8x + 5]$ (b) $k[x^2 + 8x + 5]$
(c) $k[x^2 - 5x + 8]$ (d) $k[x^2 + 5x + 8]$

24. If $x - 1$ is a factor of the polynomial $p(x) = x^3 + ax^2 + 2b$ and $a + b = 4$, then

[CBSE 2021]

- (a) $a = 5, b = -1$ (b) $a = 9, b = -5$
(c) $a = 7, b = -3$ (d) $a = 3, b = 1$

25. If α, β are the zeroes of the quadratic polynomial $p(x) = x^2 - (k + 6)x + 2(2k - 1)$, then the value of k , if

$\alpha + \beta = \frac{1}{2}\alpha\beta$, is

[CBSE 2021]

- (a) -7 (b) 7
(c) -3 (d) 3

26. If one zero of a quadratic polynomial $(kx^2 + 3x + k)$ is 2 then the value of k is [CBSE 2020]

(a) $\frac{5}{6}$ (b) $-\frac{5}{6}$
(c) $\frac{6}{5}$ (d) $-\frac{6}{5}$

27. If one of the zeroes of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is [CBSE 2020]

(a) 10 (b) -10
(c) -7 (d) -2

28. The quadratic polynomial, the sum of whose zeroes is -5 and their product is 6, is [CBSE 2020]

(a) $x^2 + 5x + 6$ (b) $x^2 - 5x + 6$
(c) $x^2 - 5x - 6$ (d) $-x^2 + 5x + 6$

Short Answer Type Questions

29. Find the zeroes of the quadratic polynomial $x^2 - 15$ and verify the relationship between the zeroes and the coefficients of the polynomial. [CBSE 2024]

30. If α and β are the zeroes of the polynomial $x^2 + x - 2$, then find the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$. [CBSE 2024]

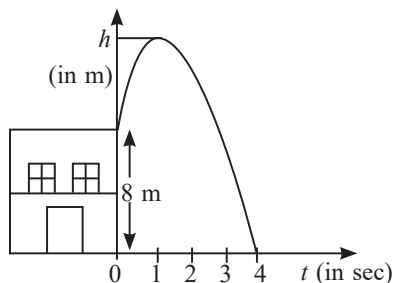
31. If α, β are the zeroes of the polynomial $p(x) = 5x^2 - 6x + 1$, then find the value of $\alpha + \beta + \alpha\beta$. [CBSE 2024]

32. If one zero of the polynomial $p(x) = 6x^2 + 37x - (k - 2)$ is reciprocal of the other, then find the value of k . [CBSE 2023]

Case Based Type Questions

Case Study Based Questions: Sukriti throws a ball upwards, from a rooftop which is 8 m high from ground level. The ball reaches to some maximum height and then returns and hit the ground. If height of the ball at time t (in sec) is represented by h (m), then equation of its path is given as $h = -t^2 + 2t + 8$

Based on above information, answer the following:



33. The maximum height achieved by ball is [CBSE 2021]

(a) 7 m (b) 8 m
(c) 9 m (d) 10 m

34. The polynomial represented by above graph is [CBSE 2021]

(a) Linear polynomial (b) Quadratic polynomial
(c) Constant polynomial (d) Cubic polynomial

35. Time taken by ball to reach maximum height is [CBSE 2021]

(a) 2 sec. (b) 4 sec.
(c) 1 sec. (d) 2 min.

36. Number of zero of the polynomial whose graph is given, is [CBSE 2021]

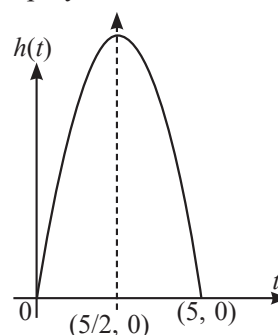
(a) 1 (b) 2
(c) 0 (d) 3

37. Zeroes of the polynomial are [CBSE 2021]

(a) 4 (b) -2, 4
(c) 2, 4 (d) 0, 4

Long Answer Type Questions

38. A ball is thrown in the air so that t seconds after it is thrown, its height h metre above its starting point is given by the polynomial $h = 25t - 5t^2$. [CBSE 2024]



Observe the graph of the polynomial and answer the following questions:

- Write zeroes of the given polynomials.
 - Find the maximum height achieved by ball.
 - After throwing upward, how much time did the ball take to reach to the height of 30 m?
 - Find the two different values of t when the height of the ball was 20 m.
39. Find the zeroes of the polynomial $4x^2 + 4x - 3$ and verify the relationship between zeroes and coefficients of the polynomial. [CBSE 2024]

40. If α and β are zeroes of the quadratic polynomial $p(x) = x^2 - 5x + 4$, then find the value of $\frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta$.

[CBSE 2024]

41. If α, β are zeroes of the quadratic polynomial $x^2 - 5x + 6$, form another quadratic polynomial whose zeroes are $1/\alpha, 1/\beta$.

[CBSE 2023]

42. If α and β are the zeroes of the polynomial $f(x) = x^2 - 4x - 5$, then find the value of $\alpha^2 + \beta^2$.

[CBSE 2020]

43. Find a quadratic polynomial whose zeroes are reciprocals of the zeroes of the polynomial $f(x) = ax^2 + bx + c, a \neq 0$.

[CBSE 2020]



Explanations

Previous Year Questions

Multiple Choice Questions

1. (b) $p(x) = 2x^2 - k\sqrt{2}x + 1$

Sum of zeroes = $\sqrt{2}$

$$\frac{-b}{a} = \sqrt{2} \Rightarrow \frac{-(-k\sqrt{2})}{2} = \sqrt{2}$$

$$k\sqrt{2} = 2\sqrt{2}$$

$$k = \frac{2\sqrt{2}}{\sqrt{2}} = 2$$

2. (a) Let assume that a and b are zeroes of polynomial $4x^2 - 5x - 6$

Sum of zeroes = $\frac{-\text{coefficient of } x}{\text{coefficient of } x^2}$

$$\Rightarrow \alpha + \beta = -\frac{(-5)}{4} = \frac{5}{4}$$

Now, it is given that zeroes of a polynomial $x^2 + px + q$ are twice the zeroes of the polynomials $4x^2 - 5x - 6$.

So, it means,

$$2\alpha \text{ and } 2\beta \text{ are zeroes of polynomial } x^2 + px + q$$

We know,

Sum of the zeroes = $\frac{-\text{coefficient of } x}{\text{coefficient of } x^2}$

$$\Rightarrow 2\alpha + 2\beta = -\frac{p}{1}$$

$$-p = 2(\alpha + \beta)$$

$$p = -2 \times \frac{5}{4} \Rightarrow p = -\frac{5}{2}$$

3. (d) A is false but R is true

A polynomial of degree $n(n > 1)$ can have at most n zeroes. If it is a quadratic polynomial i.e. the degree of the polynomial is 2, then it can have 2 distinct zeroes or 2 equal zeroes. 2 equal zeroes means that it has single (1) zero.

Therefore, even if the graph touches x -axis at only one point it can be quadratic.

4. (c) Here, $\alpha + \beta = 2\sqrt{3}$ and $\alpha\beta = 3$

then a quadratic polynomial = $x^2 - (\alpha + \beta)x + \alpha\beta$

$$= x^2 - 2\sqrt{3}x + 3$$

5. (b) $\alpha + \beta = \frac{-b}{a} = \frac{-(-5)}{6}$; $\alpha \cdot \beta = \frac{c}{a} = \frac{-4}{6}$

$$\alpha + \beta = \frac{5}{6}; \alpha \cdot \beta = \frac{-4}{6}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta} = \frac{5/6}{-4/6} = \frac{-5}{4}$$

6. (c) $\alpha + \beta = \frac{-3}{5}$, $\alpha\beta = \frac{-7}{5}$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

$$\Rightarrow \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\frac{-3}{5}}{\frac{-7}{5}} = \frac{3}{7}$$

7. (b) Given that α and β are zeroes of polynomial $2x^2 - 9x + 5$

We know,

Sum of the zeroes = $\frac{-\text{coefficient of } x}{\text{coefficient of } x^2}$

$$\Rightarrow \alpha + \beta = -\frac{(-9)}{2} = \frac{9}{2}$$

Also product of the zeroes = $\frac{\text{constant}}{\text{coefficient of } x^2}$

$$\Rightarrow \alpha\beta = \frac{5}{2}$$

Now, consider $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= \left(\frac{9}{2}\right)^2 - 2 \times \frac{5}{2} = \frac{81}{4} - 5 = \frac{81 - 20}{4} = \frac{61}{4}$$

8. (b) $\alpha - \beta = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$

$$\alpha + \beta = \frac{-b}{a} = \frac{-8}{-1} = 8$$

$$\alpha\beta = \frac{c}{a} = \frac{9}{-1} = -9$$

$$\Rightarrow \alpha - \beta = \sqrt{(8)^2 - 4(-9)} = \sqrt{64 + 36}$$

$$\alpha - \beta = 10$$

9. (b) Let k should be added to the polynomial.

$$\text{Let } P(x) = x^2 - 5x + 4 + k$$

$$P(3) = 0$$

$$\Rightarrow 3^2 - 5 \times 3 + 4 + k = 0$$

$$\Rightarrow -2 + k = 0$$

$$\Rightarrow k = 2$$

10. (b) One zero $\alpha = 2 + \sqrt{5}$ and another say β

$$\text{Sum of zeroes} = \alpha + \beta = 4$$

$$\Rightarrow 2 + \sqrt{5} + \beta = 4 \Rightarrow \beta = 2 - \sqrt{5}$$

$$\text{Polynomials, } p(x) = (x - 2 - \sqrt{5})(x - 2 + \sqrt{5})$$

$$p(x) = x^2 - 4x - 1$$

11. (b) $p(1) = 1^2 - 5(1) + 6 = 1 - 5 + 6 = 2$

$$p(4) = 4^2 - 5(4) + 6 = 16 - 20 + 6 = 2$$

$$p(1) + p(4) = 2 + 2 = 4$$

12. (d) Sum of zeroes, $\alpha + \beta = 30/k$

$$\text{Product of zeroes, } \alpha\beta = 45k/k = 45$$

$$\text{Given, } \alpha + \beta = \alpha\beta$$

$$30/k = 45 \text{ and } 30 = 45k \Rightarrow k = \frac{2}{3}$$

13. (b) We know that,

The constant polynomial 0 is called a zero polynomial.

The degree of a zero polynomial is not defined.

$\therefore$ Assertion is true.

The degree of a non-zero constant polynomial is zero.

$\therefore$ Reason is true.

Since both Assertion and Reason are true and Reason is not a correct explanation of Assertion.

The correct answer is (b).

14. (b) $p(x) = x^2 + 5x + 6$

$$p(-2) = (-2)^2 + 5(-2) + 6 = 0$$

15. (b) $x^2 - 2x - 1$ is the required polynomial.

16. (d) $\alpha + \beta = 0$ $\left[\because \alpha + \beta = \frac{-b}{a} \right]$

17. (d) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$

$$\therefore \alpha + \beta = \frac{3}{4}, \alpha\beta = \frac{-7}{4}$$

$$\Rightarrow \frac{\alpha + \beta}{\alpha\beta} = \frac{3 \times 4}{4(-7)} = \frac{-3}{7}$$

18. (b) $K[x^2 - (\text{sum of zeroes})x + (\text{product of zeroes})]$

where K is a non-zero constant.

$$\therefore p(x) = K[x^2 - 5x]$$

19. (c) $p(x) = x^2 - 5x + 6 = 0$

$$(x - 2)(x - 3) = 0$$

$$x = 2, x = 3$$

20. (a) Polynomials, $p(x) = x^2 + 99x + 127$

$$\text{Sum of zeroes} = -\frac{b}{a} = -99 = \text{negative}$$

$$\text{Product of zeroes} = \frac{c}{a} = 127 = \text{positive}$$

So, both zeroes must be negative.

21. (a) Zeroes of a polynomial $f(x)$ would be those points where the graph $f(x)$ will touch or cut the x -axis.

$$\therefore \text{Number of zeroes} = 4$$

22. (c) Graph intersect x -axis at 3 points.

23. (a) Required polynomial $= k[x^2 - 8x + 5]$

24. (b) $p(1) = 1 + a + 2b = 0$

$$\Rightarrow a + 2b = -1 \text{ and } a + b = 4$$

$$\Rightarrow b = -5 \text{ and } a = 9.$$

25. (b) $\alpha + \beta = k + 6$ and $\alpha\beta = 4k - 2$

$$\Rightarrow \alpha + \beta = \frac{\alpha\beta}{2} \Rightarrow k + 6 = 2k - 1.$$

$$\therefore k = 7$$

26. (d) 2 is a zero of polynomial $p(x) = kx^2 + 3x + k$.

$$\Rightarrow p(2) = 0$$

$$\Rightarrow k(2^2) + 3(2) + k = 0$$

$$\Rightarrow 4k + 6 + k = 0$$

$$\Rightarrow 5k = -6$$

$$\therefore k = \frac{-6}{5}$$

27. (b) Let $f(x) = x^2 + 3x + k$

$$f(2) = (2)^2 + 3(2) + k = 0$$

$$\Rightarrow 4 + 6 + k = 0 \Rightarrow k = -10$$

Hence, option (b) is correct.

28. (a) Quadratic polynomial

$$= x^2 - (\text{sum of zeroes})x + \text{product of zeroes}$$

$$= x^2 - (-5)x + 6$$

$$= x^2 + 5x + 6$$

Hence, option (a) is correct.

Short Answer Type Questions

29. Let $f(x) = x^2 - 15$

To calculate the zeroes of the given equation,
put $f(x) = 0$.

$$x^2 - 15 = 0$$

$$(x + \sqrt{15})(x - \sqrt{15}) = 0$$

$$x = \sqrt{15}, -\sqrt{15}$$

The zeroes of the given equation is $\sqrt{15}$ and $-\sqrt{15}$

$$\text{Sum of the zeroes is } \sqrt{15} + (-\sqrt{15}) = 0$$

$$\text{Product of the zeroes is } \sqrt{15} \times -\sqrt{15} = -15$$

According to the given equation,

The sum of the zeroes is,

$$\frac{-b}{a} = \frac{-(0)}{1} = 0$$

The product of the zeroes is

$$\frac{c}{a} = \frac{-15}{1} = -15$$

Hence, it is verified that.

30. Consider the given polynomial.

$$\Rightarrow x^2 + x - 2$$

Since, α and β are the zeroes of the polynomial.

$$\alpha + \beta = -1$$

$$\alpha\beta = -2$$

Since,

$$= \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$$

Therefore,

$$= \frac{(-1)^2 - 2 \times -2}{-2} = \frac{1 + 4}{-2} = \frac{-5}{2}$$

31. Roots using Vieta's formulas: $\alpha + \beta = -\frac{-6}{5} = \frac{6}{5}$

$$\text{Product of the roots: } \alpha\beta = \frac{1}{5}$$

$$\Rightarrow \alpha + \beta + \alpha\beta = \frac{6}{5} + \frac{1}{5} = \frac{7}{5}$$

32. $p(x) = 6x^2 + 37x - (k - 2)$

Let α, β be the zeroes of $p(x)$

$$\therefore \beta = \frac{1}{\alpha} \quad [\text{Given condition}]$$

$$\alpha\beta = 1 \quad \dots(i)$$

$$\text{Also, } \alpha\beta = \frac{-(k-2)}{6} \quad \dots(ii)$$

$$\text{From (i) and (ii), } \frac{-(k-2)}{6} = 1$$

$$\Rightarrow 2 - k = 6 \Rightarrow k = -4$$

Case Based Type Questions

33. (c) As, we can see from the graph maximum height is achieved at $t = 1$ s.

Height attained at $t = 1$ s

$$h = -(1)^2 + 2(1) + 8 = 9\text{m}$$

34. (b) Quadratic polynomial.

35. (c) As, we can see from the graph, ball reaches at maximum height at $t = 1$ s.

36. (b) Since, it is a quadratic polynomial so, it will have 2 zeroes.

37. (b) Zeroes of the polynomial, $h = -t^2 + 2t + 8 = 0$

$$-(t^2 - 2t - 8) = 0$$

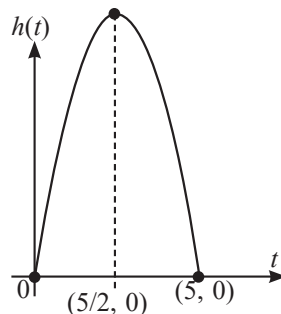
$$\Rightarrow (t - 4)(t + 2) = 0$$

$$t = 4 \text{ and } t = -2$$

Long Answer Type Questions

38. (i) Zeroes of the polynomial are $x = 0$; $x = 5$

(ii) Find the maximum height achieved by ball.
(1m)



$$h = 25t - 5t^2 \quad \dots(1)$$

Maximum height is achieved when $t = 5/2$.

$$h = 25 \times \frac{5}{2} - 5 \times \left(\frac{5}{2}\right)^2$$

$$h = \frac{125}{2} - 5 \times \frac{25}{4}$$

$$h = \frac{125}{4} \text{ or } h = 31.25 \text{ m}$$

(iii) (a) $h = 25t - 5t^2$

$$30 = 25t - 5t^2$$

$$t^2 - 5t + 6 = 0$$

$$t(t-2) - 3(t-2) = 0$$

$$(t-2) = 0; (t-3) = 0$$

$$t = 2\text{s}; t \neq 3\text{s}$$

$$\text{Hence, } t = 2 \text{ seconds}$$

(b) $h = 25t - 5t^2$

$$20 = 25t - 5t^2$$

$$5t^2 - 25t + 20 = 0$$

$$t^2 - 5t + 4 = 0$$

$$t^2 - 4t - 1t + 4 = 0$$

$$t(t-4) - 1(t-4) = 0$$

$$t = 1\text{s}; t = 4\text{s}$$

39. $4x^2 + 4x - 3 = 0$

$$4x^2 + 6x - 2x - 3 = 0$$

$$2x(2x+3) - 1(2x+3) = 0$$

$$\Rightarrow (2x+3)(2x-1) = 0 \Rightarrow x = -\frac{3}{2}, \frac{1}{2}$$

$$4x^2 + 4x - 3$$

$$a = 4, b = 4, c = -3$$

$$\alpha + \beta = -\frac{b}{a}$$

$$\frac{1}{2} + \left(\frac{-3}{2}\right) = \frac{-4}{4} \Rightarrow \frac{1-3}{2} = -1$$

$$\frac{-2}{2} = -1$$

$$-1 = -1$$

$$\text{LHS} = \text{RHS}$$

$$\alpha\beta = \frac{c}{a} \Rightarrow \frac{1}{2} \times \frac{-3}{2} = \frac{-3}{4}$$

$$-\frac{3}{4} = \frac{-3}{4}$$

$$\text{LHS} = \text{RHS}$$

40. $P(x) = x^2 - 5x + 4$

$$\alpha + \beta = \frac{-b}{a} = -\frac{(-5)}{1} = 5$$

$$\alpha\beta = \frac{c}{a} = \frac{4}{1} = 4$$

$$\frac{1}{\alpha} + \frac{1}{\beta} - 2\alpha\beta = \frac{\beta + \alpha - 2(\alpha\beta)^2}{\alpha\beta} = \frac{5 - 2(4)^2}{4} = \frac{5 - 32}{4} = \frac{-27}{4}$$

41. α, β are zeroes of the polynomial $x^2 - 5x + 6$.

$$\alpha + \beta = 5 \text{ and } \alpha\beta = 6$$

Let S and P be the sum and product of zeroes

$$\frac{1}{\alpha} \text{ and } \frac{1}{\beta}$$

$$S = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{5}{6}$$

$$P = \frac{1}{\alpha} \times \frac{1}{\beta} = \frac{1}{\alpha\beta} = \frac{1}{6}$$

Required quadratic polynomial is $k\left(x^2 - \frac{5}{6}x + \frac{1}{6}\right)$, where k is any non-zero constant.

42. α and β are zeroes of the polynomial

$$f(x) = x^2 - 4x - 5$$

$$\text{where } a = 1, b = -4, c = -5$$

$$\Rightarrow \alpha + \beta = -\frac{b}{a} = 4 \text{ and } \alpha\beta = \frac{c}{a} = -5,$$

$$\text{Now, } \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (4)^2 - 2(-5) = 16 + 10 = 26$$

43. Let α and β are the zeroes of the polynomial

$$f(x) = ax^2 + bx + c$$

$$\therefore (\alpha + \beta) = \frac{-b}{a} \quad \dots(i) \text{ and } \alpha\beta = \frac{c}{a} \quad \dots(ii)$$

According to the question, $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the zeroes of

the required quadratic polynomial

$\therefore$ Sum of zeroes of required polynomial

$$S' = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-b}{c} \quad [\text{From equation (i) and (ii)}]$$

and product of zeroes of required polynomial

$$= \frac{1}{\alpha} \times \frac{1}{\beta}$$

$$p' = \frac{1}{\alpha\beta} = \frac{a}{c} \quad [\text{From equation (ii)}]$$

$\therefore$ Equation of the required quadratic polynomial
 $= k(x^2 - S'x + P')$, where k is any non-zero constant

$$= k \left(x^2 - \left(\frac{-b}{c} \right) x + \frac{a}{c} \right) [\text{From equation (iii) and (iv)}]$$

$$= k \left(x^2 + \frac{b}{c} x + \frac{a}{c} \right)$$

$$= k'(cx + bx + a).$$



3 CHAPTER

Pair of Linear Equations in Two Variables

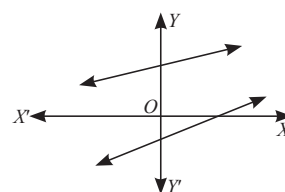


Previous Year Questions

Multiple Choice Questions

- If a pair of linear equations in two variables is consistent, then the lines represented by the two equations are: [CBSE 2024]
 (a) Always intersecting
 (b) Parallel
 (c) always coincident
 (d) Intersecting or coincident
- If $ax + by = a^2 - b^2$ and $bx + ay = 0$, then the value of $x + y$ is: [CBSE 2024]
 (a) $a^2 - b^2$ (b) $a + b$
 (c) $a - b$ (d) $a^2 + b^2$
- The pair of linear equation $x + 2y + 5 = 0$ and $-3x = 6y - 1$ has [CBSE 2024]
 (a) Unique solution
 (b) Exactly two solutions
 (c) Infinitely many solution
 (d) No solution
- Which out of the following type of straight lines will be represented by the system of equations $3x + 4y = 5$ and $6x + 8y = 7$? [CBSE 2024]
 (a) Parallel
 (b) Intersecting
 (c) Coincident
 (d) Perpendicular to each other
- The solution of the pair of linear equations $\frac{2x}{3} - \frac{y}{2} + \frac{1}{6} = 0$ and $\frac{x}{2} + \frac{2y}{3} = 3$ is: [CBSE 2024]
 (a) $x = 2, y = 3$ (b) $x = -2, y = 3$
 (c) $x = 2, y = -3$ (d) $x = -2, y = -3$
- In the given figure, graphs of two linear equations are shown. The pair of these linear equations is:

[CBSE 2024]



- consistent with unique solution.
 - consistent with infinitely many solutions.
 - inconsistent
 - inconsistent but can be made consistent by extending these lines.
- The value of k for which the system of equations $3x - y + 8 = 0$ and $6x - ky + 16 = 0$ has infinitely many solutions, is [CBSE 2024]
 (a) -2 (b) 2
 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$
 - The value of k , for which the pair of linear equations $x + y - 4 = 0$, $2x + ky - 3 = 0$ have no solution, is [CBSE 2021]
 (a) 0 (b) 2
 (c) 6 (d) 8
 - Perimeter of a rectangle whose length (l) is 4 cm more than twice its breadth (b) is 14 cm.
 The pair of linear equations representing the above information is [CBSE 2021]
 (a) $l + 4 = 2b, 2(l + b) = 14$
 (b) $l - b = 4, 2(l + b) = 14$
 (c) $l = 2b + 4, l + b = 14$
 (d) $l = 2b + 4, 2(l + b) = 14$
 - The solution of the pair of linear equation $x = -5$

and $y = 6$ is: [CBSE 2021]

- (a) $(-5, 6)$ (b) $(-5, 0)$
(c) $(0, 6)$ (d) $(0, 0)$

11. The value of k for which the pair of linear equation $3x + 5y = 8$ and $kx + 15y = 24$ has infinitely many solution is [CBSE 2021]

- (a) 3 (b) 9
(c) 5 (d) 15

12. The value of x and y satisfying the two equations $32x + 33y = 34$, $33x + 32y = 31$ respectively are: [CBSE 2021]

- (a) $-1, 2$ (b) $-1, 4$
(c) $1, -2$ (d) $-1, -4$

13. Two lines are given to be parallel. The equation of one of the lines is $3x - 2y = 5$. The equation of the second line can be [CBSE 2021]

- (a) $9x + 8y = 7$ (b) $-12x - 8y = 8$
(c) $-12x + 8y = 7$ (d) $12x + 8y = 7$

14. The value of k for which the system of equations $x + y - 4 = 0$ and $2x + ky = 3$, has no solution, is [CBSE 2020]

- (a) -2 (b) $\neq 2$
(c) 3 (d) 2

Very Short Answer Type Questions

15. If the pair of linear equations $x - y = 1$, $x + ky = 5$ has a unique solution $x = 2$, $y = 1$, then the value of k is [CBSE 2023]

- (a) -2 (b) -3
(c) 3 (d) 4

16. The pair of linear equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ has [CBSE 2023]

- (a) A unique solution
(b) Exactly two solutions
(c) Infinitely many solutions
(d) No solution

17. The pair of linear equations $2x = 5y + 6$ and $15y = 6x - 18$ represents two lines which are: [CBSE 2023]

- (a) Intersecting

- (b) Parallel
(c) Coincident
(d) Either intersecting or parallel

Short Answer Type Questions

18. The monthly incomes of A and B are in the ratio $8 : 7$ and their expenditure are in the ratio $19 : 16$. If each saves ₹ 2550 per month, find the monthly income of each. [CBSE 2024]

19. The sum of digits of a 2-digit number is 14. The number obtained by interchanging its digits exceeds the given number by 18. Find the number. [CBSE 2024]

20. (a) If $2x + y = 13$ and $4x - y = 17$, find the value of $(x - y)$. [CBSE 2024]

OR

- (b) Sum of two numbers is 105 and their difference is 45. Find the numbers. [CBSE 2024]

Case Based Type Questions

Book store shopkeeper gives books on rent for reading. He has variety of books in his store related to fiction, stories and quizzes etc. He takes a fixed charge for the first two days and an additional charge for subsequent day. Amruta paid ₹ 22 for a book and kept for 6 days; while Radhika paid ₹ 16 for keeping the book for 4 days.



Assume that the fixed charge be ₹ x and additional charge (per day) be ₹ y .

Based on the above information, answer any four of the following questions.

21. The situation of amount paid by Radhika, is algebraically represented by: [CBSE 2021]
(a) $x - 4y = 16$ (b) $x + 4y = 16$

(c) $x - 2y = 16$ (d) $x + 2y = 16$

22. The situation of amount paid by Amruta, is algebraically represented by [CBSE 2021]

(a) $x - 2y = 11$ (b) $x - 2y = 22$
(c) $x + 4y = 22$ (d) $x - 4y = 11$

23. What are the fixed charges for a book?

[CBSE 2021]

(a) ₹9 (b) ₹10
(c) ₹13 (d) ₹15

24. What are the additional charges for each subsequent day for a book? [CBSE 2021]

(a) ₹6 (b) ₹5
(c) ₹4 (d) ₹3

25. What is the total amount paid by both, if both of them have kept the book for 2 more days?

[CBSE 2021]

(a) ₹35 (b) ₹52
(c) ₹50 (d) ₹58

Long Answer Type Questions

26. Essel world is one of the India's largest amusement parks that offers a diverse range of thrilling rides, water attractions and entertainment options for visitors of all ages. The park is known for its iconic "water kingdom" section, making it a popular destination for family outings and fun filled adventure. The ticket charges for the park are ₹150 per child and ₹250 per adult.



One day, the cashier of the park found that 300 tickets were sold and an amount of ₹ 55,000 was collected.

Based on the above, answer the following questions:

- (i) If the number of children visited be x and the number of adults visited be y , then write the given situation algebraically.

- (ii) (a) How many children visited the amusement park that day?

OR

- (b) How many adults visited the amusement park that day?

- (iii) How much amount will be collected if 250 children and 100 adults visit the amusement park?

27. If three times the greater of two numbers is divided by the smaller one, we get 4 as the quotient and 3 as the remainder. Also, if seven times the smaller number is divided by greater one, we get 5 as the quotient and 1 as the remainder. Find the numbers.

[CBSE 2024]

28. (a) Using graphical method, solve the following system of equations: [CBSE 2024]

$$3x + y + 4 = 0 \text{ and } 3x - y + 2 = 0$$

OR

- (b) Tara scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 mark been deducted for each wrong answer, then Tara would have scored 50 marks. Assuming that Tara attempted all questions, find the total number of question in the test. [CBSE 2024]

29. Solve the following system of linear equation

$$7x - 2y = 5 \text{ and } 8x + 7y = 15 \text{ and verify your answer. [CBSE 2024]}$$

30. Three year ago, Rashmi was thrice as old as Nazma. Ten years later, Rashmi will be twice as old as Nazma. How old are Rashmi and Nazma now? [CBSE 2024]

31. Solve the following system of linear equations: $2p + 3q = 13$ and $5p - 4q = -2$ [CBSE 2024]

32. In a chemistry lab, there is some quantity of 50% acid solution and some quantity 25% acid solution. How much of each should be mixed to make 10 litres of 40% acid solution? [CBSE 2024]

33. (a) If we add 1 to the numerator and subtract 1 from the denominator, a fraction reduces to 1. It becomes $\frac{1}{2}$ if we only add 1 to the denominator.

What is the fraction?

[CBSE 2023]

OR

- (b) For which value of ' k ' will the following pair of linear equations have no solution? [CBSE 2023]

$$3x + y = 1$$

$$(2k - 1)x + (k - 1)y = 2k + 1$$

34. Two schools ' P ' and ' Q ' decided to award prizes to their students for two games of Hockey ₹ x per student and Cricket ₹ y per student. School ' P ' decided to award a total of ₹9,500 for the two games to 5 and 4 students respectively; while school ' Q ' decided to award ₹7,370 for the two games to 4 and 3 students respectively.



Based on the above information, answer the following questions: [CBSE 2023]

- (i) Represent the following information algebraically (in terms of x and y)
- (ii) (A) What is the prize amount for hockey?
(B) Prize amount on which game is more and by how much
- (iii) What will be the total prize amount if there are 2 students each from two games?
35. If the equations $kx - 2y = 3$ and $3x + y = 5$ represent two intersecting lines at unique point, then the value of k is _____. [CBSE 2020]



Explanations

Previous Year Questions

Multiple Choice Questions

- (a)
- (c) $ax + by = a^2 - b^2$... (i)
 $bx + ay = 0$... (ii)
 Multiplying equation (i) by b and (ii) by a ,
 $abx + b^2y = b(a^2 - b^2)$
 $abx + a^2y = 0$

$$\frac{y(b^2 - a^2) = b(a^2 - b^2)}{y = -b}$$
 [put this in equation (ii)]
 $b \times x + a(-b) = 0$
 $x = a \Rightarrow x + y = a - b.$
- (d) $\frac{1}{-3} = \frac{2}{-6} \neq \frac{5}{1}$
 $\frac{1}{-3} = \frac{-1}{3} \neq \frac{5}{1}$
 Condition of parallel lines. [No solution]
- (a) $\frac{a_1}{a_2} = \frac{3}{6} = \frac{1}{2}$
 $\frac{b_1}{b_2} = \frac{4}{8} = \frac{1}{2}, \frac{c_1}{c_2} = \frac{5}{7}$
 $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ [Parallel lines]
- (a) Correct answer is (a) $x = 2, y = 3$
 $\frac{2x}{3} - \frac{y}{2} + \frac{1}{6} = 0$... (i)
 $\frac{x}{2} + \frac{2y}{3} = 3$... (ii)
 Multiplying (i) and (ii) by 6, we get
 $4x - 3y = -1$... (iii)
 $3x - 4y = 18$... (iv)
 Multiplying (iii) by 4 and (iv) by 3 and adding we get
 $16x + 9x = -4 + 54$
 $\Rightarrow x = \frac{50}{25} = 2$
 Now, putting $x = 2$ in (iv), we have

$$3 \times 2 + 4y = 18 \Rightarrow y = \frac{18-6}{4} = 3$$

Thus, $x = 2$ and $y = 3$.

- (a)
- (b) Condition for coincident lines is
 $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$... (i)
 Given lines:
 $3x - y + 8 = 0$
 $6x - ky + 16 = 0$
 Here, $a_1 = 3, b_1 = -1, c_1 = 8$
 From equation (i)
 $\frac{3}{6} = \frac{-1}{-k} = \frac{8}{16} \Rightarrow \frac{1}{k} = \frac{1}{2}$
 $\therefore k = 2$
- (b) For no solution $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
 $\Rightarrow \frac{1}{2} = \frac{1}{k} \neq \frac{-4}{-3} \Rightarrow k = 2$
 Hence, option (b) is correct.
- (d) Perimeter, $2(l + b) = 14$... (i)
 $l = 2b + 4$... (ii)
- (a) $(-5, 6)$ is the solution of $x = -5$ and $y = 6$.
 $(-5, 6)$ is the solution of $x = -5$ and $y = 6$.
- (b) For, infinitely many solution
 $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \frac{3}{k} = \frac{5}{15} = \frac{8}{24}$
 $k = 9$
- (a) $32x + 33y = 34$... (i)
 $33x + 32y = 31$... (ii)
 Adding equation (i) and (ii) and subtracting equation (ii) and (i), we get
 $65x + 65y = 65$ or $x + y = 1$... (iii)
 and $-x + y = 3$... (iv)
 Adding equation (iii) and (iv), we get
 $y = 2$
 Substituting the value of y in equation (iii),

$$x = -1.$$

13. (c) If two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are parallel, then

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

It can only possible between $3x - 2y = 5$ and $-12x + 8y = 7$.

14. (d) For no solution; $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

$$\therefore \frac{1}{2} = \frac{1}{k} \neq \frac{-4}{-3} \Rightarrow k = 2$$

Hence, option (d) is correct.

Very Short Type Questions

15. (c) $x + ky = 5$

$$\text{At } x = 2, y = 1$$

$$2 + k = 5$$

$$\therefore k = 3$$

16. (d) $\frac{1}{-3} = \frac{2}{-6} \neq \frac{5}{1}$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

17. (c) $2x - 5y - 6 = 0$

$$-6x + 15y + 18 = 0$$

$$\frac{1}{3} = \frac{-1}{3} = \frac{-1}{3}$$

Short Answer Type Questions

18. Let monthly incomes be $8x$, $7x$ respectively....let expenses be $19y$, $16y$ respectively.....given, savings of both persons is equal to 2550.....

$$\text{So, } 8x - 19y = 7x - 16y$$

$$\Rightarrow x = 3y \quad \dots(i)$$

now, substitute (i) in

$$8x - 19y = 2550$$

$$\Rightarrow 24y - 19y = 2550$$

$$\Rightarrow 5y = 2550$$

$$\Rightarrow y = 510$$

$$\text{Using (i), } x = 3 \times 510 = 1530$$

so their monthly incomes are (1530×8) and

(1530×7) therefore, their incomes are Rs 12,240 and Rs. 10,710.

19. Let the two digit number be yx , where x is the unit's digit and y is the ten's digit.

According to question:

$$x + y = 14 \quad \dots(i)$$

$$xy - yx = 18$$

$$10x + y - (10y + x) = 18$$

$$9x - 9y = 18 \quad \dots(ii)$$

$$x - y = 2$$

$$x + y = 14$$

$$\hline 2x = 16$$

$$x = 8, y = 6$$

Hence, the number is = 68.

20. (a) Given system of equations are:

$$2x + y = 13 \quad \dots(i)$$

$$4x - y = 17 \quad \dots(ii)$$

Now adding both the equations we will have:

$$6x = 30$$

$$x = \frac{30}{6} = 5$$

Now substituting $x = 5$ in the equation (i)

$$2x + y = 13$$

$$2(5) + y = 13$$

$$10 + y = 13$$

$$y = 13 - 10$$

$$\therefore y = 3$$

Now substituting $x = 5$ and $y = 3$ in $(x - y)$

$$x - y = 5 - 3 = 2.$$

- (b) Let x and y be the two numbers

According to the questions,

$$x + y = 105 \quad \dots(i)$$

$$x - y = 45 \quad \dots(ii)$$

Adding (i) and (ii), we get

$$2x = 150$$

$$x = \frac{150}{2}$$

$$\therefore x = 75$$

Now, substituting $x = 75$ in the equation (i)

$$x + y = 105$$

$$75 + y = 105$$

$$y = 105 - 75$$

$$\therefore y = 30$$

Therefore, the two number are $x = 75$ and $y = 30$.

Case Based Type Questions

21. (d), 22. (c), 23. (b), 24. (d)

For Amruta, $x + (6 - 2)y = 22$

$$\text{i.e., } x + 4y = 22 \quad \dots(i)$$

For Radhika, $x + (4 - 2)y = 16$

$$\text{i.e., } x + 2y = 16 \quad \dots(ii)$$

Solving equation (i) and (ii),

we get $x = 10$ and $y = 3$

i.e., Fixed charges (x) = ₹10 $\dots(iii)$

and additional charges per subsequent day (y) = ₹3 $\dots(v)$

25. (c) Total amount paid for 2 more days by both

$$= (x + 4y) + 2y + (x + 2y) + 2y$$

$$= 2x + 10y$$

$$= 2 \times 10 + 10 \times 3 = ₹50$$

Long Answer Type Questions

26. (i) $x + y = 300 \quad \dots(i)$

$$150x + 250y = 55000 \quad \dots(ii)$$

(ii) (a) Solving equation (i) and (ii)

Number of children visited park (x) = 200

(b) Solving equation (i) and (ii)

Number of adults visited park (y) = 100

(iii) Amount collected = $250 \times 150 + 100 \times 250$
= ₹ 62,500

27. Let the larger number be x and the smaller one be y . We know that

$$\text{Dividend} = (\text{Divisor} \times \text{Quotient}) + \text{Remainder} \quad \dots(i)$$

When $3x$ is divided by y , we get 4 as quotient and 3 as remainder. Therefore, by using (i), we get

$$3x = 4y + 3 \Rightarrow 3x - 4y - 3 = 0 \quad \dots(ii)$$

When $7y$ is divided by x , we get 5 as quotient and 1

as remainder. Therefore, by using (i), we get

$$7y = 5x + 1 \Rightarrow 5x - 7y + 1 = 0 \quad \dots(iii)$$

Solving equation (ii) and (iii), by cross multiplication, we get

$$\frac{x}{-4-21} = \frac{-y}{3+15} = \frac{1}{-21+20}$$

$$\frac{x}{-25} = \frac{-y}{18} = -1$$

$$x = 25, y = 18.$$

28. (a) $3x + y + 4 = 0 \quad \dots(i)$

$$\text{and } 3x - y + 2 = 0 \quad \dots(ii)$$

$$a_1 = 3, b_1 = 1, c_1 = 4 \text{ and } a_2 = 3, b_2 = -1, c_2 = 2$$

$$\frac{a_1}{a_2} = \frac{3}{3} = 1, \frac{b_1}{b_2} = \frac{1}{-1} = -1, \frac{c_1}{c_2} = \frac{4}{2} = 2$$

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

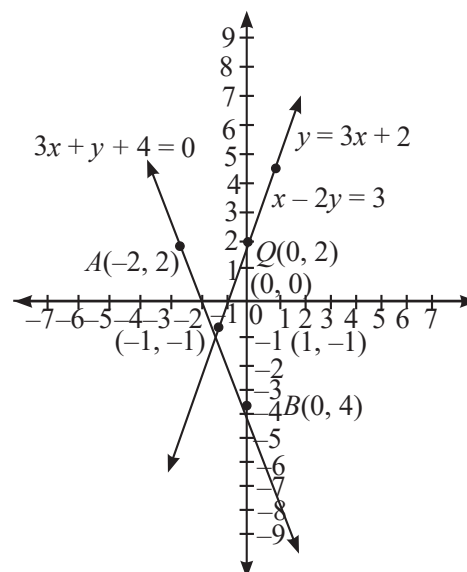
Hence, the system is consistent.

For equation (i), $y = -(3x + 4)$

x	-2	-1
y	2	-1

From equation (ii), $y = 3x + 2$

x	0	-1
$y = 3x + 2$	2	-1



Therefore, the above pair are consistent and solution is $x = -1, y = -1$

(b) Let x be the number of right answers and y be the number of wrong answers.

∴ According to the question,

$$3x - y = 40 \quad \dots(i)$$

$$4x - 2y = 50$$

$$\text{and, } 2x - y = 25 \quad \dots(ii)$$

on subtraction $x = 15$

Putting the value of x in (i)

$$3(15) - y = 40$$

$$y = 5$$

Number of right answers = 15

Number of wrong answers = 5

Total number of questions = $5 + 15 = 20$

$$29. (7x - 2y = 5) \times 7 \quad \dots(i)$$

$$(8x + 7y = 15) \times 2 \quad \dots(ii)$$

$$49x - 14y = 35$$

$$16x + 14y = 30$$

$$\hline 65x = 65$$

$$x = 1$$

$$y = \frac{7x - 5}{2} = \frac{7 - 5}{2} = 1$$

Verification: Substitute $x = 1$ and $y = 1$ in the given equation (i) and (ii), we get

In equation (i)

$$\text{L.H.S.} = 7 \times 1 - 2 \times 1 = 5$$

$$\text{R.H.S.} = 5$$

$$\text{L.H.S.} = \text{R.H.S.}$$

In equation (ii)

$$\text{L.H.S.} = 8 \times 1 + 7 \times 1 = 15$$

$$\text{R.H.S.} = 15$$

$$\text{L.H.S.} = \text{R.H.S.}$$

$$30. \text{ Let the age of Nazma 3 yrs ago be } x.$$

$$\therefore \text{ Rashmi's age} = 3x$$

$$\text{Present ages Nazma} = (x + 3) \text{ years}$$

$$\text{Rashmi} = (3x + 3) \text{ years}$$

$$\text{After 10 years Nazma} = (x + 13) \text{ years}$$

$$\text{Rashmi} = (3x + 13) \text{ years}$$

According to given question

$$3x + 13 = 2(x + 13)$$

$$3x + 13 = 2x + 26$$

$$x = 13$$

$$\therefore \text{ Present age of Nazma} = x + 3 = 16 \text{ years.}$$

$$\text{Rashmi} = 3x + 3 \text{ years.} = 3 \times 13 + 3 = 42 \text{ years.}$$

$$31. 2p + 3q = 13 \quad \dots(i)$$

$$5p - 4q = -2 \quad \dots(ii)$$

Multiplying (i) by 4 and (ii) by 3

$$8p + 12q = 52$$

$$15p - 12q = 86$$

$$\hline 23p = 46$$

$$p = 2$$

Putting the value of p in equation (ii), we get

$$5 \times 2 - 4q = -2$$

$$10 + 2 = 4q$$

$$q = \frac{12}{4} = 3$$

$$p = 2, q = 3$$

$$32. \text{ Let } x \text{ litres and } y \text{ litres be the amount of solution taken from 50\% and 25\% acid solutions respectively.}$$

As per the questions

$$50\% \text{ of } x + 25\% \text{ of } y = 40\% \text{ of } 10$$

$$\Rightarrow 0.50x + 0.25y = 4$$

$$\Rightarrow 2x + y = 16 \quad \dots(i)$$

Since, the total number is 10 litres, so

$$x + y = 10$$

Subtracting (ii) from (i), we get

$$x = 6 \text{ litres}$$

Now, putting $x = 6$ in (ii), we have

$$\Rightarrow 6 + y = 10 \Rightarrow y = 4 \text{ litres}$$

Hence, volume of 50% acid solution = 6 litres and volume of 25% acid solution = 4 litres

$$33. (a) \text{ Let required fraction be } \frac{x}{y}$$

$$\text{According to question, } \frac{x+1}{y-1} = 1$$

$$\Rightarrow x + 1 = y - 1$$

$$\Rightarrow x = y - 2 \quad \dots(i)$$

$$\text{Also, } \frac{x}{y+1} = \frac{1}{2}$$

$$\Rightarrow 2x = y + 1 \quad \dots(ii)$$

From equations (i) and (ii), we get

$$2y - 4 = y + 1$$

$$y = 5$$

$$\therefore x = 3$$

$$\text{Required fraction } \frac{x}{y} \text{ is } \frac{3}{5}.$$

OR

$$(b) 3x + y = 1$$

$$(2k - 1)x + (k - 1)y = 2k + 1$$

For no solution; $\frac{3}{2k-1} = \frac{1}{k-1} \neq \frac{1}{2k+1}$

$$\Rightarrow \frac{3}{2k-1} = \frac{1}{k-1}$$

$$\Rightarrow 2k - 1 = 3k - 3 \Rightarrow k = 2$$

Also, $\frac{1}{k-1} \neq \frac{1}{2k+1}$

$$2k + 1 \neq k - 1 \Rightarrow k \neq -2.$$

34. (i) $5x + 4y = 9500$... (i)

$4x + 3y = 7370$... (ii)

(ii) (a) Multiplying (i) by 3 and (ii) by 4; we get

$15x + 12y = 28,500$... (iii)

$16x + 12y = 29,480$... (iv)

Subtracting (iii) and (iv)

$16x + 12y = 29,480$

$15x + 12y = 28,500$

$x = 980$

Prize amount for Hockey = ₹980 per student

OR

(b) Multiplying (i) by 3 and (ii) by 4; we get

$15x + 12y = 28,500$... (iii)

$16x + 12y = 29,480$... (iv)

Subtracting (iii) from (iv)

$16x + 12y = 28,500$

$15x + 12y = 28,500$

$x = 980$

Putting $x = 980$ in (i);

$5(980) + 4y = 9500$

$\Rightarrow y = 1150$

$\therefore$ Prize amount of Hockey = ₹980 per student

prize amount of Cricket = ₹1150 per student.

$\Rightarrow$ Prize amount per student of Cricket greater by ₹170

(iii) Total prize amount if there are 2 students each from 2 games = $2(x + y)$
 $= 2(980 + 1150) = ₹4260$

35. For any real number except $k = -6$.

$kx - 2y = 3$ and $3x + y = 5$ represent lines intersecting at a unique point.

$$\Rightarrow \frac{k}{3} \neq \frac{-2}{1} \Rightarrow k \neq -6$$

For any real number except $k = -6$

The given equation represent two intersecting lines at unique point.

4

CHAPTER

Quadratic Equations



Previous Year Questions

Multiple Choice Questions

1. The value(s) of k for which the quadratic equation $5x^2 - 9kx + 5 = 0$ has real and equal roots, is/are: [CBSE 2024]

(a) $-\frac{10}{9}$ (b) $\pm \frac{9}{10}$
(c) $\frac{10}{9}$ (d) $\pm \frac{10}{9}$

2. A quadratic polynomial whose zeroes are 3 and -2 , is: [CBSE 2024]

(a) $x^2 - x - 6$ (b) $x^2 + x - 6$
(c) $2x^2 - x - 12$ (d) $x^2 + x + 6$

3. Which of the following equations has 2 as a root? [CBSE 2024]

(a) $x^2 - 4x + 5 = 0$ (b) $x^2 + 3x - 12 = 0$
(c) $2x^2 - 7x + 6 = 0$ (d) $3x^2 - 6x - 2 = 0$

4. A quadratic polynomial, the sum of whose zeroes is -5 and their product is 6, is: [CBSE 2024]

(a) $x^2 + 5x + 6$ (b) $x^2 - 5x + 6$
(c) $x^2 - 5x + 6$ (d) $-x^2 + 5x + 6$

5. The zeroes of the polynomial $3x^2 + 11x - 4$ are: [CBSE 2024]

(a) $\frac{1}{3}, 4$ (b) $-\frac{1}{3}, -4$
(c) $\frac{1}{3}, -4$ (d) $-\frac{1}{3}, 4$

6. If one zeroes of a quadratic polynomial $kx^2 + 4x + k$ is 1, then the value of k is: [CBSE 2024]

(a) 2 (b) -2
(c) 4 (d) -4

7. The roots of the quadratic equation $x^2 - 4 = 0$ is/are: [CBSE 2024]

(a) 2 only (b) $-2, 2$
(c) 4 only (d) $-4, 4$

8. Which of the following is not a quadratic equation [CBSE 2024]

(a) $(x - 2)^2 + 1 = 2x - 3$
(b) $(2x - 1)(x - 3) = (x + 5)(x - 1)$
(c) $x(x + 1) + 8 = (x + 2)(x - 2)$
(d) $2x + \frac{3}{x} = 5$

9. The roots of the quadratic equation $x^2 + 3x - 10 = 0$ are: [CBSE 2024]

(a) 5, 2 (b) $-5, 2$
(c) 5, -2 (d) $-5, -2$

10. The discriminant of the quadratic equation $x^2 - 4x + 3 = 0$ [CBSE 2024]

(a) 28 (b) -8
(c) 4 (d) 2

11. A quadratic equation whose one root is 2 and the sum of whose roots is zero, is: [CBSE 2023]

(a) $x^2 + 4 = 0$ (b) $x^2 - 2 = 0$
(c) $4x^2 - 1 = 0$ (d) $x^2 - 4 = 0$

12. Which of the following is not a quadratic equation? [CBSE 2023]

(a) $2(x - 1)^2 = 4x^2 - 2x + 1$
(b) $2x - x^2 = x^2 + 5$
(c) $(\sqrt{2}x + \sqrt{3})^2 + x^2 = 3x^2 - 5x$
(d) $(x^2 + 2x)^2 = x^4 + 3 + 4x^3$

13. The roots of the equation $x^2 + 3x - 10 = 0$ are
[CBSE 2023]

(a) 2, -5 (b) -2, 5
(c) 2, 5 (d) -2, -5

14. **Assertion (A):** If one root of the quadratic equation $4x^2 - 10x + (k - 4)$ is reciprocal of the other, then value of k is 8.

Reason (R): Roots of the quadratic equation $x^2 - x + 1 = 0$ are real.
[CBSE 2023]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A)
(c) Assertion (A) is true but Reason (R) is false
(d) Assertion (A) is false but Reason (R) is true

Short Answer Type Questions

15. The altitdde of a right angled triangle is 7 cm less than its base. If its hypotenuse is 17 cm long, then

- (a) represent the above information in the form of a quadratic equation;
(b) find the length of the sides of the triangle.

[CBSE 2024]

16. Find the zeroes of the quadratic polynomial $5x^2 + 3x - 2$ and verify the relationship between the zeroes and the co-efficients.
[CBSE 2024]

17. (a) Zeroes of the quadratic polynomial $x^2 - 3x + 2$ are α and β . Construct a quadratic polynomial whose zeroes are $2\alpha + 1$ and $2\beta + 1$.

- (b) Find the zeroes of the polynomial $4x^2 - 4x + 1$ and verify the relationship between the zeroes and the coefficients.
[CBSE 2024]

18. (a) If α, β are zeroes of the polynomial $8x^2 + 14x + 3$, then find the value of $\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$.

OR

- (b) Find a quadratic polynomial whose zeroes are -9 and 6.
[CBSE 2024]

19. (a) Using quadratic formula, find the real roots of the equation $2x^2 + 2x + 9 = 0$, if they exist.

OR

- (b) Find the value of ' k ' for which the quadratic equation $kx^2 - 2kx + 6 = 0$ has real and equal roots. Also, find the roots.
[CBSE 2024]

20. If $\frac{2}{3}$ is a root of the quadratic equation $kx^2 - x - 2 = 0$, then find the value of k .
[CBSE 2024]

21. (a) If α, β are zeroes of the quadratic polynomial $2x^2 + 7x + 5$, then, find the value of $\alpha^2 + \beta^2 + \alpha\beta$.

OR

- (b) If one zero of the quadratic polynomial $6x^2 + 37x - (p - 2)$ is reciprocal of the other, then find the value of p .
[CBSE 2024]

22. Find the zeroes of the quadratic equation polynomial $6x^2 - 7x - 3$ and verify the relationship between the zeroes and the coefficients of the polynomials. [CBSE 2024]

23. (a) Find the discriminant of the quadratic equation and hence find the nature of its roots.

[CBSE 2023]

OR

- (b) Find the roots of the quadratic equation $x^2 - x - 2 = 0$
[CBSE 2023]

24. (a) Find the sum and product of the roots of the quadratic equation $2x^2 - 9x + 4 = 0$.

[CBSE 2023]

OR

- (b) Find the discriminant of the quadratic equation $4x^2 - 5 = 0$ and hence comment on the nature of roots of the equation.
[CBSE 2023]

25. Find the value of ' p ' for which the quadratic equation $px(x - 2) + 6 = 0$ has two equal real roots.

[CBSE 2023]

26. Find the nature of the roots of the quadratic equation :
 $4x^2 - 5x - 1 = 0$
[CBSE 2022]

27. Solve the quadratic equation $x^2 + 2\sqrt{2}x - 6 = 0$ for x .
[CBSE 2022]

28. In a flight of 600 km, an aircraft was slowed due to bad weather. Its average speed for the trip was reduced by 200 km/hr and time of flight increased by 30 minutes. Find the original duration of flight.
[CBSE 2020]

29. If quadratic equation $3x^2 - 4x + k = 0$ has equal roots, then the value of k is
[CBSE 2020]

30. For what values of k , the roots of the equation $x^2 + 4x + k = 0$ are real?
[CBSE 2019]

31. Find the value of k for which the roots of the equation $3x^2 - 10x + k = 0$ are reciprocal of each other. [CBSE 2019]
32. If $x = 3$ is one root of the quadratic equation $x^2 - 2kx - 6 = 0$, then find the value of k . [CBSE 2018]
33. A plane left 30 minutes late than its scheduled time and in order to reach the destination 1500 km away in time, it had to increase its speed by 100 km/h from the usual speed. Find its usual speed. [CBSE 2018]
34. Find the value of p , for which one root of the quadratic equation $px^2 - 14x + 8 = 0$ is 6 times the other. [CBSE 2017]
35. If $ad \neq bc$, then prove that the equation $(a^2 + b^2)x^2 + 2(ac + bc)x + (c^2 + d^2) = 0$ has no real roots. [CBSE 2017]
36. If -5 is a root of the quadratic equation $2x^2 + px - 15 = 0$ and the quadratic equation $p(x^2 + x) + k = 0$ has equal roots, find the value of k . [CBSE 2016]
37. Solve the following quadratic equation for x :
 $4x^2 + 4bx - (a^2 - b^2) = 0$ [CBSE 2015]
38. Solve for x : $\sqrt{3}x^2 - 2\sqrt{2}x - 2\sqrt{3} = 0$ [CBSE 2015]
39. If the quadratic equation $px^2 - 2\sqrt{5}px + 15 = 0$ has two equal roots, then find the value of p . [CBSE 2015]

Long Answer Type Questions

40. The sum of the ages of a boy and his sister (in years) is 25 and product of their ages is 150. Find their present ages. [CBSE 2022]
23. (a) A 2-digit number is such that the product of its digits is 24. If 18 is subtracted from the number, the digits interchange their places. Find the number. [CBSE 2022]

OR

- (b) The difference of the squares of two numbers is 180. The square of the smaller number is 8 times the greater number. Find the two numbers. [CBSE 2022]
41. A two digit number is such that the product of its digits is 14. If 45 is added to the number; the digits interchange their places. Find the number. [CBSE 2020]
42. Two water taps together can fill a tank in $1\frac{7}{8}$ hours. The tap with longer diameter takes 2 hours less than the tap with smaller one to fill the tank separately. Find the time in which each tap can fill the tank separately. [CBSE 2019]
43. A motor-boat whose speed is 18 km/hr in still water take 1 hr more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream. [CBSE 2018]
44. A train travels at a certain average speed for a distance of 63 km and then travels a distance of 72 km at an average speed of 6 km/hr more than its original speed. It takes 3 hours to complete total journey, what is the original average speed? [CBSE 2018]
45. Two taps running together can fill a tank in $3\frac{1}{13}$ hours. If one tap takes 3 hours more than the other to fill the tank, then how much time will each tap take to fill the tank? [CBSE 2017]
46. A motor-boat whose speed is 24 km/h in still water takes 1 hour more to go 32 km upstream than to return downstream to the same spot. Find the speed of the stream. [CBSE 2016]
47. The diagonal of a rectangular field is 16 metres more than the shorter side. If the longer side is 14 metres more than the shorter side, then find the lengths of the sides of the field. [CBSE 2015]
48. A train travels at a certain average speed for a distance of 54 km and then travels a distance of 63 km at an average speed of 6 km/h more than the first speed. If it takes 3 hours to complete the total journey, what is its first speed? [CBSE 2015]



Explanations

Previous Year Questions

Multiple Choice Questions

1. (d) For real and equal roots

$$D = 0$$

$$b^2 - 4ac = 0$$

$$a = 5, b = -9k \text{ and } c = 5$$

$$(-9k)^2 - 4(5)(5) = 0$$

$$81k^2 - 100 = 0$$

$$k^2 = \frac{100}{81} \Rightarrow k = \pm \frac{10}{9}$$

2. (a) Roots are 3 and -2

Therefore, factors are $(x - 3)$ and $(x + 2)$

$$(x - 3)(x + 2) = x^2 + 2x - 3x - 6$$

$$= x^2 - x - 6$$

3. (c) For a quadratic equation $f(x)$ if a is root then,

$$f(a) = 0$$

$$f(x) = 2x^2 - 7x + 6$$

$$f(2) = 2(2)^2 - 7(2) + 6$$

$$= 8 - 14 + 6 = 0$$

4. (a) Quadratic polynomial is $x^2 - (\text{sum of roots})x + (\text{product of roots})$

$$= x^2 - (-5)x + 6 = 0$$

$$= x^2 + 5x + 6 = 0$$

5. (c) $f(x) = 3x^2 + 11x - 4$

$$\text{So, } 3x^2 + 11x - 4 = 0$$

$$= 3x^2 + 12x - x - 4 = 0$$

$$= 3x(x + 4) - 1(x + 4) = 0$$

$$= (3x - 1)(x + 4) = 0$$

$$x = \frac{1}{3} \text{ and } -4.$$

6. (b) Let $f(x)$ be a quadratic polynomial and if ' a ' is root then $f(a) = 0$

$$f(x) = kx^2 + 4x + k$$

$$f(1) = 0$$

$$k(1)^2 + 4(1) + k = 0$$

$$2k = -4 \Rightarrow k = -2$$

7. (b) $f(x) = x^2 - 4$

$$x^2 - 4 = 0 \Rightarrow x^2 - (2)^2 = 0$$

$$(a^2 - b^2) = (a + b)(a - b)$$

$$(x + 2)(x - 2) = 0$$

$$x = \pm 2$$

8. (c) $x(x + 1) + 8 = (x + 2)(x - 2)$

$$\Rightarrow x^2 + x + 8 = x^2 - 4$$

$$\Rightarrow x = -12$$

This is not a quadratic equation.

9. (b) $x^2 + 3x - 10 = 0$

$$\Rightarrow x^2 + 5x - 2x - 10 = 0$$

$$\Rightarrow x(x + 5) - 2(x + 5) = 0$$

$$\Rightarrow (x - 2)(x + 5) = 0$$

$$x = 2 \text{ and } -5.$$

10. (c) $f(x) = x^2 - 4x + 3$

$$a = 1, b = -4 \text{ and } c = 3$$

$$D = b^2 - 4ac$$

$$D = (-4)^2 - 4(1)(3)$$

$$\Rightarrow D = 16 - 12$$

$$D = 4$$

11. (d) Let the roots be 2 and β

$$2 + \beta = 0$$

$$\Rightarrow \beta = -2$$

$$\Rightarrow \text{Product of roots} = (2)(-2) = -4$$

$$\Rightarrow \text{Quadratic equation} = x^2 - 4 = 0$$

12. (c) $(\sqrt{2}x + \sqrt{3})^2 = x^2 = 3x^2 - 5x$

$$\Rightarrow 2x^2 + 3 + 2\sqrt{6}x + x^2 = 3x^2 - 5x$$

$$\Rightarrow 2\sqrt{6}x + 5x + 3 = 0$$

$\therefore$ Not a quadratic equation.

13. (a) $x^2 + 3x - 10 = 0$

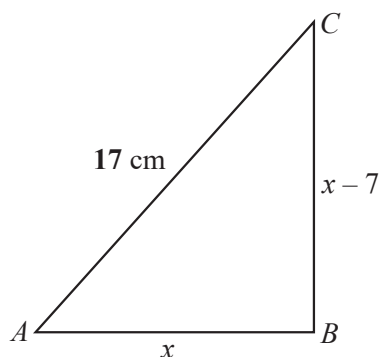
$$(x + 5)(x - 2) = 0 \Rightarrow x = -5, 2$$

14. (a) Assertion (A) is true but Reason (R) is false

Short Answer Type Questions

15. Let the base be x .

$$\text{Altitude} = (x - 7)$$



In $\triangle ABC$

$$AC^2 = AB^2 + BC^2$$

$$(17)^2 = x^2 + (x - 7)^2$$

$$289 = x^2 + x^2 + 49 - 14x$$

$$\Rightarrow 2x^2 - 14x - 240 = 0 \Rightarrow x^2 - 7x - 120 = 0$$

$$\Rightarrow x^2 - 15x + 8x - 120 = 0$$

$$\Rightarrow x(x - 15) + 8(x - 15) = 0$$

$$\Rightarrow (x + 8)(x - 15) = 0$$

$$x = -8 \text{ and } 15$$

$x \neq -8$, because side cannot be negative.

$$x = 15 \text{ cm} \Rightarrow \text{Base} = 15 \text{ cm}$$

$$\text{Altitude} = 8 \text{ cm.}$$

16. $5x^2 + 3x - 2 = 0$

$$5x^2 + 5x - 2x - 2 = 0$$

$$5x(x + 1) - 2(x + 1) = 0$$

$$(5x - 2)(x + 1) = 0$$

$$x = \frac{2}{5} \text{ and } -1$$

$$\text{Sum of roots} = \frac{-3}{5} \text{ and product of roots} = \frac{-2}{5}$$

In the given quadratic equation,

$$\text{Sum of roots} = -\frac{b}{a} = -\frac{3}{5}$$

$$\text{Product of roots} = \frac{c}{a} = -\frac{2}{5}$$

17. (a) $x^2 - 3x + 2 = 0$

$$x^2 - 2x - x + 2 = 0$$

$$x(x - 2) - 1(x - 2) = 0$$

$$(x - 1)(x - 2) = 0 \Rightarrow x = 1 \text{ and } 2$$

$$\alpha = 1 \text{ and } \beta = 2$$

Now, constructing the new quadratic polynomial.

$$2\alpha + 1 = 3$$

$$2\beta + 1 = 5$$

$$x^2 - (\text{sum of roots})x + (\text{product of roots})$$

$$x^2 - (3 + 5)x + (3 \times 5)$$

$$x^2 - 8x + 15$$

$$(b) 4x^2 - 4x + 1 = 0$$

$$4x^2 - 2x - 2x + 1 = 0$$

$$2x(2x - 1) - 1(2x - 1) = 0$$

$$(2x - 1)(2x - 1) = 0$$

$$x = \frac{1}{2} \text{ and } \frac{1}{2}$$

$$\text{Sum of roots} = 1 \text{ and product of roots} = \frac{1}{4}.$$

In the given quadratic equation

$$\text{Sum of roots} = -\frac{b}{a} = -\frac{(-4)}{4} = 1$$

$$\text{Product of roots} = \frac{c}{a} = \frac{1}{4}$$

18. (a) $8x^2 + 14x + 3$

$$\text{Sum of roots} = \frac{-b}{a} = \frac{-14}{8} = \frac{-7}{4}$$

$$\text{Product of roots} = \frac{c}{a} = \frac{3}{8}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{\frac{-7}{4}}{\frac{3}{8}} = \frac{-7}{4} \times \frac{8}{3} = \frac{-14}{3}$$

$$(b) \alpha = -9 \text{ and } \beta = 6$$

$$\alpha + \beta = -3 \text{ and } \alpha\beta = -54$$

$$x^2 - (\text{sum of roots})x + (\text{product of roots})$$

$$x^2 - (-3)x + (-54)$$

$$= x^2 + 3x - 54$$

15 (a) $2x^2 + 2x + 9 = 0$

$$a = 2, b = 2 \text{ and } c = 9$$

$$D = b^2 - 4ac = (2)^2 - 4(2)(9)$$

$$D = -68 \text{ and } D < 0$$

Hence, real roots does not exist.

$$(b) kx^2 - 2kx + 6 = 0$$

$$a = k, b = -2k \text{ and } c = 6$$

For real and equal roots

$$D = 0$$

$$D = b^2 - 4ac \Rightarrow D = (-2k)^2 - 4(k)(6) = 0$$

$$4k^2 - 24k = 0$$

$$k^2 - 6k = 0$$

$$\Rightarrow k(k - 6) = 0$$

$$\Rightarrow k = 0 \text{ and } k = 6$$

$k \neq 0$, because if $k = 0$ then it is not a quadratic equation.

$$6x^2 - 12x + 6 = 0 \Rightarrow x^2 - 2x + 1 = 0$$

$$(x - 1)^2 = 0 \Rightarrow x = 1$$

19. $f(x) = kx^2 - x - 2$

If $\frac{2}{3}$ is root then,

$$f\left(\frac{2}{3}\right) = 0$$

$$f\left(\frac{2}{3}\right) = k\left(\frac{2}{3}\right)^2 - \left(\frac{2}{3}\right) - 2 = 0$$

$$\frac{4k}{9} - \frac{2}{3} - 2 = 0 \Rightarrow \frac{4k}{9} = \frac{8}{3}$$

$$k = 6$$

20. (a) $2x^2 + 7x + 5 = 0$

$$\text{Sum of roots} = \frac{-b}{a} = \frac{-7}{2}$$

$$\text{Product of roots} = \frac{c}{a} = \frac{5}{2}$$

$$\alpha^2 + \beta^2 + \alpha\beta + \alpha\beta - \alpha\beta = \alpha^2 + \beta^2 + 2\alpha\beta - \alpha\beta \\ = (\alpha + \beta)^2 - \alpha\beta$$

Now, substituting the value,

$$\left(\frac{-7}{2}\right)^2 - \frac{5}{2} = \frac{49}{4} - \frac{5}{2} = \frac{39}{4}$$

(b) Given, $6x^2 + 37x - (p - 2)$

$$\text{Product of roots} = 1 = \frac{c}{a} \\ -\frac{(p-2)}{6} = 1$$

$$p - 2 = -6$$

$$p = -4$$

21. $6x^2 - 7x - 3 = 0$

$$6x^2 - 9x + 2x - 3 = 0$$

$$3x(2x - 3) + 1(2x - 3) = 0$$

$$(3x + 1)(2x - 3) = 0$$

$$\text{Product of roots} = -\frac{1}{2}$$

$$\text{Sum of roots} = \frac{7}{6}$$

In the given quadratic equation

$$\alpha + \beta = \frac{-b}{a} = \frac{7}{6}$$

$$\alpha\beta = -\frac{3}{6} = -\frac{1}{2}$$

22. (a) Discriminant (D) of a quadratic equation

$$ax^2 + bx + c = 0 \text{ is } b^2 - 4ac$$

$$\text{Discriminant for } 3x^2 - 2x + \frac{1}{3} = 0 \text{ is}$$

$$D = (-2)^2 - 4(3)\left(\frac{1}{3}\right) = 0$$

Hence, roots are real and equal.

OR

(b) $x^2 - x - 2 = 0$

$$\Rightarrow x^2 - 2x - x - 2 = 0$$

$$\Rightarrow x(x - 2) + 1(x - 2) = 0$$

$$\Rightarrow (x - 2) + (x + 1) = 0$$

$$\Rightarrow x - 2 = 0 \text{ and } x = -1$$

Hence, roots are 2 and -1.

23. (a) $\therefore$ For $ax^2 + bx + c = 0$

$$\text{Sum of roots} = \frac{-b}{a}$$

$$\text{Product of roots} = \frac{c}{a}$$

$$\Rightarrow \text{For } 2x^2 - 9x + 4 = 0$$

$$\therefore \text{Sum of roots} = \frac{-(-9)}{2} = \frac{9}{2}$$

$$\therefore \text{Product of roots} = \frac{4}{2} = 2$$

OR

$$\Rightarrow \text{For } ax^2 + bx + c = 0, D = b^2 - 4ac$$

$$\therefore \text{For } 4x^2 - 5 = 0$$

$$\therefore D = (0)^2 - 4(4)(-5) = 80$$

$$\therefore D > 0.$$

Roots are real and distinct.

24. For equal real roots,

Discriminant, $D = 0$

Here, equation is

$$\Rightarrow px(x-2) + 6 = 0$$

$$\therefore D = b^2 - 4ac \text{ for } ax^2 + bx + c = 0$$

$$\text{Here } D = (-2p)^2 - 4(p)(6)$$

$$\Rightarrow 0 = 4p^2 - 24p \Rightarrow 0 = 4p(p-6)$$

$$p = 0 \text{ or } p = 6$$

But $p \neq 0$ as coefficient of x^2 should be not zero.

$$p = 6$$

25. $4x^2 - 5x - 1 = 0$

$$D = b^2 - 4ac, \text{ where } a = 4, b = -5 \text{ and } c = -1$$

$$\Rightarrow D = 25 + 16 = 41 \Rightarrow D > 0$$

$\therefore$ The given equation has real and distinct roots.

26. $x^2 + 2\sqrt{2}x - 6 = 0$

$$\therefore x^2 + 3\sqrt{2}x - \sqrt{2}x - 6 = 0$$

$$x(x + 3\sqrt{2}) - \sqrt{2}(x + 3\sqrt{2}) = 0$$

$$(x + 3\sqrt{2})(x - \sqrt{2}) = 0$$

$$\Rightarrow x = -3\sqrt{2}, \sqrt{2}.$$

27. Let the duration of the flight be x hours

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{600}{x} \text{ km/h}$$

Duration of the flight due to show down

$$= x + \frac{30}{60} = x + \frac{1}{2}$$

According to question

$$\frac{600}{x} - \frac{600}{x + \frac{1}{2}} = 200 \Rightarrow \frac{3}{x} - \frac{3}{x + \frac{1}{2}} = 1$$

$$\Rightarrow \frac{3(2x+1) - 6x}{x(2x+1)} = 1 \Rightarrow \frac{6x+3-6x}{x(2x+1)} = 1$$

$$\Rightarrow \frac{3}{x(2x+1)} = 1 \Rightarrow 2x^2 + x - 3 = 0$$

$$\Rightarrow 2x^2 + 3x - 2x - 3 = 0$$

$$\Rightarrow x(2x+3) - 1(2x+3) = 0$$

$$\Rightarrow (2x^2+3)(x-1) = 0$$

$$x = 1$$

Original duration of the flight is 1 hour.

28. Quadratic equation $3x^2 - 4x + k = 0$ has equal roots

$$\Rightarrow D = b^2 - 4ac = 0, \text{ where } a = 3, b = -4 \text{ and}$$

$$c = k$$

$$\Rightarrow (-4)^2 - 4 \times 3 \times k = 0$$

$$\Rightarrow 16 - 12k = 0$$

$$\Rightarrow k = \frac{16}{12} = \frac{4}{3}$$

29. $x^2 + 4x + k = 0$

$\therefore$ Roots of given equation are real,

$$D \geq 0$$

$$\Rightarrow (4)^2 - 4 \times k \geq 0 \Rightarrow 4k \geq -16 \Rightarrow k \geq -4$$

$\therefore k$ has all real values ≤ 4

30. $3x^2 - 10x + k = 0$

$\therefore$ Roots of given equation are reciprocal of each other

Let the root be α and $\frac{1}{\alpha}$.

$$\text{Product of roots} = \frac{c}{a}$$

$$\Rightarrow \alpha \cdot \frac{1}{\alpha} = \frac{k}{3}$$

$$\therefore k = 3$$

31. $\therefore x = 3$ is one of the root $x^2 - 2kx - 6 = 0$

$$(3)^2 - 2k(3) - 6 = 0$$

$$9 - 6k - 6 = 0$$

$$3 = 6k$$

$$k = \frac{3}{6} = \frac{1}{2}$$

32. Let the usual speed of the plane be x km/hr

Time taken to cover 1500 km with usual speed

$$= \frac{1500}{x} \text{ hrs}$$

Time taken to cover 1500 km with speed of

$$x + 100 \text{ km/hr} = \frac{1500}{x + 100} \text{ hrs.}$$

$$\therefore \frac{1500}{x} = \frac{1500}{x + 100} + \frac{1}{2}$$

$$\frac{1500}{x} - \frac{1500}{x + 100} = \frac{1}{2}$$

$$1500 \left(\frac{x + 100 - x}{x(x + 100)} \right) = \frac{1}{2}$$

$$150000 \times 2 = x(x + 100)$$

$$x^2 + 100x - 300000 = 0$$

$$x^2 + 100x - 300000 = 0$$

$$x = -600 \text{ or } x = 500$$

But speed can't be negative.

Hence, usual speed 500 km/hr.

33. Quadratic equation $px^2 - 14x + 8 = 0$

Also, one root is 6 times the other

Let say one root = x

Second root = $6x$

From the equation: Sum of the roots = $+\frac{14}{p}$

$$\text{Product of roots} = \frac{8}{p}$$

$$\therefore x + 6x = \frac{14}{p}$$

$$x = \frac{2}{p} \Rightarrow 6x^2 = \frac{8}{p} \Rightarrow 6\left(\frac{2}{p}\right)^2 = \frac{8}{p}$$

$$\frac{6 \times 4}{p^2} = \frac{8}{p}$$

$$p = 3$$

34. Given $ad \neq bc$ for the equation $(a^2 + b^2)x^2 + 2(ac + bd)x + (c^2 + d^2) = 0$

For this equation not to have real roots its discriminant < 0

$$D = 4(ac + bd)^2 - 4(a^2 + b^2)(c^2 + d^2)$$

$$D = 4a^2c^2 + 4b^2d^2 + 8abcd - 4a^2c^2 - 4b^2d^2 - 4b^2c^2 - 4a^2d^2$$

$$D = -4(a^2d^2 + b^2c^2 - 2acbd)$$

$$D = -4(ad + bc)$$

Given $ad \neq bc$

$$\therefore D < 0$$

Quadratic equation has not real roots.

35. Given -5 is a root of the quadratic equation $2x^2 + px - 15 = 0$

$\therefore -5$ satisfies the given equation,

$$\therefore 2(-5)^2 + p(-5) - 15 = 0$$

$$\therefore 50 - 5p - 15 = 0$$

$$\therefore 35 - 5p = 0$$

$$\therefore 5p = 35 \Rightarrow p = 7$$

Substituting $p = 7$ in $p(x^2 + x) + k = 0$, we get

$$7(x^2 + x) + k = 0$$

$$\therefore 7x^2 + 7x + k = 0$$

The roots of the equation are equal.

$$\therefore \text{Discriminant} = b^2 - 4ac = 0$$

Here $a = 7$, $b = 7$, $c = k$

$$b^2 - 4ac = 0$$

$$\therefore (7)^2 - 4(7)(k) = 0$$

$$\therefore 49 - 28k = 0$$

$$\therefore 28k = 49$$

$$\therefore k = \frac{49}{28} = \frac{7}{4}$$

36. $4x^2 + 4bx - (a^2 - b^2) = 0$

$$\Rightarrow x^2 + bx - \left(\frac{a^2 - b^2}{4}\right) = 0$$

$$\Rightarrow x^2 + 2\left(\frac{b}{2}\right)x = \frac{a^2 - b^2}{4}$$

$$\Rightarrow x^2 + 2\left(\frac{b}{2}\right)x + \left(\frac{b}{2}\right)^2 = \frac{a^2 - b^2}{4} + \left(\frac{b}{2}\right)^2$$

$$\Rightarrow \left(x + \frac{b}{2}\right)^2 = \frac{a^2}{4} \Rightarrow x + \frac{b}{2} = \pm \frac{a}{2}$$

$$\Rightarrow x = \frac{-b}{2} \pm \frac{a}{2} \Rightarrow x = \frac{-b - a}{2}, \frac{-b + a}{2}$$

Hence, the roots are $-\left(\frac{a+b}{2}\right)$ and $\left(\frac{a+b}{2}\right)$.

37. For the given equation $\sqrt{3}x^2 - 2\sqrt{x} - 2\sqrt{3} = 0$

Comparing this equation with $ax^2 + bx + c = 0$,

We obtain $a = \sqrt{3}$, $b = -2\sqrt{2}$, $c = -2\sqrt{3}$

Now, $\sqrt{D} = \sqrt{b^2 - 4ac}$

$$= \sqrt{(-2\sqrt{2})^2 - 4(\sqrt{3})(-2\sqrt{3})}$$

$$= \sqrt{8 + 24} = \sqrt{32} = 4\sqrt{2}$$

Using quadratic formula, we obtain

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \Rightarrow x = \frac{-(-2\sqrt{2}) \pm 4\sqrt{2}}{2\sqrt{3}}$$

$$\Rightarrow x = \frac{2\sqrt{2} + 4\sqrt{2}}{2\sqrt{3}} \text{ or } \frac{2\sqrt{2} - 4\sqrt{2}}{2\sqrt{3}}$$

$$\Rightarrow x = \frac{\sqrt{2} + 2\sqrt{2}}{\sqrt{3}} \text{ or } \frac{\sqrt{2} - 2\sqrt{2}}{\sqrt{3}}$$

$$\Rightarrow x = \frac{3\sqrt{2}}{\sqrt{3}} \text{ or } x = \frac{-\sqrt{2}}{\sqrt{3}}$$

$$\Rightarrow x = \sqrt{3}\sqrt{2} \text{ or } x = \frac{-\sqrt{2}}{\sqrt{3}}$$

$$\therefore x = \sqrt{6} \text{ or } x = \frac{-\sqrt{2}}{\sqrt{3}}$$

38. Given quadratic equation is, $px^2 - 2\sqrt{5}px + 15 = 0$

Here $a = p$, $b = -2\sqrt{5}p$, $c = 15$

for real equal roots, discriminant = 0

$$\therefore b^2 - 4ac = 0$$

$$\therefore (-2\sqrt{5}p)^2 - 4p(15) = 0$$

$$\therefore 20p^2 - 60p = 0$$

$$\therefore 20p(p - 3) = 0$$

$$\therefore p = 3 \text{ or } p = 0$$

But, $p = 0$ is not possible.

$$\therefore p = 3$$

Long Answer Type Questions

39. Let age of boy be x years, then age of his sister will be $(25 - x)$ years

Product of their ages, $(x)(25 - x) = 150$

$$\Rightarrow 25x - x^2 = 150$$

$$\Rightarrow x^2 - 25x + 150 = 0$$

$$\Rightarrow (x - 15)(x - 10) = 0$$

$$\Rightarrow x = 10 \text{ and } 15.$$

Hence, their present age's are 10 years and 15 years.

40. (a) Let x be the digit at 10th place of given two digit number and y be the unit's place of given two digit number.

According to the question,

$$xy = 24 \Rightarrow y = \frac{24}{x} \quad \dots(i)$$

$$\text{and } 10x + y - 18 = 10y + x$$

$$\Rightarrow 9x - 9y = 18$$

$$\Rightarrow x - y = 2 \quad \dots(ii)$$

From equation (i) and (ii), we get

$$x - \frac{24}{x} = 2$$

$$\text{or } x^2 - 2x - 24 = 0$$

$$\text{or } x^2 - 6x + 4x - 24 = 0$$

$$\text{or } (x - 6)(x + 4) = 0$$

$$x = 6 \text{ or } x = -4$$

$$\therefore x = 6 \quad [\text{Because } x \text{ can't be negative}]$$

$$\text{From (i), } y = 4$$

$$\therefore \text{Original number is } 64.$$

OR

(b) Let x and y be the two numbers such that $x > y$

According to the question,

$$x^2 - y^2 = 180 \quad \dots(i)$$

$$\text{and } y^2 = 8x \quad \dots(ii)$$

From (i) and (ii), we get

$$x^2 - 8x - 180 = 0$$

$$\text{or } (x - 18)(x + 10) = 0$$

$$x = 18, -10$$

$$x = 18 \quad [\text{Because } x \text{ cannot be negative}]$$

From (ii)

Put $x = 18$ in equation (ii), we get

$$y^2 = 144 \text{ or } y = \pm 12$$

$$\therefore \text{Required number are } (18, 12) \text{ and } (18, -12).$$

41. Let the unit digit of the two digit number be x .

$$\therefore \text{Ten's digit will be } \frac{14}{x}$$

According to question,

$$10 \times \frac{14}{x} + x + 45 = 10x + \frac{14}{x}$$

$$\Rightarrow \frac{140}{x} + x + 45 = \frac{10x^2 + 14}{x}$$

$$\Rightarrow \frac{140 + x^2 + 45x}{x} = \frac{10x^2 + 14}{x}$$

$$\Rightarrow 9x^2 - 45x - 125 = 0$$

$$\Rightarrow 9x^2 - 63x + 18x - 126 = 0$$

$$\Rightarrow 9x(x - 7) + 18(x + 7) = 0$$

$$\Rightarrow (x - 7)(9x + 18) = 0$$

$$\Rightarrow \text{Either } x = 7 \text{ or } x = -2$$

$$\therefore x = 7 \quad [\because x \neq -2]$$

$$\therefore \text{Ten's digit} = \frac{14}{7} = 2$$

So, the number is 27.

42. Let the time in which tap with longer and smaller

diameter can fill the tank separately be x hours and y hours respectively.

According to the question

$$\frac{1}{x} + \frac{1}{y} = \frac{8}{15} \quad \dots(i)$$

$$\text{and } x = y - 2 \quad \dots(ii)$$

On substituting $x = y - 2$ from (ii) in (i), we get

$$\frac{1}{y-2} + \frac{1}{y} = \frac{8}{15} \Rightarrow \frac{y+y-2}{y^2-2y} = \frac{8}{15}$$

$$\Rightarrow 15(2y-2) = 8(y^2-2y)$$

$$\Rightarrow 30y - 30 = 8y^2 - 16y$$

$$\Rightarrow 8y^2 - 46y + 30 = 0$$

$$\Rightarrow 4y^2 - 20y - 3y + 15 = 0$$

$$\Rightarrow (4y-3)(y-5) = 0$$

$$\Rightarrow y = \frac{3}{4}, y = 5$$

Substituting values of y in (ii), we get

$$x = \frac{3}{4} - 2$$

$$x = \frac{-5}{4}$$

$$\therefore x \neq \frac{-5}{4}$$

$$x = 5 - 2$$

$$x = 3 \quad (\text{time cannot be negative})$$

Hence, the time taken by tap with longer diameter is 3 hours and the time taken by tap with smaller diameter is 5 hours, in order to fill the tank separately.

43. Let the speed of stream be x km/hr.

Now, for upstream: speed = $(18 - x)$ km/hr

$$\therefore \text{Time taken} = \left(\frac{24}{18-x} \right) \text{ hr}$$

Now, for downstream: speed = $(18 + x)$ km/hr

$$\therefore \text{Time taken} = \left(\frac{24}{18+x} \right) \text{ hr}$$

Given that,

$$\frac{24}{18-x} = \frac{24}{18+x} + 1$$

$$-1 = \frac{24}{18+x} - \frac{24}{18-x}$$

$$-1 = \frac{24[(18-x) - (18+x)]}{(18)^2 - x^2}$$

$$-1 = \frac{24[-2x]}{324 - x^2}$$

$$-324 + x^2 = -48x$$

$$x^2 + 48x - 324 = 0$$

$$x^2 + 54x - 6x - 324 = 0$$

$$(x+54)(x-6) = 0$$

$$x = -54 \text{ or } x = 6$$

$$x = -54 \text{ km/hr} \quad (\text{not possible})$$

Therefore, speed of the stream = 6 km/hr

44. Let x be the original average speed of the train for 63 km

Then, $(x + 6)$ will be the new average speed for remaining 75 km.

Total time taken to complete the journey is 3 hrs.

$$\therefore \frac{63}{x} + \frac{72}{(x+6)} = 3 \quad \left(\because \text{Time} = \frac{\text{Distance}}{\text{Speed}} \right)$$

$$\therefore \frac{63x + 378 + 72x}{x(x+6)} = 3$$

$$\Rightarrow 135x + 378 = 3x^2 + 18x$$

$$\Rightarrow x^2 - 39x - 126 = 0$$

$$\Rightarrow x - 42 \text{ or } x = -3$$

Since, speed cannot be negative.

Therefore $x = 42$ km/hr.

45. Two tap when run together fill the tank in $3\frac{1}{13}$ hrs

Say taps are A , B and

A fills the tank by itself in x hrs

B fills tank in $(x + 3)$ hrs

$$\text{Portion of tank filled by } A \text{ (in 1 hr)} = \frac{1}{x}$$

$$\text{Portion of tank filled by } B \text{ (in 1 hr)} = \frac{1}{x+3}$$

$$\text{Portion of tank filled by } A \text{ and } B \text{ (both in 1 hr)} = \frac{13}{40}$$

$$\therefore \frac{1}{x} + \frac{1}{x+3} = \frac{13}{40}$$

$$(x+3+x)40 = 13(x)(x+3)$$

$$80x + 120 = 13x^2 + 39x$$

$$\Rightarrow 13x^2 - 41x - 120 = 0$$

$$\Rightarrow 13x^2 - 65x + 24x - 120 = 0$$

$$\Rightarrow x = 5 \text{ or } \frac{-24}{13} \quad [\text{But negative value not be taken}]$$

$\therefore A$ fills tank in 5 hrs.

B fills tank in 5 hrs.

46. Let the speed of the stream be s km/h

Speed of the motor boat 24 km/h

Speed of the motor boat (upstream) = $24 - s$

Speed of the motor boat (downstream) = $24 + s$

According to the given condition

$$\frac{32}{24-s} - \frac{32}{24+s} = 1$$

$$\therefore 32 \left(\frac{1}{24-s} - \frac{1}{24+s} \right) = 1$$

$$\therefore 32 \left(\frac{24+s-24+s}{576-s^2} \right) = 1$$

$$\therefore 32 \times 2s = 576 - s^2$$

$$\therefore s^2 \times 64s - 576 = 0$$

$$\therefore (s+72)(s-8) = 0$$

$$\therefore s = -72 \text{ or } s = 8$$

Since, speed of the stream cannot be negative, the speed of the stream is 8 km/h.

47. Let l be the length of the longer side and b be the length of the shorter side.

Given that the length of the diagonal of the rectangular field is 16 metres more than the shorter side.

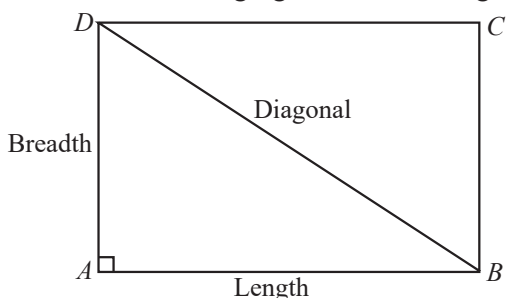
Thus, diagonal = $16 + b$

Since longer side is 14 metres more than shorter side, we have,

$$l = 14 + b$$

Diagonal is the hypotenuse of the triangle.

Consider the following figure of the rectangular field.



By applying Pythagoras Theorem in $\triangle ABD$, we have,

$$\text{Diagonal}^2 = \text{Length}^2 + \text{Breadth}^2$$

$$(16 + b)^2 = (14 + b)^2 + b^2$$

$$256 + b^2 + 32b = 196 + b^2 + 28b + b^2$$

$$\Rightarrow 256 + 32b = 196 + 28b + b^2$$

$$\Rightarrow 60 + 32b = 28b + b^2$$

$$\Rightarrow b^2 - 4b - 60 = 0$$

$$\Rightarrow b^2 - 10b + 6b - 60 = 0$$

$$\Rightarrow b(b - 10) + 6(b - 10) = 0$$

$$\Rightarrow (b + 6)(b - 10) = 0$$

$$\Rightarrow (b + 6) = 0 \text{ or } (b - 10) = 0$$

$$\Rightarrow b = -6 \text{ or } b = 10$$

As breadth cannot be negative, breadth = 10 m

Thus, length of the rectangular field = $14 + 10 = 24$ m.

48. Let x be the first speed of the train.

We know that, $\frac{\text{Distance}}{\text{Speed}} = \text{time}$

Thus, we have,

$$\frac{54}{x} + \frac{63}{x+6} = 3 \Rightarrow \frac{54(x+6) + 63x}{x(x+6)} = 3$$

$$\Rightarrow 54(x+6) + 63x = 3x(x+6)$$

$$\Rightarrow 54x + 324 + 63x = 3x^2 + 18x$$

$$\Rightarrow 117x + 324 = 3x^2 + 18x$$

$$\Rightarrow 3x^2 - 117x - 324 + 18x = 0$$

$$\Rightarrow 3x^2 + 99x - 324 = 0$$

$$\Rightarrow x^2 - 33x - 108 = 0$$

$$\Rightarrow x^2 - 36x + 3x - 108 = 0$$

$$\Rightarrow x(x - 36) + 3(x - 36) = 0$$

$$\Rightarrow x(x + 3)(x - 36) = 0$$

$$\Rightarrow (x + 3) = 0 \text{ or } (x - 36) = 0$$

$$\Rightarrow x = -3 \text{ or } x = 36$$

Speed cannot be negative. Hence, initial speed of the train is 36 km/hour.

5

CHAPTER

Arithmetic Progressions



Previous Year Questions

Multiple Choice Questions

- In an A.P., if the first term $a = 7$, n th term $a_n = 84$ and the sum of first n terms $s_n = \frac{2093}{2}$, then n is equal to: [CBSE 2024]
 (a) 22 (b) 24
 (c) 23 (d) 26
- The common difference of an A.P. in which $a_{15} - a_{11} = 48$, is [CBSE 2024]
 (a) 12 (b) 16
 (c) -12 (d) -16
- Which term of the A.P. $-29, -26, -23, \dots, 61$ is 16? [CBSE 2024]
 (a) 11^{th} (b) 16^{th}
 (c) 10^{th} (d) 31^{st}
- The common difference of the A.P.
 $\frac{1}{2x}, \frac{1-4x}{2x}, \frac{1-8x}{2x}, \dots$ is: [CBSE 2024]
 (a) $-2x$ (b) 2
 (c) 2 (d) $2x$
- The next (4^{th}) term of the A.P. $\sqrt{18}, \sqrt{50}, \sqrt{98}, \dots$ is: [CBSE 2024]
 (a) $\sqrt{128}$ (b) $\sqrt{140}$
 (c) $\sqrt{162}$ (d) $\sqrt{200}$
- The number of terms in the A.P. 3, 6, 9, 12, ..., 111 is: [CBSE 2024]
 (a) 36 (b) 40
 (c) 37 (d) 30
- The value of x for which $2x, (x + 10)$ and

$(3x + 2)$ are the three consecutive terms of an AP, is: [CBSE 2020]

- (a) 6 (b) -6
 (c) 18 (d) -18
- The first term of an AP is p and the common difference is q , then its 10^{th} term is: [CBSE 2020]
 (a) $q + 9p$ (b) $p - 9q$
 (c) $p + 9q$ (d) $2p + 9q$
 - If $k + 2, 4k - 6$ and $3k - 2$ are three consecutive terms of an A.P., then the value of k is: [CBSE 2023]
 (a) 3 (b) -3
 (c) 4 (d) -4
 - DIRECTION :** In the question, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option out of the following:
Assertion (A) : a, b, c are in A.P. if and only if $2b = a + c$.
Reason (R) : The sum of first n odd natural numbers is n^2 . [CBSE 2023]
 (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A)
 (c) Assertion (A) is true but Reason (R) is false
 (d) Assertion (A) is false but Reason (R) is true

Short Answer Type Questions

- The sum of first and eighth terms of an A.P. is 32 and their product is 60. Find the first term and

common difference of the A.P. Hence, also find the sum of its first 20 terms. [CBSE 2024]

12. In an A.P. of 40 terms, the sum of first 9 terms is 153 and the sum of last 6 terms is 687. Determine the first term and common difference of A.P. Also, find the sum of all the terms of the A.P. [CBSE 2024]

13. For what value of k will $k + 9$, $2k - 1$ and $2k + 7$ are the consecutive terms of an AP? [CBSE 2016]

14. What is the common difference of an AP in which $a_{21} - a_7 = 84$? [CBSE 2017]

15. In an AP, if the common difference (d) = -4 , and the seventh term (a_7) is 4, then find the first term [CBSE 2018]

16. How many two digit numbers are divisible by 3? [CBSE 2019]

17. The n^{th} term of an AP is $(7 - 4n)$, then what is its common difference? [CBSE 2020]

18. Which term of the AP 3, 15, 27, 39, will be 120 more than its 21st term? [CBSE 2009, 2019]

19. The first and the last term of an AP are 5 and 45 respectively. If the sum of all its terms is 400, find its common difference. [CBSE 2014, 2017]

20. In an AP, if $S_5 + S_7 = 167$ and $S_{10} = 235$, then find the AP, where S_n denotes the sum of its first n terms. [CBSE 2015]

21. The 4th term of an AP is zero. Prove that the 25th term of the AP is three times its 11th term. [CBSE 2016]

22. Which term of the progression $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \dots$ is the first negative term? [CBSE 2017]

23. Find the sum of first 8 multiples of 3. [CBSE 2018]

24. If S_n , the sum of first n terms of an AP is given by $S_n = 3n^2 - 4n$, find the n^{th} term. [CBSE 2019]

25. Show that $(a - b)$, $(a^2 + b^2)$ and $(a + b)^2$ are in A.P. [CBSE 2020]

26. (a) Which term of the A.P. 3, 8, 13, 18... is 78? [CBSE 2022]

OR

- (b) Find the common difference of an A.P. whose n^{th} term is given by $a_n = 6n - 5$. [CBSE 2022]

27. Find the sum of the first fifteen multiples of 8. [CBSE 2022]

28. (a) Which term of the A.P. $-\frac{11}{2}, -3, -\frac{1}{2}, \dots$ is $\frac{49}{2}$? [CBSE 2022]

OR

- (b) Find a and b so that the numbers $a, 7, b, 23$ are in A.P. [CBSE 2022]

29. Find the sum of first 20 terms of an A.P. whose n^{th} term is given as $a_n = 5 - 2n$. [CBSE 2022]

30. The 14th term of an AP is twice its 8th term. If its 6th term is -8 , then find the sum of its first 20 terms. [CBSE 2015]

31. If the ratio of the sum of first n terms of two AP's is $(7n + 1) : (4n + 27)$, find the ratio of their m^{th} terms. [CBSE 2016]

Long Answer Type Questions

32. (a) If the sum of first m terms of an A.P. is same as sum of its first n terms ($m \neq n$), then show that the sum of its first $(m + n)$ terms is zero. [CBSE 2024]

OR

- (b) In an A.P., the sum of three consecutive terms is 24 and the sum of their squares is 194. Find the numbers. [CBSE 2024]

33. (a) If the sum of first 7 terms of an A.P. is 49 and that of first 17 terms is 289, find the sum of its first 20 terms. [CBSE 2024]

OR

- (b) The ratio of the 10th term to its 30th term of an A.P. is 1:3 and the sum of its first six terms is 42. Find the first term and the common difference of A.P. [CBSE 2024]

34. If the sum of the first 14 terms of an A.P. is 1050 and the first term is 10, then, find the 20th term and the n^{th} term. [CBSE 2024]

OR

The first term of an A.P. is 5, the last term is 45 and the sum of all the terms is 400. Find the number of terms and the common difference of the A.P.

[CBSE 2024]

35. (a) The sum of first 15 terms of an A.P. is 750 and its first term is 15. Find its 20th term.

[CBSE 2023]

OR

- (b) Rohan repays his total loan of ₹ 1,18,000 by paying every month starting with the first instalment of ₹ 1,000. If he increases the instalment ₹ 100 by every month, what amount will be paid by him in the 30th instalment? What amount of loan has he paid after 30th instalment?

[CBSE 2023]

36. Find the 60th term of the AP 8, 10, 12 if it has a total of 60 terms and hence find the sum of its last 10 terms.

[CBSE 2015]

37. The houses in a row numbered consecutively from 1 to 49. Show that there exists a value of X such that sum of numbers of houses preceding the house numbered X is equal to sum of the numbers of houses following X .

[CBSE 2016]

38. The sum of four consecutive numbers in an AP is 32 and the ratio of the product of the first and the last term to the product of two middle terms is 7 : 15. Find the numbers.

[CBSE 2018]

39. If the sum of first four terms of an AP is 40 and that of first 14 terms is 280. Find the sum of its first n terms.

[CBSE 2019]

40. If 4 times the 4th term of an AP is equal to 18 times the 18th term, then find the 22th term.

[CBSE 2020]

OR

How many terms of the AP : 24, 21, 18,.... must be taken so that their sum is 78?

[CBSE 2020]

41. The sum of four consecutive numbers in AP is 32 and the ratio of the product of the first and last terms to the product of two middle terms is 7:15. Find the numbers.

[CBSE 2020]

OR

Solve: $1 + 4 + 7 + 10 + \dots + x = 287$

[CBSE 2020]

42. (a) Find the sum of first 51 terms of an A.P. whose second and third terms are 14 and 18, respectively.

[CBSE 2023]

OR

- (b) The first term of an A.P. is 5, the last term is 45 and the sum is 400. Find the number of terms and the common difference.

[CBSE 2023]



Explanations

Previous Year Questions

Multiple Choice Questions

1. (c) $S_n = \frac{n}{2}[a + l]$

$$\frac{2093}{2} = \frac{n}{2}[7 + 84]$$

$$n = \frac{2093}{91} = 23$$

2. (a) Let's find the common difference in the arithmetic progression (A.P.) based on the given information.

Given: $a_{15} - a_{11} = 48$

We know that the general formula for the n^{th} term of an A.P. is:

$$a_n = a_1 + (n - 1)d$$

where: a_n represents the n^{th} term.

a_1 is the first term,

d is the common difference, and

n is the position of the term.

$$a_{15} = a + 14d$$

...(i)

$$a_{11} = a + 10d$$

...(ii)

By subtracting equation (ii) from equation (i)

$$a_{15} - a_{11} = [a + 14d] - [a + 10d]$$

$$4d = 48 \Rightarrow d = 12$$

3. (b) Given: $a = -29$; $d = -26 - (-29) = 3$

Let n^{th} term will be 16

$$a_n = a + (n - 1)d$$

$$16 = -29 + (n - 1)3$$

$$n - 1 = \frac{45}{3}$$

$$n = 16$$

4. (b) Given A.P. is $\frac{1}{2x}, \frac{1-4x}{2x}, \frac{1-8x}{2x}, \dots$

Here, $a_1 = \frac{1}{2x}$, $a_2 = \frac{1-4x}{2x}$, $a_3 = \frac{1-8x}{2x}$

$$\therefore \text{Common difference } (d) = a_2 - a_1$$

$$= \frac{1-4x}{2x} - \frac{1}{2x} = \frac{1-4x-1}{2x} = \frac{-4x}{2x} = -2$$

5. (c) $\sqrt{18}, \sqrt{50}, \sqrt{98}, \dots$

$$\Rightarrow \sqrt{9 \times 2}, \sqrt{25 \times 2}, \sqrt{49 \times 2}, \dots$$

$$\Rightarrow 3\sqrt{2}, 5\sqrt{2}, 7\sqrt{2}, \dots$$

$$\text{C.d.} = 5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$$

$$\therefore \text{Next term} = 7\sqrt{2} + 2\sqrt{2} = 9\sqrt{2}$$

$$= \sqrt{81 \times 2} = \sqrt{162}$$

6. (c) Clearly, the given sequence is an A.P. with first term $a = 3$ and common difference $d = 3$. Let there be n terms in the given sequence. Then, n^{th} term = 111

$$a + (n - 1)d = 111$$

$$3 + (n - 1) \times 3 = 111$$

$$n = 37$$

Thus, the given sequence contains 37 terms.

7. (a) $2x, (x + 10), (3x + 2)$ are in A.P.

$$\therefore x + 10 - 2x = 3x + 2 - x - 10$$

$$\Rightarrow x = 6$$

Hence, option (a) is correct.

8. (c) 10^{th} term = $p + (10 - 1)q$

$$\therefore a_{10} = p + 9q$$

Hence, option (c) is correct.

9. (a) $(4k - 6) - (k + 2) = (3k - 2) - (4k - 6)$

$$\Rightarrow 3k - 8 = -k + 4$$

$$\Rightarrow 4k = 12$$

$$\Rightarrow k = 3$$

10. (b) A : $b - a = c - b$

$$\Rightarrow 2b = a + c$$

$$\text{R : Sum of first } n \text{ odd natural numbers} = n^2$$

Short Answer Type Questions

11. $a_1 + a_8 = 32$

$$= a + a + 7d = 32$$

$$\Rightarrow a + 7d = 32 - a$$

$$a(a + 7d) = 60$$

$$a(32 - a) = 60$$

$$32a - a^2 = 60$$

$$a^2 - 32a + 60 = 0$$

$$a^2 - 30a - 2a + 60 = 0$$

$$a(a - 30) - 2(a - 30) = 0$$

$$a = 30, 2$$

$$\therefore d = \frac{32 - 2a}{7} = \frac{32 - 60}{7} = -4 \text{ or } \frac{32 - 2 \times 2}{7} = 4$$

$$\therefore S_{20} = \frac{20}{2} [2 \times 30 + 19 \times (-4)] = -160$$

$$\text{and } S_{20} = \frac{20}{2} [2 \times 2 + 19(4)] = 800$$

12. $S_9 = 153$

No. of terms = 40

$$153 = \frac{9}{2} [2a + 8d]$$

$$a + 4d = 17$$

...(i)

$$S_6, \text{ from last} = 687$$

$$687 = \frac{6}{2} [2(a + 39d) + 5(-d)]$$

$$2a + 73d = 229$$

...(ii)

Solving (i) & (ii) we get

$$a = 5, d = 3$$

$$S_{40} = 20(2 \times 5 + 39 \times 3) = 2540$$

13. If $k + 9$, $2k - 1$ and $2k + 7$ are the consecutive terms of AP, then the common difference will be the same.

$$\therefore (2k - 1)(k + 9) = (2k + 7)(2k - 1)$$

$$\therefore k - 10 = 8$$

$$\therefore k = 18$$

14. Given, $a_{21} - a_7 = 84$

...(i)

In an AP $a_1, a_2, a_3, a_4, \dots$

$$a_n = a_1 + (n - 1)d$$

d = common difference

$$a_{21} = a_1 + 20d$$

...(ii)

$$a_7 = a_1 + 6d$$

...(iii)

Substituting (ii) and (iii) in (i)

$$a_1 + 20d - a_1 - 6d = 84$$

$$14d = 84$$

$$d = 6$$

$$\therefore \text{Common difference} = 6.$$

15. $a_7 = 4$

$$a + 6d = 4$$

$$(\text{as } a_n = a + (n - 1)d)$$

but $d = -4$

$$a + 6(-4) = 4$$

$$a + (-24) = 4$$

$$a = 4 + 24 = 28$$

Therefore first term $a = 28$

16. Two digit numbers divisible by 3 are 12, 15, 18, ..., 99.

$$a = 12, d = 15 - 12 = 3$$

$$\Rightarrow T = 99$$

$$\Rightarrow a + (n - 1)d = 99$$

$$\Rightarrow -12 + (n - 1)3 = 99$$

$$\Rightarrow n = 30$$

$\therefore$ Number of two digit numbers divisible by 3 are 30.

17. $T_n = 7 - 4n$

$$T_1 = 7 - 4(1) = 3$$

$$T_2 = 7 - 4(2) = 7 - 8 = -1$$

$$\therefore \text{Common difference} = T_2 - T_1 = -1 - 3 = -4.$$

18. Given an AP 3, 15, 27, 39, ...

Lets say n th term is 120 more than 21st term

$$\therefore T_n = 120 + T_{21}$$

$$a + (n - 1)d = 120 + (a + 20d)$$

$$(n - 1)12 = 120 + 20 \times 12$$

$$n - 1 = 30$$

$$\therefore 31^{\text{st}} \text{ term is 120 more than } 12^{\text{th}} \text{ term.}$$

19. Let a be the first term and d be the common difference.

Given: $a = 5, T_n = 45, S = 400$

We know: $T = a + (n - 1)d$

$$\Rightarrow 45 = 5 + (n - 1)d$$

$$\Rightarrow 40 = (n - 1)d$$

...(i)

$$\text{And } S_n = \frac{n}{2}(a + T_n)$$

$$\Rightarrow 400 = \frac{n}{2}(5 + 45) \Rightarrow \frac{n}{2} = \frac{400}{50}$$

$$\Rightarrow n = 2 \times 8 = 16$$

On substituting $n = 16$ in (i),

we get: $40 = (16 - 1)d$

$$40 = (16 - 1)d \Rightarrow 40 = (15)d$$

$$\Rightarrow d = \frac{40}{15} = \frac{8}{3}$$

Thus, the common difference is $\frac{8}{3}$.

20. $S_5 + S_7 = 167$ and $S_{10} = 235$

$$\text{Now, } S_n = \frac{n}{2} \{2a + (n-1)d\}$$

$$\therefore S_5 + S_7 = 167$$

$$\Rightarrow \frac{5}{2} \{2a + 4d\} + \frac{7}{2} \{2a + 6d\} = 167$$

$$\Rightarrow 5a + 10d + 7a + 21d = 167$$

$$\Rightarrow 12a + 31d = 167 \quad \dots(i)$$

$$\text{Also, } S_{10} = 235$$

$$\therefore \frac{10}{2} \{2a + 9d\} = 235$$

$$\Rightarrow 10a + 45d = 235$$

$$\Rightarrow 2a + 9d = 47 \quad \dots(ii)$$

Multiplying equation (ii) by 6, we get

$$12a + 54d = 282 \quad \dots(iii)$$

Subtracting (i) from (iii), we get

$$12a + 54d = 282$$

$$(-)12 + 31d = -167$$

$$\therefore 23d = 15$$

$$\therefore d = 5$$

Substituting value of d in (ii), we have

$$2a + 9(5) = 47$$

$$\Rightarrow 2a + 45 = 47 \Rightarrow 2a = 2$$

$$\Rightarrow a = 1$$

Thus, the given AP is 1, 6, 11, 16, ...

21. 4th term of an AP = $a_4 = 0$

$$\therefore a + (4-1)d = 0$$

$$\therefore a + 3d = 0$$

$$\therefore a = -3d \quad \dots(i)$$

$$25^{\text{th}} \text{ term of an AP} = a_{25}$$

$$= a + (25-1)d$$

$$= -3d + 24d = 21d$$

[From (i)]

$$3 \text{ times } 11^{\text{th}} \text{ term of an AP} = 3a_{11}$$

$$= 3[a + (11-1)d]$$

$$= 3[a + 10d] = 3[-3d + 10d] = 3 \times 7d = 21d$$

$$\therefore a_{25} = 3a_{11}$$

i.e., the 25th term of the AP is three times its 11th term.

22. Given progression $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \dots$

This is an Arithmetic progression because

Common difference

$$(d) = 19\frac{1}{4} - 20 = 18\frac{1}{2} - 19\frac{1}{4} = \dots$$

$$d = \frac{-3}{4}$$

$$\text{Any } n^{\text{th}} \text{ term } a_n = 20 + (n-1)\left(\frac{-3}{4}\right) = \frac{83-3n}{4}$$

Any term $a_n < 0$ when $83 < 3n$

$$\Rightarrow n > \frac{83}{3} \Rightarrow n = 28.$$

$\therefore$ 28th term will be the first negative term.

23. First 8 multiples of 3 are

$$3, 6, 9, 12, 15, 18, 21, 24$$

The above sequence is an AP

$$a = 3, d = 3 \text{ and last term } l = 24$$

$$S_n = \frac{n}{2}(a+l) = \frac{8}{2}[3+24] = 4(27)$$

$$S_n = 108$$

24. $S_n = 3n^2 - 4n$

Let S_{n-1} be sum of $(n-1)$ terms

$$T_n = S_n - S_{n-1}$$

$$= (3n^2 - 4n) - [3(n-1)^2 - 4(n-1)]$$

$$= (3n^2 - 4n) - [3n^2 - 6n + 3 - 4n + 4]$$

$$= 3n^2 - 4n - 3n^2 + 10n - 7]$$

$$\therefore t_n = 6n - 7$$

So, required n^{th} term = $6n - 7$.

25. Common difference must be equal

$$\therefore (a^2 + b^2) - (a-b)^2 = (a+b)^2 - (a^2 + b^2)$$

$$\Rightarrow (a^2 + b^2) - (a^2 + b^2 - 2ab) = (a^2 + b^2 + 2ab) - a^2 - b^2$$

$$\Rightarrow a^2 + b^2 - a^2 - b^2 + 2ab = a^2 + b^2 + 2ab - a^2 - b^2$$

$$\Rightarrow 2ab = 2ab$$

Hence, $(a-b)^2, (a^2 + b^2)$ and $(a+b)^2$ are in A.P.

26. (a) Given A.P. is 3, 8, 13, 18, ...

$$\text{Here, } a = 3 \text{ and } d = 8 - 3 = 5$$

$$a_n = a + (n-1)d \quad [n^{\text{th}} \text{ term}]$$

$$\Rightarrow 78 = 3 + (n-1)5$$

$$\Rightarrow \frac{75}{5} = n - 1$$

$$\Rightarrow n = 16$$

$\therefore$ 78 is 16th term of the given A.P.

(b) n^{th} term of A.P. is

$$a_n = 6n - 5 \text{ if } n = 1,$$

$$\Rightarrow a_1 = 6 - 5 = 1 \text{ if } n_2 = 1$$

$$\Rightarrow a_2 = 6 \times 2 - 5 = 7$$

$$\therefore \text{Common difference } (d) = 7 - 1 = 6$$

27. First fifteen multiples of 8 are 8, 16, 24

Here, $a = 8$ and $d = 8$

$$S_{15} = \frac{15}{2}[2 \times 8 + (15-1)8]$$

$$= \frac{15}{2}[16 + 112] = \frac{15 \times 128}{2} = 960$$

$\therefore$ Sum of first fifteen multiples of 8 is 960.

28. (a) Given A.P. $-\frac{11}{2}, -3, -\frac{1}{2}, \dots$

$$\text{Here, } a = -\frac{11}{2}, d = -3 + \frac{11}{2} = \frac{11-6}{2} = \frac{5}{2}$$

$$t_n = \frac{49}{2}$$

$$a + (n-1)d = \frac{49}{2} \text{ or } -\frac{11}{2} + (n-1)\left(\frac{5}{2}\right) = \frac{49}{2}$$

$$\text{or } -11 + 5n - 5 = 49 \Rightarrow 5n = 49 + 16$$

$$\Rightarrow 5n = 65 \Rightarrow n = \frac{65}{5} = 13 \Rightarrow n = 13.$$

OR

(b) Given $a, 7, b, 23$ are in A.P.

$$\therefore 7 - a = b - 7 = 23 - b$$

$$\Rightarrow 7 - a = b - 7$$

$$\Rightarrow a + b = 14$$

...(i)

$$\text{and } b - 7 = 23 - b$$

$$\Rightarrow 2b = 30 \Rightarrow b = 15$$

From (i)

$$a = 14 - 15 = -1$$

29. Sum of n terms A.P. if n^{th} term of A.P. is given by,

$$S_n = \frac{n}{2}[a + a_n]$$

If $n = 1$

$$a_1 = 5 - 2 = 3 \text{ and if } n = 20$$

$$a_{20} = 5 - 40 = -35$$

$$S_{20} = \frac{20}{2}[a + a_{20}] = \frac{20}{2}[3 + (-35)] = 10[-32]$$

$$S_{20} = -320$$

30. Here it is given that,

$$T_{14} = 2(T_8)$$

$$\Rightarrow a + (14-1)d = 2[a + (8-1)d]$$

$$\Rightarrow a + 13d = 2[a + 7d]$$

$$\Rightarrow a + 13d = 2a + 14d$$

$$\Rightarrow 13d - 14d = 2a - a \Rightarrow -d = a \quad \dots(i)$$

Now, it is given that its 6th term is -8 .

$$T = -8$$

$$\Rightarrow a + (6-1)d = -8$$

$$\Rightarrow a + 5d = -8$$

$$\Rightarrow 4d = -8$$

[$\because$ Using (i)]

$$\Rightarrow 4d - 8 \Rightarrow d = -2$$

Substituting this in eq. (i), we get $a = 2$.

Now, the sum of 20 terms,

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$S_{20} = \frac{20}{2}[2a + (20-1)d]$$

$$= 10[2(2) + 19(-2)] = 10[4 - 38] = -340$$

31. Let a_1, a_2 be the first terms and d_1, d_2 the common difference of the two given AP's

$$\text{Thus, we have } S_{20} = \frac{n}{2}[2a_1 + (n-1)d_1]$$

$$\text{and } S_{n'} = \frac{n}{2}[2a_2 + (n-1)d_2]$$

$$\therefore \frac{S_n}{S_{n'}} = \frac{\frac{n}{2}[2a_1 + (n-1)d_1]}{\frac{n}{2}[2a_2 + (n-1)d_2]} = \frac{2a_1 + (n-1)d_1}{2a_2 + (n-1)d_2}$$

$$\text{It is given that } \frac{S_n}{S_{n'}} = \frac{7n+1}{4n+27}$$

$$\therefore \frac{2a_1 + (n-1)d_1}{2a_2 + (n-1)d_2} = \frac{7n+1}{4n+27} \quad \dots(i)$$

To find the ratio of the m^{th} term of the two given AP's replace n by $(2m-1)$ in equation (i)

$$\therefore \frac{2a_1 + (2m-1-1)d_1}{2a_2 + (2m-1-1)d_2} = \frac{7(2m-1)+1}{4(2m-1)+27}$$

$$\therefore \frac{2a_1 + (2m-2)d_1}{2a_2 + (2m-2)d_2} = \frac{14m-7+1}{8m-4+27}$$

$$\therefore \frac{a_1 + (m-1)d_1}{a_2 + (m-1)d_2} = \frac{14m-6}{8m-23}$$

Hence, the ratio of the m^{th} term of the two AP's is $14m-6 : 8m+23$.

Long Answer Type Questions

32. (a) Let the AP be denoted as $a_1, a_2, a_3, \dots, a_n$, with a common difference d .

$$\text{Sum of first } m \text{ terms: } S_m = \frac{m}{2}(2a_1 + (m-1)d)$$

$$\text{Sum of first } n \text{ terms: } S_n = \frac{n}{2}(2a_1 + (n-1)d)$$

Given that the 2 sums are equal. Hence,

$$\frac{m}{2}(2a_1 + (m-1)d) = \frac{n}{2}(2a_1 + (n-1)d)$$

$$2a_1m + (m^2 - m)d = 2a_1n + (n^2 - n)d$$

$$\therefore 2a_1 = d(1 - m - n)$$

Sum of first $(m - n)$ terms is given by

$$S_{m+n} = \frac{m+n}{2}[2a_1 + (m+n-1)d]$$

$$\text{Substituting } 2a_1 = d(1 - m - n),$$

$$S_{m+n} = \frac{m+n}{2}[d(1 - m - n) + (m+n-1)d] = 0$$

OR

- (b) Let the three consecutive natural numbers be $n, n+1$ and $n+2$.

Therefore, by the given condition, we have:

$$n^2 + (n+1)^2 + (n+2)^2 = 194$$

$$n^2 + n^2 + 2n + 1 + n^2 + 4n + 4 = 194$$

$$3n^2 + 6n - 189 = 0$$

$$n^2 + 2n - 63 = 0$$

Splitting the middle term, we get

$$n^2 + 9n - 7n - 63 = 0$$

$$n(n+9) - 7(n-9) = 0$$

$$(n+9)(n-7) = 0$$

$$\therefore n = 7 \text{ or } n = -9$$

Since, n is a natural number, $n = 7$

Thus, the numbers are 7, 8 and 9.

33. (a) Given that, sum of the first 7 terms is 49

We know sum of n terms;

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

Where a is the first term and d is the common difference

$$\therefore S_7 = \frac{7}{2}[2a + (7-1)d]$$

$$\Rightarrow 49 = \frac{7}{2}[2a + 6d] \Rightarrow 49 = \frac{7}{2} \times 2[a + 3d]$$

$$\Rightarrow 49 = 7(a + 3d) \Rightarrow 7 = a + 3d \quad \dots(i)$$

Given that, sum of the first 17 terms is 289

$$\therefore S_{17} = \frac{17}{2}[2a + (17-1)d]$$

$$\Rightarrow 289 = \frac{17}{2}[2a + 16d]$$

$$\Rightarrow 289 = \frac{17}{2} \times 2[a + 8d]$$

$$\Rightarrow 289 = 17[a + 8d]$$

$$\Rightarrow 17 = a + 8d \quad \dots(ii)$$

Subtracting equation (ii) - (i)

$$\Rightarrow a + 8d - (a + 3d) = 17 - 7$$

$$\Rightarrow a + 8d - a - 3d = 10$$

$$\Rightarrow 5d = 10 \Rightarrow d = 2$$

Put the value of d in equation (i),

$$\Rightarrow a + 3(2) = 7$$

$$\Rightarrow a + 6 = 7$$

$$\Rightarrow a = 7 - 6 = 1$$

Therefore, $a = 1, d = 2$

We know that the sum of first n term of an A.P.

$$\text{series is given by } S_n = \frac{n}{2}[2a + (n-1)d]$$

$$\therefore S_n = \frac{n}{2}[2a + (n-1)d]$$

$$\Rightarrow S_n = \frac{n}{2}[2(1) + (n-1)2]$$

$$\Rightarrow S_n = \frac{n}{2}[2 + (n-1)2]$$

$$\Rightarrow S_n = \frac{n}{2}[2n] \Rightarrow S_n = n^2$$

Hence, sum of n terms is n^2

OR

Given the ratio of the 10th term of the 30th term is 1 : 3 in an AP

Express the ratio in terms of the terms:

$$\frac{a + 9d}{a + 29d} = \frac{1}{3}$$

Given the sum of the first 6 terms: $S_6 = 42$

$$S_6 = \frac{6}{2}(2a + 5d) = 42$$

Simplify the sum equation to get: $6(2a + 5d) = 84$

Solve the system of equation to find the term and common difference after solving, you get the first term $a = 3$ and the common difference $d = 5$.

34. Here, $S_{14} = 1050$, $n = 14$, $a = 10$

We know that $S_n = \left(\frac{n}{2}\right)[2a + (n-1)d]$

Substituting the values we have,

$$1050 = \left(\frac{14}{2}\right)[20 + 13d]$$

$$\Rightarrow 1050 = 140 + 91d \Rightarrow 910 = 91d$$

$$\Rightarrow d = 10$$

Therefore, $a_{20} = 10 + (20 - 1) \times 10 = 200$

i.e. 20th term is 200.

OR

The $n = 16$ and $d = \frac{8}{3}$

first term $a = 5$, last term $l = 45$ and $S_n = 400$

Since, $S_n = \frac{n}{2}(a + l)$

$$\Rightarrow 400 = \frac{n}{2}(5 + 45) \Rightarrow 400 = \frac{n}{2}(50)$$

$$\Rightarrow 400 = n(25) \Rightarrow n = 16$$

Since, $l = a + (n - 1)d$

$$\Rightarrow 45 = 5 + (16 - 1)d \Rightarrow 45 - 5 = 15d$$

$$\Rightarrow 40 = 15d \Rightarrow d = \frac{40}{15} \Rightarrow d = \frac{8}{3}$$

35. (a) Given : $a = 15$, $S_{15} = 750$

To find : a_{20}

Let d and n be common difference and number of terms respectively

$$\therefore S_n = \frac{n}{2}(2a + (n-1)d)$$

$$\Rightarrow S_{15} = \frac{15}{2}(2(15) + (15-1)d)$$

$$\Rightarrow 750 = \frac{15}{2}(30 + 14d)$$

$$\Rightarrow 14d + 30 = 50 \times 2 \Rightarrow 14d = 100 - 30$$

$$\Rightarrow 14d = 70 \Rightarrow d = 5$$

Also, $a_n = a + (n - 1)$

$$\Rightarrow a_{20} = 15 + (20 - 1)(5)$$

$$= 15 + 19(5) = 15 + 95 = 110.$$

OR

(b) Instalments to be paid by Rohan 1000, 1100, 1200,.....

This sequence is an A.P.

Here, $a = 1000$ and $d = 100$

To find: a_{30} and S_{30}

$$\therefore a_n = a + (n - 1)d$$

$$\Rightarrow a_{30} = 1000 + (30 - 1)100$$

$$= 1000 + 29(100)$$

$$= 1000 + 2900 = 3900$$

$$\text{Also, } S_n = \frac{n}{2}(a + a_n) \Rightarrow S_{30} = \frac{30}{2}(1000 + 3900)$$

$$\Rightarrow S_{30} = 15 \times 4900 = 73500$$

$\therefore$ In the 30th instalment, he will pay ₹3900 and he has paid ₹73500 after 30 instalments.

36. Consider the given AP 8, 10, 12,...

Here the first term is 8 and the common difference is $10 - 8 = 2$

General term of an AP is t_n is given by,

$$t_n = a + (n - 1)d$$

$$\Rightarrow t_{60} = 8 + (60 - 1) \times 2$$

$$\Rightarrow t_{60} = 8 + 59 \times 2$$

$$\Rightarrow t_{60} = 8 + 118 \Rightarrow t_{60} = 126$$

We need to find the sum of the last 10 terms.

Thus,

Sum of last 10 terms = Sum of first 60 terms

– Sum of first 50 terms

$$\Rightarrow S_n = \frac{n}{2}[2a + (n-1)d]$$

$$\Rightarrow S_{60} = \frac{60}{2}[2 \times 8 + (60-1) \times 2]$$

$$\Rightarrow S_{60} = 30[16 + 59 \times 2]$$

$$\Rightarrow S_{60} = 30[134]$$

$$\Rightarrow S_{60} = 4020$$

Similarly

$$\Rightarrow S_{50} = \frac{50}{2}[2 \times 8 + (50-1) \times 2]$$

$$\Rightarrow S_{50} = 25[16 + 49 \times 2]$$

$$\Rightarrow S_{50} = 25[114] \Rightarrow S_{50} = 2850$$

Thus the sum of last 10 terms

$$= S_{60} - S_{50} = 4020 - 2850 = 1170$$

37. Let there be a value of X such that the sum of the numbers of the houses preceding the house numbered X is equal to the sum of the numbers of the houses following it.

That is, $1 + 2 + 3 + \dots + (X - 1) = (X + 1) + (X + 2) + \dots + 49$

$$\therefore [1 + 2 + 3 + \dots + (X - 1)] = [1 + 2 + \dots + X + (X - 1) + \dots + 49 - (1 + 2 + 3 + \dots + X)]$$

$$\therefore \frac{X-1}{2}[1+X-1] = \frac{49}{2}[1+49] - \frac{X}{2}[1+X]$$

$$\therefore X(X-1) = 49 \times 50 - X(1+X)$$

$$\therefore X(X-1) = X(1+X) = 49 \times 50$$

$$\therefore X^2 - X + X + X^2 = 49 \times 50$$

$$\therefore 2X^2 = 49 \times 50$$

$$\therefore X^2 = 49 \times 25$$

$$\therefore X = 7 \times 5 = 35$$

Since X is not a fraction, the value of x satisfying the given condition exists and is equal to 35.

38. Let the numbers be $(a - 3d)$, $(a - d)$, $(a + d)$ and $(a + 3d)$

$$(a - 3d) + (a - d) + (a + d) + (a + 3d) = 32$$

$$\Rightarrow 4a = 32$$

$$a = 8$$

$$\text{Also, } \frac{(a-3d)(a+3d)}{(a-d)(a+d)} = \frac{7}{15}$$

$$\Rightarrow 15a^2 - 135d^2 = 7a^2 - 7d^2$$

$$\Rightarrow 8a^2 = 128d^2$$

$$d^2 = \frac{8a^2}{128} = \frac{8 \times 8 \times 8}{128}$$

$$d^2 = 4$$

$$d = \pm 2$$

If $d = 2$ number are: 2, 6, 10, 14

If $d = -2$ number are: 14, 10, 6, 2.

39. Let the first four terms be a , $a + d$, $a + 2d$, $a + 3d$

$$a + a + d + a + 2d + a + 3d = 40$$

$$\Rightarrow 2a + 3d = 20 \quad \dots(i)$$

$$\text{Sum of first 14 terms} = 280$$

$$\frac{n}{2}[2a + (n-1)d] = 280$$

$$\Rightarrow \frac{14}{2}[2a + 13d] = 280$$

$$\Rightarrow 2a + 13d = 40 \quad \dots(ii)$$

On subtracting (i) from (ii), we get $d = 2$

Substituting the value of d in (i)

$$a = 7$$

$$\therefore \text{Sum of } n \text{ terms} = \frac{n}{2}[2a + (n-1)d]$$

$$= \frac{n}{2}[14 + (n-1)2] = n^2 + 6n$$

40. Let the first term and common difference be a and d .

According to the question,

$$4(a + 3d) = 18 \times (a + 17d)$$

$$\Rightarrow 4a + 12d = 18a + 306d$$

$$\Rightarrow 14a + 294d = 0$$

$$\Rightarrow a + 21d = 0$$

$$\therefore a_{22} = a + 21d = 0$$

OR

Given A.P. 24, 21, 18, ...

$$\therefore \text{First term} = 24 = a$$

$$\text{and common difference} = -3 = d \quad \dots(i)$$

Let number of terms is n .

$$\therefore \text{Sum of } n \text{ terms} = \frac{n}{2}[2a + (n-1)d]$$

According to question

$$\Rightarrow 78 = \frac{n}{2}[2 \times 24 - 3(n-1)] \quad [\text{From (i) and given}]$$

$$\Rightarrow 78 = \frac{n}{2}[51 - 3n]$$

$$\Rightarrow n^2 - 17n + 52 = 0$$

$$\Rightarrow n^2 - 13n - 4n + 52 = 0$$

$$\Rightarrow n(n-13) - 4(n+13) = 0$$

$$(n-13)(n+4) = 0$$

$$n = 13, 4$$

For first 4 terms and first 13 terms in both case we get sum 78.

41. Let the four consecutive numbers in A.P. are

$$(a - 3d), (a - d), (a + d) \text{ and } (a + 3d),$$

$\therefore$ According to the question,

$$(a - 3d) + (a - d) + (a + d) + (a + 3d) = 32$$

$$\Rightarrow 4a = 32 \Rightarrow a = 8 \quad \dots(i)$$

and, according to the 2nd condition given,

$$\frac{(a-3d) \times (a+3d)}{(a-d) \times (a+d)} = \frac{7}{15}$$

$$\Rightarrow \frac{(8-3d) \times (8+3d)}{(8-d) \times (8+d)} = \frac{7}{15}$$

$$\Rightarrow \frac{64-9d^2}{64-d^2} = \frac{7}{15}$$

$$\Rightarrow 15(64-9d^2) = 7(64-d^2)$$

$$\Rightarrow 128d^2 = 512 \Rightarrow d^2 = 4 \Rightarrow d = \pm 2$$

$\therefore$ Number are 2, 6, 10 and 14 or 14, 10, 6 and 2.

OR

Here 1, 4, 7, 10, x is an A.P.

With first term $a = 1$ and common difference $d = 3$.

Let there be n terms in the A.P. Then,

$x = n^{\text{th}}$ term

$$\Rightarrow x = 1 + (n-1) \times 3 = 3n-2 \quad \dots(i)$$

Now, $1 + 4 + 7 + 10 + \dots + x = 287$

$$\Rightarrow \frac{n}{2}[1+x] = 287 \quad \left[S_n = \frac{n}{2}(a+l) \right]$$

$$\Rightarrow \frac{n}{2}[1+3n-2] = 287$$

$$\Rightarrow 3n^2 - n - 574$$

$$\Rightarrow 3n - n - 574 = 0$$

$$\Rightarrow 3n^2 - 42n + 41n - 574 = 0$$

$$\Rightarrow 3n(n-14) + 41(n-14) = 0$$

$$\Rightarrow (n-14)(3n+41) = 0$$

$$\Rightarrow n-14 = 0 \quad [\because 3n+41 \neq 0]$$

$$\Rightarrow n = 14$$

Putting $n = 14$ in eqn (i), we get

$$x = 3 \times 14 - 2 \Rightarrow x = 40$$

42. (a) Let first term and common difference of A.P. be a and d respectively

$$a_2 = 14 \text{ and } a = 18$$

$$a + d = 14 \quad \dots(i)$$

$$a + 2d = 18 \quad \dots(ii)$$

Subtracting (i) from (ii), we get

$$d = 4$$

Putting $d = 4$ in equation (i), we get

$$a = 10$$

$$S_{51} = \frac{51}{2}[2(10) + (50)(4)] = \frac{51}{2}[220]$$

OR

- (b) Let d and n be the common difference and number of terms of A.P. respectively

$$a = a + (n-1)d$$

$$45 = 5 + (n-1)d$$

$$\Rightarrow 40 = (n-1)d \quad \dots(i)$$

$$n \text{ and } S = \frac{n}{2}[a + a_n]$$

$$400 = \frac{n}{2}[5+45]$$

$$400 = 25n \Rightarrow n = 16 \quad \dots(ii)$$

From (i) and (ii), we get

$$(16-1)d = 40$$

$$15d = 40 \quad \dots(ii)$$

$$d = \frac{8}{3}$$

Hence, number of terms and common difference

is 16 and $\frac{8}{3}$ respectively.



6

CHAPTER

Triangles

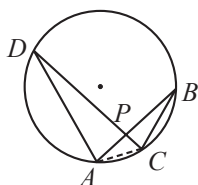


Previous Year Questions

Multiple Choice Questions

1. AB and CD are two chords of a circle intersecting at P . Choose the correct statement from the following:

[CBSE 2024]



- (a) $\triangle ADP \sim \triangle CBA$ (b) $\triangle ADP \sim \triangle BPC$
(c) $\triangle ADP \sim \triangle BCP$ (d) $\triangle ADP \sim \triangle CBP$

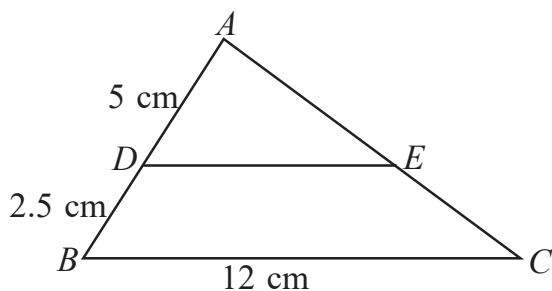
2. The perimeters of two similar triangles ABC and PQR are 56 cm and 48 cm respectively. $\frac{PQ}{AB}$ is equal to

[CBSE 2024]

- (a) $\frac{7}{8}$ (b) $\frac{6}{7}$
(c) $\frac{7}{6}$ (d) $\frac{8}{7}$

3. In the given figure $\triangle ABC$ is shown. DE is parallel to BC . If $AD = 5$ cm, $DB = 2.5$ cm and $BC = 12$ cm, then DE is equal to

[CBSE 2024]

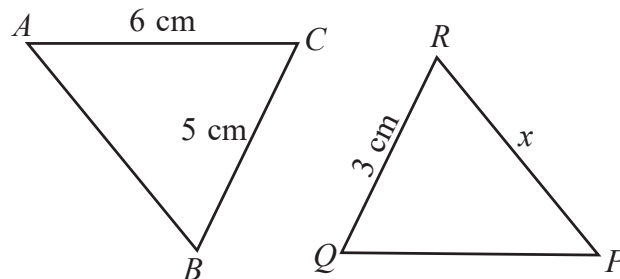


- (a) 10 cm (b) 6 cm
(c) 8 cm (d) 7.5 cm

4. If the diagonals of a quadrilateral divide each other proportionally, then it is a: [CBSE 2024]

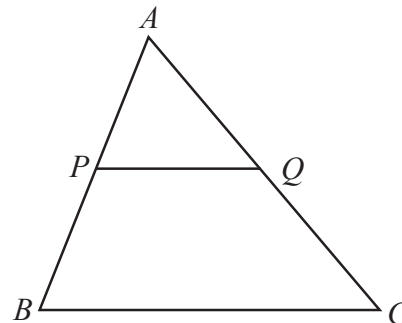
- (a) parallelogram
(b) rectangle
(c) square
(d) trapezium

5. In the given figure, $\triangle ABC \sim \triangle QPR$. If $AC = 6$ cm, $BC = 5$ cm, $QR = 3$ cm and $PR = x$, then the value of x is [CBSE 2023]



- (a) 3.6 cm (b) 2.5 cm
(c) 10 cm (d) 3.2 cm

6. In ABC , $PQ \parallel BC$, if $PB = 6$ cm, $AP = 4$ cm, $AQ = 8$ cm, find the length of AC . [CBSE 2023]



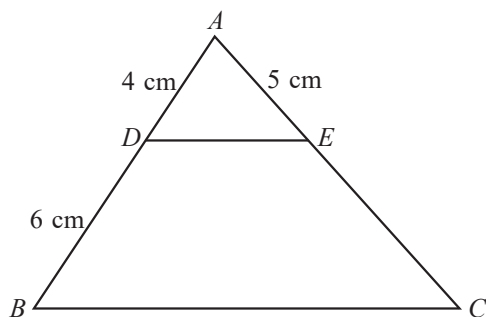
- (a) 12 cm (b) 20 cm

- (c) 6 cm (d) 14 cm

7. If $\triangle ABC \sim \triangle DEF$ and $\angle A = 47^\circ$, $\angle E = 83^\circ$, then $\angle C$ is equal [CBSE 2023]

- (a) 47° (b) 50°
(c) 83° (d) 130°

8. In $\triangle ABC$, $DE \parallel BC$, $AD = 4$ cm, $DB = 6$ cm, and $AE = 5$ cm. Then length of EC is [CBSE 2021]

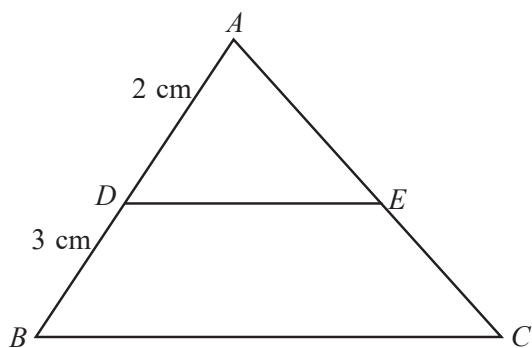


- (a) 7 cm (b) 6.5 cm
(c) 7.5 cm (d) 8 cm

9. Which of the following is a correct statement? [CBSE 2021]

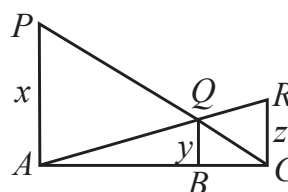
- (a) Two congruent figures are always similar
(b) Two similar figures are always congruent
(c) All rectangles are similar
(d) The polygons having same number of sides are similar

10. In $\triangle ABC$, $DE \parallel BC$, $AD = 2$ cm, $DB = 3$ cm, $DE:BC$ is equal to [CBSE 2021]



- (a) 2 : 3 (b) 2 : 5
(c) 1 : 2 (d) 3 : 5

11. If fig., PA , QB and RC are each perpendicular to AC . If $x = 8$ cm and $z = 6$ cm, then y is equal to [CBSE 2021]



- (a) $\frac{56}{7}$ cm (b) $\frac{7}{56}$ cm
(c) $\frac{25}{7}$ cm (d) $\frac{24}{7}$ cm

12. In $\triangle ABC$ and $\triangle DEF$, $\angle F = \angle C$, $\angle B = \angle E$ and $AB = \frac{1}{2}DE$. Then, the two triangles are [CBSE 2021]

- (a) Congruent, but not similar
(b) Similar, but not congruent
(c) Neither congruent nor similar
(d) Congruent as well as similar

13. In a $\triangle ABC$, $A = x^\circ$, $B = (3x - 2)^\circ$, $C = y^\circ$. Also, $\angle C - \angle B = 9^\circ$. The sum of the greatest and the smallest angles of this triangle is [CBSE 2021]

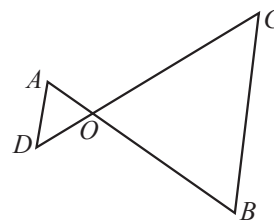
- (a) 107° (b) 135°
(c) 155° (d) 145°

Short Answer Type Questions

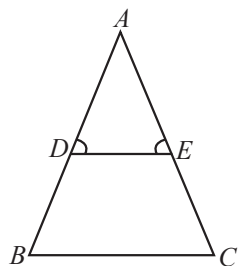
14. **Assertion (A):** A line drawn parallel to any one side of a triangle intersect the other two sides in the same ratio.

Reason (R): Parallel lines cannot be drawn to any side of a triangle. [CBSE 2024]

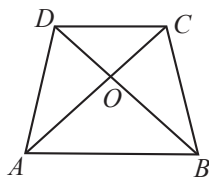
15. In the given figure, $OA \cdot OB = OC \cdot OD$. Prove that $\triangle AOD \sim \triangle COB$. [CBSE 2024]



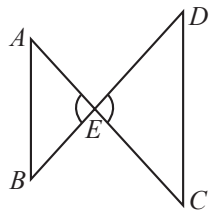
16. In the given figure, $\angle D = \angle E$ and $\frac{AD}{DB} = \frac{AE}{EC}$. Prove that $\triangle ABC$ is isosceles. [CBSE 2024]



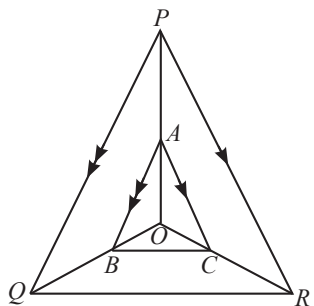
17. In a $\triangle ABC$, D and E are points on the sides AB and AC respectively such that $BD = CE$. If $\angle B = \angle C$, then show that $DE \parallel BC$. [CBSE 2024]
18. If $\triangle ABC \sim \triangle DEF$ and $AB = 4$ cm, $DE = 6$ cm, $EF = 9$ cm and $FD = 12$ cm, find the perimeter of $\triangle ABC$. [CBSE 2024]
19. Diagonal AC and BD of a trapezium $ABCD$ intersect at O , where $AB \parallel DC$. If $\frac{DO}{OB} = \frac{1}{2}$, then show that $AB = 2CD$. [CBSE 2024]



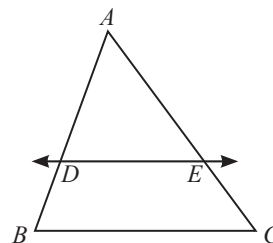
20. In the given figure $\frac{EA}{EC} = \frac{EB}{ED}$, prove that $\triangle EAB \sim \triangle ECD$. [CBSE 2024]



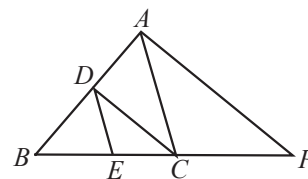
21. Through the mid-point M of the side CD of a parallelogram $ABCD$, the line BM is drawn intersecting AC in L and AD produced in E . Prove that $EL = 2BL$. [CBSE 2024]
22. In the adjoining figure, A , B and C are points on OP , OQ and OR respectively such that $AB \parallel PQ$ and $AC \parallel PR$. Show that $BC \parallel QR$. [CBSE 2023]



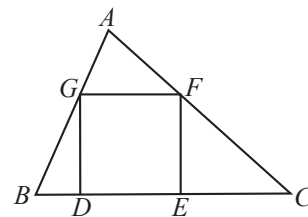
23. The perimeters of two similar triangles are 25 cm and 15 cm respectively. If one side of the first triangle is 9 cm, then the corresponding side of second triangle is _____. [CBSE 2020]
24. In figure, in $\triangle ABC$, $DE \parallel BC$ such that $AD = 2.4$ cm, $AB = 3.2$ cm and $AC = 8$ cm, then what is the length of AE ? [CBSE 2020]



25. In figure $DE \parallel AC$ and $DC \parallel AP$ prove that $\frac{BE}{EC} = \frac{BC}{CP}$ [CBSE 2020]



26. In figure, $DEFG$ is a square in a triangle ABC right angled at A .

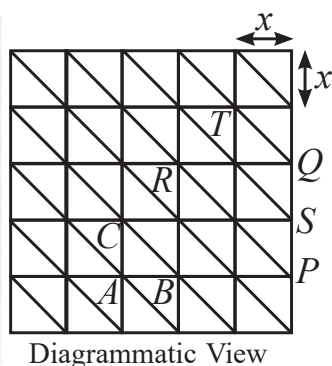


- Prove that [CBSE 2020]
(i) $\triangle AGF \sim \triangle DBG$ (ii) $\triangle AGF \sim \triangle EFC$

27. The lengths of the diagonals of a rhombus are 30 cm and 40 cm. Find the side of the rhombus. [CBSE 2020]
28. In $\triangle LMN$, $\angle L = 50^\circ$ and $\angle N = 60^\circ$. If $\triangle LMN \sim \triangle PQR$, then find $\angle Q$. [CBSE 2020]

Case Based Type Questions

- 29.



Diagrammatic View

Quilts are available in various colours and design. Geometric design includes shapes like squares, triangles, rectangles, hexagons, etc.

One such design is shown above. Two triangles are highlighted, $\triangle ABC$ and $\triangle PQR$.

Based on above information, answer the following questions:

- (i) Which of the following criteria is not suitable for $\triangle ABC$ to be similar to $\triangle QRP$? [CBSE 2021]

- (a) SAS (b) AAA
(c) SSS (d) RHS

- (ii) If each square is of length x unit, then length BC is equal to [CBSE 2021]

- (a) $x\sqrt{2}$ unit (b) $2x$ unit
(c) $2\sqrt{x}$ unit (d) $x\sqrt{x}$ unit

- (iii) Ratio $BC : PR$ is equal to [CBSE 2021]

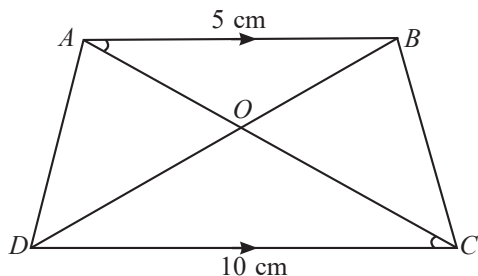
- (a) 2 : 1 (b) 1 : 4
(c) 1 : 2 (d) 4 : 1

- (iv) Which of the following is not true [CBSE 2021]

- (a) $\triangle TQS \sim \triangle PQR$ (b) $\triangle CBA \sim \triangle STQ$
(c) $\triangle BAC \sim \triangle PQR$ (d) $\triangle PQR \sim \triangle ABC$

30. A farmer has a field in the shape of trapezium, whose map with scale 1 cm = 20 m, is given:

The field is divided into four parts by joining the opposite vertices.



Based on above information, answer the following

questions:

- (i) The two triangular regions AOB and COD are [CBSE 2021]

- (a) Similar by AA criterion
(b) Similar by SAS criterion
(c) Similar by RHS criterion
(d) Not similar

- (ii) If the ratio of the perimeter of $\triangle AOB$ to the perimeter of $\triangle COD$ would have been 1 : 4, then [CBSE 2021]

- (a) $AB = 2CD$ (b) $AB = 4CD$
(c) $CD = 2AB$ (d) $CD = 4AB$

- (iii) If in $\triangle AOD$ and BOC , $\frac{AO}{BC} = \frac{AD}{BO} = \frac{OD}{OC}$, [CBSE 2021]

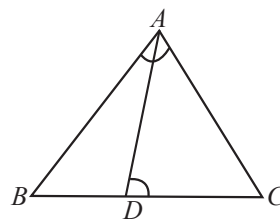
- (a) $\triangle AOD \sim \triangle BOC$ (b) $\triangle AOD \sim \triangle BCO$
(c) $\triangle ADO \sim \triangle BCO$ (d) $\triangle ODA \sim \triangle OCB$

Long Answer Type Questions

31. $ABCD$ is a trapezium with $AB \parallel DC$ and the diagonals intersect each other at the point O .

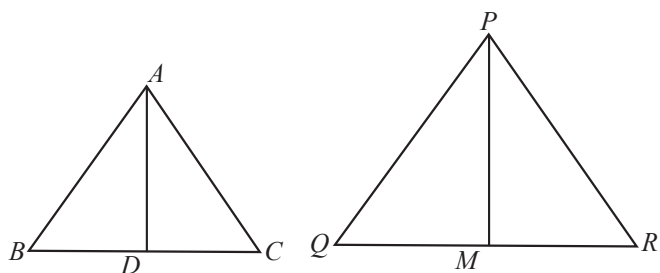
Show that $\frac{AO}{BO} = \frac{CO}{DO}$ [CBSE 2024]

32. D is a point on the side BC of $\triangle ABC$ such that $\triangle ADC = \triangle BAC$. Show that $AC^2 = BC \times DC$. [CBSE 2024]



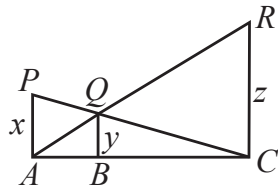
33. Sides AB and BC and median AD of a $\triangle ABC$ are respectively proportional to sides PQ and QR and median PM of $\triangle PQR$. show that $\triangle ABC \sim \triangle PQR$.

[CBSE 2024]



34. $ABCD$ is a trapezium with $AB \parallel DC$. AC and BD intersect at E . If $\triangle AED \sim \triangle BEC$, then prove that $AD = BC$. [CBSE 2024]

35. In the given figure PA , QB and RC are each perpendicular to AC . If $AP = x$, $BQ = y$ and $CR = z$, then prove that $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$ [CBSE 2024]



36. Observe the figures given below carefully and answer the questions:

[CBSE 2023]

Figure A

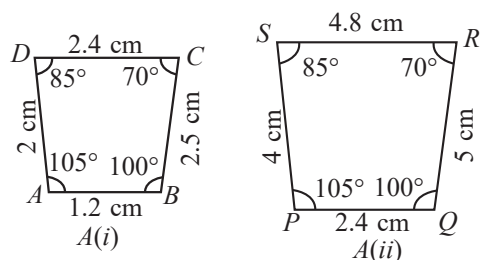


Figure B

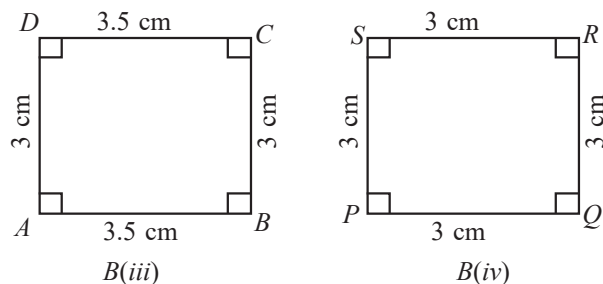
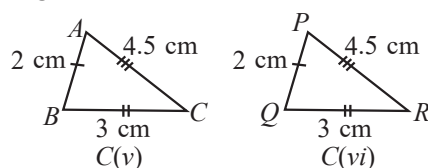


Figure C



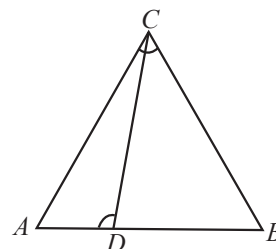
- Name the figure(s) wherein two figures are similar.
- Name the figure(s) wherein the figures are congruent.
- (A) Prove that congruent triangles are also similar but not the converse.

OR

(B) What more is least needed for two similar triangles to be congruent? [CBSE 2023]

37. In the given figure, $\angle ADC = \angle BCA$; prove that $\triangle ACB \sim \triangle ADC$. Hence find BD if $AC = 8$ cm and $AD = 3$ cm.

[CBSE 2023]





Explanations

Previous Year Questions

Multiple Choice Questions

1. (d) In $\triangle ADP$ and $\triangle CBP$

$$\angle A = \angle C \quad [\text{angles of the same segment}]$$

$$\angle D = \angle B \quad [\text{angles of the same segment}]$$

$$\triangle ADP \sim \triangle CBP \quad [\text{AA Postulate}]$$

2. (b) Perimeter of $\triangle ABC = 56$ cm

$$\text{Perimeter of } \triangle PQR = 48 \text{ cm}$$

$$\frac{PQ}{AB} = \frac{48}{56} = \frac{6}{7}$$

3. (c)

$$DE \parallel BC$$

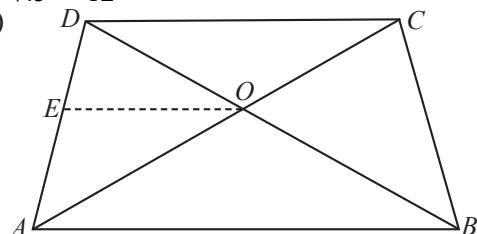
$$DB = 2.5 \text{ cm}, BC = 12 \text{ cm}, AD = 5 \text{ cm}$$

$$\text{Since, } DE \parallel BC$$

$$\Rightarrow \frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$$

$$\Rightarrow \frac{5}{7.5} = \frac{DE}{12} \Rightarrow DE = 8 \text{ cm}$$

4. (d)



$$\frac{AO}{BO} = \frac{CO}{DO} \quad (\text{Given})$$

$$\Rightarrow \frac{AO}{CO} = \frac{BO}{DO} \quad \dots(i)$$

$$\text{In } \triangle ABD, EO \parallel AB \quad (\text{Construction})$$

$$\therefore \frac{AE}{ED} = \frac{BO}{DO} \quad (\text{By BPT}) \dots(ii)$$

From equation (i) and (ii)

$$\frac{AE}{ED} = \frac{AO}{CO} \Rightarrow EO \parallel DC \quad (\text{Converse of BPT})$$

$$\text{But } EO \parallel AB \quad (\text{Construction})$$

$$\therefore EO \parallel AB$$

$$\Rightarrow \text{In quad } ABCD, AB \parallel DC$$

$ABCD$ is a trapezium.

5. (b) $\triangle ABC \sim \triangle QPR$

$$\frac{AO}{BO} = \frac{CO}{DO} \quad (\text{Given})$$

$$\Rightarrow \frac{AC}{BC} = \frac{QR}{PR} \Rightarrow \frac{6}{5} = \frac{3}{x}$$

$$\Rightarrow x = 2.5 \text{ cm}$$

$$6. (b) \frac{AP}{AB} = \frac{AQ}{AC} \quad [\text{By BPT}]$$

$$\Rightarrow \frac{4}{10} = \frac{8}{AC}$$

$$AC = 20 \text{ cm.}$$

$$7. (b) \angle E = \angle B = 83^\circ$$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow \angle C = 180^\circ - 47^\circ - 83^\circ = 50^\circ$$

$$8. (c) \text{Applying B.P.T. in } \triangle ABC, DE \parallel BC$$

$$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{4}{6} = \frac{5}{EC}$$

$$\Rightarrow EC = 7.5 \text{ cm}$$

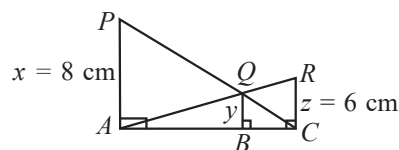
$$9. (a) \text{Two congruent figures are always similar.}$$

$$10. (b) DE \parallel BC$$

$$\therefore \frac{DE}{BC} = \frac{AD}{AB}$$

$$\Rightarrow DE:BC = 2:5$$

$$11. (d)$$



$$\frac{y}{x} = \frac{BC}{AC} \quad [\text{By BPT}]$$

$$\Rightarrow \frac{y}{x} = \frac{AB + BC}{BC} = \frac{AB}{BC} + 1 \quad \dots(i)$$

$$\text{and } \frac{z}{y} = \frac{AC}{AB} = 1 + \frac{BC}{AB} = 1 + \frac{BC}{AB} = 1 + \frac{y}{x-y} \quad [\text{By BPT}]$$

[From (i)]

$$\Rightarrow \frac{6}{y} = 1 + \frac{y}{8-y} \Rightarrow 8y = 48 - 6y \Rightarrow y = \frac{24}{7} \text{ cm}$$

$$12. (b) \text{Here } \angle F = \angle C, \angle B = \angle E \text{ and } AB = \frac{1}{2} DE$$

Since, AB and DE are not equal.

So, $\triangle ABC \sim \triangle DEF$.

13. (a) $y^\circ - (3x - 2)^\circ = 9^\circ$... (i)
 $3x^\circ - y^\circ = -7$ and $x^\circ + (3x - 2)^\circ + y^\circ = 180^\circ$
 $\Rightarrow 4x^\circ + y^\circ = 182^\circ$... (ii)
 $\Rightarrow x^\circ = \frac{182^\circ - 7^\circ}{7} = 25^\circ$ and $y^\circ = 82^\circ$
 $\Rightarrow \angle A = 25^\circ$ and $\angle B = 73^\circ$ and $\angle C = 82^\circ$
 $\therefore$ Sum of greatest and smallest angles
 $= 82^\circ + 25^\circ = 107^\circ$

Short Answer Type Questions

14. Assertion is correct but reason is not true.

15. Here, we are given,

$$OA \times OB = OC \times OD$$

Dividing both sides by $OB \times OC$, we get,

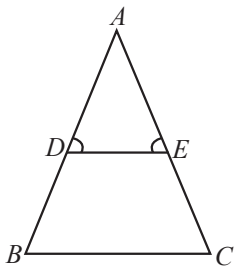
$$\frac{OA}{OC} = \frac{OD}{OB}$$

$$\text{Also, } \angle AOD = \angle BOC$$

So, by side angle side similarity rule,

$$\triangle AOD \sim \triangle COB$$

- 16.



$$\frac{AD}{DB} = \frac{AE}{EC} \quad (\text{Given})$$

Therefore, $DE \parallel BC$

(converse of Basic Proportionality theorem)

So, $\angle D = \angle B$ and $\angle E = \angle C$ (Corresponding angles)

... (i)

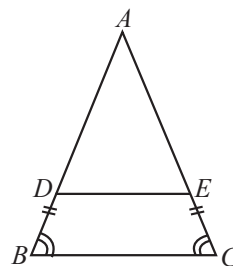
But $\angle D = \angle E$ (Given)

Therefore, $\angle B = \angle C$ [From (i)]

So, $AB = AC$ (Sides opposite to equal angles)

i.e., $\triangle ABC$ is an isosceles triangle.

17. Given is a $\triangle ABC$ such that $\angle B = \angle C$ and $BD = CE$.



To prove: $DE \parallel BC$

We know that, sides opposite to equal angles of a triangle are equal.

$$AB = AC [\because \angle B = \angle C]$$

$$AD + DB = AE + EC$$

$$\text{Also, } BD = CE$$

$$\therefore AD = AE$$

$$\text{Hence, } \frac{AD}{DB} = \frac{AE}{EC}$$

According to the converse of basic proportionality theorem, if a line divides any two sides of a triangle in the same ratio, then the line must be parallel to the third side.

Therefore, by using the converse of the basic proportionality theorem, we get $DE \parallel BC$.

18. Given: $\triangle ABC \sim \triangle DEF$

$$\text{So, } \frac{AB}{ED} = \frac{BC}{EF} = \frac{AC}{DF} \Rightarrow \frac{4}{6} = \frac{BC}{9} = \frac{AC}{12}$$

Considering the first two terms,

$$\frac{4}{6} = \frac{BC}{9} \Rightarrow \frac{4 \times 9}{6} = BC \Rightarrow BC = 6 \text{ cm}$$

Considering the last two terms,

$$\frac{BC}{9} = \frac{AC}{12} \Rightarrow \frac{12 \times 6}{9} = AC \Rightarrow AC = 8 \text{ cm}$$

$$\text{Perimeter of } \triangle ABC = AB + BC + AC$$

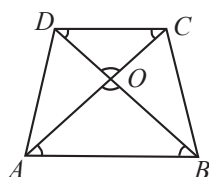
$$\text{Perimeter of } \triangle ABC = 4 + 6 + 8$$

$$\text{Perimeter of } \triangle ABC = 18 \text{ cm}$$

Hence, perimeter of $\triangle ABC$ is 18 cm.

19. Given: $AB \parallel DC$

$$\frac{DO}{OB} = \frac{1}{2}$$



To show: $AB = 2CD$

In $\triangle COD$ and $\triangle AOB$

$$\angle COD = \angle AOB$$

[Alternate Interior angle between two parallel lines]

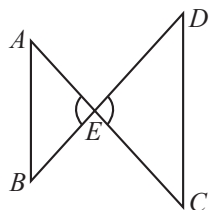
By AA similarity criteria $\triangle COD \sim \triangle AOB$

$$\frac{OD}{OB} = \frac{CD}{AB}$$

$$\frac{1}{2} = \frac{CD}{AB}$$

$$AB = 2CD$$

20. It is given that $\frac{EA}{EC} = \frac{EB}{ED}$.



Converse of basic proportionality theorem states that if any line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side.

$$\text{It is given that } \frac{EA}{EC} = \frac{EB}{ED}$$

$$\text{So, } \frac{EA}{EB} = \frac{EC}{ED}$$

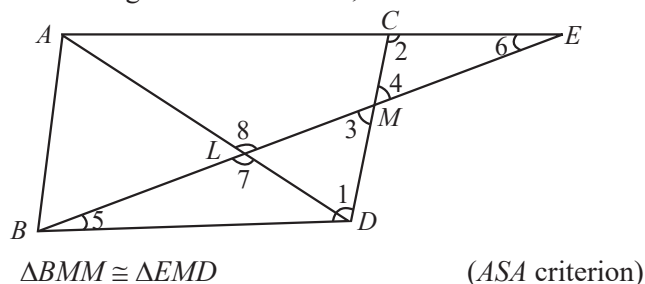
In $\triangle EAB$ and $\triangle ECD$

$$\frac{EA}{EB} = \frac{EC}{ED}$$

$$\angle AEB = \angle DEC \quad (\text{Vertically opposite angles})$$

$$\Rightarrow \triangle EAB \sim \triangle ECD \quad (\text{By SAS similarity criterion})$$

21. In a triangle BMC and DME ,



$$BC = ED \quad (\text{Corresponding parts of congruent triangle}) \quad \dots(i)$$

$$BC = AD \quad (\text{opposite sides of a parallelogram}) \dots(ii)$$

Adding (i) and (ii), we get

$$2BC = DE + AD$$

$$\Rightarrow 2BC = AE$$

$$\Rightarrow \frac{BC}{AE} = \frac{1}{2}$$

$\dots(iii)$

Now in $\triangle BCL$ and $\triangle EAL$

$$\angle 5 = \angle 6 \quad (\text{Alternate } \angle \text{s as } BC \parallel AE)$$

$$\angle 7 = \angle 8 \quad (\text{Vertically opposite angles})$$

$$\triangle BCL \sim \triangle EAL \quad (AA \text{ corollary})$$

$$\frac{BC}{AE} = \frac{BL}{EL} \quad (\text{corresponding sides of similar triangles are proportional}) \quad \dots(iv)$$

$$\frac{1}{2} = \frac{BL}{EL} \quad [\text{from (iii) and (iv)}]$$

$$EL = 2BL \text{ hence proved.}$$

22. In PQR , $AC \parallel PR$

$$\frac{OA}{AP} = \frac{OC}{CR} \quad \dots(i) \quad [BPT]$$

In $\triangle OPQ$, $AB \parallel PQ$

$$\frac{OA}{AP} = \frac{OB}{BQ} \quad \dots(ii) \quad [BPT]$$

From (i) and (ii) we get

$$\frac{OC}{CR} = \frac{OB}{BQ}$$

So, by converse of BPT

$$BC \parallel QR$$

23. Let perimeters of two similar triangles be P_1 and P_2 and their corresponding sides be a_1 and a_2 .

$$\therefore \frac{P_1}{P_2} = \frac{a_1}{a_2} \Rightarrow \frac{25}{15} = \frac{9}{a_2} \Rightarrow a_2 = 5.4 \text{ cm}$$

24. $\therefore DE \parallel BC$

$$\therefore \frac{AD}{AB} = \frac{AE}{AC} \Rightarrow \frac{2.4}{3.2} = \frac{AE}{8}$$

$$\therefore AE = \frac{24}{32} \times 8 = 6 \text{ cm}$$

25. In $\triangle BAC$; $DE \parallel AC$

$$\frac{BE}{EC} = \frac{BD}{DA} \quad \dots(i)$$

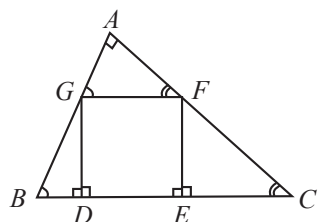
In $\triangle BAP$; $DC \parallel AP$

$$\frac{BC}{CP} = \frac{BD}{DA} \quad \dots(ii)$$

From (i) and (ii), we have

$$\frac{BE}{EC} = \frac{BC}{CP} \text{ hence proved.}$$

26. $DEFG$ is a square and $\triangle ABC$ is a right triangle right angled at A .



- (i) In $\triangle AGF$ and $\triangle DBG$

$$\angle A = \angle D = 90^\circ \text{ and } \angle AGF = \angle GBD$$

($\because GF \parallel BC \Rightarrow$ Corresponding angles)

By AA similarity

$$\triangle AGF \sim \triangle DBG$$

- (ii) In $\triangle AGF$ and $\triangle EFC$

$$\angle A = \angle E = 90^\circ$$

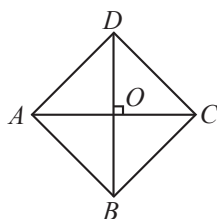
$$\angle AFG = \angle ECF$$

($\because GF \parallel BC \Rightarrow$ Corresponding angles)

By AA similarity

$$\triangle AGF \sim \triangle EFC \text{ Hence Proved.}$$

27. Length of the diagonals of a rhombus are 30 cm and 40 cm



$$\text{i.e., } BD = 30 \text{ cm } AC = 40 \text{ cm}$$

$$\therefore OD = OB = 15 \text{ cm}$$

In $\triangle AOD$,

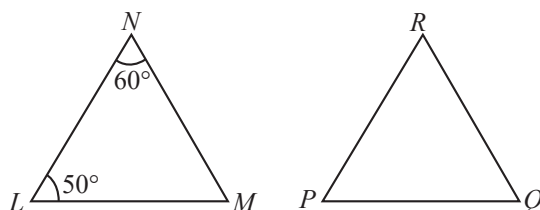
$$OA^2 + OD^2 = AD^2$$

$$20^2 + (15)^2 = AD^2$$

$$AD = 25 \text{ cm}$$

$$\text{Side of rhombus} = 25 \text{ cm}$$

28. Given $\triangle LMN \sim \triangle PQR$



In similar triangles, corresponding angles are equal.

$$\therefore \angle L = \angle P$$

$$\angle M = \angle Q$$

$$\angle N = \angle R$$

In $\triangle LMN$,

$$\angle L + \angle M + \angle N = 180^\circ$$

$$\angle M = 180^\circ - 50^\circ - 60^\circ$$

$$\angle M = 70^\circ$$

$$\therefore \angle Q = 70^\circ$$

Case Based Type Questions

29. (i) (d) Because, according to criteria of similarity RHS similarity is not possible

$$(ii) (a) BC = \sqrt{AC^2 + AB^2} = x\sqrt{2} \text{ units}$$

$$(iii) (c) \frac{BC}{PR} = \frac{x\sqrt{2}}{2x\sqrt{2}} = \frac{1}{2}$$

$$\therefore BC : PR = 1 : 2$$

- (iv) (d) $\angle A = 90^\circ$ and $\angle P = 45^\circ$

$$\therefore \triangle PQR \text{ is not similar to } \triangle ABC.$$

30. (i) (a) $\triangle AOB \sim \triangle COD$ [By AA similarity]

$$(ii) (d) \frac{\text{Perimeter of } \triangle AOB}{\text{Perimeter of } \triangle COD} = \frac{AB}{CD} \text{ or } \frac{AB}{CD} = \frac{1}{4}$$

$$(iii) (b) \text{ If } \frac{AO}{BC} = \frac{AD}{BO} = \frac{OD}{OC} \text{ or } \triangle AOD \sim \triangle BCO$$

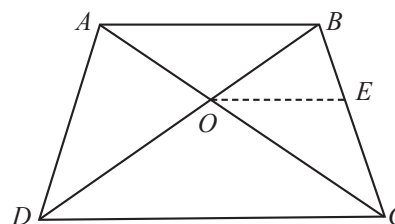
Long Answer Type Questions

31. Given: $ABCD$ is a trapezium and $AB \parallel DC$

$$\text{To prove: } \frac{AO}{BO} = \frac{CO}{DO}$$

Construction: Draw $OE \parallel CD$ such that E lies on BC .

Proof: In $\triangle BDC$



By basic proportionality Theorem,

$$\frac{BO}{OD} = \frac{BE}{EC}$$

...(i)

Now, In $\triangle BAC$,

By basic proportionality Theorem,

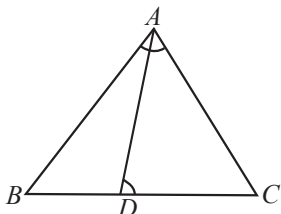
$$\frac{AO}{OC} = \frac{BE}{EC} \quad \dots(ii)$$

From (i) and (ii)

$$\frac{AO}{OC} = \frac{BO}{OD}$$

$$\text{i.e. } \frac{AO}{BO} = \frac{CO}{DO}$$

32.



In $\triangle ADC$ and $\triangle BAC$

$$\angle ADC = \angle BAC \quad (\text{Given})$$

$\angle C$ is common

$\therefore$ by AA criteria of Similarity, $\triangle ADC \sim \triangle BAC$

$$\Rightarrow \frac{AD}{BA} = \frac{DC}{AC} = \frac{AC}{BC} \Rightarrow \frac{DC}{AC} = \frac{AC}{BC}$$

$$\therefore CA^2 = CB \cdot CD$$

33. Given AD and PM are medians of $\triangle ABC$ and $\triangle PQR$,

$$\therefore BD = \frac{1}{2}BC \text{ and } QM = \frac{1}{2}QR \quad \dots(i)$$

Given that,

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM} \quad \dots(ii)$$

From (i) and (ii),

$$\frac{AB}{PQ} = \frac{BD}{QM} = \frac{AD}{PM} \quad \dots(iii)$$

In $\triangle ABD$ and $\triangle PQM$

$$\frac{AB}{PQ} = \frac{BD}{QM} = \frac{AD}{PM} \quad [\text{From (iii)}]$$

$\therefore$ By SSS criterion of proportionality $\triangle ABD \sim \triangle PQM$

$$\therefore \angle B = \angle Q \quad (\text{Corresponding sides of similar triangles}) \quad \dots(iv)$$

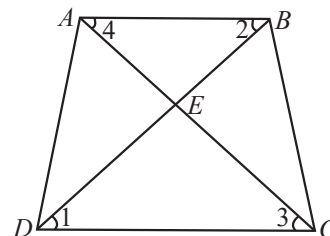
$\therefore$ In $\triangle ABC$ and $\triangle PQR$

$$\frac{AB}{PQ} = \frac{BC}{QR} \quad [\text{From (ii)}]$$

$$\angle B = \angle Q \quad [\text{From iv}]$$

$\therefore$ By SAS criterion of proportionality $\triangle ABC \sim \triangle PQR$

34. It is given that $\triangle AED \sim \triangle BEC$



$$\frac{AE}{BE} = \frac{ED}{EC} = \frac{AD}{BC} \quad \dots(i)$$

(Corresponding side proportional)

In $\triangle ABE$ and $\triangle CDE$,

$$\angle AEB = \angle CED \quad [\text{Vertically opposite angles}]$$

$$\angle EAB = \angle ECD \quad [\text{Alternate angles}]$$

$$\therefore \triangle ABE \sim \triangle CDE \quad [\text{AA similarity}]$$

$$\Rightarrow \frac{AB}{CD} = \frac{EB}{ED} = \frac{AE}{EC}$$

[Corresponding side are proportional]

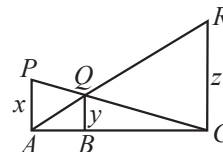
$$\frac{EC}{ED} = \frac{AE}{EB} \quad \dots(ii)$$

From (i) and (ii), we get

$$\frac{AD}{BC} = \frac{ED}{EC} \Rightarrow \frac{AD}{BC} = 1 \quad (ED = EC)$$

$\therefore AD = BC$ proved.

35. It is given that $\triangle AED \sim \triangle BEC$



$$\angle CAP = \angle CBQ = 90^\circ$$

(common angle)

So by AA, $\triangle CAP \sim \triangle CBQ$

$$\text{Hence } \frac{BQ}{AP} = \frac{BC}{AC}$$

$$\frac{y}{x} = \frac{BC}{AC} \quad \dots(i)$$

Now consider $\triangle ACR$ and $\triangle ABQ$

$$\angle ACR = \angle ABQ = 90^\circ$$

(common angle)

So by AA, $\triangle ACR \sim \triangle ABQ$

$$\text{Hence } \frac{BQ}{CR} = \frac{AB}{AC}$$

$$\frac{y}{z} = \frac{AB}{AC} \quad \dots(ii)$$

Adding (i) and (ii)

$$\frac{y}{x} + \frac{y}{z} = \frac{BC}{AC} + \frac{AB}{AC}$$

$$y\left(\frac{1}{x} + \frac{1}{z}\right) = \frac{BC + AB}{AC}$$

$$y\left(\frac{1}{x} + \frac{1}{z}\right) = \frac{AC}{AC}$$

$$y\left(\frac{1}{x} + \frac{1}{z}\right) = 1$$

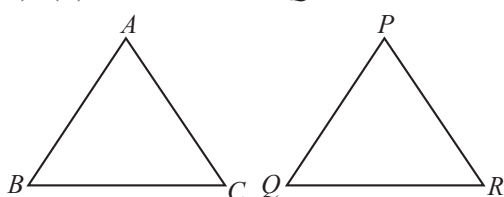
$$\frac{1}{x} + \frac{1}{z} = \frac{1}{y} \text{ Hence proved.}$$

36. (i)

Figure A and figure C

(ii) Figure C

(iii) (A) Let $\triangle ABC \cong \triangle PQR$



By CPCT,

$$\angle ABC = \angle PQR, \angle BAC = \angle QPR$$

Using AA similarity criterion,

$$\triangle ABC \sim \triangle PQR$$

Conserve may not be true, for example

Let ABC and PQR be two triangles such that

$$AB = 3 \text{ cm}, BC = 4 \text{ cm}, AC = 5 \text{ cm}$$

$$PQ = 6 \text{ cm}, QR = 8 \text{ cm}, PR = 10 \text{ cm}$$

$$\text{Hence, } \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = \frac{1}{2}$$

$$\triangle ABC \sim \triangle PQR \quad [\text{By SSS similarity criterion}]$$

But $\triangle ABC \not\cong \triangle PQR$ as $AB \neq PQ$.

OR

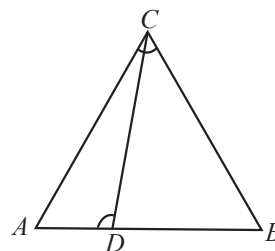
(B) For two similar triangles to be congruent, ratio of length of corresponding sides should be 1 : 1.

i.e. if $\triangle ABC \sim \triangle PQR$, then

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = 1$$

$$\Rightarrow \triangle ABC \cong \triangle PQR$$

37. In $\triangle ACB$ and $\triangle ADC$



$$\angle CAB = \angle CAD \quad [\text{Common}]$$

$$\angle ADC = \angle BCA \quad [\text{Given}]$$

$$\therefore \triangle ACB \sim \triangle ADC \quad [\text{By AA similarity criterion}]$$

$$\frac{AC}{AD} = \frac{BC}{CD} = \frac{AB}{AC}$$

$$\Rightarrow AC \times AC = AD \times AB$$

$$\Rightarrow 8 \times 8 = 3(AB) \Rightarrow AB = \frac{64}{3}$$

$$\therefore BD = AB - AD = \frac{64}{3} - 3 = \frac{55}{3}$$

$$\therefore BD = \frac{55}{3} \text{ cm}$$

7

CHAPTER

Coordinate Geometry



Previous Year Questions

Multiple Choice Questions

1. Point P divides the line segment joining the points $A(4, -5)$ and $B(1, 2)$ in the ratio $5 : 2$. Co-ordinates of point P are: [CBSE 2024]

- (a) $\left(\frac{5}{2}, \frac{-3}{2}\right)$ (b) $\left(\frac{11}{7}, 0\right)$
(c) $\left(\frac{13}{7}, 0\right)$ (d) $\left(0, \frac{13}{7}\right)$

2. If the distance between the points $(3, -5)$ and $(x, -5)$ is 15 units, then the values of x are: [CBSE 2024]

- (a) 12, -18 (b) -12, 18
(c) 18, 5 (d) -9, -12

3. The centre of a circle is at $(2, 0)$. If one end of a diameter is at $(6, 0)$, then the other end is at: [CBSE 2024]

- (a) $(0, 0)$ (b) $(4, 0)$
(c) $(-2, 0)$ (d) $(-6, 0)$

4. The fourth vertex D of a parallelogram $ABCD$ whose three vertices are $A(-2, 3)$, $B(6, 7)$ and $C(8, 3)$ is: [CBSE 2024]

- (a) $(0, 1)$ (b) $(0, -1)$
(c) $(-1, 0)$ (d) $(1, 0)$

5. **Assertion (A):** Mid-point of a line segment divides the line segment in the ratio $1 : 1$.

Reason (R): The ratio in which the point $(-3, k)$ divides the line segment joining the points $(-5, 4)$ and $(-2, 3)$ is $1 : 2$. [CBSE 2024]

6. **Assertion (A):** The point which divides the line

segment joining the points $A(1, 2)$ and $B(-1, 1)$ internally in the ratio $1 : 2$ is $\left(\frac{-1}{3}, \frac{5}{3}\right)$.

Reason (R): The coordinates of the point which divides the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the ratio $m_1 : m_2$ are

$$\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right) \quad [\text{CBSE 2024}]$$

7. The distance of the point $(-6, 8)$ from origin is [CBSE 2023]

- (a) 6 (b) -6
(c) 8 (d) 10

8. The distance of the point $(-1, 7)$ from x -axis is [CBSE 2023]

- (a) -1 (b) 7
(c) 6 (d) $\sqrt{50}$

9. The origin divides the line segment AB joining the points $A(1, -3)$ and $B(-3, 9)$ in the ratio: [CBSE 2021]

- (a) $3 : 1$ (b) $1 : 3$
(c) $2 : 3$ (d) $1 : 1$

10. A circle of radius 3 units is centred at $(0, 0)$. Which of the following points lie outside the circle? [CBSE 2021]

- (a) $(-1, -1)$ (b) $(0, 3)$
(c) $(1, 2)$ (d) $(3, 1)$

11. The mid-point of line segment joining the points $(-3, 9)$ and $(-6, -4)$ is [CBSE 2021]

- (a) $\left(\frac{-3}{2}, \frac{-13}{2}\right)$ (b) $\left(\frac{9}{2}, \frac{-5}{2}\right)$

(c) $\left(\frac{-9}{2}, \frac{5}{2}\right)$ (d) $\left(\frac{9}{2}, \frac{5}{2}\right)$

12. If $A(3, \sqrt{3})$, $B(0, 0)$ and $C(3, k)$ are the three vertices of an equilateral triangle ABC , then the value of k is [CBSE 2021]

(a) 2 (b) -3
(c) $-\sqrt{3}$ (d) $-\sqrt{2}$

13. Three vertices of a parallelogram $ABCD$ are $A(1, 4)$, $B(-2, 3)$ and $C(5, 8)$. The ordinate of the fourth vertex D is [CBSE 2021]

(a) 8 (b) 9
(c) 7 (d) 6

14. Points $A(-1, y)$ and $B(5, 7)$ lie on a circle with centre $O(2, -3y)$. The values of y are [CBSE 2021]

(a) 1, -7 (b) -1, 7
(c) 2, 7 (d) -2, -7

15. The ratio in which the line $3x + y - 9 = 0$ divides the line segment joining the points $(1, 3)$ and $(2, 7)$ is [CBSE 2021]

(a) 3 : 2 (b) 2 : 3
(c) 3 : 4 (d) 4 : 3

16. If $A(4, -2)$, $B(7, -2)$ and $C(7, 9)$ are the vertices of a $\triangle ABC$, then $\triangle ABC$ is [CBSE 2021]

(a) Equilateral triangle
(b) Isosceles triangle
(c) Right angled triangle
(d) Isosceles right angled triangle

17. The line segment joining the points $P(-3, 2)$ and $Q(5, 7)$ is divided by the y -axis in the ratio [CBSE 2021]

(a) 3 : 1 (b) 3 : 4
(c) 3 : 2 (d) 3 : 5

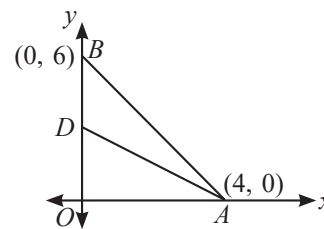
18. A point $(x, 1)$ is equidistant from $(0, 0)$ and $(2, 0)$. Then the value of x is: [CBSE 2021]

(a) 1 (b) 0
(c) 2 (d) $1/2$

19. The ratio in which the point $(4, 0)$ divides the line segment joining the points $(4, 6)$ and $(4, -8)$ is [CBSE 2021]

(a) 1 : 2 (b) 3 : 4
(c) 4 : 3 (d) 1 : 1

20. The vertices of a triangle OAB are $O(0, 0)$, $A(4, 0)$ and $B(0, 6)$. The median AD is drawn on OB . The length AD is [CBSE 2021]



(a) $\sqrt{52}$ units (b) 5 units
(c) 25 units (d) 10 units

21. The distance between the points $(a\cos\theta + b\sin\theta, 0)$ and $(0, a\sin\theta - b\cos\theta)$, is [CBSE 2020]

(a) $a^2 + b^2$ (b) $a^2 - b^2$
(c) $\sqrt{a^2 + b^2}$ (d) $\sqrt{a^2 - b^2}$

22. If the point $P(k, 0)$ divides the line segment joining the points $A(2, -2)$ and $B(-7, 4)$ in the ratio 1 : 2, then the value of k is [CBSE 2020]

(a) 1 (b) 2
(c) -2 (d) -1

23. Distance of point $P(3, 4)$ from x -axis is [CBSE 2020]

(a) 3 units (b) 4 units
(c) 5 units (d) 1 unit

24. If the distance between the points $A(4, p)$ and $B(1, 0)$ is 5 units, then the value(s) of p is (are) [CBSE 2020]

(a) 4 only (b) -4 only
(c) ± 4 (d) 0

Short Answer Type Questions

25. $XOYZ$ is a rectangle with vertices $X(-3, 0)$, $O(0, 0)$, $Y(0, 4)$ and $Z(x, y)$. The length of its each diagonal is: [CBSE 2024]

26. Find the type of triangle ABC formed whose vertices are $A(1, 0)$, $B(-5, 0)$ and $C(-2, 5)$. [CBSE 2024]

27. (a) In what ratio is the line segment joining the points $(3, -5)$ and $(-1, 6)$ divided by the line $y = x$? [CBSE 2024]

OR

- (b) $A(3, 0)$, $B(6, 4)$ and $C(-1, 3)$ are vertices of a triangle ABC . E is the mid-point of AC . Calculate the length of median BE . [CBSE 2024]

28. (a) Find a relation between x and y such that the point $P(x, y)$ is equidistant from the points $A(7, 1)$ and $B(3, 5)$. [CBSE 2024]

OR

(b) Points $A(-1, y)$ and $B(5, 7)$ lie on a circle with centre $O(2, -3y)$ such that AB is a diameter of the circle. Find the value of y . Also, find the radius of the circle. [CBSE 2024]

29. (a) Find the ratio in which the line segment joining the points $(5, 3)$ and $(-1, 6)$ is divided by y -axis.

[CBSE 2024]

OR

(b) $P(-2, 5)$ and $Q(3, 2)$ are two points. Find the coordinates of the point R on line segment PQ such that $PR = 2QR$. [CBSE 2024]

30. (a) Find the ratio in which the point $P(-4, 6)$ divides the line segment joining the points $A(-6, 10)$ and $B(3, -8)$. [CBSE 2024]

OR

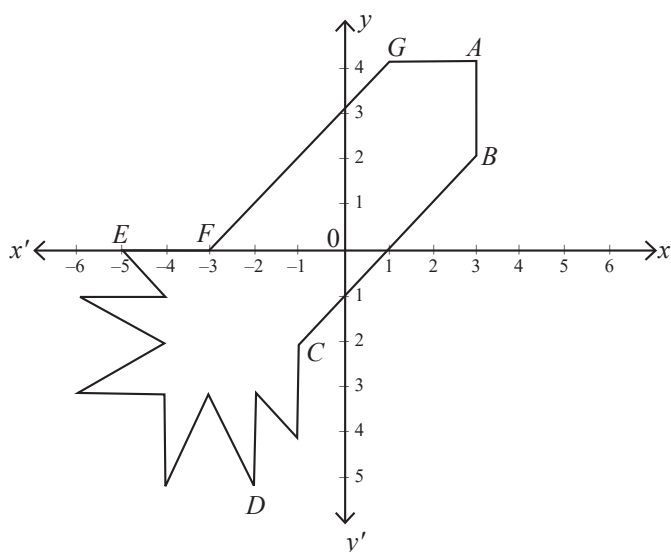
30. (b) Prove that the points $(3, 0)$, $(6, 4)$ and $(-1, 3)$ are the vertices of an isosceles triangle.

[CBSE 2024]

Case Based Type Questions

31. Ryan, from a very young age, was fascinated by the twinkling of stars and the vastness of space. He always dreamt of becoming an astronaut one day. So he started to sketch his own rocket designs on the graph sheet. One such design is given below:

[CBSE 2024]



Based on the above, answer the following questions:

- (i) Find the mid-point of the segment joining F and G .
(ii) (a) What is the distance between the points A and C ?

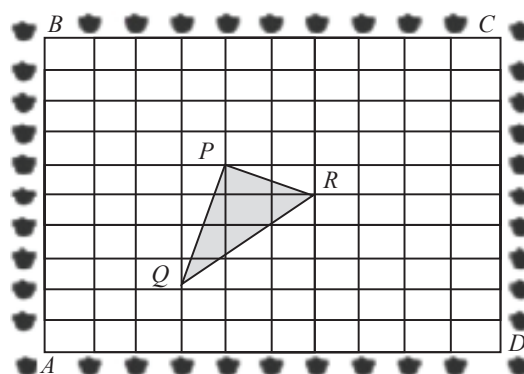
OR

(b) Find the coordinates of the point which divides the line segment joining the points A and B in the ratio $1 : 3$ internally.

- (iii) What are the coordinates of the point D ?

32. A garden is in the shape of a square. The gardener grew saplings of Ashoka tree on the boundary of the garden at the distance of 1 m from each other. He wants to decorate the garden with rose plants. He chose a triangular region inside the garden to grow rose plants. In the above situation, the gardener took help from the students of class 10. They made a chart for it which looks like the given figure.

[CBSE 2024]



Based on the above, answer the following questions:

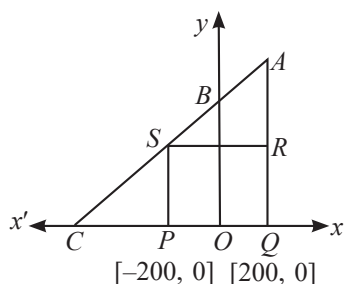
- (i) If A is taken as origin, what are the coordinates of the vertices of ΔPQR ?
(ii) (a) Find distances PQ and QR .

OR

(b) Find the coordinates of the point which divides the line segment joining points P and R in the ratio $2 : 1$ internally.

- (iii) Find out if ΔPQR is an isosceles triangle.

33. Jagdish has a field which is in the shape of a right angled triangle AQC . He wants to leave a space in the form of a square $PQRS$ inside the field for growing wheat and the remaining for growing vegetables (as shown in the figure). In the field, there is a pole marked as O .



Based on the above information, answer the following questions:

- (i) Taking O as origin, coordinates of P are $(-200, 0)$ and of Q are $(200, 0)$. $PQRS$ being a square, what are the coordinates of R and S ?

[CBSE 2023]

- (ii) (a) What is the area of square $PQRS$?

[CBSE 2023]

OR

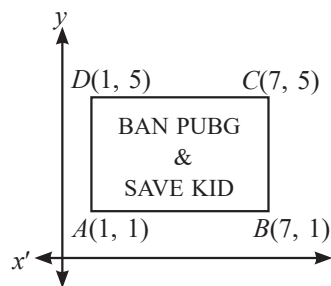
- (b) What is the length of diagonal PR in square $PQRS$?

[CBSE 2023]

- (iii) If S divides CA in the ratio $K : 1$, what is the value of K , where point A is $(200, 800)$?

[CBSE 2023]

34. Use of mobile screen for long hours makes your eye sight weak and give you headaches. Children who are addicted to play “PUBG” can get easily stressed out. To raise social awareness about ill effects of playing PUBG, a school decided to start ‘BAN PUBG’ campaign, in which students are asked to prepare campaign board in the shape of a rectangle. One such campaign board made by class X student of the school is shown in the figure.



Based on the above information, answer the following questions:

- (i) Find the coordinates of the point of intersection of diagonals AC and BD . [CBSE 2023]
 (ii) Find the length of the diagonal AC . [CBSE 2023]
 (iii) (a) Find the area of the campaign Board $ABCD$. [CBSE 2023]

OR

- (b) Find the ratio of the length of side AB to the length of the diagonal AC . [CBSE 2023]

Long Answer Type Questions

35. (a) Find the ratio in which the point $\left(\frac{8}{5}, y\right)$ divides the line segment joining the points $(1, 2)$ and $(2, 3)$. Also, find the value of y . [CBSE 2024]

OR

36. (b) $ABCD$ is a rectangle formed by the points $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$, P , Q , R and S are mid-points of sides AB , BC , CD and DA respectively. Show that diagonals of the quadrilateral $PQRS$ bisect each other. [CBSE 2024]
 37. (a) Find the coordinates of the point which divides the join of $A(-1, 7)$ and $B(4, -3)$ in the ratio $2 : 3$. [CBSE 2023]

OR

- (b) If the points $A(2, 3)$, $B(-5, 6)$, $C(6, 7)$ and $D(p, 4)$ are the vertices of a parallelogram $ABCD$, find the value of p . [CBSE 2023]



Explanations

Previous Year Questions

Multiple Choice Questions

- (c) $\left(\frac{13}{7}, 0\right)$
- (b) $-12, 18$
- (c) $(-2, 0)$
- (b) $(0, -1)$
- (c) Assertion (A) is true but Reason (R) is false.
- (d) Assertion (A) is false, but Reason (R) is true.
- (d) Required distance
 $= \sqrt{(-6)^2 + (8)^2} = \sqrt{100} = 10 \text{ units}$
- (b) Distance of point $(-1, 7)$ from x -axis is 7.
- (b) Let $(0, 0)$ divides the line segment AB in $k : 1$

$$\therefore \frac{1-3k}{k+1} = 0 \text{ and } \frac{-3+9k}{k+1} = 0$$

$$\Rightarrow k = \frac{1}{3}$$

Required ratio = $1 : 3$

- (d) Let $O(0, 0)$ be centre of circle.
 and $A(-1, -1)$, $B(0, 3)$, $C(1, 2)$, $D(3, 1)$
 $OA = \sqrt{(0+1)^2 + (0+1)^2} = \sqrt{2} \text{ units}$
 $OB = \sqrt{(0+0)^2 + (3-0)^2} = 3 \text{ units}$
 $OC = \sqrt{(1-0)^2 + (2-0)^2} = \sqrt{5} \text{ units}$
 $OD = \sqrt{(3-0)^2 + (1-0)^2} = \sqrt{10} \text{ units} > 3 \text{ units}$
 So, $(3, 1)$ lies outside the circle.

- (c) Let $C(x, y)$ be the mid-point
 Applying mid-point formula

$$\begin{array}{ccc} \bullet & \bullet & \bullet \\ A(-3, 9) & C(x, y) & B(-6, -4) \end{array}$$

$$x = \frac{-3-6}{2} = \frac{-9}{2}$$

$$y = \frac{9-4}{2} = \frac{5}{2}$$

So, mid-point is $\left(\frac{-9}{2}, \frac{5}{2}\right)$.

- (c) A, B and C are the vertices of an equilateral triangle, then $AB = BC$

$$\sqrt{(3-0)^2 + (\sqrt{3}-0)^2} = \sqrt{(3-0)^2 + (k-0)^2}$$

$$\Rightarrow \sqrt{9+3} = \sqrt{9+k^2}$$

Squaring both sides,

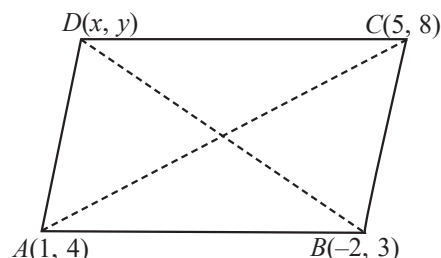
$$12 = 9 + k^2$$

$$k^2 = 3$$

$$\text{or } k = \pm\sqrt{3}$$

- (b) $ABCD$ is a parallelogram

Hence, O is the mid-point of both AC and BD .



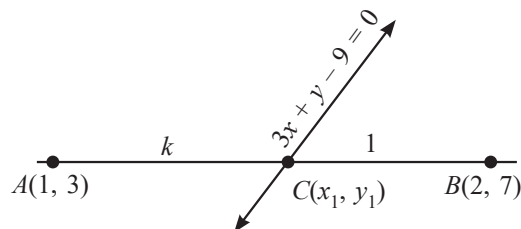
$\therefore$ For ordinate of point D ,

$$\frac{y+3}{2} = \frac{4+8}{2} \Rightarrow y = 9$$

- (b) $\sqrt{(2+1)^2 + (y+3y)^2} = \sqrt{(5-2)^2 + (7+3y)^2}$
 $\Rightarrow 9 + 16y^2 = 9 + 49 + 9y^2 + 42y$
 $\Rightarrow 7y^2 - 42y - 49 = 0$
 $\Rightarrow y^2 - 6y - 7 = 0$
 $\Rightarrow (y-7)(y+1) = 0$
 $\therefore y = 7, -1$

- (c) Let $3x + y - 9 = 0$ divides the line segment formed by joining the point $A(1, 3)$ and $B(2, 7)$ in $k : 1$

(i.e. at point C)



Now, $x_1 = \frac{2k+1}{k+1}$ and $y_1 = \frac{7k+3}{k+1}$

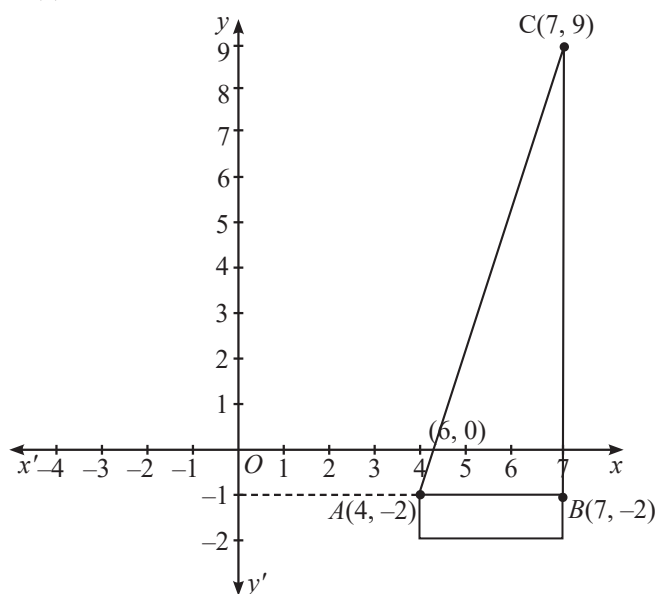
Point C lies on $3x + y - 9 = 0$, then

$$3\left(\frac{2k+1}{k+1}\right) + \left(\frac{7k+3}{k+1}\right) - 9 = 0$$

$$\Rightarrow 6k + 3 + 7k + 3 - 9k - 9 = 0$$

$$\Rightarrow 4k - 3 = 0 \Rightarrow k = \frac{3}{4}$$

16. (c)

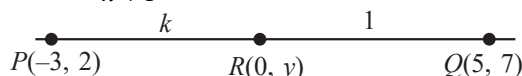


$\therefore \triangle ABC$ is a right angled triangle.

17. (d) Any point on y-axis is of the form $(0, y)$

Let R divides PQ in the ratio $k : 1$

$$\therefore 0 = \frac{5k + 1(-3)}{k + 1}$$



$$\Rightarrow 5k - 3 = 0 \text{ or } k = \frac{3}{5}$$

18. (a) $(x - 0)^2 + (1 - 0)^2 = (x - 2)^2 + (1 - 0)^2$

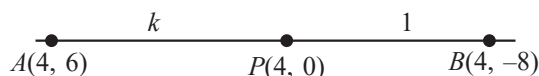
$$x^2 + 1 = x^2 + 4 - 4x + 1$$

$$x = 1$$

19. (b) Let $P(4, 0)$ divides $A(4, 6)$ and $B(4, -8)$ in $k : 1$

Applying section formula

$$\therefore 4 = \frac{k(4) + 1(4)}{k + 1}$$



$$0 = \frac{-8k + 1(6)}{k + 1} \Rightarrow k = \frac{3}{4} \text{ or } 3 : 4$$

20. (b) $OD = \frac{OB}{2} = 3$ units

$$OA = 4 \text{ units}$$

[Given]

$$\therefore AD = \sqrt{OD^2 + OA^2} \quad [\because \angle AOD = 90^\circ]$$

$$AD = 5 \text{ units}$$

21. (c) Distance between $A(a \cos \theta + b \sin \theta, 0)$ and $B(0, a \sin \theta - b \cos \theta)$ is

$$\begin{aligned} AB &= \sqrt{((a \cos \theta + b \sin \theta) - 0)^2 + (0 - (a \sin \theta - b \cos \theta))^2} \\ &= \sqrt{(a \cos \theta + b \sin \theta)^2 + (a \sin \theta - b \cos \theta)^2} \\ &= \sqrt{a^2 + b^2} \end{aligned}$$

Option (c) is correct.

22. (d) $\therefore k = \frac{(1 \times -7) + (2 \times 2)}{1 + 2}$ [Using section formula]

$$k = -1$$

Hence, option (d) is correct.

23. (b) Distance of point $(3, 4)$ from x-axis is its y-coordinate

24. (c) $A(4, p)$, $B(1, 0)$

$$AB = 5$$

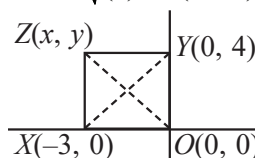
$$\therefore \sqrt{(4-1)^2 + (p-0)^2} = 5$$

$$\Rightarrow 9 + p^2 = 25$$

$$\Rightarrow p^2 = 16 \Rightarrow p = \pm 4$$

Short Answer Type Questions

25. $XY = \sqrt{(3)^2 + (4-0)^2} = 5 = OZ$



Length of diagonal = 5 units

26. $A(1, 0)$ $B(-5, 0)$ $C(-2, 5)$

$$AB = \sqrt{(-5-1)^2 + (0-0)^2} = 6$$

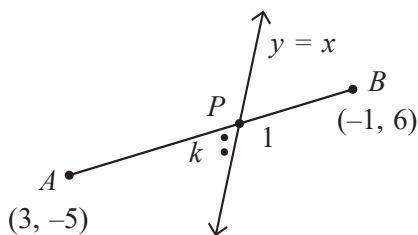
$$BC = \sqrt{(-5+2)^2 + (0-5)^2} = \sqrt{34}$$

$$CA = \sqrt{(1+2)^2 + (0-5)^2} = \sqrt{34}$$

$$\therefore BC = CA$$

So, $\triangle ABC$ is isosceles.

27. (a)



Let the required ratio be $K : 1$

Coordinates of point P are $\left(\frac{-K+3}{K+1}, \frac{6K-5}{K+1}\right)$

Point P lies on lines $y = x \Rightarrow \frac{-K+3}{K+1} = \frac{6K-5}{K+1}$

Solving, we get $K = \frac{8}{7}$
 $\therefore$ Required ratio is $8 : 7$

OR

27. (b) Mid-point of AC is $E(1, \frac{3}{2})$

Length of median BE

$$= \sqrt{(6-1)^2 + (4-\frac{3}{2})^2} = \sqrt{\frac{125}{4}} \text{ or } \frac{5\sqrt{5}}{2}$$

28. (a) $PA = PB$

$$\Rightarrow PA^2 = PB^2$$

$$(x-7)^2 + (y-1)^2 = (x-3)^2 + (y-5)^2$$

$$\Rightarrow -8x + 8y + 16 = 0 \text{ or } x - y - 2 = 0$$

OR

28. (b) Centre $O(2, -3y)$ is the mid point of AB

$$\therefore \frac{y+7}{2} = -3y$$

$$\Rightarrow y = -1$$

$$\text{Radius} = OB = \sqrt{(5-2)^2 + (7-3)^2} = 5$$

29. (a) Let the line segment intersects y -axis at $(0, y)$.

Let the required ratio be $k : 1$

$$\therefore 0 = \frac{(-1)k + 5(1)}{k+1}$$

$$\Rightarrow k = 5$$

Hence, ratio is $5 : 1$

OR

29. (b) Let coordinates of R be (x, y) ,

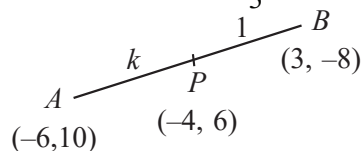
$$PR : RQ = 2 : 1$$

$$x = \frac{2(3)+1(-2)}{2+1} = \frac{4}{3}$$

$$y = \frac{2(2)+1(5)}{2+1} = 3$$

$\therefore$ Coordinates of the point $R(\frac{4}{3}, 3)$

30. (a)



Let the ratio be $k : 1$

$$-4 = \frac{3k-6}{k+1}$$

$$\Rightarrow k = \frac{2}{7}$$

$\therefore$ Required ratio is $2 : 7$

OR

30. (b) Let $A(3, 0)$, $B(6, 4)$, $C(-1, 3)$

$$AB = \sqrt{(3-6)^2 + (0-4)^2} = 5$$

$$BC = \sqrt{(6+1)^2 + (4-3)^2} = \sqrt{50}$$

$$CA = \sqrt{(3+1)^2 + (0-3)^2} = 5$$

As, $AB = AC$

$\Rightarrow ABC$ is an isosceles triangle.

Case Based Type Questions

31. (i) Mid-point of FG is $\left(\frac{-3+1}{2}, \frac{0+4}{2}\right) = (-1, 2)$

$$(ii) (a) AC = \sqrt{(-1-3)^2 + (-2-4)^2} \\ = \sqrt{52} \text{ or } 2\sqrt{13}$$

OR

(b) The coordinates of required point are

$$\left(\frac{1 \times 3 + 3 \times 3}{1+3}, \frac{1 \times 2 + 3 \times 4}{1+3}\right) \text{ i.e. } \left(3, \frac{7}{2}\right)$$

(iii) $D(-2, -5)$

32. (i) $P(4, 6)$, $Q(3, 2)$, $R(6, 5)$

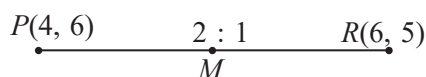
$$(ii) PQ = \sqrt{(4-3)^2 + (6-2)^2}$$

$$PQ = \sqrt{17}$$

$$QR = \sqrt{(6-3)^2 + (5-2)^2} = \sqrt{9+9} = 3\sqrt{2}$$

OR

(b) The coordinate of required point are



(text missing) i.e. $\left(\frac{16}{3}, \frac{16}{3}\right)$

$$(iii) PQ = \sqrt{(4-3)^2 + (6-2)^2} = \sqrt{17}$$

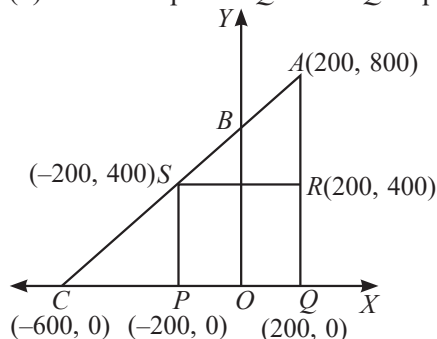
$$QR = \sqrt{(3-6)^2 + (2-5)^2} = \sqrt{18} = 3\sqrt{2}$$

$$PR = \sqrt{(4-6)^2 + (6-5)^2} = \sqrt{5}$$

$$PQ \neq QR \neq PR$$

33. (i) Coordinates of R are (200, 400)
 $[\because \text{Abscissa of } R = \text{Abscissa of } Q \text{ and side of square } PQRS = 400 \text{ units}]$
 Similarly, coordinates of S are (-200, 400)

- (ii) (a) Area of square PQRS = PQ^2 sq. units



$$400 \times 400 = 160000 \text{ sq. units}$$

$$(b) \sqrt{(200+200)^2 + (400)^2} = 400\sqrt{2} \text{ units}$$

- (iii) Here,
 $\Rightarrow$ Coordinate of C are (-600, 0),
 Coordinate of A are (200, 800),
 And Coordinate of S are (-200, 400)
 $\Rightarrow (-200, 400) = \left(\frac{200K + (-600)}{K+1}, \frac{800K + 0}{K+1}\right)$
 [Using section formula]
 $\Rightarrow 400K + 400 = 800K + 0$
 [Comparing y-coordinates]
 $\Rightarrow 400K = 400 \Rightarrow K = 1$

34. ABCD is a rectangle where A(1, 1), B(7, 1), C(7, 5) and D(1, 5)

- (i) Coordinates of the point of intersection of diagonals AC and BD = $\left(\frac{7+1}{2}, \frac{1+5}{2}\right) = (4, 3)$

- (ii) Length of diagonal AC = $\sqrt{(7-1)^2 + (5-1)^2}$
 $= \sqrt{16+36} = \sqrt{52} = 2\sqrt{13} \text{ units}$

- (iii) (a) AB = 6 units = CD
 BC = 4 units = AD

Area of compaign Board ABCD

$$= 6 \times 4 = 24 \text{ sq. units}$$

OR

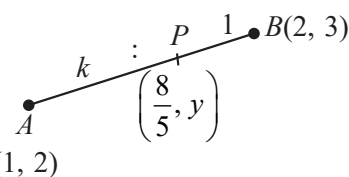
- (b) Ratio of the length of side AB to the length of the diagonal AC = $\frac{AB}{AC}$

$$= \frac{6}{\sqrt{(5-1)^2 + (7-1)^2}} = \frac{5}{\sqrt{16+36}}$$

$$= \frac{6}{\sqrt{52}} = \frac{6}{2\sqrt{13}} = \frac{3\sqrt{13}}{13} = 3\sqrt{13}:13$$

Long Answer Type Questions

35. (a)



Let AP : PB = k : 1

$$\therefore \frac{2k+1}{k+1} = \frac{8}{5}$$

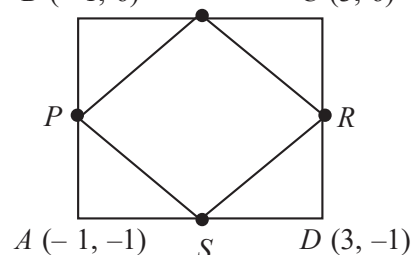
$$\Rightarrow k = \frac{3}{2}$$

$\therefore$ required ratio is 3 : 2.

$$y = \frac{3 \times 3 + 2 \times 2}{3 + 2} = \frac{13}{5}$$

OR

36. (b) B(-1, 6) Q C(3, 6)



Co-ordinates of point P are $\left(\frac{-1-1}{2}, \frac{-1+6}{2}\right)$
 i.e. $\left(-1, \frac{5}{2}\right)$

Co-ordinates of point Q are $\left(\frac{-1+3}{2}, \frac{6+6}{2}\right)$
 i.e. (1, 6)

Co-ordinates of point R are $\left(\frac{3+3}{2}, \frac{6-1}{2}\right)$
 i.e. $\left(3, \frac{5}{2}\right)$

Co-ordinates of point S are $\left(\frac{-1+3}{2}, \frac{-1-1}{2}\right)$

i.e. $(1, -1)$

Co-ordinates of mid point of diagonal QS are

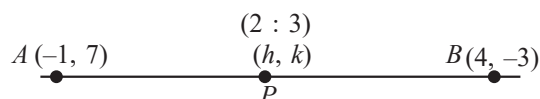
$$\left(\frac{1+1}{2}, \frac{6-1}{2}\right) \text{ i.e. } \left(1, \frac{5}{2}\right)$$

Co-ordinates of mid point of diagonal PR are

$$\left(\frac{-1+3}{2}, \frac{\frac{5}{2}+\frac{5}{2}}{2}\right) \text{ i.e. } \left(1, \frac{5}{2}\right)$$

Since, coordinates of mid point of QS = coordinates of mid point of PR . Therefore, diagonals PR and QS bisect each other.

37. (a) Let the point be $P(h, k)$



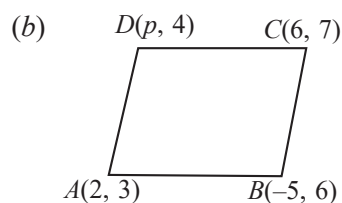
using section formula,

$$h = \frac{2(4) + 3(-1)}{2 + 3} = \frac{5}{5} = 1$$

$$k = \frac{2(-3) + 3(7)}{2 + 3} = 3$$

$$P(h, k) = P(1, 3)$$

OR



$ABCD$ is a prallelogram

$$\Rightarrow AB = CD$$

$$\Rightarrow \sqrt{(6-3)^2 + (-5-2)^2} = \sqrt{(7-4)^2 + (6-p)^2}$$

$$\Rightarrow 9 + 49 = 9 + (6-p)^2$$

$$\Rightarrow 6-p = \pm 7$$

$$\Rightarrow p = 6-7 \text{ and } p = 6+7$$

$$\Rightarrow p = -1, 13$$

$$\Rightarrow p = -1 \text{ (Not possible as in this case } AD \neq BC)$$

$$\Rightarrow p = 13$$

8

CHAPTER

Introduction to Trigonometry



Previous Year Questions

Multiple Choice Questions

1. In a right-angled $\triangle ABC$, $A = 90^\circ$ and $AB = AC$. The value of $\sin C$ is:

(a) 0
(b) $\frac{\sqrt{3}}{2}$
(c) $\frac{1}{2}$
(d) $\frac{1}{\sqrt{2}}$

2. If $\cos A = \frac{1}{2}$, then $\tan A$ is equal to:

(a) $\frac{1}{\sqrt{3}}$
(b) $\sqrt{3}$
(c) 3
(d) $\frac{3}{2}$

3. If $\sin A = \frac{3}{5}$, then value of $\cot A$ is:

(a) $\frac{3}{4}$
(b) $\frac{4}{3}$
(c) $\frac{4}{5}$
(d) $\frac{5}{4}$

4. $\frac{1 + \tan^2 A}{1 + \cot^2 A}$ is equal to:

(a) $\sec^2 A$
(b) -1
(c) $\cot^2 A$
(d) $\tan^2 A$

5. $\frac{2\tan 30^\circ}{1 - \tan^2 30^\circ}$ is equal to:

(a) $\cos 60^\circ$
(b) $\sin 60^\circ$
(c) $\tan 60^\circ$
(d) $\sin 30^\circ$

6. $\left(\frac{1}{\tan^2 \theta} - \frac{1}{\sin^2 \theta} \right)$ is equal to:

(a) 1
(b) -1
(c) $\sec^2 \theta$
(d) $\sin^2 \theta$

7. $\left(\frac{5}{\cot^2 \theta} - \frac{5}{\cos^2 \theta} \right)$ is equal to:

(a) 1
(b) 5
(c) -5
(d) 0

8. If $\tan^2 \theta = 3$, where θ is an acute angle, then the value of θ is:

(a) 30°
(b) 60°
(c) 0°
(d) 45°

9. If $\sin \theta = \frac{1}{3}$, then $\sec \theta$ is equal to:

(a) $\frac{2\sqrt{2}}{3}$
(b) $\frac{3}{2\sqrt{2}}$
(c) 3
(d) $\frac{1}{\sqrt{3}}$

10. For what value of θ , $\sin^2 \theta + \sin \theta + \cos^2 \theta$ is equal to 2?

(a) 45°
(b) 0°
(c) 90°
(d) 30°

11. $\left(\frac{2}{3} \sin 0^\circ - \frac{4}{5} \cos 0^\circ \right)$ is equal to: [CBSE 2023]

(a) $\frac{2}{3}$
(b) $-\frac{4}{5}$
(c) 0
(d) $-\frac{2}{15}$

12. Which of the following is true for all values of θ ($0^\circ \leq \theta \leq 90^\circ$)? [CBSE 2023]

(a) $\cos^2 \theta - \sin^2 \theta = 1$
(b) $\operatorname{cosec}^2 \theta - \sec^2 \theta = 1$
(c) $\sec^2 \theta - \tan^2 \theta = 1$
(d) $\cot^2 \theta - \tan^2 \theta = 1$

13. The value of $(\tan^2 45^\circ - \cos^2 60^\circ)$ is [CBSE 2021]

(a) $1/2$
(b) $1/4$
(c) $3/2$
(d) $3/4$

14. Which of the following is not defined [CBSE 2021]

- (a) $\sec 0^\circ$ (b) $\operatorname{cosec} 90^\circ$
(c) $\tan 90^\circ$ (d) $\cot 90^\circ$

15. In a right-angled triangle PQR $\angle Q = 90^\circ$. If $\angle P = 45^\circ$, then value of $\tan P - \cos^2 R$ is

[CBSE 2021]

- (a) 0 (b) 1
(c) $1/2$ (d) $3/2$

16. If $\tan \theta = \frac{2}{3}$, then the value of $\sec \theta$ is

[CBSE 2021]

- (a) $\frac{\sqrt{13}}{3}$ (b) $\frac{\sqrt{5}}{3}$
(c) $\sqrt{\frac{13}{3}}$ (d) $\frac{3}{\sqrt{13}}$

17. If $\sin \theta - \cos \theta = 0$, then the value of θ is

[CBSE 2021]

- (a) 30° (b) 45°
(c) 90° (d) 0°

18. $\frac{1}{1+\sin \theta} + \frac{1}{1-\sin \theta}$ can be simplified to get

[CBSE 2021]

- (a) $2\cos^2 \theta$ (b) $\frac{1}{2}\sec^2 \theta$
(c) $\frac{2}{\sin^2 \theta}$ (d) $2\sec^2 \theta$

19. $(1 + \tan^2 A)(1 + \sin A)(1 - \sin A)$ is equal to

[CBSE 2021]

- (a) $\frac{\cos^2 A}{\sec^2 A}$ (b) 1
(c) 0 (d) 2

20. If $\cot \theta = \frac{1}{\sqrt{3}}$, the value of $\sec^2 \theta + \operatorname{cosec}^2 \theta$ is

[CBSE 2021]

- (a) 1 (b) $\frac{40}{9}$
(c) $\frac{38}{9}$ (d) $5\frac{1}{3}$

21. In $\triangle ABC$ right angled at B , $\sin A = \frac{7}{25}$, then the value of $\cos C$ is

[CBSE 2021]

- (a) $\frac{7}{25}$ (b) $\frac{24}{25}$
(c) $\frac{7}{24}$ (d) $\frac{24}{7}$

22. Given that $\sec \theta = \sqrt{2}$, the value of $\frac{1+\tan \theta}{\sin \theta}$ is

[CBSE 2021]

- (a) $2\sqrt{2}$ (b) $\sqrt{2}$
(c) $3\sqrt{2}$ (d) 2

23. If θ is an acute angle and $\tan \theta + \cot \theta = 2$, then the value of $\sin^3 \theta + \cos^3 \theta$ is

[CBSE 2021]

- (a) 1 (b) $\frac{1}{2}$
(c) $\frac{\sqrt{2}}{2}$ (d) $\sqrt{2}$

24. If $a \cot \theta + b \operatorname{cosec} \theta = p$ and $b \cot \theta + a \operatorname{cosec} \theta = q$, then $p^2 - q^2 =$

[CBSE 2021]

- (a) $a^2 - b^2$ (b) $b^2 - a^2$
(c) $a^2 + b^2$ (d) $b - a$

25. If $\sec \theta + \tan \theta = p$, then $\tan \theta$ is

[CBSE 2021]

- (a) $\frac{p^2+1}{2p}$ (b) $\frac{p^2-1}{2p}$
(c) $\frac{p^2-1}{p^2+1}$ (d) $\frac{p^2+1}{p^2-1}$

Short Answer Type Questions

26. Evaluate : $4\sin^2 60^\circ \tan^2 45^\circ - 2\sec^2 30^\circ \tan^2 60^\circ$

27. (a) Prove that $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta$

28. If $\cos A = \frac{5}{13}$, then verify that

$$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} = \cos A + \sin A$$

29. Prove that $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$

30. Evaluate : $\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ - \cot 45^\circ$

31. Prove that:

$$(\tan A + \sec A)^2 + (\tan A - \sec A)^2 = 2 \left[\frac{1 + \sin^2 A}{1 - \sin^2 A} \right]$$

32. Evaluate: $\sin^2 30^\circ + \cos^2 45^\circ - \cos 0^\circ \cdot \tan 45^\circ$

33. Evaluate: $\sin A \cos B + \cos A \sin B$; if $A = 30^\circ$ and $B = 45^\circ$

34. Prove that $(\cot \theta - \operatorname{cosec} \theta)^2 = \frac{1 - \cos \theta}{1 + \cos \theta}$

35. Prove that $\frac{1 + \tan^2 A}{1 + \cot^2 A} = \sec^2 A - 1$ [CBSE 2023]

36. If $\sin\alpha = \frac{1}{2}$, then find the value of $(3\cos\alpha - 4\cos^3\alpha)$.
[CBSE 2023]

37. (a) Evaluate :

$$\frac{5}{\cot^2 30^\circ} + \frac{1}{\sin^2 60^\circ} - \cot^2 45^\circ + 2\sin^2 90^\circ$$

[CBSE 2023]

OR

(b) If θ is an acute angle and $\sin\theta = \cos\theta$, find the value of $\tan^2\theta + \cot^2\theta - 2$. [CBSE 2023]

38. If $\sin\theta + \cos\theta = \sqrt{3}$, then prove that $\tan\theta + \cot\theta = 1$.
[CBSE 2020]

39. If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \frac{1}{\sqrt{3}}$, $A > B$, then the value of A is [CBSE 2020]

40. If $5\tan\theta = 3$, then what is the value of $\left(\frac{5\sin\theta - 3\cos\theta}{4\sin\theta + 3\cos\theta}\right)$? [CBSE 2020]

41. The value of $\left(\sin^2\theta + \frac{1}{1+\tan^2\theta}\right) =$. [CBSE 2020]

OR

The value of $(1 + \tan^2\theta)(1 - \sin\theta)(1 + \sin\theta) =$ [CBSE 2020]

42. Prove that : $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$. [CBSE 2019]

43. Prove that : $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$. [CBSE 2019]

44. Prove that $\frac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A$ [CBSE 2018]

Long Answer Type Questions

45. (a) Prove that $\frac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A$ [CBSE 2023]

OR

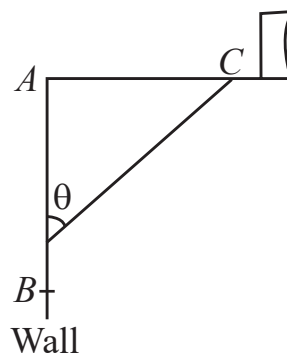
(b) Prove that $\sec A(1 - \sin A)(\sec A + \tan A) = 1$

[CBSE 2023]

46. Prove that $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}} = \sec\theta - \tan\theta$. [CBSE 2020]
OR

Prove that $\frac{\tan^2\theta}{1+\tan^2\theta} + \frac{\cot^2\theta}{1+\cot^2\theta} = 1$ [CBSE 2020]

47. The rod AC of a TV disc antenna is fixed at right angles to the wall AB and a rod CD is supporting the disc as shown in Figure. If $AC = 1.5$ m long and $CD = 3$ m, find (i) $\tan\theta$ (ii) $\sec\theta + \operatorname{cosec}\theta$
[CBSE 2020]



48. Prove that $(1 + \tan A - \sec A) \times (1 + \tan A + \sec A) = 2\tan A$ [CBSE 2020]

OR

Prove that $\frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta - 1} + \frac{\operatorname{cosec}\theta}{\operatorname{cosec}\theta + 1} = 2\sec^2\theta$ [CBSE 2020]

49. If $4\tan\theta = 3$, evaluate $\left(\frac{4\sin\theta - \cos\theta + 1}{4\sin\theta + \cos\theta - 1}\right)$ [CBSE 2018]

50. Prove that $\frac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = \frac{1}{\sec A - \tan A}$ [CBSE 2019]

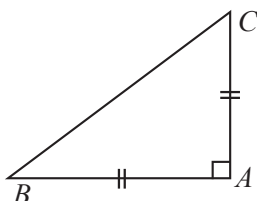


Explanations

Previous Year Questions

Multiple Choice Questions

1.



Let $AC = x$, then $AB = x$

In $\triangle BAC$,

$$BC^2 = AB^2 + AC^2$$

$$BC^2 = x^2 + x^2 = 2x^2$$

$$BC = \sqrt{2}x$$

$$\sin C = \frac{AB}{BC} = \frac{x}{\sqrt{2}x}$$

$$\boxed{\sin C = \frac{1}{\sqrt{2}}}$$

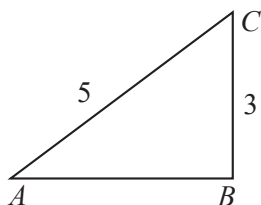
2. $\cos A = \frac{1}{2}$

$$\cos A = \cos 60^\circ$$

$$A = 60^\circ$$

$$\tan 60^\circ = \sqrt{3}$$

3. In $\triangle ABC$,



$$AC^2 = AB^2 + BC^2$$

$$25 = AB^2 + 9$$

$$AB = 4$$

$$\cot A = \frac{AB}{BC} = \frac{4}{3}$$

4. $\frac{1 + \tan^2 A}{1 + \cot^2 A}$

$$= \frac{1 + \frac{\sin^2 A}{\cos^2 A}}{1 + \frac{\cos^2 A}{\sin^2 A}} = \frac{\frac{\cos^2 A + \sin^2 A}{\cos^2 A}}{\frac{\sin^2 A + \cos^2 A}{\sin^2 A}}$$

$$= \frac{1}{\cos^2 A} \cdot \sin^2 A = \tan^2 A$$

5. $\frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ}$

$$\Rightarrow \frac{\frac{2}{\sqrt{3}}}{1 - \left(\frac{1}{\sqrt{3}}\right)^2} = \frac{\frac{2}{\sqrt{3}}}{1 - \frac{1}{3}} = \frac{2}{\sqrt{3}} \times \frac{3}{2} = \frac{3}{\sqrt{3}}$$

rationalising the denominator

$$\frac{3}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \sqrt{3} = \tan 60^\circ$$

6. $\frac{1}{\tan^2 \theta} - \frac{1}{\sin^2 \theta}$

$$= \frac{\cos^2 \theta}{\sin^2 \theta} - \frac{1}{\sin^2 \theta} = \frac{\cos^2 \theta - 1}{\sin^2 \theta}$$

$$= \frac{-(1 - \cos^2 \theta)}{\sin^2 \theta} = \frac{-\sin^2 \theta}{\sin^2 \theta} = -1$$

7. $\frac{5}{\cot^2 \theta} - \frac{5}{\cos^2 \theta}$

$$5 \left[\frac{\sin^2 \theta}{\cos^2 \theta} - \frac{1}{\cos^2 \theta} \right] = 5 \left[\frac{\sin^2 \theta - 1}{\cos^2 \theta} \right]$$

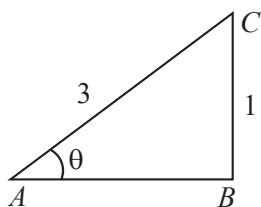
$$= -5 \left[\frac{1 - \sin^2 \theta}{\cos^2 \theta} \right] = -5 \frac{\cos^2 \theta}{\cos^2 \theta} = -5$$

8. $\tan^2 \theta = 3$

$$\tan \theta = \sqrt{3} = \tan 60^\circ$$

$$\theta = 60^\circ$$

9. $\sin \theta = \frac{1}{3}$



$$(AC)^2 = (AB)^2 + (BC)^2$$

$$(3)^2 = AB^2 + 1$$

$$AB^2 = 8$$

$$AB = 2\sqrt{2}$$

$$\sec\theta = \frac{AC}{AB} = \frac{3}{2\sqrt{2}}$$

$$10. \sin^2\theta + \sin\theta + \cos^2\theta = 2$$

$$1 + \sin\theta = 2$$

$$\sin\theta = 1 = \sin 90^\circ$$

$$\sin\theta = \sin 90^\circ$$

$$\theta = 90^\circ$$

$$11. (b) \frac{2}{3} \times 0 - \frac{4}{5} \times 1 = -\frac{4}{5}$$

$$12. (c) \sec^2\theta - \tan^2\theta = 1$$

$$13. (d) \tan^2 45^\circ - \cos^2 60^\circ = 1^2 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}$$

$$14. (c) \tan 90^\circ = \frac{\sin 90^\circ}{\cos 90^\circ} = \frac{1}{0} = \text{Not defined}$$

$$15. (c) \angle R = 180^\circ - 90^\circ - 45^\circ = 45^\circ$$

$$\tan P - \cos^2 R = \tan 45^\circ - \cos^2 45^\circ$$

$$= 1 - \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

$$16. (a) \sec\theta = \sqrt{1 + \tan^2\theta} = \frac{\sqrt{13}}{3} \quad \left[\because \tan\theta = \frac{2}{3} \right]$$

$$17. (b) \sin\theta - \cos\theta = 0 \Rightarrow \sin\theta = \cos\theta$$

$$\Rightarrow \tan\theta = 1 \Rightarrow \theta = 45^\circ$$

$$18. (d) \frac{1}{1 + \sin\theta} + \frac{1}{1 - \sin\theta} = \frac{1 - \sin\theta}{1 - \sin^2\theta} + \frac{1 + \sin\theta}{1 - \sin^2\theta}$$

$$= \frac{1 - \sin\theta}{\cos^2\theta} + \frac{1 + \sin\theta}{\cos^2\theta}$$

$$= \frac{2}{\cos^2\theta} = 2\sec^2\theta$$

$$19. (b) (1 + \tan^2 A)(1 + \sin A)(1 - \sin A)$$

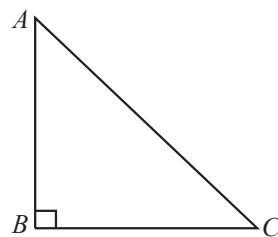
$$= \sec^2 A \times \cos^2 A = 1$$

$$20. (d) \sec^2\theta + \operatorname{cosec}^2\theta$$

$$= 1 + \tan^2\theta + 1 + \cot^2\theta$$

$$= 2 + \frac{1}{3} + 3 = 5\frac{1}{3}$$

$$21. (a) \sin A = \frac{7}{25} = \frac{BC}{AC}$$



i.e. $BC = 7a$ and $AC = 25a$, where a is any non-zero positive constant.

In $\triangle ABC$

$$\cos C = \frac{BC}{AC} = \frac{7}{25}$$

$$22. (a) \tan\theta = \pm\sqrt{\sec^2\theta - 1} = \pm\sqrt{2 - 1} = \pm 1$$

$$\text{and } \sin\theta = \pm\sqrt{1 - \frac{1}{\sec^2\theta}} = \pm\sqrt{1 - \frac{1}{2}} = \pm\sqrt{\frac{1}{2}}$$

$$\Rightarrow \frac{1 + \tan\theta}{\sin\theta} = \frac{(1 \pm 1)(\pm\sqrt{2})}{1} = 2\sqrt{2} \text{ or } 0$$

$$23. (c) \tan\theta + \cot\theta = 2$$

$$\Rightarrow \tan^2\theta + 1 = 2\tan\theta \quad \left[\because \cot\theta = \frac{1}{\tan\theta} \right]$$

$$\Rightarrow (\tan\theta + 1)^2 = 0 \Rightarrow \tan\theta - 1 = 0$$

$$\Rightarrow \tan\theta = 1 = \tan 45^\circ$$

$$\theta = 45^\circ$$

$$\therefore \sin^3\theta + \cos^3\theta = \sin^3 45^\circ + \cos^3 45^\circ$$

$$\left[\because \sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}} \right]$$

$$= \frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$24. (b) a\tan\theta + b\operatorname{cosec}\theta = p \quad \dots(i)$$

$$b\cos\theta + a\operatorname{cosec}\theta = q \quad \dots(ii)$$

Squaring both the equations and subtracting,

$$p^2 - q^2 = (a\cot\theta + b\operatorname{cosec}\theta)^2 - (b\cot\theta + a\operatorname{cosec}\theta)^2$$

$$= (a^2\cot^2\theta + b^2\operatorname{cosec}^2\theta + 2ab\cot\theta\operatorname{cosec}\theta) - (b^2\cot^2\theta + a^2\operatorname{cosec}^2\theta + 2ab\cot\theta\operatorname{cosec}\theta)$$

$$\Rightarrow (a^2 - b^2)(\cot^2\theta - \operatorname{cosec}^2\theta)$$

$$\Rightarrow (b^2 - a^2) \quad [\because \operatorname{cosec}^2\theta - \cot^2\theta = 1]$$

$$25. (b) \sec\theta + \tan\theta = p$$

$$\Rightarrow \sec \theta - \tan \theta = \frac{1}{p} \quad [\sec^2 \theta - \tan^2 \theta = 1]$$

$$\Rightarrow 2 \tan \theta = p - \frac{1}{p} \Rightarrow \tan \theta = \frac{p^2 - 1}{2p}$$

Short Answer Type Questions

26. $4 \sin^2 60^\circ \tan^2 45^\circ - 2 \sec^2 30^\circ \tan^2 60^\circ$

$$= 4 \left[\frac{\sqrt{3}}{2} \right]^2 (1)^2 - 2 \left[\frac{2}{\sqrt{3}} \right]^2 [\sqrt{3}]^2$$

$$4 \times \frac{3}{4} - 2 \times \frac{4}{3} \times 3$$

$$= 3 - 8 = -5$$

27. $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta$

L.H.S

$$\Rightarrow \sin^2 \theta + \operatorname{cosec}^2 \theta + 2 + \cos^2 \theta + \sec^2 \theta + 2 = 7 + \tan^2 \theta + \cot^2 \theta$$

$\Rightarrow$ rearranging the terms,

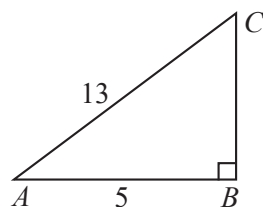
$$= 4 + (\sin^2 \theta + \cos^2 \theta) + \operatorname{cosec}^2 \theta + \sec^2 \theta = 7 + \tan^2 \theta + \cot^2 \theta.$$

$$= 5 + (1 + \cot^2 \theta) + (1 + \tan^2 \theta) = 7 + \tan^2 \theta + \cot^2 \theta.$$

$$= 7 + \tan^2 \theta + \cot^2 \theta = 7 + \tan^2 \theta + \cot^2 \theta$$

L.H.S. = R.H.S.

28.



$$\cos A = \frac{5}{13}$$

In $\triangle ABC$,

$$AC^2 = AB^2 + BC^2$$

$$(13)^2 = (5)^2 + (BC)^2$$

$$BC^2 = 169 - 25 = 144$$

$$BC = 12$$

$$\sin A = \frac{BC}{AC} = \frac{12}{13}$$

$$\tan A = \frac{BC}{AB} = \frac{12}{5} \Rightarrow \cot A = \frac{AB}{BC} = \frac{5}{12}$$

$$\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} = \cos A + \sin A$$

$$\frac{\frac{5}{13}}{1 - \frac{12}{5}} + \frac{\frac{12}{13}}{1 - \frac{5}{12}} = \frac{5}{13} + \frac{12}{13}$$

$$= \frac{\frac{5}{13}}{\frac{5-12}{5}} + \frac{\frac{12}{13}}{\frac{12-5}{12}} = \frac{17}{13}$$

$$= \frac{5}{13} \times \frac{5}{-7} + \frac{12}{13} \times \frac{12}{7}$$

$$= \frac{-25}{91} + \frac{144}{91} = \frac{119}{91}$$

$$\frac{17}{13} = \frac{17}{13}$$

L.H.S. = R.H.S.

29. $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$

$$\frac{\frac{\sin \theta}{\cos \theta}}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{1 - \frac{\sin \theta}{\cos \theta}}$$

$$= \frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} + \frac{\cos^2 \theta}{\sin \theta (\cos \theta - \sin \theta)}$$

$$= \frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} - \frac{\cos^2 \theta}{\sin \theta (\sin \theta - \cos \theta)}$$

$$= \frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta \cos \theta (\sin \theta - \cos \theta)}$$

$$= \frac{(\sin \theta - \cos \theta)(\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta)}{\sin \theta \cos \theta (\sin \theta - \cos \theta)}$$

$$[a^3 - b^3 = (a - b)(a^2 + b^2 + ab)]$$

$$= \frac{1 + \sin \theta \cos \theta}{\sin \theta \cos \theta} = \operatorname{cosec} \theta \sec \theta + 1.$$

L.H.S. = R.H.S.

30. $\frac{1}{2} \times \frac{1}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} - 1$

$$= \frac{1}{4} + \frac{3}{4} - 1 = 0$$

$$\begin{aligned}
 31. &= \left[\frac{\sin A}{\cos A} + \frac{1}{\cos A} \right]^2 + \left[\frac{\sin A}{\cos A} - \frac{1}{\cos A} \right]^2 \\
 &= \frac{(1 + \sin^2 A)^2}{\cos^2 A} + \frac{(\sin A - 1)^2}{\cos^2 A} \\
 &= \frac{1 + \sin^2 A + 2\sin^2 A + \sin^2 A + 1 - 2\sin A}{1 - \sin^2 A}
 \end{aligned}$$

$$= \frac{2 + 2\sin^2 A}{1 - \sin^2 A} = 2 \left[\frac{1 + \sin^2 A}{1 - \sin^2 A} \right]$$

L.H.S. = R.H.S.

$$\begin{aligned}
 32. &= \left(\frac{1}{2} \right)^2 + \left(\frac{1}{\sqrt{2}} \right)^2 - 1 \cdot 1 \\
 &= \frac{1}{4} + \frac{1}{2} - 1 = -\frac{1}{4}
 \end{aligned}$$

$$33. A = 30^\circ \text{ and } B = 45^\circ$$

$$\Rightarrow \sin A \cos B + \cos A \sin B$$

$$\Rightarrow \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ$$

$$\Rightarrow \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

34. Solving L.H.S.

$$\begin{aligned}
 &\left[\frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta} \right]^2 \\
 &= \frac{(\cos \theta - 1)^2}{\sin^2 \theta} = \frac{(\cos \theta - 1)^2}{1 - \cos^2 \theta} \\
 &= \frac{(\cos \theta - 1)(\cos \theta - 1)}{(1 + \cos \theta)(1 - \cos \theta)} = \frac{-(1 - \cos \theta)(\cos \theta - 1)}{(1 + \cos \theta)(1 - \cos \theta)} \\
 &= \frac{1 - \cos \theta}{1 + \cos \theta}
 \end{aligned}$$

L.H.S. = R.H.S.

35. Taking L.H.S.

$$\frac{1 + \tan^2 A}{1 + \cot^2 A} = \frac{1 + \frac{\sin^2 A}{\cos^2 A}}{1 + \frac{\cos^2 A}{\sin^2 A}}$$

$$\begin{aligned}
 &\frac{\cos^2 A + \sin^2 A}{\sin^2 A + \cos^2 A} = \frac{\sin^2 A}{\cos^2 A} \quad [\because \sin^2 A + \cos^2 A = 1] \\
 &= \frac{\sin^2 A}{\cos^2 A} \\
 &= \tan^2 A = \sec^2 A - 1 = \text{R.H.S.}
 \end{aligned}$$

$$36. \sin = \frac{1}{2} \Rightarrow \alpha = 30^\circ$$

$$\Rightarrow \cos \alpha = \frac{\sqrt{3}}{2}$$

$$\Rightarrow 3\cos \alpha - 4\cos^3 \alpha = 3\left(\frac{\sqrt{3}}{2}\right) - 4\left(\frac{\sqrt{3}}{2}\right)^3$$

$$= 3\left(\frac{\sqrt{3}}{2}\right) - \left(4 \times \frac{3\sqrt{3}}{8}\right) = 3\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{3\sqrt{3}}{2}\right) = 0$$

$$37. (a) \frac{5}{\cot^2 30^\circ} + \frac{1}{\sin^2 60^\circ} - \cot^2 45^\circ + 2\sin^2 90^\circ$$

$$= \frac{5}{(\sqrt{3})^2} + \frac{1}{\left(\frac{\sqrt{3}}{2}\right)^2} = (1)^2 + 2(1)^2$$

$$= \frac{5}{3} + \frac{4}{3} - 1 + 2 = 3 - 1 + 2 = 4$$

OR

$$(b) \sin \theta = \cos \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = \tan \theta = 1 \Rightarrow \theta = 45^\circ$$

$$\therefore \tan^2 45^\circ + \cot^2 45^\circ - 2 = 1^2 + 1^2 - 2 = 0$$

$$38. \sin \theta + \cos \theta = \sqrt{3}$$

On squaring both sides, we get

$$\Rightarrow \sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta = 3$$

$$\Rightarrow 1 + 2\sin \theta \cos \theta = 3$$

$$\Rightarrow \sin \theta \cos \theta = 1$$

Now,

$$\tan \theta + \cot \theta = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1}{2} = 1$$

$$39. \tan(A + B) = \sqrt{3}$$

$$\Rightarrow A + B = 60^\circ \quad \dots(i)$$

$$\text{Also, } \tan(A - B) = \frac{1}{\sqrt{3}}$$

$$\Rightarrow A - B = 30^\circ \quad \dots(ii) [\because A > B]$$

On adding (i) and (ii), we get

$$2A = 90^\circ \Rightarrow A = 45^\circ.$$

$$40. \tan \theta = \frac{3}{5}$$

$$\text{Now, } \frac{5 \sin \theta - 3 \cos \theta}{4 \sin \theta + 3 \cos \theta} = \frac{5 \tan \theta - 3}{4 \tan \theta + 3}$$

[Dividing numerator and denominator by $\cos \theta$]

$$\frac{3-3}{12+15} = 0$$

$$41. \sin^2 \theta + \frac{1}{\sec^2 \theta} = 2 \sin^2 \theta + \cos^2 \theta = 1$$

(using $\sec^2 \theta - \tan^2 \theta = 1$)

OR

$$(1 + \tan^2 \theta)(1 - \sin^2 \theta) \Rightarrow \sec^2 \theta \times \cos^2 \theta = 1$$

$$42. \text{L.H.S.: } (\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 \\ = \sin^2 \theta + \operatorname{cosec}^2 \theta + 2 + \cos^2 \theta + \sec^2 \theta + 2$$

$$\left[\because \sin \theta = \frac{1}{\operatorname{cosec} \theta} \text{ and } \cos \theta = \frac{1}{\sec \theta} \right]$$

$$= (\sin^2 \theta + \cos^2 \theta) + (1 + \cot^2 \theta) + (1 + \tan^2 \theta) + 4$$

$$[\cos^2 \theta + \sin^2 \theta = 1]$$

$$= 1 + 1 + 1 + 4 + \tan^2 \theta + \cot^2 \theta$$

$$[\because \operatorname{cosec}^2 \theta + 1 + \cot^2 \theta \text{ and } \sec^2 \theta = 1 + \tan^2 \theta]$$

$$= 7 + \tan^2 \theta + \cot^2 \theta = \text{R.H.S.}$$

$$43. \text{L.H.S.: } \left(1 + \frac{\cos A}{\sin A} - \frac{1}{\sin A}\right) \left(1 + \frac{\sin A}{\cos A} + \frac{1}{\cos A}\right)$$

$$= \left(\frac{\sin A + \cos A - 1}{\sin A}\right) \left(\frac{\cos A + \sin A + 1}{\cos A}\right)$$

$$= \frac{(\sin A + \cos A)^2 - (1)^2}{\sin A \cdot \cos A}$$

$$= \frac{\sin^2 A + \cos^2 A + 2 \sin A \cdot \cos A - 1}{\sin A \cdot \cos A}$$

$$[\because \sin^2 A + \cos^2 A = 1]$$

$$= 2 = \text{R.H.S.}$$

$$44. \text{L.H.S.} = \frac{\sin A - 2 \sin^3 A}{2 \cos^3 A - \cos A} = \frac{\sin A [1 - 2 \sin^2 A]}{\cos A [2 \cos^2 A - 1]}$$

$$= \frac{\sin A}{\cos A} \left(\frac{\sin^2 A + \cos^2 A - 2 \sin^2 A}{\cos^2 A - \sin^2 A - \cos^2 A} \right)$$

$$[\because \sin^2 A + \cos^2 A = 1]$$

$$= \tan A \left(\frac{\cos^2 A - 2 \sin^2 A}{\cos^2 A - \sin^2 A} \right) = \tan A = \text{R.H.S.}$$

Long Answer Type Questions

$$45. (a) \text{ To prove: } \frac{\sin A - 2 \sin^3 A}{2 \cos^3 A - \cos A} = \tan A$$

$$\text{LHS} = \frac{\sin A - 2 \sin^3 A}{2 \cos^3 A - \cos A} = \tan A$$

Dividing numerator and denominator by $\cos^3 A$, we get

$$\text{LHS} = \frac{\frac{\sin A}{\cos A \cdot \cos^2 A} - \frac{2 \sin^3 A}{\cos^3 A}}{\frac{2 \cos^3 A}{\cos^3 A} - \frac{\cos A}{\cos^3 A}}$$

$$= \frac{\sec^2 A \tan A - 2 \tan^3 A}{2 - \sec^2 A} = \frac{\tan A (\sec^2 A - 2 \tan^2 A)}{2 - 1 - \tan^2 A}$$

$$[\because \sec^2 A - \tan^2 A = 1]$$

$$= \frac{\tan A (1 - \tan^2 A)}{1 - \tan^2 A} = \tan A = \text{R.H.S.}$$

OR

$$(b) \text{ To prove: } \sec A (1 - \sin A) (\sec A + \tan A) = 1$$

$$\text{L.H.S.} = \sec A (1 - \sin A) (\sec A + \tan A)$$

$$= \sec A (1 - \sin A) \left(\frac{1}{\cos A} + \frac{\sin A}{\cos A} \right)$$

$$= \frac{\sec A}{\cos A} (1 - \sin A) (1 + \sin A) = \frac{1}{\cos^2 A} (1 - \sin^2 A)$$

$$= \frac{1}{\cos^2 A} \cdot \cos^2 A = 1 = \text{RHS} [\because \sin^2 A + \cos^2 A = 1]$$

$$46. \text{L.H.S.} = \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \sqrt{\frac{(1 - \sin \theta)(1 - \sin \theta)}{(1 + \sin \theta)(1 - \sin \theta)}}$$

[On rationalisation]

$$= \frac{1 - \sin \theta}{\cos \theta}$$

$$[\because 1 - \sin^2 \theta = \cos^2 \theta]$$

$$= \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} = (\sec \theta - \tan \theta)$$

$$\text{L.H.S.} = \text{R.H.S.}$$

OR

$$\text{L.H.S.} = \frac{\tan^2 \theta}{1 + \tan^2 \theta} + \frac{\cot^2 \theta}{1 + \cot^2 \theta} = \frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{\cos^2 \theta}{\sec^2 \theta}} + \frac{\frac{\cos^2 \theta}{\sin^2 \theta}}{\frac{\sin^2 \theta}{\operatorname{cosec}^2 \theta}}$$

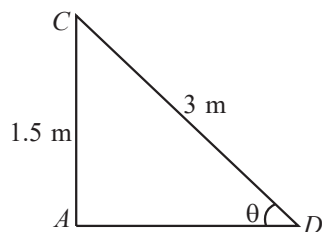
$$[\sec^2 \theta = 1 + \tan^2 \theta, \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta]$$

$$= \frac{\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta}}{\frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}} \left[\because \sec^2 \theta = \frac{1}{\cos^2 \theta}, \operatorname{cosec}^2 \theta = \frac{1}{\sin^2 \theta} \right]$$

$$= \sin^2 \theta + \cos^2 \theta = 1$$

$$\text{L.H.S.} = \text{R.H.S.}$$

$$47. AD = \sqrt{9 - 2.25} = \sqrt{6.75} = \frac{3\sqrt{3}}{2}$$



$$\therefore \tan \theta = \frac{CA}{AD} = \frac{1.5}{\frac{3\sqrt{3}}{2}} \times \frac{2}{1} = \frac{1}{\sqrt{3}}$$

$$\sec \theta + \operatorname{cosec} \theta = \frac{CD}{AD} + \frac{CD}{CA}$$

$$= 3 \left[\frac{1 \times 2}{3\sqrt{3}} + \frac{1}{1.5} \right] = 3 \left[\frac{2}{3\sqrt{3}} + \frac{2}{3} \right]$$

$$= 6 \left[\frac{1 + \sqrt{3}}{3\sqrt{3}} \right] = \frac{2(\sqrt{3} + 1)}{\sqrt{3}} = \frac{2}{3}(3 + \sqrt{3})$$

$$48. \text{L.H.S.} = (1 + \tan A - \sec A) \times (1 + \tan A + \sec A)$$

$$\because (x - y)(x + y) = x^2 - y^2$$

$$\text{here } x = 1 + \tan A, y = \sec A$$

$$\text{L.H.S.} = (1 + \tan A)^2 - (\sec A)^2$$

$$= 1 + \tan^2 A + 2\tan A - \sec^2 A$$

$$= \sec^2 A + 2\tan A - \sec^2 A \quad (1 + \tan^2 A = \sec^2 A)$$

$$= 2\tan A = \text{R.H.S.}$$

OR

$$\text{L.H.S.} = \frac{\operatorname{cosec} \theta}{\operatorname{cosec} \theta - 1} + \frac{\operatorname{cosec} \theta}{\operatorname{cosec} \theta + 1}$$

$$= \operatorname{cosec} \theta \left(\frac{1}{\operatorname{cosec} \theta - 1} + \frac{1}{\operatorname{cosec} \theta + 1} \right)$$

$$= \operatorname{cosec} \theta \left(\frac{\operatorname{cosec} \theta + 1 + \operatorname{cosec} \theta - 1}{(\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1)} \right)$$

$$= \operatorname{cosec} \theta \left(\frac{2\operatorname{cosec} \theta}{(\operatorname{cosec}^2 \theta - 1)} \right) = \frac{2\operatorname{cosec}^2 \theta}{\cot^2 \theta}$$

$$[\because 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta \Rightarrow \operatorname{cosec}^2 \theta - 1 = \cot^2 \theta]$$

$$= \frac{2 \times \frac{1}{\sin^2 \theta}}{\frac{\cos^2 \theta}{\sin^2 \theta}} \left[\because \operatorname{cosec} \theta = \frac{1}{\sin \theta}, \cot \theta = \frac{\cos \theta}{\sin \theta} \right]$$

$$= \frac{2}{\cos^2 \theta} = 2\sec^2 \theta = \text{R.H.S.}$$

$$49. \text{Given that, } \tan \theta = \frac{3}{4}$$

$$\therefore \tan^2 \theta = \frac{9}{16}$$

$$\text{We know that, } \sec^2 \theta = 1 + \tan^2 \theta$$

$$\therefore \sec^2 \theta = 1 + \frac{9}{16} = \frac{25}{16}$$

$$\Rightarrow \sec \theta = \frac{5}{4}$$

$$\text{Now, } \left(\frac{4\sin \theta - \cos \theta + 1}{4\sin \theta + \cos \theta - 1} \right) = \left(\frac{\frac{4\sin \theta}{\cos \theta} - \frac{\cos \theta}{\cos \theta} + \frac{1}{\cos \theta}}{\frac{4\sin \theta}{\cos \theta} + \frac{\cos \theta}{\cos \theta} - \frac{1}{\cos \theta}} \right)$$

$$= \frac{4\tan \theta - 1 + \sec \theta}{4\tan \theta + 1 - \sec \theta} = \frac{3 - 1 + \frac{5}{4}}{3 + 1 - \frac{5}{4}}$$

$$= \frac{2 + \frac{5}{4}}{4 - \frac{5}{4}} = \frac{\frac{8+5}{4}}{\frac{16-5}{4}} = \frac{13}{11}$$

$$50. \text{L.H.S.} = \frac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = \frac{\tan A - 1 + \sec A}{\tan A + 1 - \sec A}$$

(Dividing numerator & denominator by $\cos A$)

$$= \frac{(\tan A + \sec A) - 1}{(\tan A + \sec A) - 1}$$

$$= \frac{\{(\tan A + \sec A) - 1\}(\tan A - \sec A)}{\{(\tan A - \sec A) + 1\}(\tan A - \sec A)}$$

$$= \frac{(\tan^2 A - \sec^2 A) - (\tan A - \sec A)}{\{\tan A - \sec A + 1\}(\tan A - \sec A)}$$

$$= \frac{-1 - \tan A + \sec A}{(\tan A - \sec A + 1)(\tan A - \sec A)}$$

$$= \frac{-1(\tan A - \sec A + 1)}{(\tan A - \sec A + 1)(\tan A - \sec A)}$$

$$= \frac{1}{\sec A - \tan A} = \text{R.H.S.}$$

9

CHAPTER

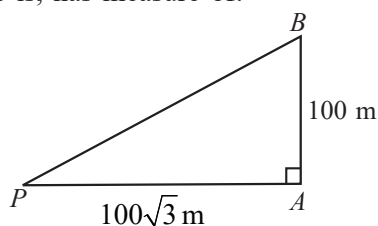
Some Applications of Trigonometry



Previous Year Questions

Multiple Choice Questions

- If a pole 6 m high casts a shadow $2\sqrt{3}$ m long on the ground, then sun's elevation is: [CBSE 2024]
(a) 30° (b) 45°
(c) 60° (d) 90°
- A lamp post 9 m high casts a shadow $3\sqrt{3}$ m long on the ground. The Sun's elevation at this moment is: [CBSE 2024]
(a) 60° (b) 90°
(c) 45° (d) 30°
- A tree casts a shadow 7 m long on the ground, when the angle of elevation of the Sun is 45° . The height of the tree is: [CBSE 2024]
(a) $7\sqrt{3}$ m (b) $(7/3)\sqrt{3}$ m
(c) 7 m (d) 3.5 m
- The angle subtended by a vertical pole of height 100m at a point on the ground $100\sqrt{3}$ m from the base is, has measure of: [CBSE 2024]



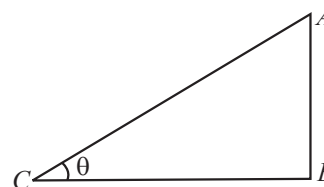
- (a) 90° (b) 60°
(c) 45° (d) 30°
- From a point on the ground, which is 30m away from the foot of a vertical tower, the angle of elevation of the top of the tower is found to be 60° . The height (in metres) of the tower is:

[CBSE 2024]

- (a) $10\sqrt{3}$ (b) $30\sqrt{3}$
(c) 60 (d) 30
- The angle of elevation of the top of a 15 m high tower at a point $15\sqrt{3}$ away from the base of the tower is: [CBSE 2023]
(a) 30° (b) 45°
(c) 60° (d) 90°

Short Answer Type Questions

- The ratio of the length of a vertical rod and the length of its shadow is $1:\sqrt{3}$. Find the angle of elevation of the sun at that moment. [CBSE 2020]
- If a tower 30 m high, casts a shadow $10\sqrt{3}$ m long on the ground, then what is the angle of elevation of the sun? [CBSE 2017]
- A ladder leaning against a wall makes an angle of 60° with the horizontal. If the foot of the ladder is 2.5 m away from the wall, find the length of the ladder. [CBSE 2016]
- In the following figure, a tower AB is 20 m high and BC, its shadow on the ground is $20\sqrt{3}$ m long. Find the Sun's altitude. [CBSE 2015]



Case Based Type Questions

- Case Study Based Question : Qutub Minar, located in South Delhi, India, was built in the year 1193. It

is 72 m high tower. Working on a school project, Charu and Daljeet visited the monument. They used trigonometry to find their distance from the tower. Observe the picture given below. Points C and D represent their positions on the ground in line with the base of tower, the angles of elevation of top of the tower (Point A) are 60° and 45° from points C and D respectively. [CBSE 2022]

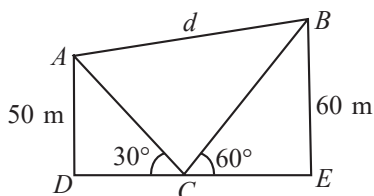


- (i) Based on above information, draw a welllabelled diagram.
- (ii) Find the distances CD , BC and BD .
(use $\sqrt{3} = 1.73$)

12. Case Study Based Question:

Kite Festival

Kite festival is celebrated in many countries at different times of the year. In India, every year 14th January is celebrated as International Kite Day. On this day many people visit India and participate in the festival by flying various kinds of kites. The picture given below, shows three kites flying together.



In figure, the angles of elevation of two kites

(Points A and B) from the hands of a man (Point C) are found to be 30° and 60° respectively. Taking $AD = 50$ m and $BE = 60$ m, find [CBSE 2022]

- (i) The lengths of strings used (take them straight) for kites A and B as shown in the figure.
- (ii) The distance ' d ' between these two kites

Long Answer Type Questions

13. A pole 6m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point P on the ground is 60° and the angle of depression of the point P from the top of the tower is 45° . Find the height of the tower and the distance of point P from the foot of the tower. (Use $\sqrt{3} = 1.73$) [CBSE 2024]
14. From the top of a 45 m high light house, the angles of depression of two ships, on the opposite side of it, are observed to be 30° and 60° . If the line joining the ships passes through the foot of the light house, find the distance between the ships. (Use $\sqrt{3} = 1.73$) [CBSE 2024]
15. From a point on the ground, the angles of elevation of the bottom and the top of transmission tower fixed at the top of a 20m high building are 45° and 60° respectively. Find the height of the tower. [CBSE 2024]
16. Two pillars of equal lengths stand on either side of a road which is 100 m wide, exactly opposite to each other. At a point on the road between the pillars, the angles of elevation of the tops of the pillars are 60° and 30° . Find the length of each pillar and distance of the point on the road from the pillars. (Use $3 = 1.732$) [CBSE 2024]
17. From a point on a bridge across the river, the angles of depressions of the banks on opposite sides of the river are 30° and 60° respectively. If the bridge is at a height of 4 m from the banks, find the width of the river. [CBSE 2024]
18. The shadow of a tower standing on a level ground is found to be 40 m longer when the Sun's altitude is 45° than when it was 60° . Find the height of the tower and the length of original shadow. (use $\sqrt{3} = 1.73$) [CBSE 2024]
19. (a) The shadow of a tower standing on a level

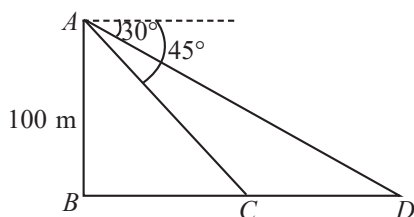
ground is found to be 40 m longer when the Sun's altitude is 30° than when it was 60° . Find the height of the tower. [CBSE 2023]

OR

- (b) From the top of a 7 m high building the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45° . Determine the height of the tower. [CBSE 2023]

20. A straight highway leads to the foot of a tower. A man standing on the top of the 75 m high tower observes two cars at angles of depression of 30° and 60° , which are approaching the foot of the tower. If one car is exactly behind the other on the same side of the tower, find the distance between the two cars. (use $\sqrt{3} = 1.73$) [CBSE 2023]

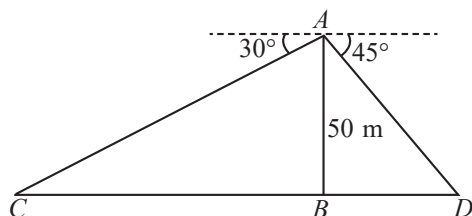
21. (a) As observed from the top of a light house 100 m above sea level, the angle of depression of a ship, sailing directly towards it, changes from 30° to 45° . Determine the distance travelled by the ship during this time. (Use $\sqrt{3} = 1.73$) [CBSE 2022]



OR

- (b) At a point on level ground, the angle of elevation of a vertical tower is, found to be α such that $\tan \alpha = \frac{1}{3}$. After walking 100 m towards the tower, the angle of elevation β becomes such that $\tan \beta = \frac{3}{4}$. Find the height of the tower. [CBSE 2022]

22. In Figure, AB is tower of height 50 m. A man standing on its top, observes two cars on the opposite sides of the tower with angles of depression 30° and 45° respectively. Find the distance between the two cars. [CBSE 2022]



23. The angle of elevation of the top of a building from the foot of a tower is 30° . The angle of elevation of the top of the tower from the foot of the building is 60° . If the tower is 60 m high, find the height of the building. [CBSE 2020]

24. A vertical tower stands on a horizontal plane and is surmounted by a vertical flag-staff of height 6 m. At a point on the plane, the angle of elevation of the bottom and top of the flag-staff are 30° and 45° respectively. Find the height of the tower. (Take $\sqrt{3} = 1.73$) [CBSE 2020]

25. A man in a boat rowing away from a light house 100 m high takes 2 minutes to change the angle of elevation of the top of the light house from 60° to 30° . Find the speed of the boat in metres per minute. [Use $\sqrt{3} = 1.732$] [CBSE 2019]

26. As observed from the top of a 100 m high light house from the sea-level, the angles of depression of two ships are 30° and 45° . If one ship is exactly being the other on the same side of the light house, find the distance between the two ships. [Use $\sqrt{3} = 1.732$] [CBSE 2018]

27. An aeroplane is flying at a height of 300 m above the ground. Flying at this height, the angles of depression from the aeroplane of two points on both banks of a river in opposite directions are 45° and 30° respectively. Find the width of the river. [Use $\sqrt{3} = 1.732$] [CBSE 2017]

28. On a straight line passing through the foot of a tower, two points C and D are at distances of 4 m and 16 m from the foot respectively. If the angles of elevation from C and D of the top of the tower are complementary, then find the height of the tower. [CBSE 2017]

29. The angle of elevation of the top Q of a vertical tower PQ from a point X on the ground is 60° . From a point Y , 40 m vertically above X , the angle of elevation of the top Q of tower is 45° . Find the height of the tower PQ and the distance PX . (Use $\sqrt{3} = 1.73$) [CBSE 2016]

30. A man standing on the deck of a ship, which is 10 m above water level, observes the angle of elevation of the top of a hill as 60° and the angle of depression of the base of the hill as 30° . Find the distance of the hill from the ship and the height of the hill. [CBSE 2016]

31. The angle of elevation of an aeroplane from point A on the ground is 60° . After flight of 15 seconds, the angle of elevation changes to 30° . If the aeroplane is flying at a constant height of $1500\sqrt{3}$ m, find the speed of the plane in km/hr. [CBSE 2015]



Explanations

Previous Year Questions

Multiple Choice Questions

1. (c) Given height of pole = 6m
length of shadow = $2\sqrt{3}$ m
Let ' θ ' be elevation of sun then

$$\tan \theta = \frac{AB}{BC} = \frac{6}{2\sqrt{3}}$$

$$\tan \theta = \sqrt{3}$$

$$\tan \theta = \tan 60^\circ$$

$$\text{then } \theta = 60^\circ$$

$$\therefore \text{Angle of elevation of sun is } 60^\circ.$$

2. (a) 60°

$$\text{Height of the pole} = 9 \text{ m}$$

$$\text{Length of shadow cast on ground} = 3\sqrt{3} \text{ m}$$

Let " θ " be angle of elevation".

According to the question,

$$\tan \theta = \frac{9}{3\sqrt{3}}$$

$$\tan \theta = \frac{3}{\sqrt{3}}$$

$$= \sqrt{3}$$

$$\therefore \tan \theta = 60^\circ$$

3. (c) 7 m

Let the length of the shadow be x m

Now, in the $\triangle ABC$,

$$\Rightarrow \tan 45^\circ = \frac{AB}{BC} = \frac{7}{x}$$

$$\Rightarrow 1 = \frac{7}{x}$$

$$\Rightarrow x = 7 \text{ m}$$

$$\therefore \text{Length of shadow} = 7 \text{ m}$$

Hence, the answer is 7m.

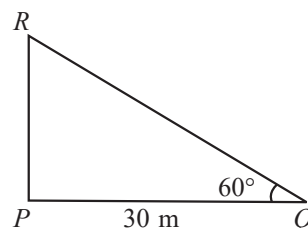
4. (a) $\tan \theta = \frac{\text{Perpendicular}}{\text{Base}}$

$$= \frac{\text{Height of pole}}{\text{Distance of pole from point } C}$$

$$\Rightarrow \tan \theta = \frac{100}{100\sqrt{3}} = \frac{1}{\sqrt{3}} = \tan 30^\circ$$

$$\Rightarrow \theta = 30^\circ$$

5. (b) $30\sqrt{3}$



PR is the tower, and Q is the point of observation at a distance of 30m from the base of the tower.

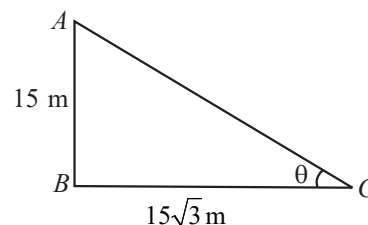
In $\triangle PQR$,

$$\tan 60^\circ = \frac{PR}{30 \text{ m}}$$

$$\therefore PR = 30\sqrt{3} \text{ m.}$$

$$\therefore \text{Height of the tower is } 30\sqrt{3} \text{ m.}$$

6. (a)



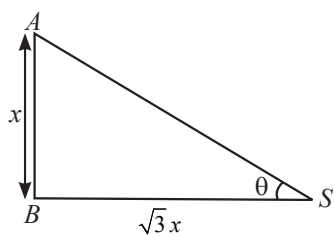
$$\tan \theta = \frac{15}{15\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\therefore \theta = 30^\circ$$

Short Answer Type Questions

7. In $\triangle ABC$

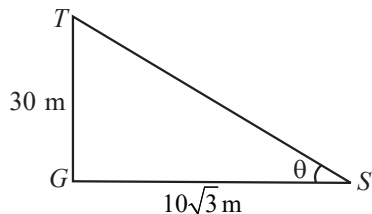
$$\tan \theta = \frac{AB}{BC} \Rightarrow \tan \theta = \frac{x}{\sqrt{3}x}$$



$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3}}$$

$$\therefore \theta = 30^\circ$$

8. Angle of elevation of sun = $\angle GST = \theta$



Hence of tower $TG = 30$ m

Length of shadow $GS = 10\sqrt{3}$ m

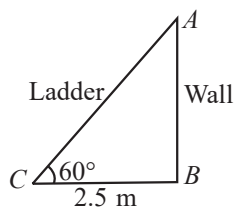
$\triangle TGS$, is a right triangle.

$$\therefore \tan \theta = \frac{30}{10\sqrt{3}}$$

$$\tan \theta = \sqrt{3}$$

$$\theta = 60. \quad [\because \tan 60 = \sqrt{3}]$$

9. Let AC be the ladder and AB be the wall.



The ladder makes an angle of 60° with the horizontal.

In $\triangle ABC$,

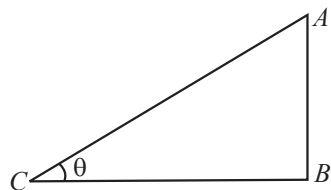
$$\cos 60^\circ = \frac{BC}{AC} \Rightarrow \frac{1}{2} = \frac{2.5}{AC}$$

Given $BC = 2.5$ m,

$$\Rightarrow AC = 5\text{m}$$

Hence, length of the ladder is $AC = 5\text{m}$.

10. Let AB the tower and BC be its shadow



$$AB = 20, BC = 20\sqrt{3}$$

In $\triangle ABC$,

$$\tan \theta = \frac{AB}{BC}$$

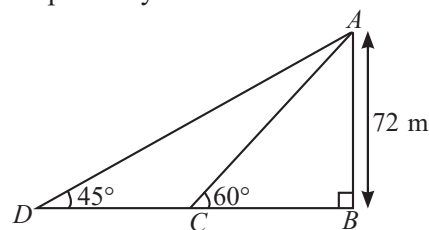
$$\tan \theta = \frac{20}{20\sqrt{3}}$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = 30^\circ$$

Case Based Type Questions

11. (i) Let positions of Charu and Daljeet be C and D respectively.



Charu is nearer to Qutub Minar as its angle of elevation is greater.

- (ii) In $\triangle ABC$,

$$\tan 60^\circ = \frac{AB}{BC} \Rightarrow \sqrt{3} = \frac{72}{BC}$$

$$BC = \frac{72}{\sqrt{3}} = 24\sqrt{3} = 41.52\text{ m}$$

$$\tan 45^\circ = \frac{AB}{BD} = \frac{72}{BD}$$

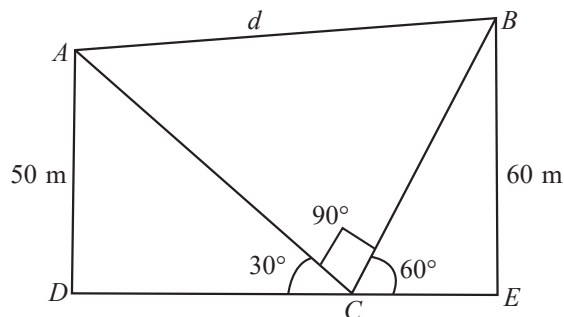
$$\Rightarrow BD = 72\text{ m} \quad [\tan 45^\circ = 1]$$

$$CD = BD - BC$$

$$CD = (72 - 41.52)\text{ m} = 30.48\text{ m}$$

12. (i) As from the figure, length of strings are AC and BC .

$$AD = 50\text{ m}, BE = 60\text{ m}$$



$$\text{In } \triangle ADC, \sin 30^\circ = \frac{AD}{AC} \Rightarrow \frac{1}{2} = \frac{50}{AC}$$

$$\Rightarrow AC = 100 \text{ m}$$

$$\text{In } \triangle BCE, \sin 60^\circ = \frac{BE}{BC} \Rightarrow \frac{\sqrt{3}}{2} = \frac{60}{BC}$$

$$\Rightarrow BC = 40\sqrt{3} \text{ m}$$

(ii) As from the figure,

we can see that $\angle ACB = 90^\circ$

Applying Pythagoras theorem in $\triangle ACB$, we get

$$\begin{aligned} d &= \sqrt{AC^2 + BC^2} \\ &= \sqrt{(100)^2 + (40\sqrt{3})^2} = 20\sqrt{37} \text{ m} \end{aligned}$$

Long Answer Type Questions

13. Let height of the tower is 'x' and distance of point P from the foot of tower is 'y'.

In $\triangle ABP$,

$$\tan 45^\circ = \frac{x}{y}$$

$$1 = \frac{x}{y}$$

$$x = y$$

In $\triangle ACP$,

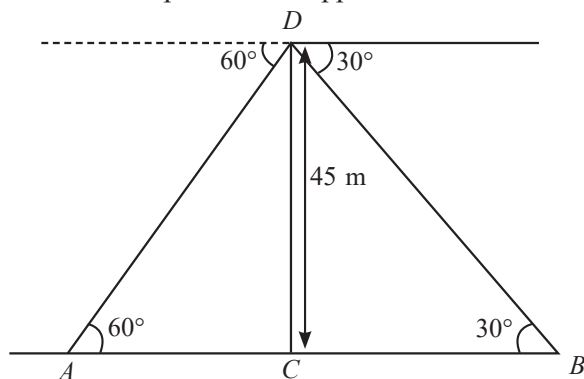
$$\tan 60^\circ = \frac{x+6}{y}$$

$$\sqrt{3} = \frac{x+6}{x}$$

$$x(\sqrt{3} - 1) = 6$$

$$x = \frac{6}{0.73} = 8.21$$

14. In the given figure CD is the height of light house and two ships 'A' & 'B' opposite side of it.



Now, in the $\triangle ACD$

$$\tan 60^\circ = \frac{CD}{AC} \Rightarrow \sqrt{3} = \frac{45}{AC}$$

$$\Rightarrow AC = 15\sqrt{3}$$

...(i)

Now, in the $\triangle BCD$

$$\tan 30^\circ = \frac{CD}{BC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{45}{BC} \Rightarrow BC = 45\sqrt{3}$$

Distance between ships 'A' and 'B' = AB

$$\Rightarrow AB = AC + BC$$

$$\Rightarrow AB = 15\sqrt{3} + 45\sqrt{3} \Rightarrow 60\sqrt{3}$$

$$\Rightarrow AB = 103.80 \text{ m.}$$

15. Let DC be the tower and BC be the building. Then, $\angle CAB = 45^\circ$, $\angle DAB = 60^\circ$, $BC = 20 \text{ m}$

Let height of the tower, $DC = h \text{ m}$.

In right $\triangle ABC$,

$$\tan 45^\circ = \frac{BC}{AB}$$

$$1 = \frac{20}{AB}$$

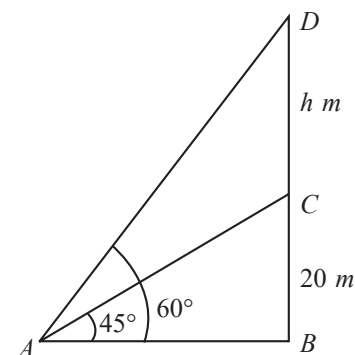
$$AB = 20 \text{ m}$$

In right $\triangle ABD$,

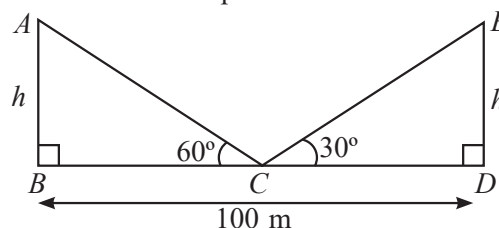
$$\tan 60^\circ = \frac{BD}{AB}$$

$$\sqrt{3} = \frac{h+20}{20}$$

$$h = 20(\sqrt{3} - 1) \text{ m.}$$



16. Let AB and ED be two pillars each of height h metres. Let C be a point on the road BD such that



$BC = x$ metres. Then $CD = (100 - x)$ metres. Given $\angle ACB = 60^\circ$ and $\angle ECD = 30^\circ$

$$\text{In } \triangle ABC, \tan 60^\circ = \frac{AB}{BC}$$

$$\Rightarrow \sqrt{3} = \frac{h}{x} \Rightarrow h = \sqrt{3}x$$

...(i)

$$\text{In } \triangle ECD, \tan 30^\circ = \frac{ED}{CD}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{100 - x} \Rightarrow h = \sqrt{3}(100 - x)$$

...(ii)

∴ Substitute, the value of h from (i) in (ii), we get

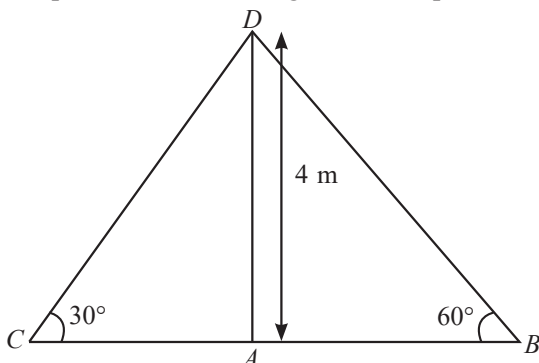
$$\sqrt{3}x \times \sqrt{3} = 100 - x \Rightarrow 3x = 100 - x$$

$$\Rightarrow 4x = 100 \Rightarrow x = 25\text{m}$$

$$\therefore h = (\sqrt{3} \times 25) = 25 \times 1.732\text{m} = 43.3\text{m}$$

∴ The required point is at a distance of 25m from the pillar B and the height of each pillar is 43.3m.

17.



The height of the bridge = 4m

In $\triangle ABC$,

$$\tan 60^\circ = \frac{AD}{AB}$$

$$\Rightarrow \frac{\sqrt{3}}{1} = \frac{4}{AB} \Rightarrow AB = \frac{4}{\sqrt{3}}$$

In $\triangle ACD$,

$$\tan 30^\circ = \frac{AD}{AC}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{4}{AC} \Rightarrow AC = 4\sqrt{3}$$

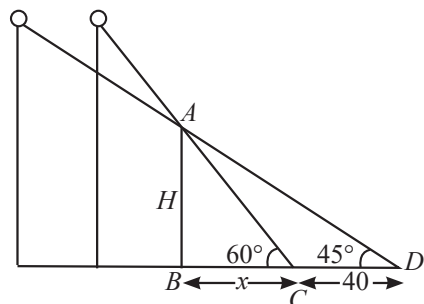
∴ The width of the river $BC = AC + AB$

$$= 4\sqrt{3} + \frac{4}{\sqrt{3}} = \frac{12 + 4}{\sqrt{3}} = \frac{16}{\sqrt{3}}$$

$$= \frac{16\sqrt{3}}{3}\text{m}.$$

18. According to question,

Shadow of tower at 45° elevation is 40 m more than shadow of tower at 60° .



Let us assume that the length of shadow due to 60°

elevation be x .

Now, it is given that $CD = 40$

So, in $\triangle ABD$

$$\tan 45^\circ = \frac{H}{x + 40} \Rightarrow H = x + 40 \quad \dots(1)$$

and in $\triangle ABC$

$$\tan 60^\circ = \frac{H}{x} \Rightarrow \sqrt{3} = \frac{H}{x} \Rightarrow H = x\sqrt{3} \quad \dots(2)$$

from (1) and (2)

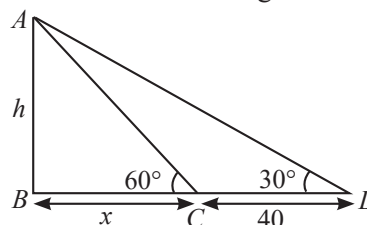
$$x + 40 = x\sqrt{3}$$

$$\Rightarrow \sqrt{3}x - x = 40$$

$$\Rightarrow x = \frac{40}{\sqrt{3} - 1} = 54.8 \text{ m (Take } \sqrt{3} = 1.73)$$

$$\text{and } H = 40 + 54.8 = 94.8 \text{ m}$$

19. (a) Let AB be tower of height h m



$$\text{In } \triangle ABC, \tan 60^\circ = \frac{AB}{BC}$$

$$\sqrt{3} = \frac{h}{x} \Rightarrow x = \frac{h}{\sqrt{3}} \quad \dots(i)$$

$$\text{In } \triangle ABD, \tan 30^\circ = \frac{AB}{BD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x + 40}$$

$$\Rightarrow x + 40 = \sqrt{3}h$$

$$\therefore \frac{h}{\sqrt{3}} + 40 = \sqrt{3}h \quad [\text{From (i)}]$$

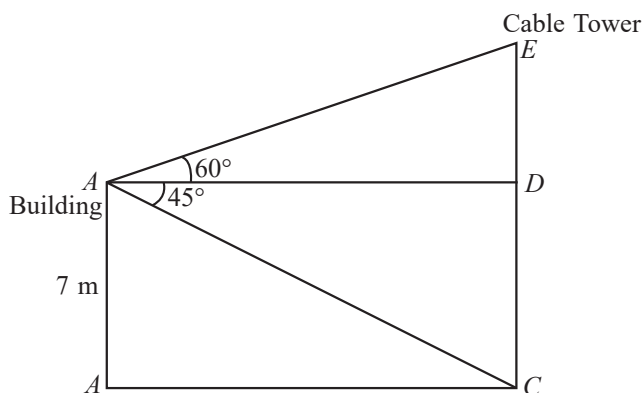
$$\Rightarrow h + 40\sqrt{3} = 3h \Rightarrow 2h = 40\sqrt{3}$$

$$\Rightarrow h = 20\sqrt{3}$$

∴ The height of the tower is $20\sqrt{3}$ m

OR

(b) $AB = CD = 7$ m



In $\triangle ADC$,

$$\tan 45^\circ = \frac{CD}{AD} \Rightarrow 1 = \frac{7}{AD}$$

$$\Rightarrow AD = 7 \text{ m}$$

In $\triangle ADE$,

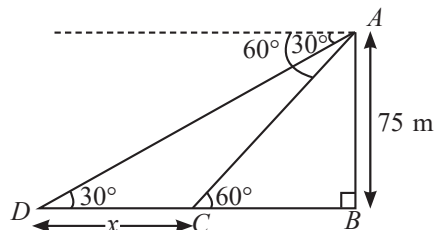
$$\tan 60^\circ = \frac{ED}{AD} \Rightarrow \sqrt{3} = \frac{ED}{7}$$

$$\Rightarrow ED = 7\sqrt{3} \text{ m}$$

The total height of the cable tower = $CD + ED$

$$= (7 + 7\sqrt{3}) \text{ m} = 7(1 + \sqrt{3}) \text{ m}$$

20.



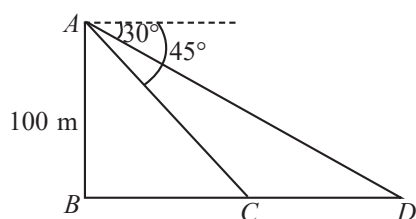
In $\triangle ABC$:

$$\tan 60^\circ = \frac{AB}{BC} \Rightarrow BC = \frac{75}{\sqrt{3}} \quad \dots(i)$$

In $\triangle ABD$:

$$\begin{aligned} \tan 30^\circ &= \frac{AB}{BC + DC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{75}{BC + x} \\ \Rightarrow BC + x &= 75\sqrt{3} \Rightarrow x = 75\sqrt{3} - \frac{75}{\sqrt{3}} \quad [\text{From (i)}] \\ \Rightarrow x &= \frac{75 \times 2}{\sqrt{3}} = \frac{75 \times 2 \times \sqrt{3}}{3} \Rightarrow x = 86.5 \text{ m} \end{aligned}$$

21. (a)



In $\triangle ABC$

$$\frac{AB}{BC} = \tan 45^\circ = 1$$

$$\Rightarrow AB = BC = 100 \text{ m} \quad \dots(i)$$

In $\triangle ABD$,

$$\frac{AB}{BD} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\Rightarrow BD = AB \times \sqrt{3} = 100\sqrt{3} \text{ m} \quad \dots(ii)$$

$$\therefore CD = BD - BC$$

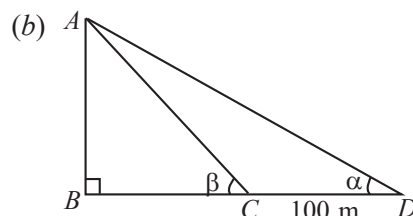
$$= (100\sqrt{3} - 100) \text{ m} \quad [\text{From (i) and (ii)}]$$

$$= 100(\sqrt{3} - 1) \text{ m} = 100(1.73 - 1) \text{ m}$$

$$= 100 \times 0.73 \text{ m} = 73 \text{ m}$$

$\therefore$ Ship will travel 73 m during the given time.

OR



Let AB represents the tower. Observer is moving from D to C .

In $\triangle ABC$,

$$\tan \beta = \frac{AB}{BC} = \frac{3}{4} \quad \dots(i)$$

In $\triangle ABD$,

$$\tan \alpha = \frac{AB}{BD} = \frac{1}{3} \quad \dots(ii)$$

From (i) and (ii), we get

$$BC = \frac{4AB}{3} \text{ and } BD = 3AB$$

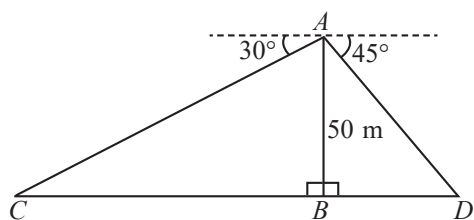
$$\Rightarrow CD = BD - BC$$

$$\Rightarrow 100 = 3AB - \frac{4AB}{3}$$

$$\Rightarrow 100 = \frac{9AB - 4AB}{3}$$

$$\Rightarrow 300 = 5AB \Rightarrow AB = 60 \text{ m.}$$

22. $\angle ACB = 30^\circ$



and $\angle ADB = 45^\circ$ [From figure]

Distance between two cars

$$= CD = BC + BD \quad [\text{from figure}] \dots(i)$$

Now, In $\triangle ABC$

$$\tan 30^\circ = \frac{AB}{BC} = \frac{50}{BC}$$

$$\text{or } BC = \frac{50}{\tan 30^\circ} = 50\sqrt{3} \text{ m}$$

and In $\triangle ABD$,

$$\tan 45^\circ = \frac{AB}{BD} = \frac{50}{BD}$$

$$BD = \frac{50}{1}$$

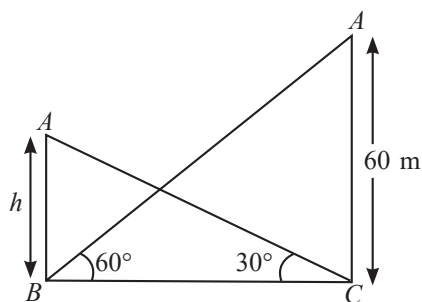
$$BD = 50 \text{ m}$$

From equation (i), we get

$$CD = BC + BD = 50\sqrt{3} + 50 = 50(\sqrt{3} + 1) \text{ m}$$

23. Let $AB = h$ m be the height of building and CD be height of tower.

$$\therefore CD = 60 \text{ m}$$



$$\text{In } \triangle BDC, \tan 60^\circ = \frac{CD}{BC}$$

$$\Rightarrow BC = \frac{60}{\sqrt{3}} = 20\sqrt{3} \text{ m} \quad \dots(i)$$

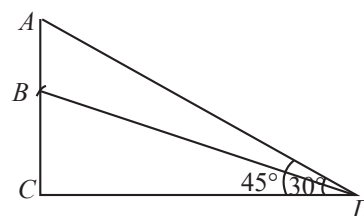
In $\triangle ABC$,

$$\tan 30^\circ = \frac{AB}{BC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{AB}{20\sqrt{3}} \quad [\text{From (i)}]$$

$$\Rightarrow AB = 20 \text{ m}$$

$$\therefore \text{Height of building} = 20 \text{ m.}$$

24. AB = height of flag staff = 6 m



Let BC = height of tower = h m

In $\triangle BCD$,

$$\frac{BC}{CD} = \tan 30^\circ$$

$$\Rightarrow \frac{h}{CD} = \frac{1}{\sqrt{3}} \Rightarrow CD = h\sqrt{3} \quad \dots(i)$$

In $\triangle ACD$,

$$\frac{AC}{CD} = \tan 45^\circ$$

$$\Rightarrow \frac{h+6}{CD} = 1 \Rightarrow h = CD - 6$$

$$\Rightarrow h = h\sqrt{3} - 6 \Rightarrow h(\sqrt{3} - 1) = 6$$

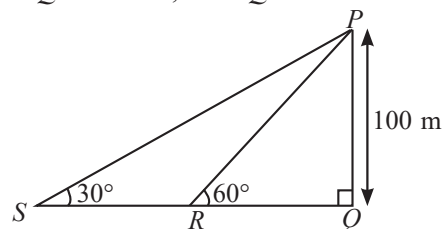
$$\Rightarrow h = \frac{6}{\sqrt{3} - 1} \Rightarrow h = 3(\sqrt{3} + 1)$$

$$h = 3 \times 2.73$$

$$h = 8.19 \text{ m}$$

25. Let the light house be PQ and the boat changes its position from R to S .

Here $PQ = 100$ m, $\angle PRQ = 60^\circ$ and $\angle PSR = 30^\circ$



In $\triangle PQR$

$$\tan 60^\circ = \frac{PQ}{QR} = \frac{100}{QR} \Rightarrow QR = \frac{100\sqrt{3}}{3} \text{ m} \quad \dots(i)$$

In $\triangle PQS$,

$$\tan 30^\circ = \frac{PQ}{QS} \Rightarrow \frac{1}{\sqrt{3}} = \frac{100}{QS}$$

$$\Rightarrow QS = 100\sqrt{3} \text{ m}$$

$$\therefore RS = QS - QR = 100\sqrt{3} - \frac{100\sqrt{3}}{3} = \frac{200\sqrt{3}}{3}$$

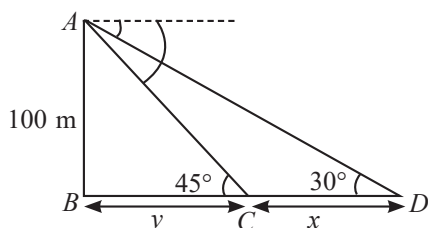
$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$= \frac{200\sqrt{3}}{3 \times 2} = \frac{100\sqrt{3}}{3} = 57.73 (\text{approx.})$$

(Using $\sqrt{3} = 1.732$)

= 57.73 m/min.

26. Let ships are at distance x from each other.



In $\triangle ABC$

$$\tan 45^\circ = \frac{AB}{y} = 1$$

$$y = 100 \text{ m} \quad \dots(i)$$

In $\triangle ABD$

$$\tan 30^\circ = \frac{AB}{BD} = \frac{100}{x+y} = \frac{1}{\sqrt{3}}$$

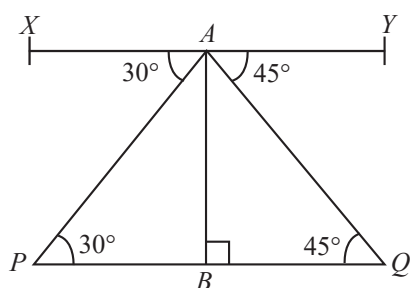
$$\Rightarrow x+y = 100\sqrt{3} \quad \dots(ii)$$

$$x = 100\sqrt{3} - y = 100\sqrt{3} - 100 = 100(\sqrt{3} - 1)$$

$$x = 100(1.732 - 1) = 100 \times 0.732 = 73.2 \text{ m}$$

$\therefore$ Ships are 73.2 meters apart.

27. Given aeroplane is at height at 300 m



$\therefore AB = 300 \text{ m}$ and $XY \parallel PQ$

Angle of depression of the two points P and Q are 30° and 45° respectively.

$$\angle XAP = 30^\circ \text{ and } \angle YAQ = 45^\circ$$

$$\angle XAP = \angle APB = 30^\circ \quad [\text{Alternate interior angles}]$$

$$\angle YAQ = \angle AQB = 45^\circ$$

In $\triangle PAB$,

$$\tan 30^\circ = \frac{AB}{PB}$$

$$PB = 300\sqrt{3} \text{ m}$$

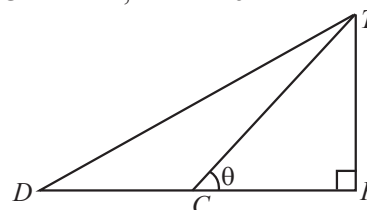
In $\triangle AQB$,

$$\tan 45^\circ = \frac{AB}{BQ} \Rightarrow 1 = \frac{300}{BQ} \Rightarrow BQ = 300$$

$\therefore$ Width of the river = $PB + BQ$

$$= 300(1 + \sqrt{3}) \text{ m}$$

28. Given $CF = 4 \text{ m}$, $DF = 16$



$$\angle TCF + \angle TDF = 90^\circ$$

Let say $\angle TCF = \theta$

$$\angle TDF = 90^\circ - \theta$$

In a right angled triangle TCF

$$\tan \theta = \frac{TF}{CF} = \frac{TF}{4}$$

$$TF = 4 \tan \theta$$

In $\triangle TDF$

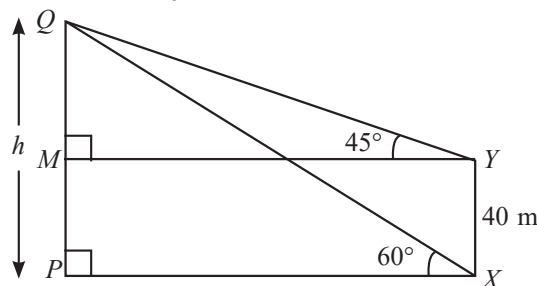
$$\tan(90^\circ - \theta) = \frac{TF}{16}$$

$$TF = 16 \cot \theta$$

Multiply (i) and (ii), we get

$$(TF)^2 = 64 \Rightarrow TF = 8 \text{ m} \Rightarrow \text{Height of tower} = 8 \text{ m}$$

29. $MP = YX = 40 \text{ m}$



$$\therefore QM = h - 40$$

In right angled $\triangle QMY$,

$$\tan 45^\circ = \frac{QM}{MY} \Rightarrow 1 = \frac{h-40}{PX} \quad (MY = PX)$$

$$\therefore PX = h - 40$$

In right angled $\triangle QPX$,

$$\tan 60^\circ = \frac{QP}{PX} \Rightarrow \sqrt{3} = \frac{QP}{PX}$$

$$PX = \frac{h}{\sqrt{3}} \quad \dots(iii)$$

From (i) and (ii), we get

$$h - 40 = \frac{h}{\sqrt{3}}$$

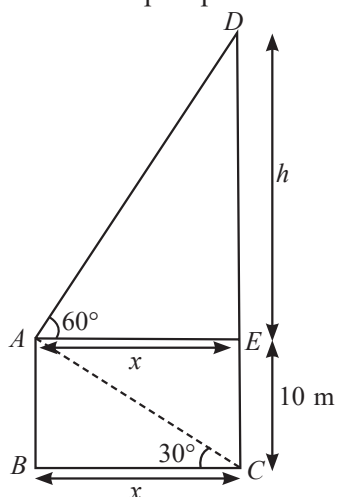
$$\therefore \sqrt{3}h - 40\sqrt{3} = h$$

$$\therefore \sqrt{3}h - h = 40\sqrt{3}$$

$$\therefore 1.73h - h = 40(1.73) \Rightarrow h = 94.79 \text{ m}$$

Thus, PQ is 94.79 m and $PX = 94.79 - 40 = 54.79$.

30. Let CD be the hill and suppose the man is standing on the deck of a ship at point A .



The angle of depression of the base C of the hill CD observed from A is 30° and the angle of elevation of the top D of the hill CD observed from A is 60° .

$$\therefore \angle EAD = 60^\circ \text{ and } \angle BCA = 30^\circ$$

In $\triangle AED$,

$$\tan 60^\circ = \frac{DE}{EA}$$

$$\sqrt{3} = \frac{h}{x}$$

$$h = \sqrt{3}x$$

...(i)

In $\triangle ABC$,

$$\tan 30^\circ = \frac{AB}{BC}$$

$$\frac{1}{\sqrt{3}} = \frac{10}{x}$$

$$x = 10\sqrt{3}$$

Substituting $x = 10\sqrt{3}$ in equation (i), we get

$$h = \sqrt{3} \times 10\sqrt{3} = 10 + 3 = 30$$

$$DE = 30 \text{ m}$$

$$CD = CE + ED = 10 + 30 = 40 \text{ m}$$

Thus, the distance of the hill from the ship is $10\sqrt{3}$ m and the height of the hill is 40 m.

31. Let BC be the height at which the aeroplane is observed from point A

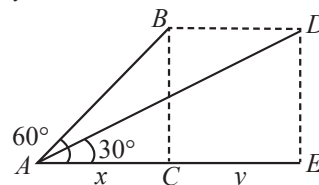
$$\text{Then, } BC = 1500\sqrt{3}$$

In 15 seconds, the aeroplane moves from point B to D .

B and D are the points where the angles of elevation 60° and 30° are formed respectively.

Let $AC = x$ metres and $CE = y$ metres

$$AE = x + y$$



In $\triangle CBA$,

$$\tan 60^\circ = \frac{BC}{AC}$$

$$\sqrt{3} = \frac{1500\sqrt{3}}{x}$$

$$x = 1500 \text{ m}$$

...(i)

In $\triangle ADE$

$$\tan 30^\circ = \frac{DE}{AE}$$

$$\frac{1}{\sqrt{3}} = \frac{1500\sqrt{3}}{x + y}$$

$$\therefore x + y = 1500 \times 3 = 4500$$

$$\therefore 1500 + y = 4500$$

$$\therefore y = 3000 \text{ m}$$

...(ii)

We know that, the aeroplane moves from point B to D in 15 seconds and the distance covered is 3000 metres.

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{Speed} = \frac{3000}{15}$$

$$\text{Speed} = 200 \text{ m/s}$$

$$\text{Converting it to km/hr} = 200 \times \frac{18}{5} = 720 \text{ km/hr}$$



Previous Year Questions

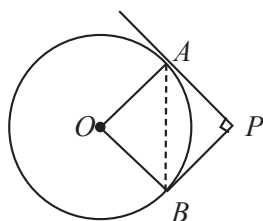
Multiple Choice Questions

1. If two tangents inclined at an angle of 60° are drawn to a circle of radius 5 cm, then the length of each tangent is : [CBSE 2024]

- (a) $\frac{5\sqrt{3}}{2}$ cm (b) 10 cm
(c) $\frac{5}{\sqrt{3}}$ cm (d) $5\sqrt{3}$ cm

2. In the given figure, tangents PA and PB to the circle centred at O , from point P are perpendicular to each other. If $PA = 5$ cm, then length of AB is equal to:

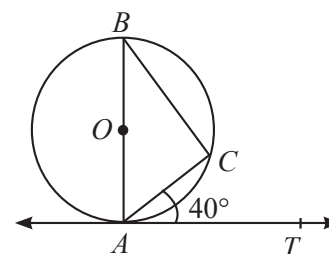
[CBSE 2024]



- (a) 5 cm (b) $5\sqrt{2}$ cm
(c) $2\sqrt{5}$ cm (d) 10 cm

3. In the given figure, AT is tangent to a circle centred at O . If $\angle CAT = 40^\circ$, then $\angle CBA$ is equal to

[CBSE 2024]

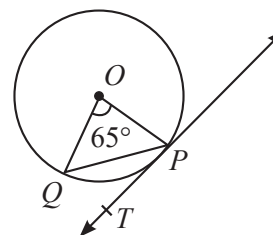


- (a) 70° (b) 50°
(c) 65° (d) 40°

4. In the given figure, PT is tangent to a circle with centre O . Chord PQ subtends an angle of 65° at the centre. The measure of $\angle QPT$ is:

[CBSE 2024]

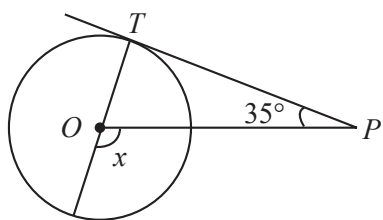
- (a) 65° (b) 57.5°
(c) 67.5° (d) 32.5°



5. Maximum number of common tangents that can be drawn to two circles intersecting at two distinct points is: [CBSE 2024]

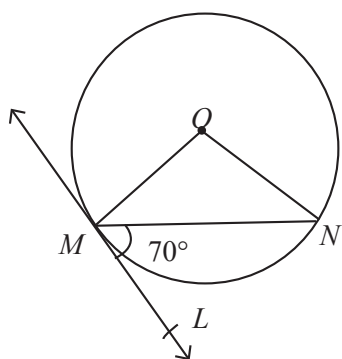
- (a) 4 (b) 3
(c) 2 (d) 1

6. In the given figure, if PT is a tangent to a circle with centre O and $\angle TPO = 35^\circ$, then the measure of $\angle x$ is: [CBSE 2024]



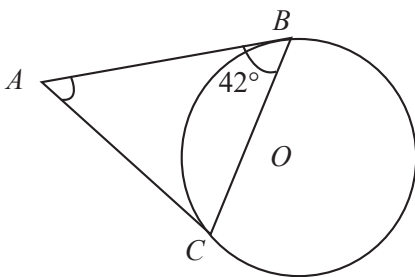
- (a) 110° (b) 115°
(c) 120° (d) 125°

7. In the given figure, O is the centre of the circle. MN is the chord and the tangent ML at point M makes an angle of 70° with MN . The measure of $\angle MON$ is: [CBSE 2024]



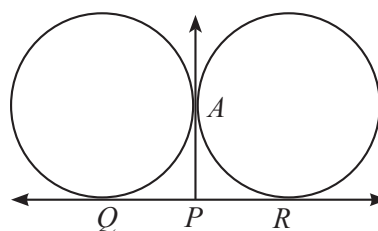
- (a) 120° (b) 140°
(c) 70° (d) 90°

8. In the given figure, AB and AC are tangents to the circle. If $\angle ABC = 42^\circ$, then the measure of $\angle BAC$ is: [CBSE 2024]



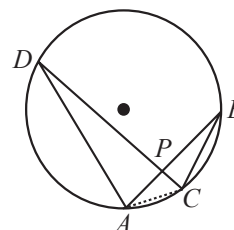
- (a) 96° (b) 42°
(c) 106° (d) 86°

9. In the given figure, QR is a common tangent to the two given circles touching externally at A . The tangent at A meets QR at P . If $AP = 4.2$ cm, then the length of QR is: [CBSE 2024]



- (a) 4.2 cm (b) 2.1 cm
(c) 8.4 cm (d) 6.3 cm

10. AB and CD are two chords of a circle intersecting at P . Choose the correct statement from the following: [CBSE 2024]



- (a) $\triangle ADP \sim \triangle CBA$ (b) $\triangle ADP \sim \triangle BPC$
(c) $\triangle ADP \sim \triangle BCP$ (d) $\triangle ADP \sim \triangle CBA$

Directions: In Q. No. 11 and 12 a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

- (a) Both, Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).
(b) Both, Assertion (A) and Reason (R) are true but Reason (R) is not correct explanation for Assertion (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

11. Assertion (A): The tangents drawn at the end points of a diameter of a circle, are parallel.

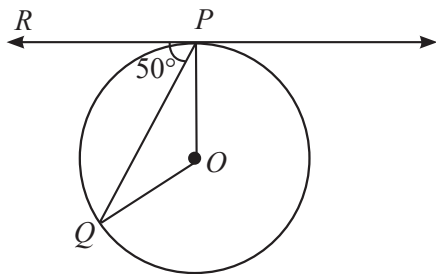
Reason (R): Diameter of a circle is the longest chord. [CBSE 2024]

12. Assertion (A): A tangent to a circle is perpendicular to the radius through the point of contact. Reason (R): The lengths of tangents drawn from an external point to a circle are equal. [CBSE 2023]

13. In the given figure, O is the centre of the circle and PQ is the chord, If the tangent PR at P makes an angle of 50° with PQ , then the measure of $\angle POQ$

is

[CBSE 2023]



- (a) 50° (b) 40°
(c) 100° (d) 130°

14. How many tangents can be drawn to a circle from a point on it? [CBSE 2023]

- (a) One (b) Two
(c) Infinite (d) Zero

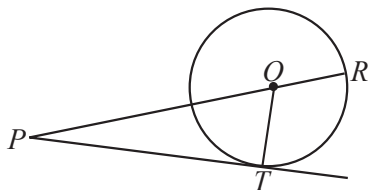
15. The length of the tangent from an external point A to a circle, of radius 3 cm, is 4 cm. The distance of A from the centre of the circle is: [CBSE 2023]

- (a) 7 cm (b) 5 cm
(c) $\sqrt{7}$ cm (d) 25 cm

16. The length of tangent drawn to a circle of radius 9 cm from a point 41 cm away from the centre is [CBSE 2023]

- (a) 40 cm (b) 9 cm
(c) 41 cm (d) 50 cm

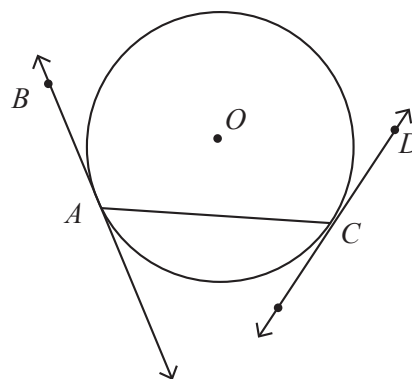
17. In the given figure, on a circle of radius 7 cm, tangent PT is drawn from a point P such that $PT = 24$ cm. If O is the centre of the circle, then the length of PR is [CBSE 2020]



- (a) 30 cm (b) 28 cm
(c) 32 cm (d) 25 cm

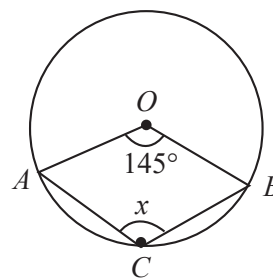
Short Answer Type Questions

18. In the given figure, AB and CD are tangents to a circle centred at O . Is $\angle BAC = \angle DCA$? Justify your answer. [CBSE 2024]

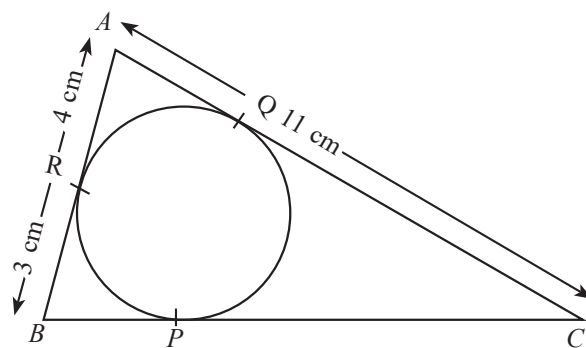


19. Prove that the tangents drawn at the end points of a chord of a circle makes equal angles with the chord. [CBSE 2024]

20. In the given figure, O is the centre of the circle. If $\angle AOB = 145^\circ$, then find the value of x . [CBSE 2024]



21. In the given figure, $\triangle ABC$ is circumscribing a circle. Find the length of BC , if $AR = 4$ cm, $BR = 3$ cm and $AC = 11$ cm. [CBSE 2024]

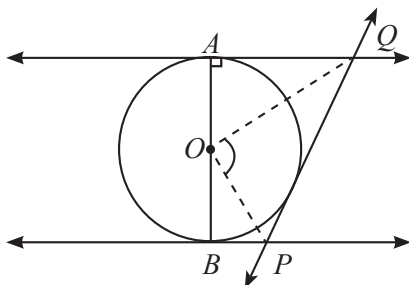


22. If two tangents inclined at an angle of 60° are drawn to a circle of radius 3 cm, then find the length of each tangent. [CBSE 2024]

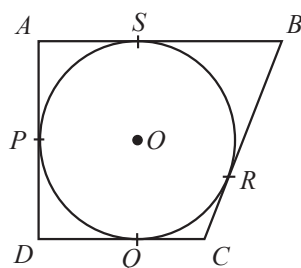
23. From an external point P , two tangents PA and PB are drawn to a circle with centre O . At a point E on the circle, a tangent is drawn which intersects PA

and PB at C and D respectively. If $PA = 10$ cm, find the perimeter of ΔPCD . [CBSE 2024]

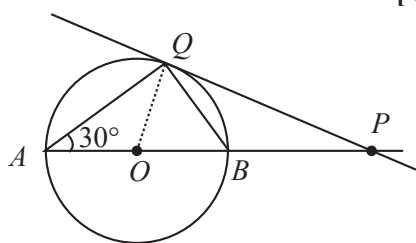
24. (A) In the given figure, AB is a diameter of the circle with centre O . AQ , BP and PQ are tangents to the circle. Prove that $\angle POQ = 90^\circ$. [CBSE 2024]



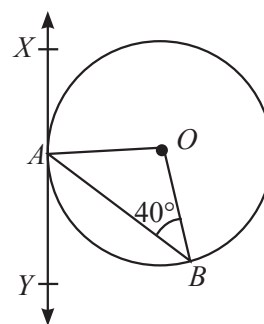
25. A circle with centre O and radius 8 cm is inscribed in a quadrilateral $ABCD$ in which P , Q , R , S are the points of contact as shown. If AD is perpendicular to DC , $BC = 30$ cm and $BS = 24$ cm, then find the length DC . [CBSE 2024]



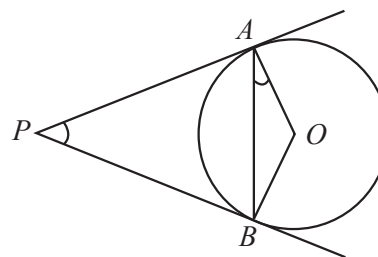
26. In the given figure, PQ is tangent to a circle centred at O and $\angle BAQ = 30^\circ$; show that $BP = BQ$. [CBSE 2024]



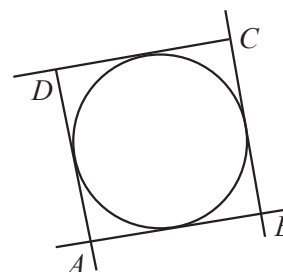
27. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle. [CBSE 2023]
28. Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line-segment joining the points of contact at the centre. [CBSE 2023]
29. In figure, XAY is a tangent to the circle centered at O . If $\angle ABO = 40^\circ$, then find $m\angle BAY$ and $m\angle AOB$. [CBSE 2022]



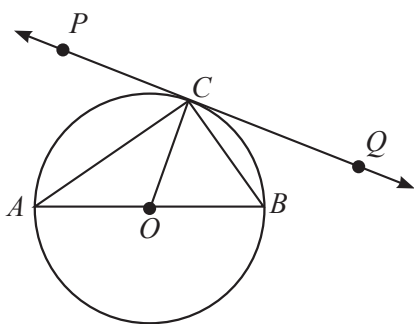
30. PA and PB are tangents drawn to the circle with centre O as shown in the figure. Prove that $\angle APB = 2\angle OAB$. [CBSE 2023]



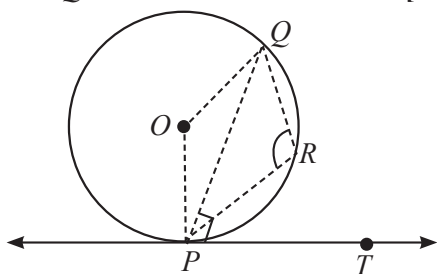
31. In figure, a circle touches all the four sides of a quadrilateral $ABCD$. If $AB = 6$ cm, $BC = 9$ cm and $CD = 8$ cm, then find the length of AD . [CBSE 2020]



32. Prove that the tangents drawn at the end points of a chord of a circle make equal angles with the chord. [CBSE 2017]
33. A circle touches all the four sides of a quadrilateral $ABCD$. Prove that $AB + CD = BC + DA$. [CBSE 2017]
34. If the angle between two tangents drawn from an external point P to a circle of radius a and centre of O , is 60° , then find the length of OP . [CBSE 2017]
35. Prove that the length of the tangents drawn from an external point to a circle are equal. [CBSE 2014, 2015, 2016, 2017]
36. In the figure, PQ is a tangent at a point C to a circle with centre O . If AB is a diameter and $\angle CAB = 30^\circ$, find $\angle PCA$. [CBSE 2016]

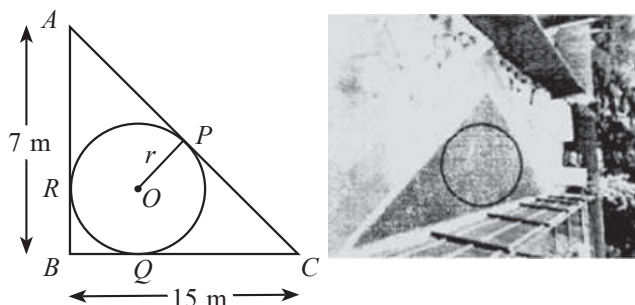


37. In the following figure, PQ is a chord of a circle with centre O and PT is a tangent. If $\angle QPT = 60^\circ$, find $\angle PRQ$ [CBSE 2015]



Case Based Type Questions

38. A backyard is in the shape of a triangle ABC with right angle at B . $AB = 7$ m and $BC = 15$ m. A circular pit was dug inside it such that it touches the walls AC , BC and AB at P , Q and R respectively such that $AP = x$ m.



Based on the above information, answer the following questions:

- Find the length of AR in terms of x .
- Write the type of quadrilateral $BQOR$.
- (a) Find the length PC in terms of x and hence find the value of x .

OR

- (b) Find x and hence find the radius of circle.

Long Answer Type Questions

39. A circle is inscribed in a right-angled triangle ABC , right-angled at B . If $BC = 7$ cm and $AB = 24$ cm, find the radius of the circle. [CBSE 2024]
40. In the given figure, AB , BC , CD and DA are tangents to the circle with centre O forming a quadrilateral $ABCD$.

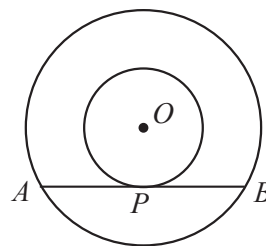
Show that $\angle AOB + \angle COD = 180^\circ$ [CBSE 2024]

41. (a) Prove that the lengths of tangents drawn from an external point to a circle are equal.

[CBSE 2023]

OR

- (b) Two concentric circles with centre O are of radii 3 cm and 5 cm. Find the length of chord AB of the larger circle which touches the smaller circle at P . [CBSE 2023]

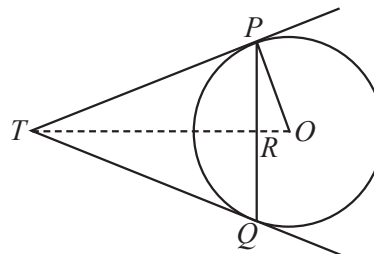


42. (a) Prove that a parallelogram circumscribing a circle is a rhombus. [CBSE 2022]

OR

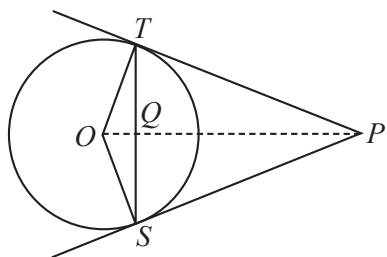
- (b) Prove that the perpendicular at the point of contact to the tangent to a circle passes through the centre of the circle. [CBSE 2022]

43. In figure, PQ is a chord of length 8 cm of a circle of radius 5 cm. The tangents at P and Q meet at a point T . Find the length of TP . [CBSE 2022, 2019]

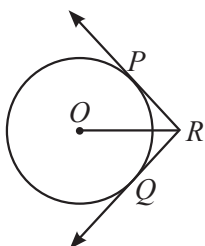


44. In figure, from an external point P , two tangents PT and PS are drawn to a circle with centre O and

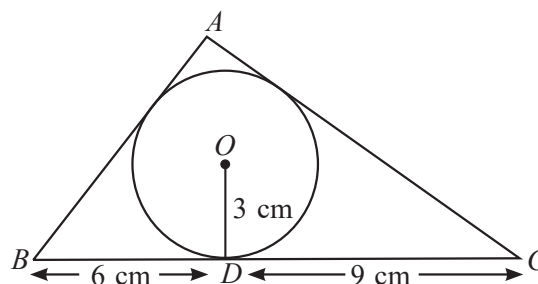
radius r . If $OP = 2r$, show that $\angle OTS = \angle OST = 30^\circ$.
[CBSE 2016]



45. In the following figure, two tangents PQ and RP are drawn from an external point R to the circle with centre O , if $\angle PRQ = 120^\circ$, then prove that $OR = PR + RQ$.
[CBSE 2015]



46. In figure, a $\triangle ABC$ is drawn to circumscribe a circle of radius 3 cm, such that the segments BD and DC are respectively of lengths 6 cm and 9 cm. If the area of $\triangle ABC$ is 54 cm, then find the lengths of sides AB and AC .
[CBSE 2015]



47. Prove that the tangent drawn at the midpoint of an arc of a circle is parallel to the chord joining the end points of the arc.
[CBSE 2015]

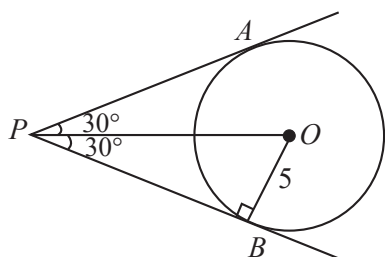


Explanations

Previous Year Questions

Multiple Choice Questions

1. (d)



OP bisect the $\angle APB$

$$\tan 30^\circ = \frac{OB}{PB}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{5}{PB}$$

$$\Rightarrow PB = 5\sqrt{3}$$

Hence, length of tangent = $5\sqrt{3}$ cm.

2. (b) $PA = PB = 5$ cm

$$\text{In } \triangle ABP, AB = \sqrt{PA^2 + PB^2} = 5\sqrt{2} \text{ cm.}$$

3. (d) $\angle OAT = 90^\circ$

$$\Rightarrow \angle OAC = 90^\circ - 40^\circ = 50^\circ$$

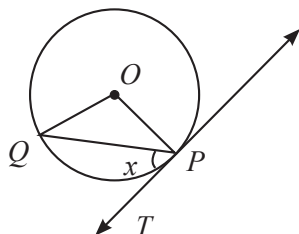
$$\angle ACB = 90^\circ$$

$$\therefore \angle ACB + \angle CBA + \angle BAC = 180^\circ$$

$$90^\circ + \angle CBA + 50^\circ = 180^\circ$$

$$\Rightarrow \angle CBA = 40^\circ$$

4. (d)



$$\text{Let } \angle QPT = x$$

$$\angle OPQ = 90^\circ - x$$

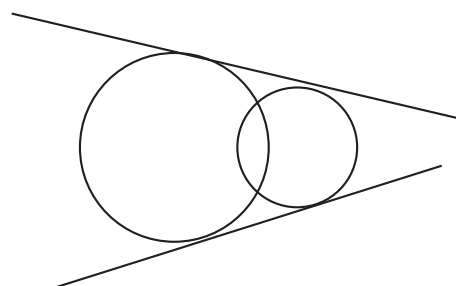
$$\angle OQP = 90^\circ - x$$

$$[OP = OQ]$$

$$\text{So, } 65^\circ + 90^\circ - x + 90^\circ - x = 180^\circ$$

$$\Rightarrow x = \frac{65^\circ}{2} = 32.5^\circ$$

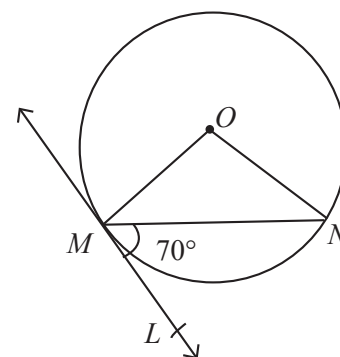
5. (c) 2



6. (d) $x = 90^\circ + 35^\circ = 125^\circ$

[By exterior angle theorem]

7. (b)



$$\angle OMN = 90^\circ - 70^\circ = 20^\circ$$

$$OM = ON$$

$$\text{So, } \angle ONM = \angle OMN = 20^\circ$$

$$\angle MON = 180^\circ - (20^\circ + 20^\circ) = 140^\circ$$

8. (a) Tangents makes equal angle with a chord.

$$\text{So, } \angle ACB = 42^\circ$$

$$\angle BAC = 180^\circ - (42^\circ + 42^\circ) = 96^\circ$$

9. (c) Since A is the point of tangency for both circles, the lengths of the tangents from P to each point of tangency (which are Q and R respectively) will be equal. Thus, $PQ = PR = AP$.

Therefore, the length of QR,

$$QR = PQ + PR$$

$$QR = 4.2 + 4.2$$

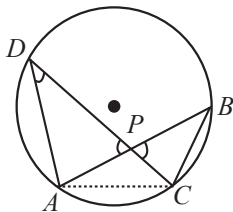
$$QR = 8.4 \text{ cm}$$

10. (d)

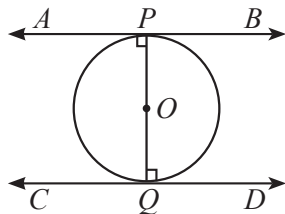
$$\angle ADC = \angle ABC$$

$$\angle APD = \angle CPB$$

$$\Rightarrow \triangle ADP \sim \triangle CBP$$



11. (b)



$$OP \perp AB \text{ and}$$

$$OQ \perp CD$$

$$\therefore \angle APO = \angle OQD = 90^\circ$$

$$\text{Hence, } AB \parallel CD$$

Reason (R) is true but not a correct explanation.

12. (b) Both assertion (A) and reason (R) are true but reason (R) does not give the correct explanation of assertion (A).

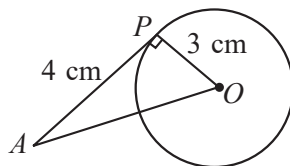
13. (c) $\angle OPQ = \angle OPR - \angle QPR$
 $= 90^\circ - 50^\circ = 40^\circ$

$$\text{Also } \angle OPQ = \angle OQP = 40^\circ$$

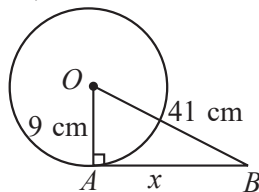
$$\angle POQ = 180^\circ - 40^\circ - 40^\circ = 100^\circ$$

14. (a) One tangent can be drawn to a circle from a point on it.

15. (b) $\angle OPA = 90^\circ$
 $\Rightarrow OA^2 = 3^2 + 4^2$
 $\Rightarrow OA = 5 \text{ cm}$



16. (a) In $\triangle OAB$,



$$OA^2 + AB^2 = OB^2$$

$$\Rightarrow 9^2 + x^2 = 41^2$$

$$\Rightarrow x^2 = 41^2 - 9^2 = 1600$$

$$\Rightarrow x = 40 \text{ cm}$$

17. (c) In $\triangle POT$,

$$(OP)^2 = (OT)^2 + (PT)^2$$

$$\Rightarrow OP^2 = (7)^2 + (24)^2$$

$$\Rightarrow OP^2 = (25)^2 \Rightarrow OP = 25 \text{ cm}$$

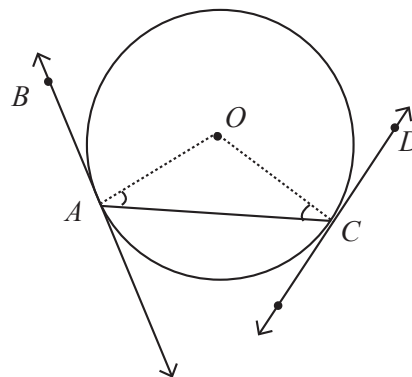
$$\therefore PR = OP + OR$$

$$= 25 + 7 = 32 \text{ cm}$$

Hence option (c) is correct.

Short Answer Type Questions

18.



$$OA = OC$$

$$\Rightarrow \angle OAC = \angle OCA \text{ and } \angle OAB = \angle OCD = 90^\circ$$

$$\Rightarrow \angle BAC = \angle DCA.$$

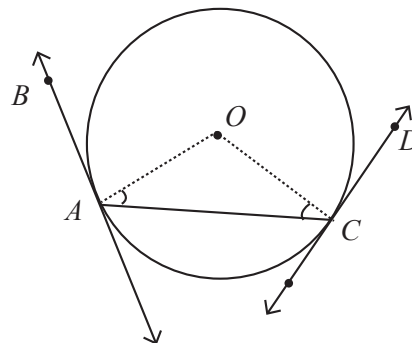
Hence proved.

19. Given:

Let a circle with centre O, AB and CD are two tangents at A and C. AC is a chord.

To prove: $\angle BAC = \angle DCA$

Construction: Join OA and OC.



$$\text{Proof: } OA = OC$$

$$\Rightarrow \angle OAC = \angle OCA$$

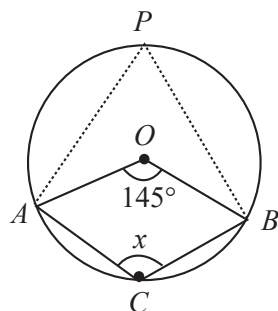
$$\text{and } \angle OAB = \angle OCD = 90^\circ$$

$$\Rightarrow \angle BAC = \angle DCA.$$

Hence proved.

20. Take a point P on the circle.

$$\text{Now, } \angle APB = \frac{145^\circ}{2}$$



$PACB$ is a cyclic quadrilateral

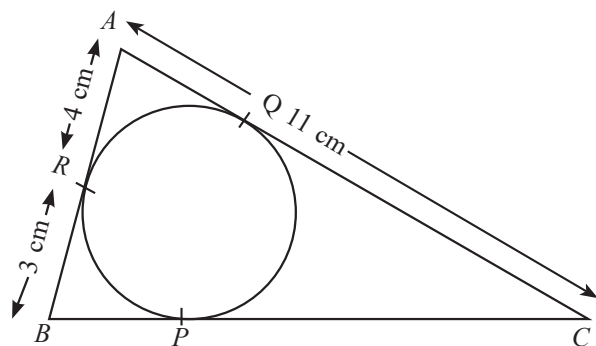
$$\text{So, } \angle P + x = 180^\circ$$

$$x = 180^\circ - \frac{145^\circ}{2}$$

$$= 180^\circ - 72.5^\circ$$

$$= 107.5^\circ$$

21.



$$AR = AQ$$

$$\Rightarrow AQ = 4 \text{ cm}$$

$$QC = AC - AQ$$

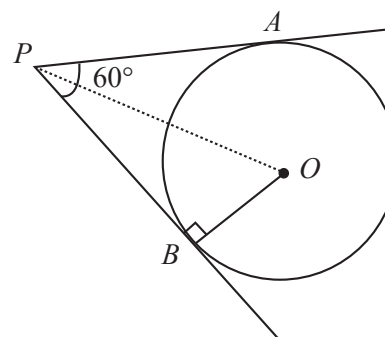
$$= 11 \text{ cm} - 4 \text{ cm} = 7 \text{ cm}$$

$$PC = QC = 7 \text{ cm}$$

$$BP = BR = 3 \text{ cm}$$

$$\Rightarrow BC = BP + PC = 3 \text{ cm} + 7 \text{ cm} \quad \boxed{BC = 10 \text{ cm}}$$

22.



OP bisect the $\angle APB$

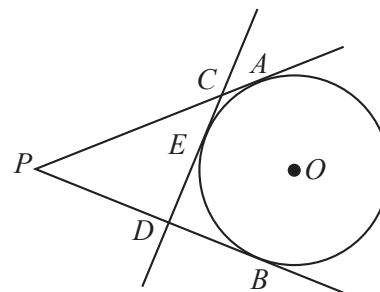
$$\text{So, } \angle OPB = 30^\circ$$

$$\tan 30^\circ = \frac{OB}{PB}$$

$$\frac{1}{\sqrt{3}} = \frac{3}{PB} \quad [OB = 3 \text{ cm}]$$

$$\Rightarrow \boxed{PB = 3\sqrt{3} \text{ cm}}$$

23.



$$\text{Perimeter of } \triangle PCD = PC + CE + ED + PD$$

$$CE = CA \quad \text{and}$$

$$ED = DB$$

$$\text{So, perimeter} = PC + CA + DB + PD$$

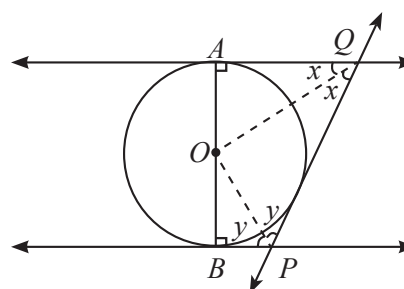
$$= PA + PB$$

$$= 2PA$$

$$[\because PA = PB]$$

$$\text{Perimeter} = 2 \times 10 \text{ cm} = 20 \text{ cm.}$$

24. OQ bisect $\angle AQP$ and similarly OP bisect $\angle BPQ$.



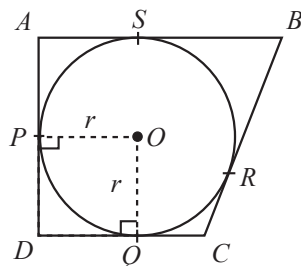
$$\text{Let } \angle AQO = \angle PQO = x, \text{ and}$$

$$\begin{aligned}\angle BPO &= \angle QPO = y \\ \Rightarrow 2x + 2y &= 180^\circ \Rightarrow x + y = 90^\circ \\ \Rightarrow \angle POQ &= 180^\circ - (x + y) \\ &= 180^\circ - 90^\circ = 90^\circ\end{aligned}$$

25. $BR = BS = 24$ cm

$$QC = RC = BC - BR$$

$$QC = 30 - 24 = 6$$
 cm

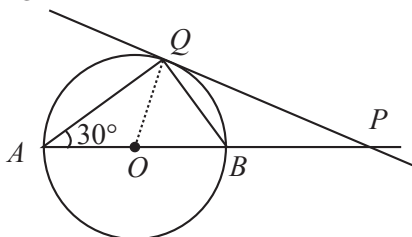


$$DQ = 8$$
 cm (radius)

$$DC = DQ + QC$$

$$= 8 \text{ cm} + 6 \text{ cm} = 14 \text{ cm}$$

26. Join OQ



$$\angle AQB = 90^\circ \quad [\text{angle made by semi circle}]$$

$$\Rightarrow \angle ABQ = 60^\circ \quad [\because OQ = OB]$$

$$OQ \perp PQ$$

$$\Rightarrow \angle BQP = \angle OQP - \angle OQB$$

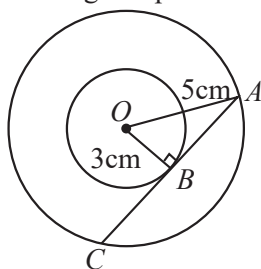
$$= 90^\circ - 60^\circ = 30^\circ$$

$$\angle BPQ = 30^\circ \quad [\text{By exterior angle property}]$$

$$\Rightarrow \angle BQP = \angle BPQ = 30^\circ$$

$$\Rightarrow \boxed{BQ = BP} \text{ . Hence proved.}$$

27. $\angle OBA = 90^\circ$ [A tangent to the circle is perpendicular to the radius through the point of contact]



$$OA^2 = OB^2 + AB^2$$

$$\Rightarrow (5)^2 = (3)^2 + AB^2$$

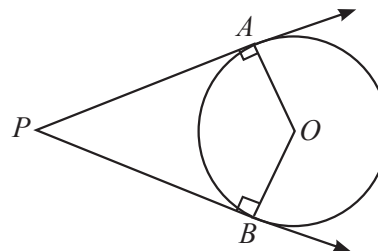
$$\Rightarrow AB^2 = 25 - 9$$

$$\Rightarrow AB = 4$$

$$\Rightarrow \text{Length of chord } AC = 2AB = 2(4) = 8 \text{ cm.}$$

The length of the chord of the larger circle which touches the smaller circle is 8 cm.

28. Given: A circle with centre O , PA and PB are tangents to the circle at A and B .



To prove: $\angle APB + \angle AOB = 180^\circ$

Proof: We know that radius is perpendicular to tangent at point of contact.

$$\Rightarrow OA \perp PA \text{ and } OB \perp PB$$

$$\Rightarrow \angle PAO = \angle PBO = 90^\circ$$

In quadrilateral $PBOA$,

$$\Rightarrow \angle APB + \angle PBO + \angle BOA + \angle OAP = 360^\circ$$

[Angle sum property of quadrilateral]

$$\Rightarrow \angle APB + 90^\circ + \angle BOA + 90^\circ = 360^\circ$$

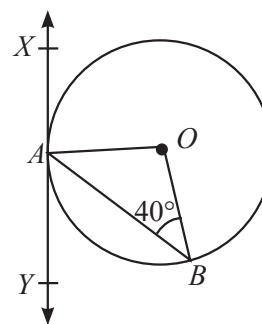
$$\Rightarrow \angle APB + \angle AOB = 180^\circ$$

Hence proved.

29. In $\triangle OAB$

$$OA = OB$$

$$\therefore \angle OAB = \angle OBA = 40^\circ \quad [\because OA = OB]$$



Since XAY is tangent to the circle.

$$\therefore \angle OAY = 90^\circ$$

[$\because$ The tangent to a circle is perpendicular to the radius through the point of contact]

$$\therefore \angle BAY + \angle OAB = 90^\circ$$

$$\angle BAY = 90^\circ - 40^\circ$$

$$\angle BAY = 50^\circ$$

Further in $\triangle ABO$,

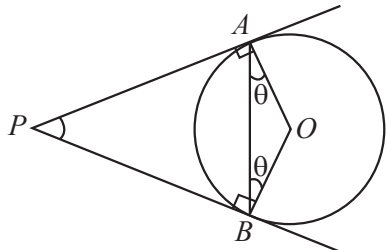
$$\angle OAB + \angle OBA + \angle AOB = 180^\circ$$

$$\angle AOB = 180^\circ - 80^\circ = 100^\circ.$$

30. Let $\angle OAB = \angle OBA = \theta$ [$\because OA = OB = \text{radius}$]

Also $OA \perp PA$

$$\therefore \angle AOB = 180^\circ - 2\theta \quad [\text{Angle sum property}]$$



In $OAPB$,

By angle sum property

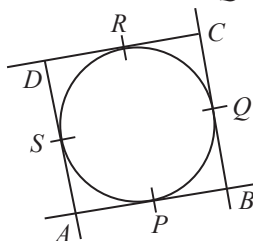
$$90^\circ + \angle APB + 90^\circ + \angle AOB = 360^\circ$$

$$90^\circ + \angle APB + 90^\circ + 180 - 2\theta = 360^\circ$$

$$\Rightarrow \angle APB = 2\angle OAB.$$

31. $\therefore$ Tangent from external point are equal in length.

Circle touches the sides at P, Q, R, S



$$\therefore AP = AS \quad \dots(i)$$

$$BP = BQ \quad \dots(ii)$$

$$CR = CQ \quad \dots(iii)$$

$$DR = DS \quad \dots(iv)$$

Adding equations (i) + (ii), (iii) and (iv)

$$AP + BP + CR + DR = AS + BQ + CQ + DS$$

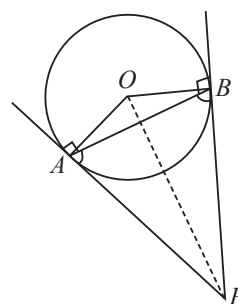
$$AB + CD = AD + BC$$

$$6 + 8 = AD + 9$$

$$AD = 14 - 9 = 5 \text{ cm}$$

32. AB is the chord

We know that $OA = OB$ [radii]



$$\angle OBP = \angle OAP = 90^\circ$$

$$OP = OP$$

[Common]

In $\triangle OAP$ and $\triangle OBP$

By RHS congruency

$$\triangle OBP \cong \triangle OAP$$

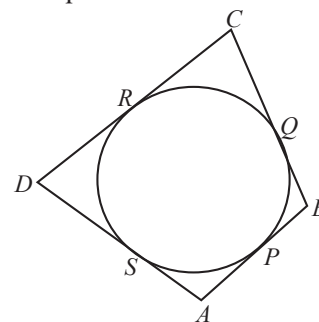
$$\therefore \text{By CPCT, } BP = AP$$

In $\triangle ABP$, $BP = AP$

Angles opposite to equal sides are equal

$$\therefore \angle BAP = \angle ABP.$$

33. $ABCD$ is the quadrilateral



Circle touches the sides at P, Q, R, S

For the circles AS & AP are tangents

$$\therefore AS = AP \quad \dots(i)$$

Similarly,

$$BP = BQ \quad \dots(ii)$$

$$CQ = CR \quad \dots(iii)$$

$$RD = DS \quad \dots(iv)$$

$$\text{Now, } AB + CD = AP + PB + CR + RD \quad \dots(v)$$

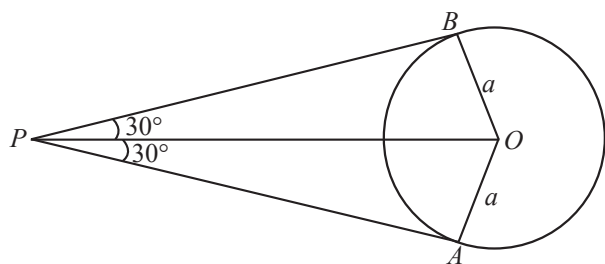
$$\text{and } BC + AD = BQ + QC + DS + AS \quad \dots(vi)$$

$$BC + AD = BP + CR + RD + AP$$

[using (i), (ii), (iii), (iv)]

$$\therefore AB + CD = BC + AD \quad [\text{using (v)}]$$

34. Given that $\angle BPA = 60^\circ$



$$OB = OA = a \quad [\text{radii}]$$

$$PA = PB \quad [\text{length of tangent are equal}]$$

$$OP = OP \quad [\text{Common}]$$

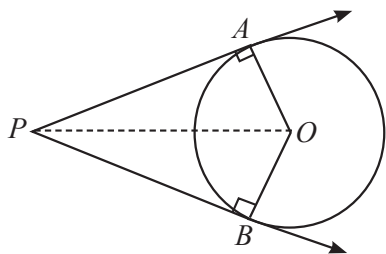
$\therefore \triangle PBO$ and $\triangle PAO$ are congruent
[By SSS criterion of congruency]

$$\therefore \angle BPO = \angle OPA = \frac{60^\circ}{2} = 30^\circ$$

$$\text{In } \triangle PBO, \sin 30^\circ = \frac{a}{OP} = \frac{1}{2} \quad (\because OB \perp BP)$$

$$OP = 2a \text{ units}$$

35. Let AP and BP be the two tangents to the circle with centre O .



To Prove: $AP = BP$

Proof: In $\triangle AOP$ and $\triangle BOP$,

$$OA = OB \quad [\text{radii of the same circle}]$$

$$\angle OAP = \angle OBP = 90^\circ$$

[since tangent at any point of a circle is perpendicular to the radius through the point of contact]

$$OP = OP \quad [\text{common}]$$

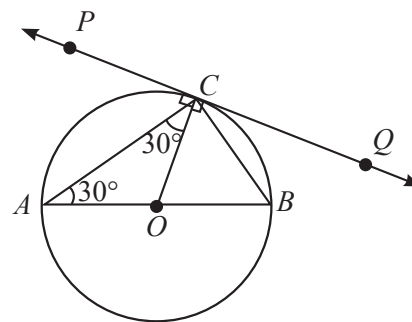
$\therefore \triangle AOP \cong \triangle BOP$ [by R.H.S. congruence criterion]

$$\therefore AP = BP$$

[corresponding parts of congruent triangles]

Hence, the length of the tangents drawn from an external point to a circle are equal.

36. In $\triangle ACO$,



$$OA = OC \quad [\text{Radii of the same circle}]$$

$\therefore \triangle ACO$ is an isosceles triangle

$$\angle CAB = 30^\circ \quad [\text{Given}]$$

$$\therefore \angle CAO = \angle ACO = 30^\circ$$

[angles opposite to equal sides of an isosceles triangle are equal]

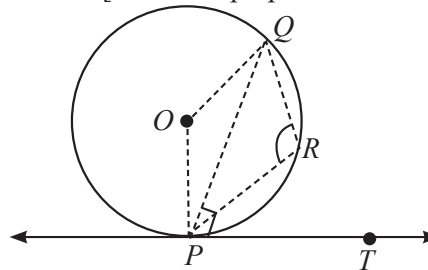
$$\angle PCO = 90^\circ$$

[Radius drawn at the point of contact is perpendicular to the tangent]

$$\text{Now } \angle PCA = \angle PCO - \angle ACO$$

$$\therefore \angle PCA = 90^\circ - 30^\circ = 60^\circ$$

37. $\angle OPT = 90^\circ$ [Radius is perpendicular to the tangent]



$$\text{So, } \angle OPQ = \angle OPT - \angle QPT$$

$$= 90^\circ - 60^\circ = 30^\circ$$

$$\angle POQ = 180^\circ - 2\angle QPO = 180^\circ - 60^\circ = 120^\circ$$

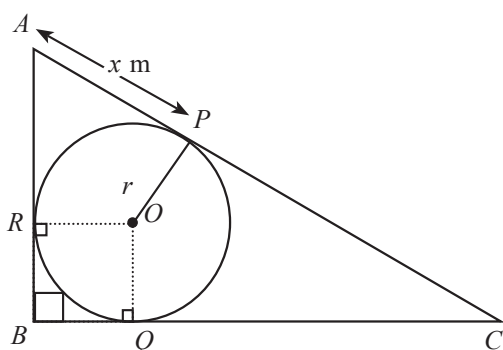
$$\text{Reflex } \angle POQ = 360^\circ - 120^\circ = 240^\circ$$

$$\angle PRQ = \frac{1}{2} \text{ reflex } \angle POQ = \frac{1}{2} \times 240^\circ = 120^\circ$$

$$\therefore \angle PRQ = 120^\circ$$

Case Based Type Questions

38. (i) $AR = AP = x \text{ m.}$



- (ii) $OR \perp AB$ and $OQ \perp BC$
and $\angle RBQ = 90^\circ$

So, $BQOR$ is a square.

- (iii) (a) $PC = QC = BC - BQ$

$$PC = 15 - r$$

$$\text{and } r = RB = 7 - AR$$

$$\Rightarrow r = 7 - x$$

$$\text{So, } PC = 15 - (7 - x)$$

$$\text{Now, } AC = \sqrt{AB^2 + BC^2}$$

$$= \sqrt{7^2 + 15^2} = \sqrt{274}$$

$$AC = \sqrt{274} = x + 8 + x$$

$$\Rightarrow 8 + 2x = 16.55 \text{ (approx.)}$$

$$x = 4.27 \text{ m.}$$

OR

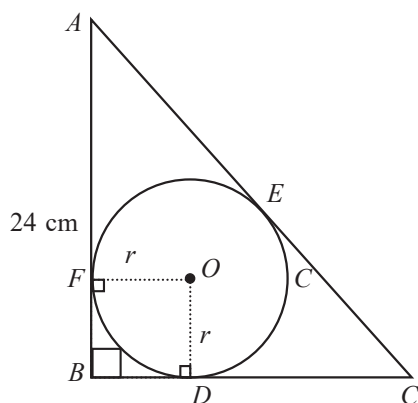
- (iii) (b) $r = RB = 7 - x$

$$= 7 - 4.27$$

$$r = 2.73 \text{ m (approx.)}$$

Long Answer Type Questions

39.



Let radius = r cm

$AB \perp OF$ and

$BC \perp OD$

$$\Rightarrow BF = BD = r$$

Tangents drawn from an external point to the circle are equal.

$$AE = AF = 24 - r \quad \dots(i) \text{ and}$$

$$CE = CD = 7 - r \quad \dots(ii)$$

$$AC = AE + CE$$

$$24 - r + 7 - r \quad [\text{from (i) and (ii)}]$$

$$AC = \sqrt{AB^2 + BC^2} = 31 - 2r$$

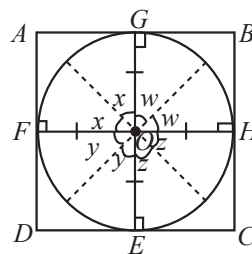
$$\Rightarrow \sqrt{24^2 + 7^2} = 31 - 2r$$

$$\Rightarrow 25 = 31 - 2r$$

$$\Rightarrow r = 3$$

Hence, radius of circle = 3 cm.

40. In $\triangle AOG$ and $\triangle AOF$,



$$OG = OF = \text{radius,}$$

$$OA = OA = (\text{Common})$$

$$AG = AF = (\text{tangents})$$

$$\therefore \triangle AOG \cong \triangle AOF \text{ (by SSS)}$$

$$\Rightarrow \angle AOG = \angle AOF = x \text{ (by cpct)}$$

Similarly,

$$\angle DOF = \angle DOE = y, \angle COE = \angle COH = z$$

$$\text{and } \angle BOH = \angle BOG = w$$

$$\Rightarrow 2(x + y + z + w) = 360^\circ$$

$$\Rightarrow x + y + z + w = 180^\circ$$

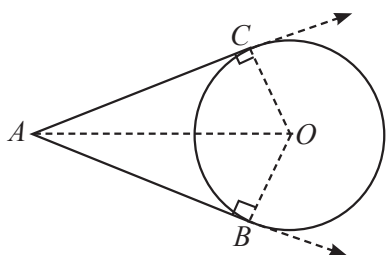
$$\Rightarrow (w + x) + (y + z) = \angle AOB + \angle COD = 180^\circ$$

41. (a) The lengths of two tangents drawn from an external point to a circle are equal.

Given: AB and AC are two tangents from a point A to a circle $C(O, r)$.

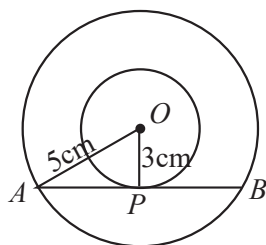
To Prove: $AB = AC$

Construction: Join OA , OB and OC .



Proof: In $\triangle OBA$ and $\triangle OCA$,
 $OB = OC$ [Radii of the same circle]
 $OA = OA$ [Common side]
 $\angle OBA \cong \angle OCA = 90^\circ$
 [Each 90° because tangent is perpendicular to radius at the point of contact]
 $\triangle OBA \cong \triangle OCA$ [By R.H.S. congruency]
 $AB = AC$ [CPCT]

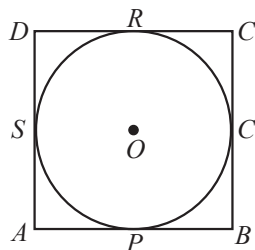
(b) In $\triangle OAP$



$OP \perp AB$
 $OA^2 - OP^2 = AP^2$
 $5^2 - 3^2 = AP^2$
 $16 = AP^2$
 $\therefore AP = 4 \text{ cm.}$
 Length of chord $AB = 2AP = 8 \text{ cm}$

42. (a) A circle with centre O .

A parallelogram $ABCD$ touching the circle at Points P, Q, R and S .



To Prove: $ABCD$ is a rhombus

Proof: A rhombus is a parallelogram with all sides equal

In parallelogram $ABCD$

$ABCD$ and $BC = AD$

We know that the lengths of tangents from an external point are equal

$$\therefore AP = AS \quad \dots(i)$$

$$BP = BQ$$

$$CQ = CR \quad \dots(ii)$$

$$DR = DS \quad \dots(iii)$$

Adding (i), (ii), (iii) and (iv), we get

$$\Rightarrow AP + BP + CR + DR = AS + BQ + CQ + DS$$

$$\Rightarrow AB + (CR + DR) = AS + BQ + CQ + DS$$

$$\Rightarrow AB + CD = (AS + DS) + (BQ + CQ)$$

$$\Rightarrow AB = CD = AD + BC$$

$$\Rightarrow CD + CD = BC + BC$$

$$[\because AB = CD \text{ and } AD = BC]$$

$$\Rightarrow CD = BC$$

$$\therefore AB = CD = BC = AD$$

All sides are equal

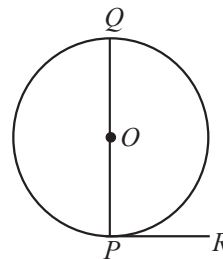
Hence, $ABCD$ is a rhombus

OR

(b) Let, O is the centre of the given circle. A segment PR has been drawn touching the circle at point P .

Draw $QP \perp RP$ at point P , such that point Q lies on the circle

Draw $QP \perp RP$ at point P , such that point Q lies on the circle.



$$\angle OPR = 90^\circ \quad [\text{Radius Tangent}]$$

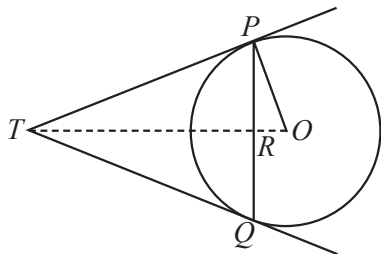
$$\text{Also, } \angle QPR = 90^\circ \quad [\text{given}]$$

$$\angle OPR = \angle QPR$$

Now, the above case is possible only when centre O lies on the line QP .

Hence, perpendicular at the point of contact to the tangent to a circle passes through the centre of the circle.

43.



In $\triangle ORP$ and $\triangle OPT$,
 $\angle ORP = \angle OPT$ [Each 90°]
 $\angle POR = \angle POT$ [Common]
 $\therefore \triangle ORP \sim \triangle OPT$ [By AA similarity]

$$\therefore \frac{OR}{OP} = \frac{PR}{PT}$$

In $\triangle PQR$,
 $OP^2 = OR^2 + PR^2$ [By pythagoras theorem]

$$\therefore (5)^2 = OR^2 + (4)^2 \quad \left[\because PR = \frac{PQ}{2} \right]$$

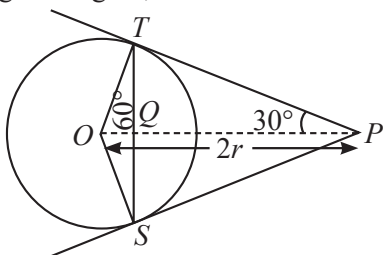
$$\Rightarrow OR = 3 \text{ cm}$$

From (i)

$$\frac{3}{5} = \frac{4}{PT}$$

$$\therefore PT = \frac{20}{3} \text{ cm}$$

44. In the given figure,



$$OP = 2r \quad \text{[Given]}$$

$$\angle OTP = 90^\circ$$

[radius drawn at the point of contact is perpendicular to the tangent]

In $\triangle OPT$,

$$\sin \angle OPT = \frac{OT}{OP} = \frac{r}{2r} = \frac{1}{2} = \sin 30^\circ$$

$$\angle OPT = 30^\circ$$

$$\angle TOP = 60^\circ$$

$\therefore \triangle OPT$ is a $30^\circ - 60^\circ - 90^\circ$, right triangle.

In $\triangle OTS$,

$$OT = OS \quad \text{[Radii of the same circle]}$$

$\therefore \triangle OTS$ is an isosceles triangle.

$$\therefore \angle OTS = \angle OST$$

[Angles opposite to equal sides of an isosceles triangle are equal]

In $\triangle OTQ$ and $\triangle OSQ$

$$OS = OT$$

$$\angle OTQ = \angle OSQ$$

[Angles opposite to equal sides of an isosceles triangle are equal]

$$\therefore \triangle OTQ = \triangle OSQ \quad \text{[By S.A.S.]}$$

$$\therefore \angle TOQ = \angle SOQ = 60^\circ \quad \text{[C.A.C.T.]}$$

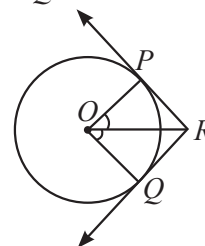
$$\therefore \angle TOS = 120^\circ$$

$$[\angle TOS = \angle TOQ + \angle SOQ = 60^\circ + 60^\circ = 120^\circ]$$

$$\therefore \angle OST + \angle OST = 180^\circ - 120^\circ = 60^\circ$$

$$\therefore \angle OTS + \angle OST = 60^\circ \div 2 = 30^\circ$$

45. Given that $\angle PRQ = 120^\circ$



We know that the line joining the centre and the external point is the angle bisector of angle between the tangents.

$$\text{Thus, } \angle PRO = \angle QRO = \frac{120^\circ}{2} = 60^\circ$$

Also we know that lengths of tangents from an external point are equal.

$$\text{Thus, } PR = RQ$$

Join OP and OQ .

Since OP and OQ are the radii from the centre O .

$$OP \perp PR \text{ and } OQ \perp RQ.$$

Thus, $\triangle OPR$ and $\triangle OQR$ are right angled congruent triangle.

$$\text{Hence, } \angle POR = 90^\circ - \angle PRO = 90^\circ - 60^\circ = 30^\circ$$

$$\angle QOR = 90^\circ - \angle QRO = 90^\circ - 60^\circ = 30^\circ$$

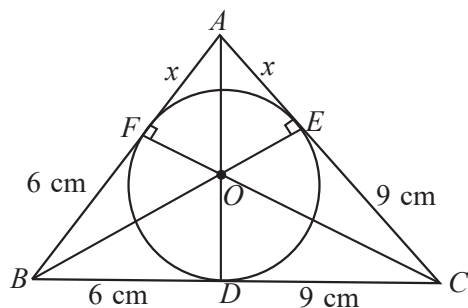
$$\sin \angle QRO = \sin 30^\circ = \frac{1}{2}$$

$$\frac{PR}{OR} = \frac{1}{2}$$

$$\text{Thus, } \Rightarrow OR = 2PR$$

$$\Rightarrow OR = PR + PR \Rightarrow OR = PR + QR$$

46. Let the given circle touch the sides AB and AC of the triangle at point F and E respectively and the length of the line segment AF be x .



Now, it can be observed that;

$$BF = BD = 6 \text{ cm} \quad (\text{tangents from point } B)$$

$$CE = CD = 9 \text{ cm} \quad (\text{tangents from point } C)$$

$$AE = AF = x \quad (\text{tangents from point } A)$$

$$AB = AF + FB = x + 6$$

$$BC = BD + DC = 6 + 9 = 15$$

$$CA = CE + EA = 9 + x$$

$$2s = AB + BC + CA = x + 6 + 15 + 9 + x = 30 + 2x$$

$$s = 15 + x$$

$$s - a = 15 + x - 15 = x$$

$$s - b = 15 + x - (x + 9) = 6$$

$$s - c = 15 + x - (6 + x) = 9$$

$$\text{Area of } \triangle ABC = \sqrt{s(s-a)(s-b)(s-c)}$$

$$54 = \sqrt{(15+x)(x)(6)(9)}$$

$$54 = 3\sqrt{6(15x+x^2)}$$

$$18 = \sqrt{6(15x+x^2)}$$

$$324 = 6(15x+x^2)$$

$$54 = 15x + x^2$$

$$x^2 + 15x - 54 = 0$$

$$x^2 + 18x - 3x - 54 = 0$$

$$x(x+18) - 3(x+18) = 0$$

$$(x+18)(x-3) = 0$$

As distance cannot be negative, $x = 3 \text{ cm}$

$$AC = 3 + 9 = 12 \text{ cm}$$

$$AB = AF + FB = 6 + x = 6 + 3 = 9 \text{ cm.}$$

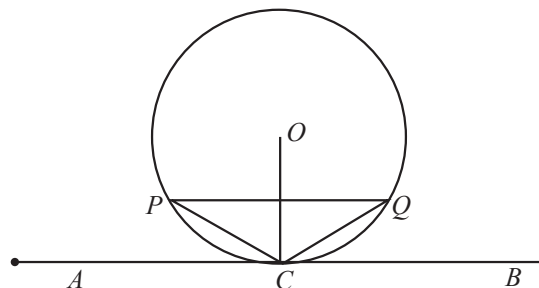
47. In the figure, C is the midpoint of the minor arc PQ , O is the centre of the circle and AB is tangent to the circle through point C .

We have to show the tangent drawn at the midpoint of the arc PQ of a circle is parallel to the chord joining the end points of the arc PQ .

We will show $PQ \parallel AB$.

It is given that C is the midpoint point of the arc PQ . So, arc PC = arc CQ .

$$\Rightarrow PC = CQ$$



This shows that $\triangle PQC$ is an isosceles triangle.

Thus, the perpendicular bisector of the side PQ of $\triangle PQC$ passes through vertex C .

The perpendicular bisector of a chord passes through the centre of the circle.

So the perpendicular bisector of PQ passes through the centre O of the circle.

Thus perpendicular bisector of PQ passes through the points O and C .

$$\Rightarrow PQ \perp OC$$

AB is the tangent to the circle through the point C on the circle.

$$\Rightarrow AB \perp OC$$

The chord PQ and the tangent AB of the circle are perpendicular to the same line OC .

$$\therefore PQ \parallel AB.$$



Previous Year Questions

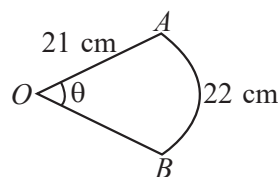
Multiple Choice Questions

- Perimeter of a sector of a circle whose central angle is 90° and radius 7 cm is: [CBSE 2024]
 (a) 35 cm (b) 11 cm
 (c) 22 cm (d) 25 cm
- The area of the sector of a circle of radius 12 cm is $60\pi \text{ cm}^2$. The central angle of this sector is: [CBSE 2024]
 (a) 120° (b) 6°
 (c) 75° (d) 150°
- Assertion (A):** If the circumference of a circle is 176 cm, then its radius is 28 cm.
Reason (R): Circumference = $2\pi \times$ radius of a circle. [CBSE 2024]
 (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
 (c) Assertion (A) is true but Reason (R) is false.
 (d) Assertion (A) is false but Reason (R) is true
- The length of an arc of a circle with radius 12 cm is 10π cm. The angle subtended by the arc at the centre of the circle, is: [CBSE 2024]
 (a) 120° (b) 6°
 (c) 75° (d) 150°
- The length of the arc of a circle of radius 14 cm which subtends an angle of 60° at the centre of the circle is [CBSE 2023]
 (a) $\frac{44}{3}$ cm (b) $\frac{88}{3}$ cm
 (c) $\frac{308}{3}$ cm (d) $\frac{616}{3}$ cm

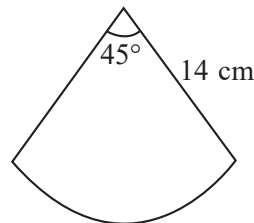
- If the radius of a semi-circular protractor is 7 cm, then its perimeter is [CBSE 2023]
 (a) 11 cm (b) 14 cm
 (c) 22 cm (d) 36 cm

- What is the area of a semi-circle of diameter 'd'? [CBSE 2023]
 (a) $\frac{1}{16}\pi d^2$ (b) $\frac{1}{4}\pi d^2$
 (c) $\frac{1}{8}\pi d^2$ (d) $\frac{1}{2}\pi d^2$

- A circular arc of length 22 cm subtends an angle θ at the centre of the circle of radius 21 cm. The value of θ is [CBSE 2021]



- The perimeter of the sector of a circle of radius 14 cm and central angle 45° is [CBSE 2021]



- The area of a quadrant of a circle where the circumference of circle is 176 m, is [CBSE 2021]
 (a) 2464 m^2 (b) 1232 m^2
 (c) 616 m^2 (d) 308 m^2

11. The minute hand of a clock is 84 cm long. The distance covered by the tip of minute hand from 10:10 AM to 10:25 AM is [CBSE 2021]
- (a) 44 cm (b) 88 cm
(c) 132 cm (d) 176 cm

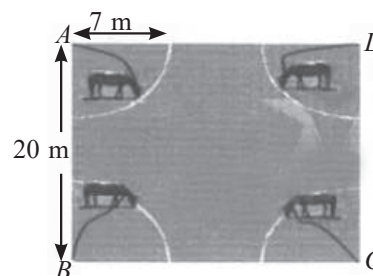
Short Answer Type Questions

12. An arc of a circle of radius 21 km subtends an angle of 60° at the centre. Find: [CBSE 2024]
- (i) The length of the arc.
(ii) The area of the minor segment of the circle made by the corresponding chord.
13. The perimeter of a certain sector of a circle of radius 5.6 m is 20.0 m. Find the area of the sector.
- (i) The length of the arc.
(ii) The area of the minor segment of the circle made by the corresponding chord. [CBSE 2024]
14. (a) The minute hand of a clock is 14 cm long. Find the area on the face of the clock described by the minute hand in 5 minutes. [CBSE 2024]
(b) Find the length of the arc of a circle which subtends an angle of 60° at the centre of the circle of radius 42 cm. [CBSE 2024]
15. An arc of a circle of radius 10 cm subtends a right angle at the centre of the circle. Find the area of the corresponding major sector. [Use $\pi = 3.14$] [CBSE 2024]
16. Find the area of the sector of a circle of radius 7 cm and of central angle 90° . Also, find the area of corresponding major sector. [CBSE 2023]
17. A piece of wire 22 cm long is bent into the form of an arc of a circle subtending an angle of 60° at its centre. Find the radius of the circle. [Use $\pi = \frac{22}{7}$] [CBSE 2020]
18. Find the area of the minor segment of a circle of radius 14 cm, when its central angle is 60° . Also find the area of the corresponding major segment [Use $\pi = \frac{22}{7}$] [CBSE 2015]

Long Answer Type Questions

19. A stable owner has four horses. He usually tie these horses with 7 m long rope to pegs at each corner of a square shaped grass field of 20 m length, to graze in his farm. But tying with rope sometimes

results in injuries to his horses, so he decided to build fence around the area so that each horse can graze. [CBSE 2024]



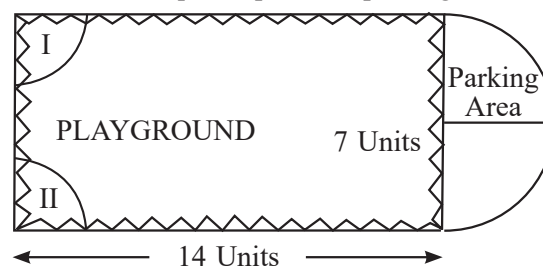
Based on the above, answer the following questions:

- (i) Find the area of the square shaped grass field
(ii) (a) Find the area of the total field in which these horses can graze.

OR

- (b) If the length of the rope of each horse is increased from 7 m to 10 m, find the area grazed by one horse. (Use $\pi = 3.14$)
(iii) What is area of the field that is left ungrazed, if the length of the rope of each horse is 7 cm?

20. Governing council of a local public development authority of Dehradun decided to build an adventurous playground on the top of a hill, which will have adequate space for parking.



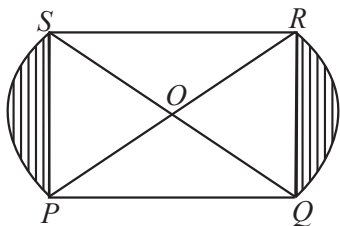
After survey, it was decided to build rectangular playground, with a semi-circular area allotted for parking at one end of the playground. The length and breadth of the rectangular playground are 14 units and 7 units, respectively. There are two quadrants of radius 2 units on one side for special seats. Based on the above information, answer the following questions:

- (i) What is the total perimeter of the parking area? [CBSE 2023]
(ii) (a) What is the total area of parking and the two quadrants? [CBSE 2023]

OR

- (b) What is the ratio of area of playground to the area of parking area? [CBSE 2023]
(iii) Find the cost of fencing the playground and parking area at the rate of ₹2 per unit. [CBSE 2023]

21. In the following figure, $PQRS$ is square lawn with side $PQ = 42$ metres. Two circular flower beds are there on the sides PS and QR with centre at O , the intersections of its diagonals. Find the total area of the two flower beds (shaded parts). [CBSE 2015]





Explanations

Previous Year Questions

Multiple Choice Questions

1. (d) Perimeter of sector

$$= \frac{90^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 7 + 2 \times 7 = 25 \text{ cm}$$

2. (d) $\frac{\theta}{360^\circ} \times \pi \times 12 \times 12 = 60\pi$

$$\theta = \frac{60 \times 360}{12 \times 12} = 150^\circ$$

3. (a) **Assertion:** $2 \times \frac{22}{7} \times r = 176 \text{ cm}$

$$\Rightarrow r \times \frac{176 \times 7}{2 \times 22} = 28 \text{ cm}$$

Hence, assertion is true.

Reason: Circumference $= 2\pi r$

Hence, both the statements are true and Reason (R) is the correct explanation of Assertion (A).

4. (d) $\frac{\theta}{360} \times 2 \times \pi \times 12 = 10\pi$

$$\theta = \frac{10 \times 360}{2 \times 12} = 150^\circ$$

5. (a) Length of arc $= \frac{60^\circ}{360^\circ} \times 2\pi \times 14 = \frac{2\pi \times 14}{6} = \frac{44}{3} \text{ cm}$

6. (d) Perimeter of protractor $= \pi r + 2r$

$$= \frac{22}{7} \times 7 + 2 \times 7 = 22 + 14 = 36 \text{ cm}$$

7. (c) Area of semicircle $= \frac{\pi \left(\frac{d}{2}\right)^2}{2} = \frac{1}{8} \pi d^2$

8. (c) $\frac{2\pi r \times \theta}{360^\circ} = 22$

$$\frac{2 \times 22}{7} \times \frac{21 \times \theta}{360^\circ} = 22$$

$$\therefore \theta = 60^\circ$$

9. (d) Perimeter of the sector

$$= 2\pi R \times \left(\frac{\theta}{360^\circ}\right) + 2R$$

$$= \frac{45^\circ}{360^\circ} \times 2\pi R + 2R = 39 \text{ cm}$$

$$= \frac{45^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 14 + 2 \times 14$$

10. (c) Area of quadrant $= \frac{\pi r^2}{4}$

$$= \frac{22 \times 28 \times 28}{7 \times 4} = 616 \text{ m}^2 \quad [\because 2\pi r = 176 \text{ m}]$$

$$\left[r = \frac{176 \times 7}{2 \times 22} = 28 \text{ m} \right]$$

$$= 12 \times \frac{22}{7} \times \frac{14}{6} = \frac{44}{3}$$

11. (c) Angle made by minute hand of a clock in 1 minute $= 6^\circ$

Angle made by minute hand of the clock between 10:10 am to 10:28 am.

$$(\text{i.e. 15 minutes}) = 15 \times 6^\circ = 90^\circ$$

$$\text{Distance covered } (l) = \frac{90^\circ}{180^\circ} \times \frac{22}{7} \times 84$$

$$\left[\because l = \frac{\theta}{180^\circ} \times \pi r \right]$$

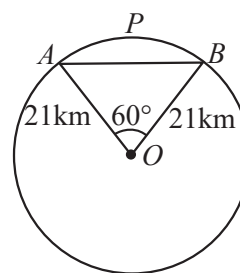
Short Answer Type Questions

12. $r = 21 \text{ cm}$, $\theta = 60^\circ$

(i) Length of arc $APB = \frac{\theta}{360^\circ} \times 2\pi r$

$$= \frac{60^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 21 = 22 \text{ km}$$

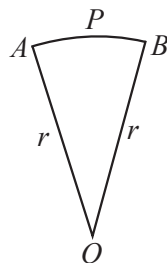
(ii) Area of minor segment = Area of sector $APBO$
– Area of $\triangle AOB$



$$\begin{aligned}
 &= \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 21 \times 21 - \frac{1}{2} \times 21 \times 21 \times \sin 60^\circ \\
 &= 231 - \frac{1}{2} \times 21 \times 21 \times \frac{\sqrt{3}}{2} \\
 &= (231 - 110.25\sqrt{3}) \text{ cm}^2 = 40.05 \text{ cm}^2 \text{ (approx.)}
 \end{aligned}$$

13. $2r + \frac{2\pi r\theta}{360} = 20$

$$\Rightarrow 2 \times 5.6 + \frac{2\pi \times 5.6 \times \theta}{360} = 20$$



$$\Rightarrow 11.2 + 2 \times \frac{22}{7} \times 5.6 \times \frac{\theta}{360} = 20$$

Solving we get $\theta = 90^\circ$.

(i) Length of arc $= \frac{90^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 5.6 = 8.8 \text{ m}$

(ii) Area of minor segment

$$\begin{aligned}
 &= \frac{22}{7} \times 5.6 \times 5.6 \times \frac{90^\circ}{360^\circ} - \frac{1}{2} \times 5.6 \times 5.6 \times \sin 90^\circ \\
 &= 24.64 - 15.68 = 8.96 \text{ m}^2
 \end{aligned}$$

14. (a) Angle subtended in 5 min. $= 30^\circ$

$$\begin{aligned}
 \text{Area described by minute hand} &= \frac{30}{360^\circ} \times \frac{22}{7} \times 14 \times 14 \\
 &= \frac{154}{3} \text{ cm}^2 \text{ or } 51.33 \text{ cm}^2 \text{ approx.}
 \end{aligned}$$

(b) Length of arc $= 2 \times \frac{22}{7} \times 42 \times \frac{60}{360} = 44 \text{ cm}$

15. Area of circle $= 3.14 \times 10 \times 10 = 314 \text{ cm}^2$

$$\text{Area of minor sector} = \frac{3.14 \times 10 \times 10 \times 90^\circ}{360^\circ} = 78.5 \text{ cm}^2$$

$$\text{Area of major sector} = 314 - 78.5 = 235.5 \text{ cm}^2$$

16. Area of sector $= \frac{\pi r^2 \theta}{360^\circ}$

$$= \frac{22}{7} \times \frac{7 \times 7 \times 90^\circ}{360^\circ} = 38.5 \text{ cm}^2$$

Area of corresponding major sector

$= \text{Area of circle} - \text{Area of minor sector}$

$$= \pi r^2 - 38.5$$

$$\begin{aligned}
 &= \left(\frac{22}{7} \times 7 \times 7 \right) - 38.5 \\
 &= (154 - 38.5) \text{ cm}^2 = 115.5 \text{ cm}^2
 \end{aligned}$$

17. Length of arc $= 22 \text{ cm}$

$$\Rightarrow \frac{2\pi r\theta}{360^\circ} = 22$$

$$\Rightarrow 2 \times \frac{22}{7} \times r \times \frac{60^\circ}{360^\circ} = 22$$

$$\Rightarrow r = \frac{22 \times 7 \times 6}{2 \times 22} \Rightarrow r = 21 \text{ cm}$$

18. Radius of the circle $= 14 \text{ cm}$

Central angle, $\theta = 60^\circ$,

Area of the minor segment

$$\begin{aligned}
 &= \frac{\theta}{360^\circ} \times \pi r^2 - \frac{\sqrt{3}}{4} r^2 \\
 &= \frac{60^\circ}{360^\circ} \times \pi (14)^2 - \frac{\sqrt{3}}{4} \times 14^2 \\
 &= \frac{1}{6} \times \frac{22}{7} \times 14 \times 14 - \sqrt{3} \times (7)^2 = \frac{22 \times 14}{3} - 49\sqrt{3} \\
 &= \frac{22 \times 14}{3} - \frac{147\sqrt{3}}{3} = \frac{308 - 147\sqrt{3}}{3} \text{ cm}^2
 \end{aligned}$$

$\therefore$ Area of the major segment

$$\begin{aligned}
 &= \pi (14)^2 - \left(\frac{308 - 147\sqrt{3}}{3} \right) \text{ cm}^2 \\
 &= 616 - \frac{1}{3} [308 - 147\sqrt{3}] = \frac{(1540 + 147\sqrt{3})}{3} \text{ cm}^2
 \end{aligned}$$

Long Answer Type Questions

19. (i) Area of square shaped grass field $= 400 \text{ m}^2$

(ii) (a) Area of total field that horses can graze

$$= 4 \times \frac{1}{4} \times \frac{22}{7} \times 7 \times 7 = 154 \text{ m}^2$$

OR

(b) Area grazed by one horse

$$= \frac{1}{4} \times 3.14 \times 10 \times 10 = 78.5 \text{ m}^2$$

(iii) Area of the field left ungrazed $= \text{area of square field} - \text{area of field in which horses can graze.}$

Area of the field left ungrazed

$$= 400 - 154 = 246 \text{ m}^2$$

20. (i) Here

Radius of parking, $r = \frac{7}{2} = 3.5$ units

$$\begin{aligned}\text{Perimeter of parking area} &= 2r + \pi r \\ &= 2 \times \frac{7}{2} + \frac{22}{7} \times \frac{7}{2} \\ &= 7 + 11 = 18 \text{ units}\end{aligned}$$

(ii) (a) Radius of quadrant $r' = 2$ units

Radius of parking area, $r = \frac{7}{2}$ units

$$\begin{aligned}\therefore \text{Required area} &= \frac{\pi r^2}{2} + 2 \times \frac{\pi r'^2}{4} \\ &= \frac{\pi}{2}(r^2 + r'^2) \\ &= \frac{22}{7 \times 2} \left(\left(\frac{7}{2} \right)^2 + (2)^2 \right) = \frac{11}{7} \left(\frac{49}{4} + 4 \right) \\ &= \frac{11 \times 65}{28} = 25.54 \text{ square units (approx.)}\end{aligned}$$

OR

(b) Area of playground $= l \times b$ – Area of 2 quadrant
 $= 14 \times 7 = 98$ square units.

$$= 98 - \frac{2}{4} \times \frac{22}{7} \times 2 \times 2 = 98 - \frac{44}{7}$$

$$\text{Area of parking area} = \frac{\pi r^2}{2}$$

$$= \frac{\frac{22}{7} \times \frac{7}{2} \times \frac{7}{2}}{2} = \frac{77}{4} \text{ square units}$$

Required ratio

$$= \frac{98 - \frac{44}{7}}{\frac{77}{4}} = \frac{\frac{686 - 44}{7}}{\frac{77}{4}} = \frac{642}{7} \times \frac{4}{77} \Rightarrow 2568 : 539$$

(iii) Length of fencing required $= 2l + b + \pi r$

$$= 2(14) + 7 + \frac{22}{7} \times \frac{7}{2}$$

$$= 28 + 7 + 11 = 46 \text{ units}$$

$$\text{Cost of fencing} = 46 \times 2 = ₹92.$$

21. PQRS is a square

So each side is equal and angle between the adjacent sides is a right angle.

Also the diagonals perpendicularly bisector each other.

In $\triangle PQR$ using pythagoras theorem,

$$PR^2 = PQ^2 + QR^2$$

$$PR^2 = (42)^2 + (42)^2$$

$$PR = 42\sqrt{2} \text{ m}$$

$$OR = \frac{1}{2}PR = \frac{42}{\sqrt{2}} = OQ$$

From the figure we can see that the radius of flower bed ORQ is OR.

$$\text{Area of sector } ORQ = \frac{1}{4}\pi r^2 = \frac{1}{4}\pi \left(\frac{42}{\sqrt{2}} \right)^2$$

$$\text{Area of the } \triangle ROQ = \frac{1}{2} \times RO \times OQ$$

$$= \frac{1}{2} \times \frac{42}{\sqrt{2}} \times \frac{42}{\sqrt{2}} = \left(\frac{42}{2} \right)^2$$

Area of the flower bed ORQ

= Area of sector ORQ – Area of the $\triangle ROQ$

$$= \frac{1}{4}\pi \left(\frac{42}{\sqrt{2}} \right)^2 - \left(\frac{42}{2} \right)^2 = \left(\frac{42}{\sqrt{2}} \right)^2 \left[\frac{\pi}{4} - \frac{1}{2} \right]$$

$$= 251.37 \text{ cm}^2$$

Area of the flower bed ORQ = Area of the flower bed OPS = 251.37 cm²

Total area of the flower beds

= Area of the flower bed ORQ + Area of the flower bed OPS.

$$= 251.37 + 251.37 = 502.74 \text{ m}^2$$



Previous Year Questions

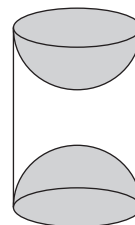
Multiple Choice Questions

- A solid sphere is cut into two hemispheres. The ratio of the surface areas of sphere to that of two hemispheres taken together, is: [CBSE 2024]
 (a) 1 : 1 (b) 1 : 4
 (c) 2 : 3 (d) 3 : 2
- The volume of the largest right circular cone that can be carved out from a solid cube of edge 2 cm is: [CBSE 2024]
 (a) $\frac{4\pi}{3} \text{ cm}^3$ (b) $\frac{5\pi}{3} \text{ cm}^3$
 (c) $\frac{8\pi}{3} \text{ cm}^3$ (d) $\frac{2\pi}{3} \text{ cm}^3$
- The ratio of total surface area of a solid hemisphere to the square of its radius is: [CBSE 2024]
 (a) $2\pi : 1$ (b) $4\pi : 1$
 (c) $3\pi : 1$ (d) $1 : 4\pi$
- Assertion (A):** Two cubes each of edge length 10 cm are joined together. The total surface area of newly formed cuboid is 1200 cm^2 .
Reason (R): Area of each surface of a cube of side 10 cm is 100 cm^2 . [CBSE 2024]
 (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
 (c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

Short Answer Type Questions

- A wooden article was made by scooping out a hemisphere from each end of a solid cylinder, as shown in the figure. If the height of the cylinder is 5.8 cm and its base is of radius 2.1 cm, find the total surface area of the article. [CBSE 2024]



Case Based Type Questions

- The word 'circus' has the same root as 'circle'. In a closed circular area, various entertainment acts including human skill and animal training are presented before the crowd.

A circus tent is cylindrical upto a height of 8 m and conical above it. The diameter of the base is 28 m and total height of tent is 18.5 m. [CBSE 2024]



Based on the above information, answer the following questions:

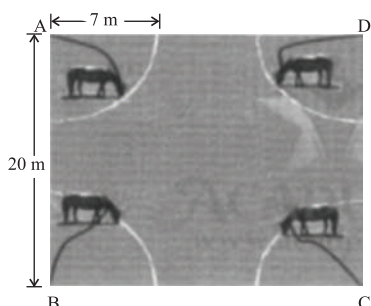
- Find slant height of the conical part.
- Determine the floor area of the tent.

(iii) (a) Find area of the cloth used for making tent.

OR

(b) Find total volume of air inside an empty tent.

7. A stable owner has four horses. He usually tie these horses with 7 m long rope to pegs at each corner of a square shaped grass field of 20 m length, to graze in his farm. But tying with rope sometimes results in injuries to his horses, so he decided to build fence around the area so that each horse can graze. [CBSE 2024]



Based on the above information, answer the following questions:

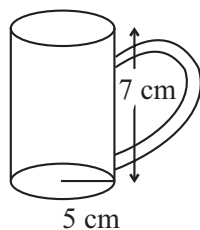
- Find the area of the square shaped grass field.
- (a) Find the area of the total field in which these horses can graze.

OR

(b) If the length of the rope of each horse is increased from 7 m to 10 m, find the area grazed by one horse. (Use $\pi = 3.14$)

(iii) What is area of the field that is left ungrazed, if the length of the rope of each horse is 7 m?

8. Tamper-proof tetra-packed milk guarantees both freshness and security. This milk ensures uncompromised quality, preserving the nutritional values within and making it a reliable choice for health-conscious individuals. [CBSE 2024]



500 mL milk is packed in a cuboidal container

of dimensions $15 \text{ cm} \times 8 \text{ cm} \times 5 \text{ cm}$. These milk packets are then packed in cuboidal cartons of dimensions $30 \text{ cm} \times 32 \times 15 \text{ cm}$.

Based on the above given information, answer the following questions:

- Find the volume of the cuboidal carton.
- (a) Find the total surface area of a milk packet.

OR

(b) How many milk packets can be filled in a carton?

(iii) How much milk can the cup (as shown in the figure) hold?

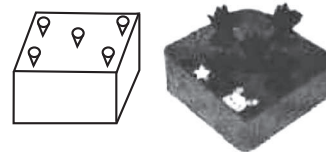
9. Case Study Based Question:

A solid cuboidal toy is made of wood. It has five cone shaped cavities to hold toy carrots.

The dimensions of the toy are

$$= 10 \text{ cm} \times 10 \text{ cm} \times 8 \text{ cm}.$$

Each cone carved out has Radius = 2.1 cm and Height = 6 cm. [CBSE 2022]



- Find the volume of wood carved out to make five conical cavities.
- Find the volume of the wood in the final product.

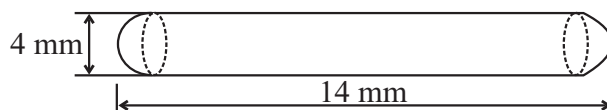
Long Answer Type Questions

11. The difference between the outer and inner radii of a hollow right circular cylinder of length 14 cm is 1 cm. If the volume of the metal used in making the cylinder is 176 cm^3 , find the outer and inner radii of the cylinder. [CBSE 2024]

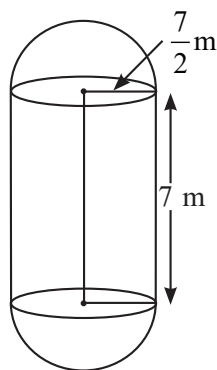
- 12 (a) A solid iron pole consists of a solid cylinder of height 200 cm and base diameter 28 cm, which is surmounted by another cylinder of height 50 cm and radius 7 cm. Find the mass of the pole, given that 1 cm^3 of iron has approximately 8 g mass. [CBSE 2024]

OR

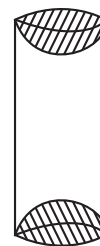
- 12 (b) A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 4 mm, find its surface area. Also, find its volume. [CBSE 2024]



13. The boilers are used in thermal power plants to store water and then used to produce steam. One such boiler consists of a cylindrical part in middle and two hemispherical parts at its both ends. Length of the cylindrical part is 7 m and radius of cylindrical part is $\frac{7}{2}$ m. Find the total surface area and the volume of the boiler. Also, find the ratio of the volume of cylindrical part to the volume of one hemispherical part. [CBSE 2023]



14. From a solid cylinder of height 20 cm and diameter 12 cm, a conical cavity of height 8 cm and radius 6 cm is hallowed out. Find the total surface area of the remaining solid. [CBSE 2023]
15. A solid is in the shape of a cone surmounted on a hemisphere. The radius of each of them being 3.5 cm and the total height of the solid is 9.5 cm. Find the volume of the solid. [CBSE 2020]
16. A wooden article was made by scooping out a hemisphere from each end of a solid cylinder, as shown in figure. If the height of the cylinder is 10 cm and its base is of radius 3.5 cm. Find the total surface area of the article: [CBSE 2018]



17. A sphere of diameter 12 cm, is dropped in a right circular cylindrical vessel, partly filled with water. If the sphere is completely submerged in water, the water level in the cylindrical vessel rises by $3\frac{5}{9}$ cm. Find the diameter of the cylindrical vessel. [CBSE 2016]

18. Due to heavy floods in a state, thousands were rendered homeless. 50 schools collectively offered to the state government to provide place and the canvas for 1500 tents to be fixed by the governments and decided to share the whole expenditure equally. The lower part of each tent is cylindrical of base radius 2.8 m and height 3.5 m, with conical upper part of same base radius but of height 2.1 m. If the canvas used to make the tents costs ₹ 120 per sq. m, find the amount shared by each school to set up the tents. What value is generated by the above problem? (Use $\pi = \frac{22}{7}$) [CBSE 2016]

19. Due to sudden floods, some welfare associations jointly requested the government to get 100 tents fixed immediately and offered to contribute 50% of the cost. If the lower part of each tent is of the form of a cylinder of diameter 4.2 m and height 4 m with the conical upper part of same diameter but height 2.8 m, and the canvas to be used costs ₹100 per sq. m, find the amount, the association will have to pay. What values are shown by these association?

$$\left[\text{Use: } \pi = \frac{22}{7} \right]$$

[CBSE 2015]



Explanations

Previous Year Questions

Multiple Choice Questions

1. (c) 2 : 3
2. (d) $\frac{2\pi}{3} \text{ cm}^3$
3. (c) $3\pi : 1$
4. (d) Assertion (A) is not true but Reason (R) is true.

Short Answer Type Questions

5. CSA of cylinder = $2 \times \frac{22}{7} \times 2.1 \times 5.8$
 $= 76.56 \text{ cm}^2$
 CSA of two hemisphere = $4 \times \frac{22}{7} \times 2.1 \times 2.1$
 $= 55.44 \text{ cm}^2$
 Total Surface Area of article = $76.56 + 55.44 = 132 \text{ cm}^2$

Case Based Type Questions

6. (i) Height of conical part = $18.5 - 8 = 10.5 \text{ m}$
 Radius of conical part = 14 m
 Slant height = $\sqrt{(10.5)^2 + (14)^2} = 17.5 \text{ m}$
 (ii) Floor area = $\frac{22}{7} \times 14 \times 14 = 616 \text{ m}^2$
 (iii) (a) Area of cloth used
 $= 2 \times \frac{22}{7} \times 14 \times 8 + \frac{22}{7} \times 14 \times 17.5$
 $= 1474 \text{ m}^2$
OR
 (b) Volume of air inside the tent
 $= \frac{22}{7} \times 14 \times 14 \times 8 + \frac{1}{3} \times \frac{22}{7} \times 14 \times 14 \times 10.5$
 $= 7084 \text{ m}^3$
 7. (i) Area of square shaped grass field = 400 m^2
 (ii) (a) Area of total field that horses can graze
 $= 4 \times \frac{1}{4} \times \frac{22}{7} \times 7 \times 7 = 154 \text{ m}^2$
OR
 (b) area grazed by one horse
 $= \frac{1}{4} \times 3.14 \times 10 \times 10 = 78.5 \text{ m}^2$

- (iii) Area of the field left ungrazed = area of square field – area of field in which horses can graze.
 Area of the field left ungrazed = $400 - 154$
 $= 246 \text{ m}^2$

8. (i) Volume of cuboidal carton = $30 \times 32 \times 15$
 $= 14400 \text{ cm}^3$
 (ii) (a) Total surface area of milk packet
 $= 2(15 \times 8 + 8 \times 5 + 5 \times 15) = 470 \text{ cm}^2$

OR

- (b) Number of milk packets in carton =
 $\frac{30 \times 32 \times 15}{15 \times 8 \times 5} = 24$
 (iii) Capacity of the cup = $\frac{22}{7} \times 5 \times 5 \times 7$
 $= 550 \text{ cm}^3$ or 550 ml .
9. (i) Dimensions of cuboid = $10 \text{ cm} \times 10 \text{ cm} \times 8 \text{ cm}$
 Dimensions of cone,
 Radius, $R = 2.1 \text{ cm}$
 Height, $H = 6 \text{ cm}$
 Volume of wood carved out
 $= \text{Volume of 5 cones} = \frac{1}{3}(\pi)R^2H \times 5$
 $= 5 \times \frac{1}{3} \times \frac{22}{7} \times (2.1)^2 \times 6 = 138.6 \text{ cm}^3$
 (ii) Volume of the wood in the final product =
 Volume of cuboid – Volume of wood carved out
 $= (10 \times 10 \times 8 - 138.6) \text{ cm}^3 = 661.4 \text{ cm}^3$

Long Answer Type Questions

11. Let outer radius be $r_2 \text{ cm}$ and inner radius be $r_1 \text{ cm}$.
 $\therefore r_2 - r_1 = 1 \text{ cm}$... (i)
 Volume of metal used = 176 cm^3
 $\Rightarrow \frac{22}{7} \times 14 \times (r_2^2 - r_1^2) = 176$
 $\Rightarrow r_2 + r_1 = 4$... (ii)
 Solving (i) and (ii), we get
 $r_2 = \frac{5}{2}$ or 2.5 , $r_1 = \frac{3}{2}$ or 1.5
 Therefore, outer radius = 2.5 cm and inner radius = 1.5 cm
- 12 (a) Radius of lower cylinder = 14 cm
 Volume of pole

$$= \frac{22}{7} \times 14 \times 14 \times 200 + \frac{22}{7} \times 7 \times 7 \times 50$$

$$= 130900 \text{ cm}^3$$

$$\text{Mass of the pole} = 8 \times 130900$$

$$= 1047200 \text{ gm or } 1047.2 \text{ kg}$$

OR

- 12 (b) Radius of hemisphere = radius of cylinder = 2 mm

$$\text{Length of cylindrical part} = 14 - 4 = 10 \text{ mm.}$$

$$\text{Surface area of the capsule} = \text{CSA of cylinder} + 2(\text{CSA of hemisphere})$$

$$= \frac{22}{7} \times 2 \times 10 + 2 \times 2 \times \frac{22}{7} \times 2 \times 2 = 176 \text{ mm}^2$$

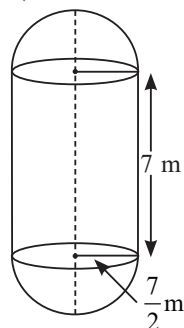
$$\text{Volume of the capsule}$$

$$= \text{volume of cylinder} + 2(\text{volume of hemisphere})$$

$$= 2 \times \frac{22}{7} \times 2 \times 2 \times 10 + 2 \times \frac{2}{3} \times \frac{22}{7} \times 2 \times 2 \times 2$$

$$= 40\pi + \frac{32}{3}\pi = \frac{152\pi}{3} \text{ mm}^3$$

13. Total surface area of the boiler = 2 × (curved surface area of hemisphere) + (curved surface area of cylindrical part)



$$= \left[2 \left\{ 2 \times \frac{22}{7} \times \left(\frac{7}{2} \right)^2 \right\} + \left\{ 2 \times \frac{22}{7} \times \left(\frac{7}{2} \right) \times 7 \right\} \right]$$

$$= 154 + 154 = 308 \text{ m}^2$$

$$\text{Total volume of boiler} = 2 \times \text{volume of hemispherical part} + \text{volume of cylinder}$$

$$= 2 \times \frac{2}{3} \times \frac{22}{7} \times \left(\frac{7}{2} \right)^3 + \frac{22}{7} \times \left(\frac{7}{2} \right)^2 \times 7$$

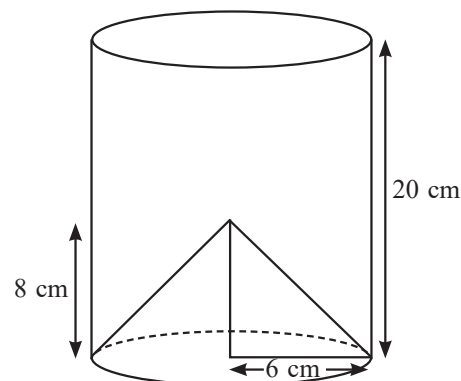
$$= \frac{11 \times 49}{3} + \frac{11 \times 49}{2} = \frac{539}{3} + \frac{539}{2}$$

$$= \frac{1078 + 1617}{6} = \frac{2695}{6} = 449.16 \text{ m}^3$$

$$\text{Required ratio} = \frac{\text{Volume of cylinder}}{\text{Volume of one hemispherical part}}$$

$$= \frac{\pi \left(\frac{7}{2} \right)^2 \times 7}{\frac{2}{3} \times \pi \times \left(\frac{7}{2} \right)^3} = 3:1$$

14. Surface area of remaining solid = Total surface area of cylinder – Area of base + curved surface area of cone



Height of cylinder (H) = 20 cm, height of cone (h) = 8 cm. Radius of base of cylinder (r) = 6 cm and radius of base of cone (r) = 6 cm.

$$\Rightarrow \text{Surface area of remaining solid}$$

$$= 2\pi rH + 2\pi r^2 - \pi r^2 + \pi rl$$

$$\text{where } l = \text{slant height of cone}$$

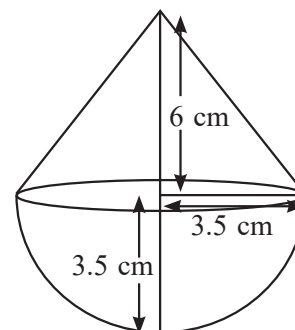
$$= 2 \times \frac{22}{7} \times 6 \times 20 + \frac{22}{7} \times 6 \times 6 + \frac{22}{7} \times 6 \times 10$$

$$[\because l = \sqrt{6^2 + 8^2} = 10 \text{ cm}]$$

$$= \frac{22}{7} \times 6[40 + 6 + 10]$$

$$= \frac{22}{7} \times 6 \times 56 = 1056 \text{ cm}^2$$

15. According to the question, we get following figure.



$\therefore$ Volume of solid = Volume of cone + volume of hemisphere

$$\Rightarrow \text{Volume} = \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$$

$$\Rightarrow \text{Volume} = \frac{1}{3}\pi(3.5)^2 \times 6 + \frac{2}{3}\pi(3.5)^3$$

$$\Rightarrow \text{Volume} = \frac{1}{3}\pi(3.5)^2[6 + 3.5 \times 2]$$

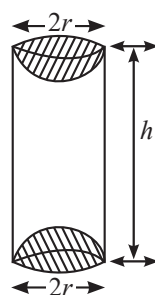
$$\Rightarrow \text{Volume} = \frac{1}{3} \times \frac{22}{7} \times \frac{49}{4} \times 13$$

$$\Rightarrow \text{Volume} = \frac{1}{3} \times \frac{2002}{4} = \frac{1001}{6}$$

$$\Rightarrow \text{Volume} = 166\frac{5}{6}\text{cm}^3$$

$$\therefore \text{Volume of solid} = 166\frac{5}{6}\text{cm}^3$$

16. Let r be the radius of the base of the cylinder and h be its height. Then,



Total surface area of the article = curved surface area of the cylinder + 2 (Curved surface area of a hemisphere)

$$= 2\pi rh + 2 \times 2\pi r^2$$

$$= 2\pi r(h + 2r)$$

$$= 2 \times \frac{22}{7} \times 3.5(10 + 2 \times 3.5)\text{cm}^2$$

$$= 22 \times 17\text{cm}^2 = 374\text{cm}^2$$

17. Radius of sphere = $r = 6\text{cm}$

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi \times (6)^3 = 288\pi\text{cm}^3$$

Let R be the radius of cylindrical vessel.

Rise in the water level of cylindrical vessel.

$$= h = 3\frac{5}{9}\text{cm} = \frac{32}{9}\text{cm}$$

Increase in volume of cylindrical vessel

$$= \pi R^2 h = \pi R^2 \times \frac{32}{9} = \frac{32}{9}\pi R^2$$

Now, volume of water displaced by the sphere is equal to volume of sphere

$$\therefore \frac{32}{9}\pi R^2 = 288\pi$$

$$\therefore R^2 = \frac{288 \times 9}{32} = 81$$

$$\therefore R = 9\text{cm}$$

$$\therefore \text{Diameter of the cylindrical vessel} = 2 \times R$$

$$= 2 \times 9 = 18\text{cm}$$

18. Height of conical upper part = 2.1 m, and radius = 2.8 m
(Slant height of cone)² = $2.1^2 + 2.8^2$
 $= 4.41 + 7.84$

$$\text{Slant height of cone} = \sqrt{12.25} = 3.5\text{m}$$

The canvas used for each tent

Curved surface area of cylindrical base + curved surface area of conical upper part

$$= 2\pi rh + \pi rl = \pi r(2h + l)$$

$$= \frac{22}{7} \times 2.8(7 + 3.5) = \frac{22}{7} \times 2.8 \times (10.5) = 92.4\text{m}^2$$

So, the canvas used for one tent is 92.4m^2

Thus, the canvas used for 1500 tents.

$$= (92.4 \times 1500)\text{m}^2$$

Canvas used to make the tent cost ₹120 per sq. m

So, canvas used to make 1500 tents will cost

$$= ₹92.4 \times 1500 \times 120$$

The amount shared by each school to set up the tents

$$= \frac{92.4 \times 1500 \times 120}{50} = 332640$$

The amount shared by each school to set up the tents is 3,32,640.

The value of helping others in times of troubles is generated from the problem.

19. Diameter of the tent = 4.2 m

Radius of the tent, $r = 2.1\text{m}$

Height of the cylindrical part of tent, $h_{\text{cylinder}} = 4\text{m}$

Height of the conical part, $h_{\text{cone}} = 2.8\text{m}$

$$= \sqrt{h_{\text{cone}}^2 + r^2} = \sqrt{2.8^2 + 2.1^2}$$

$$= \sqrt{2.8^2 + 2.1^2} = 3.5\text{m}$$

Curved surface area of the cylinder = $2\pi rh$

$$= 2 \times \frac{22}{7} \times 2.1 \times 4$$

$$= 22 \times 0.3 \times 8 = 52.8\text{m}^2$$

Curved surface area of the conical tent

$$= \pi rl = \frac{22}{7} \times 2.1 \times 3.5 = 23.1\text{m}^2$$

Total area of cloth required for building one tent
= Curved surface area of the cylinder + Curved
surface area of the conical tent

$$= 52.8 + 23.1 = 75.9 \text{ m}^2$$

$$\text{Cost of building one tent} = 75.9 \times 100 = ₹7590$$

$$\text{Total cost of 100 tents} = 7590 \times 100 = ₹7,59,000$$

Cost borne by the association

$$= \frac{50}{100} \times 7,59,000 = ₹ 3,79,500$$

It shows the helping nature, unit and cooperativeness
of the associations.



Previous Year Questions

Multiple Choice Questions

- For some data $x_1, x_2, \dots, x_n$ with respective frequencies $f_1, f_2, \dots, f_n$ the value of $\sum f_i(x - \bar{x})$ is equal to: [CBSE 2024]
 (a) $n\bar{x}$ (b) 1
 (c) $\sum f_i$ (d) 0
- The middle most observation of every data arranged in order is called: [CBSE 2024]
 (a) mode (b) median
 (c) mean (d) deviation
- If the value of each observation in a data is increased by 2, then median of the new data [CBSE 2024]
 (a) increases by 2 (b) increases by $2n$
 (c) remains the same (d) decreases by 2
- The mean of five observations is 15. If the mean of first three observations is 14 and that of the last three observations is 17, then the third observation is [CBSE 2024]
 (a) 20 (b) 19
 (c) 18 (d) 17
- If the difference between mode and median of a data is 24, then the difference of its median and mean is: [CBSE 2024]
 (a) 12 (b) 24
 (c) 8 (d) 36
- For the data 2, 9, $x + 6$, $2x + 3$, 5, 10, 5; if the mean is 7, then the value of x is: [CBSE 2024]
 (a) 9 (b) 6
 (c) 5 (d) 3
- If a certain variable x divides a statistical data arranged in order into two equal parts; then the value of x is called the: [CBSE 2024]

- (a) mean (b) median
 (c) mode (d) range

- The mean and median of a statistical data are 21 and 23 respectively. The mode of the data is: [CBSE 2024]
 (a) 27 (b) 22
 (c) 17 (d) 23
- The annual rainfall record of a city for 66 days is given in the following table: [CBSE 2024]

Rainfall (in cm)	Number of days
0-10	22
10-20	10
20-30	8
30-40	15
40-50	5
50-60	6

The difference of upper limits of modal and median classes is:

- (a) 10 (b) 15
 (c) 20 (d) 30
- If the mean and median of a data are 10 and 11 respectively, then mode of the data is: [CBSE 2024]
 (a) 12 (b) 8
 (c) 20 (d) 13
 - If for a distribution, $\sum_{i=1}^n f_i x_i = 132 + 5p$, $\sum_{i=1}^n f_i = 20$ and the mean of the distribution is 8.1, then the value of p is: [CBSE 2024]

- (a) 3 (b) 6
(c) 4 (d) 5

12. The following distribution gives the daily income of 50 workers of a factory: [CBSE 2024]

Income (in ₹)	Number of workers
400-424	12
425-449	14
450-474	8
475-499	6
500-524	10

The lower limit of the modal class is:

- (a) 10-20 (b) 20-30
(c) 30-40 (d) 40-50
13. The median group in the following frequency distribution is: [CBSE 2024]

Class	Frequency
0-10	5
10-20	8
20-30	20
30-40	15
40-50	7
50-60	5

- (a) 10-20 (b) 20-30
(c) 30-40 (d) 40-50
14. For the following distribution :

Class	0-50	5-10	10-15	15-20	20-25
Frequency	10	15	12	20	9

The sum of lower limits of median class and modal class is: [CBSE 2023]

- (a) 15 (b) 25
(c) 30 (d) 35

Short Answer Type Questions

15. In a test, the marks obtained by 100 students (out of 50) are given below: [CBSE 2024]

Marks obtained	Number of students
0-10	12
10-20	23
20-30	34

30-40	25
40-50	6

Find the mean marks of the students.

16. The distribution below gives the weights of 30 students of a class. Find the median weight of the students:

Weight in kg Number of Students [CBSE 2023]

Weight (in cm)	Number of students
40-45	2
45-50	3
50-55	8
55-60	6
60-65	6
65-70	3
70-75	2

17. Find the mode of the following frequency distribution: [CBSE 2022]

Classes	Frequency
20-30	25
30-40	30
40-50	45
50-60	42
60-70	35

18. If mode of the following frequency distribution is 55, then find the value of x . [CBSE 2022]

Classes	Frequency
0-15	10
15-30	7
30-45	x
45-60	15
60-75	10
75-90	12

19. Find the mean of the following frequency distribution: [CBSE 2022]

Classes	Frequency
10-15	4
15-20	10
20-25	5
25-30	6
30-35	5

20. The median of following frequency distribution is 25. Find the value of x . [CBSE 2022]

Classes	Frequency
0-10	6
10-20	9
20-30	10
30-40	8
40-50	x

21. Heights of 50 students of Class X of a school are recorded and following data is obtained: [CBSE 2022]

Height (in cm)	Number of students
130-135	4
135-140	11
140-145	12
145-150	7
150-155	10
155-160	6

Find the median height of the students.

22. Find the mean of the following data: [CBSE 2020]

Classes	Frequency
0-20	20
20-40	35
40-60	52
60-80	44
80-100	38
100-120	31

23. Find the mean of the following distribution:

Classes	Frequency
3-5	5
5-7	10
7-9	10
9-11	7
11-13	8

OR

Find the mode of the following data: [CBSE 2020]

Classes	Frequency
0-20	6
20-40	8
40-60	10
60-80	12
80-100	6
100-120	5
120-140	3

24. Find the mode of the following frequency distribution. [CBSE 2019]

Classes	Frequency
0-10	8
10-20	10
20-30	10
30-40	16
40-50	12
50-60	6
60-70	7

Case Based Type Questions

25. BINGO is game of chance. The host has 75 balls numbered 1 through 75. Each player has a BINGO card with some numbers written on it. The participant cancels the number on the card when called out a number written on the ball selected at random. Whosoever cancels all the numbers on his/her card, says BINGO and wins the game.



The table given below, shows the data of one such game where 48 balls were used before Tara said 'BINGO'.

Number announced	Number of times
0-15	8
15-30	9
30-45	10
45-60	12
65-75	9

Base on the above information, answer the following:

[CBSE 2024]

- Write the median class.
- When first ball was picked up, what was the probability of calling out an even number?
- (a) Find median of the given data.

OR

- Find mode of the given data.

Long Answer Type Questions

26. The following table shows the ages of the patients admitted in a hospital during a year: [CBSE 2024]

Age (in years)	Number of patients
5-15	6
15-25	11
25-35	21
35-45	23
45-55	14
55-65	5

Find the mode and mean of the given above.

27. Consider the following distribution of hourly wages of 50 workers of a factory: [CBSE 2024]

Age (in years)	Number of patients
100-120	12
120-140	14
140-160	8
160-180	6
180-200	10

Find the mean and the median of the above data.

28. The monthly expenditure on milk in 200 families of a Housing Society is given below:

Monthly Expenditure (in ₹)	Number of families
1000-1500	24
1500-2000	40
2000-2500	33
2500-3000	x
3000-3500	30
3500-4000	22
4000-4500	16
4500-5000	7

Find the value of x and also, find the median and mean expenditure on milk. [CBSE 2023]

29. (a) The mean of the following frequency distribution is 25. Find the value of f .

[CBSE 2022]

Classes	Frequency
0-10	5
10-20	18
20-30	15
30-40	f
40-50	6

OR

- (b) Find the mean of the following data using assumed mean method: [CBSE 2022]

Classes	Frequency
0-5	8
5-10	7
10-15	10
15-20	13
20-25	12

30. The following table gives production yield per hectare (in quintals) of wheat of 100 farms of a village:

Production yield / hect.	No. of farms
40-45	4
45-50	6
50-55	16
55-60	20
60-65	30
65-70	24

Change the distribution to 'a more than' type distribution and draw its give. [CBSE 2020]

OR

The median of the following data is 525. Find the values of x and y , if total frequency is 100:

[CBSE 2020]

Classes	Frequency
0-100	2
100-200	5
200-300	x
300-400	12
400-500	17
500-600	20
600-700	y
700-800	9
800-900	7
900-1000	4

31. If the median of the following frequency distribution is 32.5. Find the values of f_1 and f_2 . [CBSE 2019]

Classes	Frequency
0-10	f_1
10-20	5
20-30	9
30-40	12

40-50	f_2
50-60	3
60-70	2
Total	40

32. The table below shown the salaries of 280 persons:
[CBSE 2018]

Salary (In thousand)	No. of Person
5-10	49
10-15	133
15-20	63
20-25	15
25-30	6
30-35	7
35-40	4
40-45	2
45-50	1

Calculate the median salary of the data.

33. The mean of the following distribution is 18. Find the frequency f of the class 19 – 21. [CBSE 2018]

Classes	Frequency
11-13	3
13-15	6
15-17	9
17-19	13
19-21	f
21-23	5
23-25	4

34. The following distribution gives the daily income of 50 workers of a factory:

Daily income (In)	Number of workers
100-120	12
120-140	14
140-160	8
160-180	6
180-200	10

Convert the distribution above to a less than type cumulative frequency distribution and draw its ogive.

[CBSE 2018]





Explanations

Previous Year Questions

Multiple Choice Questions

1. (d) 0

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

$$\therefore \sum f_i x_i = \bar{x} \sum f_i = 0$$

$$\sum f_i (x_i - \bar{x}) = 0$$

2. (b) The median is the middlemost observation of every data arranged in order (ascending or descending).

3. (a)

4. (c) It is given that mean of 5 observations is 15.

$$\therefore \text{The sum of observation} = 15 \times 5 = 75.$$

$$\therefore \text{It is given that mean of the first 3 observations is 14.}$$

$$\therefore \text{The sum of first three observations} = 14 \times 3 = 42.$$

$$\Rightarrow \text{Given that mean of the last 3 observations is 17.}$$

$$\therefore \text{The sum of the last three observations} = 17 \times 3 = 51$$

$$\therefore \text{The third observation} = (42 + 51) - 75$$

$$= 93 - 75 = 18.$$

5. (a) Mode - Median = 24

$$\therefore \text{Mode} = 24 + \text{Median}$$

$$\text{But mode} = 3 \text{ median} - 2 \text{ mean}$$

$$\therefore 3 \text{ median} - 2 \text{ mean} = 24 + \text{median}$$

$$3 \text{ median} - \text{median} - 2 \text{ mean} = 24$$

$$\Rightarrow 2 \text{ median} - 2 \text{ mean} = 24$$

$$\Rightarrow \text{Median} - \text{Mean} = 12.$$

6. (d) $2 + 9 + x + 6 + 2x + 3 + 3 + 5 + 10 + 5$

$$= 40 + 3x$$

$$\text{Mean} = \frac{\text{Sum of all data points}}{\text{Number of data points}}$$

$$7 = \frac{40 + 3x}{7}$$

$$7 \times 7 = 40 + 3x$$

$$49 = 40 + 3x$$

$$x = 3$$

7. (b) The median divides statistical data arranged in order into two equal parts.

8. (a) Mode = 3 Median - 2 Mean

$$= 3(23) - 2(21)$$

$$= 69 - 42 = 27$$

9. (c) Modal class is 0-10

So, the upper limit of modal class is 10

Median class will be 20-30

Upper limit of median class is 30

So, the difference between upper limits of modal and median classes is 20.

10. (d) Mode = 3 Median - 2 Mean

$$= 3(11) - 2(10) = 33 - 20 = 13$$

11. (b) Given: $\sum_{i=1}^n f_i x_i = 132 + 5p$, $\sum_{i=1}^n f_i = 20$

$$8.1 = \frac{(132 + 5p)}{20}$$

$$162 = 132 + 5p$$

$$p = 6$$

12. (c) Modal class will be 425-449

But for a regular class interval class interval will be 424.5 - 449.5

Hence lower limit of modal class will be 424.5

13. (b)

Class	Frequency	C.f.
0-10	5	5
10-20	8	13
20-30	20	33
30-40	15	48
40-50	7	55
50-60	5	60
	$\Sigma f = 60$	

Here, $N = \Sigma f = 60$

$$\therefore \frac{N}{2} = \frac{60}{2} = 30$$

So, the median class is 20-30.

14. (b) Median class = 10-15

Median class = 10-15

$$\therefore \text{Required sum} = 10 + 15 = 25$$

Short Answer Type Questions

15.

Marks obtained	Number of students	x_i	xf_i
0-10	12	5	60
10-20	23	15	345
20-30	34	25	850
30-40	25	35	875
40-50	6	45	270
	$\Sigma f = 100$		2400

$$\text{Mean} = \frac{\Sigma x_i f_i}{\Sigma f_i} = \frac{2400}{100} = 24$$

16.

Weight (in cm)	Number of students	Cumulative frequency
40-45	2	2
45-50	3	5
50-55	8	13 = Cf
55-60	6	19
60-65	6	25
65-70	3	28
70-75	2	30

Total $N = 100$ Here, $\frac{N}{2} = \frac{30}{2} = 15$, which lies in the class 55-60 $l = 55$, $h = 60 - 55 = 5$, $Cf = 13$, $f = 6$

$$\begin{aligned} \text{Median} &= l + \left\{ \frac{\frac{N}{2} - Cf}{f} \right\} \times h \\ &= 55 + \left\{ \frac{15 - 13}{6} \right\} \times 5 = 55 + \left\{ \frac{5}{3} \right\} = 56.66 (\text{approx.}) \end{aligned}$$

$$17. \text{ Mode} = l + \frac{f_m - f_1}{(2f_m - f_1 - f_2)} \times h$$

where f_m = frequency of modal class f_1 = frequency preceding frequency of modal class f_2 = frequency succeeding frequency of modal class $\Rightarrow f_m = 45$ $f_1 = 30$, $f_2 = 42$, $h = 10$, $l = 40$

$$\begin{aligned} \therefore \text{Mode} &= 40 + \left(\frac{45 - 30}{92 - 72} \right) \times 10 \\ &= 40 + \left(\frac{15}{18} \times 10 \right) = 40 + \left(\frac{150}{18} \right) \\ &= 40 + 8.33 = 48.33 \end{aligned}$$

18. Mode = 55 $\Rightarrow$ Modal class is 45 – 60where f_m = frequency of modal class f_1 = frequency preceding frequency of modal class f_2 = frequency succeeding frequency of modal class $\therefore l = 45$, $f_m = 15$, $f_1 = x$, $f_2 = 10$, $h = 15$

$$\text{Mode} = l + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$

$$55 = 45 + \left(\frac{15 - x}{30 - x - 10} \right) \times 15$$

$$\Rightarrow 2(20 - x) = 3(15 - x)$$

$$\Rightarrow (40 - 2x) = (45 - 3x) \Rightarrow x = 5$$

19.

Classes	Frequency	Class Marks (x_i)	Product ($f_i x_i$)
10-15	4	12.5	50.00
15-20	10	17.5	175.00
20-25	5	22.5	112.50
25-30	6	27.5	165.00
30-35	5	32.5	162.50
Total	N = 30		$\Sigma f_i x_i = 665.00$

$$\text{Mean } (\bar{x}) = \frac{1}{N} \sum_{i=1}^k f_i x_i$$

$$\begin{aligned} &= \frac{\sum_{i=1}^k f_i x_i}{N} = \frac{665.0}{30} = 22.17 (\text{approx.}) \end{aligned}$$

20.

Classes	Frequency	C.f.
0-10	6	6
10-20	9	15
20-30	10	25
30-40	8	33
40-50	x	$33 + x$

Median = 25

 $\Rightarrow$ Median class is 20 – 30 $\Rightarrow f = 10$, $c.f. = 15$, $N = 33 + x$, $h = 10$ and $l = 20$

$$\text{Median} = l + \left\{ \frac{\frac{N}{2} - cf}{f} \right\} \times h$$

$$\Rightarrow 25 = 20 + \left\{ \frac{\frac{33+x}{2} - 15}{10} \right\} \times 10$$

$$\Rightarrow 5 = \left(\frac{33+x-30}{2} \right) \Rightarrow 5 = 3 + x$$

$$\therefore x = 7$$

21.

Height (in cm)	Number of students (f_i)	Cumulative frequency
130-135	4	4
135-140	11	15
140-145	12	27
145-150	7	34
150-155	10	44
155-160	6	50

$N = 50$, so $\frac{N}{2} = 25$. So, median class lies in the class 140 – 145, then $l = 140$, $c.f. = 15$, $f = 12$, $h = 5$

$$\text{Median} = l + \frac{\left(\frac{N}{2} - c.f.\right)}{f} \times h$$

$$= 140 + \left(\frac{25 - 15}{12}\right) 5 = 144.166\ldots$$

Median height of students = 144.17 (approx.)

22.

Classes	x_i	f_i	$d_i = x_i - A$		$f_i x_i$
0-20	10	20	10 – 50 = –40	–2	–40
20-40	30	35	30 – 50 = –20	–1	–25
40-60	50 = A	52	50 – 50 = 0	0	0
60-80	70	44	70 – 50 = 20	1	41
80-100	90	38	90 – 50 = 40	2	76
100-120	110	31	110 – 50 = 60	3	93
		$\Sigma f_i = 220$			$f_i x_i = 138$

$$\bar{X} = A + \frac{\Sigma f_i d_i}{\Sigma f_i} \times h$$

$$= 50 + \frac{138}{220} \times 20 = 50 + 12.55 = 62.55$$

23.

Classes	Mid-value (x_i)	Frequency (f_i)	$(f_i x_i)$
3-5	4	5	20
5-7	6	10	60
7-9	8	10	80
9-11	10	7	70
11-13	12	8	90
Total		$\Sigma f_i = 40$	$\Sigma f_i x_i = 326$

$$\therefore \text{Mean} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{326}{40} = 8.15$$

OR

Here, the maximum frequency is 12 and the corresponding class is 60-80. So, 60-80 is the modal class such that $l = 60$, $h = 20$, $f_0 = 12$, $f_1 = 10$ and $f_2 = 6$.

$$\therefore \text{Mode} = 60 + \left(\frac{12 - 10}{2 \times 12 - 10 - 6}\right) \times 20$$

$$= 60 + \frac{2}{8} \times 20 = 60 + 5 = 65$$

24.

Classes	Frequency
0-10	8
10-20	10
20-30	10 $\rightarrow f_0$
30-40	16 $\rightarrow f_1$
40-50	12 $\rightarrow f_2$
50-60	6
60-70	7

Here, 30 – 40 is the modal class, and $l = 30$, $h = 10$

$$\therefore \text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$= 30 + \left(\frac{16 - 10}{2 \times 16 - 10 - 12}\right) \times 10$$

$$= 30 + \frac{6}{10} \times 10 = 30 + 6 = 36$$

Case Based Type Questions

25. (i)

Number announced	Number of times	C.f.
0-15	8	8
15-30	9	17
30-45	10	27
45-60	12	39
65-75	9	48
	$\Sigma f = 48$	

C.f. just greater than $\frac{48}{2}$ i.e 24 is 27

$\therefore$ Median class = 30 – 45

(ii) Required Probability = $\frac{37}{75}$

$$\begin{aligned} \text{(iii) (a) Median} &= l + \left\{ \frac{\frac{N}{2} - cf}{f} \right\} \times h \\ &= 30 + 15 \times \frac{24-17}{10} = 30 + 15 \times 0.7 \\ &= 30 + 10.5 = 40.5. \end{aligned}$$

OR

(b) Modal class 45-60

$$\begin{aligned} \text{Mode} &= \ell + h \left\{ \frac{f_k - f_{k-1}}{2f_k - f_{k-1} - f_{k+1}} \right\} \\ &= 45 + 15 \left\{ \frac{12-10}{24-10-9} \right\} = 45 + 15 \times \frac{2}{5} = 51 \end{aligned}$$

Long Answer Type Questions

26. Mode: The class which has the highest frequency.

In this case, class interval 35-45 is the modal class.

Now, the lower limit of modal class,

$$l = 35, h = 10, f_1 = 23, f_0 = 21, f_2 = 14$$

We know that,

$$\begin{aligned} \text{Mode} &= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h \\ &= 35 + \left(\frac{23-21}{2(23)-21-14} \right) \times 10 \\ &= 35 + \frac{2}{11} \times 20 = 36.8 \end{aligned}$$

Let's take assumed mean, A as 30

Class interval = 10

Age (in years)	Mid value (x_i)	Frequency (f_i)		$f_i u_i$
5-15	10	6	-2	-12
15-25	20	11	-1	-11
25-35	30	21	0	0
35-45	40	23	1	23
45-55	50	14	2	28
55-65	60	5	3	15
Total		80		43

$$\therefore u_i = \frac{x_i - A}{h} = \frac{x_i - 30}{10}$$

$$\begin{aligned} \bar{x} &= A + h \frac{\sum f_i u_i}{\sum f_i} = 30 + 10 \times \frac{43}{80} \\ &= 30 + 5.375 = 35.375 \approx 35.37 \end{aligned}$$

Hence, mode = 36.8 years,

Mean = 35.37 years.

27. Let the assumed mean be $A = 120$ and $h = 20$.

Age (in years)	Mid value (x_i)	Number of patients	$d_i = x_i - A = x_i - 150$		$f_i u_i$
100-120	110	12	-40	-2	-24
120-140	130	14	-20	-1	-14
140-160	150	8	0	0	0
160-180	170	6	20	1	6
180-200	190	10	40	2	20
		$\sum f_i = 50$			$\sum f_i u_i = -12$

We know that mean, $\bar{X} = A + h \left(\frac{\sum f_i u_i}{\sum f_i} \right)$

Now, we have $N = \sum f_i = 50$, $\sum f_i u_i = -12$, $h = 20$
 $A = 150$.

Putting the values in the above formula, we have

$$\begin{aligned} \bar{X} &= A + \frac{\sum f_i u_i}{\sum f_i} = 150 + 20 \left(\frac{-12}{50} \right) \\ &= 150 - \frac{240}{50} = 150 - 4.8 = 145.2 \end{aligned}$$

Hence, the mean daily wage of workers is Rs. 145.20

$\Rightarrow$ We have, $N = 50$. Then $\frac{N}{2} = 25$.

$\Rightarrow$ Median class is 120 - 140

l = lower limit of the modal class

h = size of the class intervals

f = frequency of the modal class

f_1 = frequency of the class preceding the modal class

f_2 = frequency of the class succeed the modal class.

so, $l = 120$, $h = 20$, $N = 50$, $cf = 12$, $f = 14$.

$$\text{Median} = l + \frac{\frac{N}{2} - cf}{f} \times h$$

$$\Rightarrow \text{Median} = 120 + \frac{25-12}{14} \times 20$$

$$\Rightarrow \text{Median} = 120 + \frac{130}{7}$$

$$\therefore \text{Median} = 120 + 18.6 = 138.6$$

28.

Monthly Expenditure (in ₹)	Number of families f_i	Class mark (x)	$d_i = x_i - A$	$f_i d_i$	Cumulative frequency
1000-1500	24	1250	-1500	-20000	24
1500-2000	40	1750	-1000	-40000	64
2000-2500	33	2250	-500	-16500	97
2500-3000	x	2750 $= A$	0	0	125
3000-3500	30	3250	500	15000	156
3500-4000	22	28	1000	22000	177
4000-4500	16	30	1500	24000	193
4500-5000	7		2000	14000	200
	$\Sigma f_i = 200$			$\Sigma f_i d_i = -17500$	

Here, $24 + 40 + 33 + x + 30 + 22 + 16 + 7 = 200$

$$\Rightarrow x + 172 = 200 \Rightarrow x = 28$$

Now, mean, $\bar{x} = A + \frac{\Sigma f_i d_i}{\Sigma f_i}$

$$= 2750 + \frac{(-17500)}{200} = 2750 - 87.5 = 2662.5$$

Also, $\frac{N}{2} = \frac{200}{2} = 100$

Median class = 2500 – 3000

Here, $l = 2500$, $cf = 97$, $f = 28$, $h = 500$

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

$$= 2500 + \frac{100 - 97}{28} \times 500 = 2500 + \frac{375}{7}$$

$$= 2500 + 53.57 = 2553.57 \text{ (approx.)}$$

29. (a)

Classes	Class Mark (x_i)	Frequency (f_i)	$f_i x_i$
0-10	5	5	25
10-20	15	18	270
20-30	25	15	375
30-40	35	f	$35f$
40-50	45	6	270
		$\Sigma f_i = 44 + f$	$\Sigma f_i x_i = 940 + 35f$

$$\text{Mean } (\bar{x}) = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{940 + 35f}{44 + f}$$

$$\Rightarrow 25 = \frac{940 + 35f}{44 + f} \Rightarrow f = 16$$

OR

Classes	Frequency f_i	Class mark (x_i)	$d_i = x_i - a$	$f_i d_i$
0-5	8	2.5	-10	-80
5-10	7	7.5	-5	-30
10-15	10	12.5 = a	0	0
15-20	13	17.5	5	65
20-25	12	22.5	10	120
Total	$N = 50$			$\Sigma f_i d_i = 70$

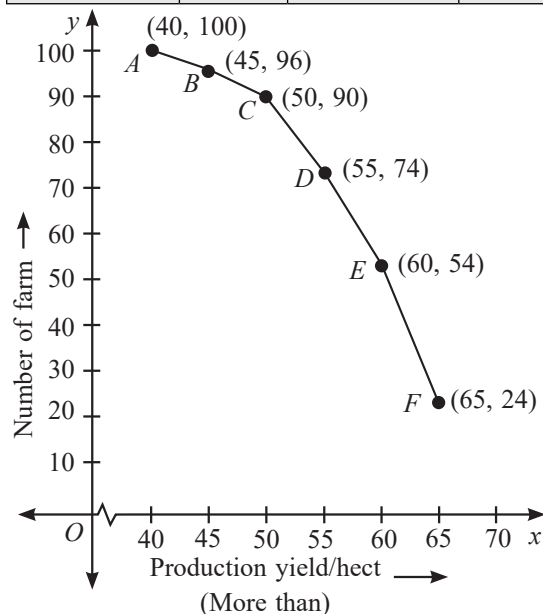
Let assume mean be $a = 12.5$ and $N = 50$

$$\therefore \bar{x} = a + \frac{1}{N} \Sigma f_i d_i$$

$$= 12.5 + \frac{1}{50} \times 70 = 12.5 + 1.4 = 13.9$$

30.

Production yield/hect	Number of farms	Production yield more than/hect	Cumulative frequency
40-45	4	40	100
45-50	6	45	96
50-55	16	50	90
55-60	20	55	74
60-65	30	60	54
65-70	24	65	24



OR

Classes	Frequency	C.f.
0-100	2	2
100-200	5	7
200-300	x	$7 + x$

300-400	12	$19 + x$
400-500	17	$36 + x$
500-600	20	$56 + x$
600-700	y	$56 + x + y$
700-800	9	$65 + x + y$
800-900	7	$72 + x + y$
900-1000	4	$76 + x + y = N$

Here $N = 100$

$$\Rightarrow 76 + x + y = 100$$

$$x + y = 24 \quad \dots(i)$$

$$\text{Median} = 525$$

$$\text{Median Class} = 500 - 600$$

$$l = 500, h = 100, f = 20$$

$$c.f. = 36 + x$$

$$\text{Median} = l + \left[\frac{\frac{N}{2} - c.f.}{f} \right] \times h$$

$$\Rightarrow 525 = 500 + \left[\frac{50 - 36 - x}{20} \right] \times 100$$

$$\Rightarrow 25 = (14 - x)5 \Rightarrow 14 - x = 5 \Rightarrow x = 9.$$

Now, from (i)

$$9 + y = 24$$

$$y = 15$$

31.

Classes	Frequency	Cumulative frequency
0-10	f_1	f_1
10-20	5	$5 + f_1$
20-30	9	$14 + f_1$
30-40	12	$26 + f_1$
40-50	f_2	$26 + f_1 + f_2$
50-60	3	$29 + f_1 + f_2$
60-70	2	$31 + f_1 + f_2$
Total	$n = 40$	

$$f_1 + 5 + 9 + 12 + f_2 + 3 + 2 = 40$$

$$f_1 + f_2 = 40 - 31 = 9 \quad \dots(i)$$

$$\text{Median} = 32.5 \quad [\text{Given}]$$

$$\therefore \text{Median class is } 30 - 40$$

$$l = 30, h = 10, c.f. = 14 + f_1, f = 12$$

$$\text{Median} = l + \left[\frac{\frac{N}{2} - c.f.}{f} \right] \times h$$

$$32.5 = 30 + \left[\frac{\frac{40}{2} - (14 + f_1)}{12} \right] \times 10$$

$$2.5 = \frac{10}{20} (20 - 14 - f_1)$$

$$3 = 6 - f_1$$

$$f_1 = 3$$

On putting in (i)

$$f_1 + f_2 = 9$$

$$f_2 = 9 - 3 = 6 \quad [\because f_1 = 3]$$

32.

Class	Frequency	Cumulative frequency
5-10	49	49
10-15	133	182
15-20	63	245
20-25	15	260
25-30	6	266
30-35	7	273
35-40	4	277
40-45	2	279
45-50	1	280

Let $N =$ total frequency

$$\therefore \text{We have } N = 280$$

$$\therefore \frac{N}{2} = \frac{280}{2} = 140$$

The cumulative frequency just greater than $\frac{N}{2}$ is 182 and the corresponding class is 10 - 15.

thus, 10 - 15 is the median class such that $l = 10$, $f = 133$, $F = 49$ and $h = 5$.

$$\text{Median} = l + \left[\frac{\frac{N}{2} - F}{f} \right] \times h = 10 + \left(\frac{140 - 49}{133} \right) \times 5 = 13.42$$

33.

Classes	Mid Values x_i	Frequency f_i	$d_i = x_i - 18$		$f_i d_i$
11-13	12	3	-6	-3	-9
13-15	14	6	-4	-2	-12
15-17	16	9	-2	-1	-9
17-19	18	13	0	0	0
19-21	20	f	2	1	f
21-23	22	5	4	2	10
23-25	24	4	6	3	12
		$\Sigma f_i = 40 + f$			

$$\Sigma f_i d_i = f - 8$$

We have $h = 2$; $A = 18$, $N = 40 + f$, $\Sigma f_i d_i = f - 8$

$$\bar{X} = 18$$

$$\therefore \text{Mean} = A + h \left\{ \frac{1}{N} \Sigma f_i d_i \right\}$$

$$18 = 18 + 2 \left\{ \frac{1}{40 + f} (f - 8) \right\}$$

$$\frac{2(f - 8)}{40 + f} = 0$$

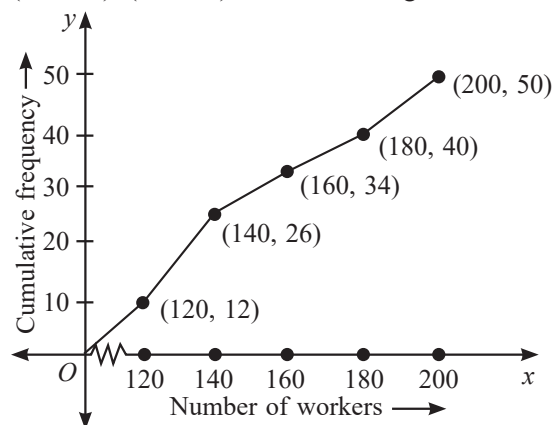
$$f - 8 = 0$$

$$f = 8$$

34.

Daily income (In)	Frequency	Income less than	Cumulative frequency
100-120	12	120	12
120-140	14	140	26
140-160	8	160	34
160-180	6	180	40
180-200	10	200	50

Using these values we plot the points (120, 12), (160, 34), (180, 40), (200, 50) on the axes to get less than ogive





Previous Year Questions

Multiple Choice Questions

1. A box contains cards numbered 6 to 50. A card is drawn at random from the box. The probability that the drawn card has a number which is a perfect square, is: [CBSE 2024]

(a) $\frac{5}{44}$ (b) $\frac{1}{9}$
(c) $\frac{1}{11}$ (d) $\frac{7}{45}$

2. Two friends were born in the year 2000. The probability that they have the same birthday is: [CBSE 2024]

(a) $\frac{1}{365}$ (b) $\frac{364}{365}$
(c) $\frac{1}{366}$ (d) $\frac{365}{366}$

3. If the probability of a player winning a game is 0.79, then the probability of his losing the same game is: [CBSE 2024]

(a) 1.79 (b) 0.31
(c) 0.21% (d) 0.21

4. Two dice are rolled together. The probability of getting sum of numbers on the two dice as 2, 3 or 5, is: [CBSE 2024]

(a) $\frac{7}{36}$ (b) $\frac{11}{36}$
(c) $\frac{5}{36}$ (d) $\frac{4}{9}$

5. From the data 1, 4, 7, 9, 16, 21, 25, if all the even numbers are removed, then the probability of getting at random a prime number from the remaining is: [CBSE 2024]

(a) $\frac{2}{5}$ (b) $\frac{1}{5}$
(c) $\frac{1}{7}$ (d) $\frac{2}{7}$

6. Which of the following is not probability of an event? [CBSE 2024]

(a) 0.89 (b) 52%
(c) $\frac{1}{13}\%$ (d) $\frac{1}{0.89}$

7. One card is drawn at random from a well shuffled deck of 52 playing cards. The probability that it is a red ace card, is: [CBSE 2024]

(a) $\frac{1}{13}$ (b) $\frac{1}{26}$
(c) $\frac{1}{52}$ (d) $\frac{1}{2}$

8. Two dice are rolled together. The probability of getting the sum of the two numbers to be more than 10, is: [CBSE 2024]

(a) $\frac{1}{9}$ (b) $\frac{1}{6}$
(c) $\frac{7}{12}$ (d) $\frac{1}{12}$

9. The probability of throwing a number greater than 2 with a fair die is: [CBSE 2024]

(a) $\frac{2}{3}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) $\frac{5}{6}$

10. The probability of guessing the correct answer to a certain test question is $\frac{x}{6}$. If the probability of not guessing the correct answer to this question is $\frac{2}{3}$, then the value of x is: [CBSE 2024]

(a) 2 (b) 3
(c) 4 (d) 6

11. If a digit is chosen at random from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9; then the probability that this digit is an odd prime number is: [CBSE 2024]

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{4}{9}$ (d) $\frac{5}{9}$

12. **Assertion (A):** In a cricket match, a batsman hits a boundary 9 times out of 45 balls he plays. The probability that in a given ball, he does not hit the boundary is $\frac{4}{5}$.

Reason (R): $P(E) + P(\text{not } E) = 1$ [CBSE 2024]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A)
(c) Assertion (A) is true but Reason (R) is false
(d) Assertion (A) is false but Reason (R) is true
13. Two coins are tossed simultaneously. The probability of getting at most one tail is: [CBSE 2024]

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) $\frac{3}{4}$ (d) 1

14. A bag contains 3 red balls, 5 white balls and 7 black balls. The probability that a ball drawn from the bag at random will be neither red nor black is:

[CBSE 2024]

(a) $\frac{1}{3}$ (b) $\frac{1}{5}$
(c) $\frac{7}{15}$ (d) $\frac{8}{15}$

15. The probability of getting a bad egg in a lot of 400 eggs is 0.045. The number of good eggs in the lot is: [CBSE 2024]

(a) 18 (b) 180
(c) 382 (d) 220

16. For an event E , if $P(\bar{E}) + P(E) = q$, then the value of $q^2 - 4$ is: [CBSE 2024]

(a) -3 (b) 3
(c) 5 (d) -5

17. Let E be an event such that $P(\text{not } E) = \frac{1}{5}$, then $P(E)$ is equal to [CBSE 2023]

(a) $\frac{1}{5}$ (b) $\frac{2}{5}$
(c) 0 (d) $\frac{4}{5}$

18. From a well-shuffled deck of 52 cards, a card is drawn at random. What is the probability of getting king of hearts? [CBSE 2023]

(a) $\frac{1}{52}$ (b) $\frac{1}{26}$
(c) $\frac{1}{13}$ (d) $\frac{12}{13}$

19. A card is drawn at random from a well-shuffled pack of 52 cards. The probability that the card drawn is not an ace is [CBSE 2023]

(a) $\frac{1}{13}$ (b) $\frac{9}{13}$
(c) $\frac{4}{13}$ (d) $\frac{12}{13}$

20. A bag contains 5 red balls and n green balls. If the probability of drawing a green ball is three times that of a red ball, then the value of n is [2023]

(a) 18 (b) 15
(c) 10 (d) 20

21. **Assertion (A):** The probability that a leap year has 53 Sundays is $\frac{2}{7}$.

Reason (R): The probability that a non-leap year has 53 Sundays is $\frac{5}{7}$. [CBSE 2023]

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A)
 (c) Assertion (A) is true but Reason (R) is false
 (d) Assertion (A) is false but Reason (R) is true
22. Two coins are tossed together. The probability of getting exactly one head is [CBSE 2021]
 (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
 (c) $\frac{3}{4}$ (d) 1
23. If $P(E) = 0.65$, then the value of $P(\text{not } E)$ is [CBSE 2021]
 (a) 1.65 (b) 0.25
 (c) 0.65 (d) 0.35
24. A bag contains 16 red balls 8 green balls and 6 blue balls. One ball is drawn at random. The probability that it is blue is [CBSE 2021]
 (a) $\frac{1}{6}$ (b) $\frac{1}{5}$
 (c) $\frac{1}{30}$ (d) $\frac{5}{6}$
25. The probability of happening of an event is 0.02. The probability of not happening of the event is [CBSE 2021]
 (a) 0.02 (b) 0.80
 (c) 0.98 (d) $\frac{49}{100}$
26. For an event E , $P(E) + P(\bar{E}) = x$, then the value of $x^3 - 3$ is [CBSE 2021]
 (a) -2 (b) 2
 (c) 1 (d) -1
27. The probability that the drawn card from a pack of 52 cards is neither an ace nor a spade is [CBSE 2021]
 (a) $\frac{9}{13}$ (b) $\frac{35}{52}$
 (c) $\frac{10}{13}$ (d) $\frac{19}{26}$
28. Which of the following cannot be the probability of an event? [CBSE 2021]
 (a) 0.01 (b) 3%
 (c) $\frac{16}{17}$ (d) $\frac{17}{16}$

29. A dice is rolled twice. The probability that 5 will not come up either time is [CBSE 2021]

- (a) $\frac{11}{36}$ (b) $\frac{1}{3}$
 (c) $\frac{13}{36}$ (d) $\frac{25}{36}$

30. The probability that a number selected at random from the numbers 1, 2, 3, ..., 15 is a multiple of 4 is [CBSE 2020]

- (a) $\frac{4}{15}$ (b) $\frac{2}{15}$
 (c) $\frac{1}{15}$ (d) $\frac{1}{5}$

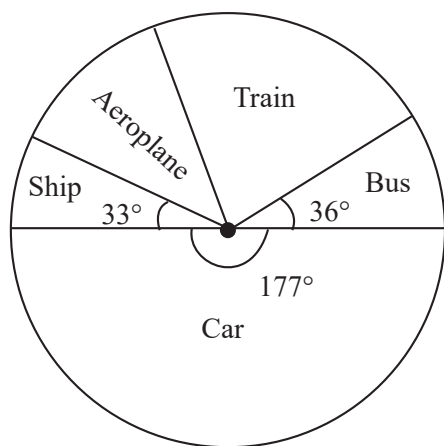
Short Answer Type Questions

31. In a pack of 52 playing cards one card is lost. From the remaining cards, a card is drawn at random. Find the probability that the drawn card is queen of heart, if the lost card is a black card. [CBSE 2024]
32. The king, queen and ace of clubs and diamonds are removed from a deck of 52 playing cards and the remaining cards are shuffled. A card is randomly drawn from the remaining cards. Find the probability of getting: [CBSE 2024]
 (i) a card of clubs.
 (ii) a red coloured card.
33. One card is drawn at random from a well shuffled deck of 52 cards. Find the probability that the card drawn:
 (i) is queen of hearts;
 (ii) is not a jack.
34. If a fair coin is tossed twice, find the probability of getting 'atmost one head'. [CBSE 2023]
35. A die is rolled once. Find the probability of getting: [CBSE 2023]
 (i) An even prime number.
 (ii) A number greater than 4
 (iii) An odd number.
36. If a pair of dice is thrown once, then what is the probability of getting a sum of 8? [CBSE 2020]
37. A letter of English alphabet is chosen at random. What is the probability that the chosen letter is a consonant. [CBSE 2020]
38. If a number x is chosen at random from the numbers -3, -2, -1, 0, 1, 2, 3. What is probability that $x^2 \leq 4$? [CBSE 2020]

39. A game consists of tossing a coin 3 times and noting the outcome each time. If getting the same result in all the tosses is a success, find the probability of losing the game. [CBSE 2019]
40. A die is thrown once. Find the probability of getting a number which (i) is a prime number (ii) lies between 2 and 6. [CBSE 2019]
41. Two different dice are tossed together. Find the probability.
(i) of getting doublet
(ii) of getting a sum 10, of the numbers on the two dice. [CBSE 2018]
42. An integer is chosen at random between 1 and 100. Find the probability that it is
(i) divisible by 8
(ii) not divisible by 8 [CBSE 2018]
43. The probability of selecting a rotten apple randomly from a heap of 900 apples is 0.18. What is the number of rotten apples in the heap? [CBSE 2017]
44. A card is drawn at random from a well shuffled pack of 52 playing cards. Find the probability of getting neither a red card nor a queen. [CBSE 2016]
45. Two different dice are tossed together. Find the probability that the product of the two numbers on the top of the dice is 6. [CBSE 2015]

Case Based Type Questions

46. In a survey on holidays, 120 people were asked to state which type of transport they used on their last holiday. The following pie chart shows the results of the survey. [CBSE 2024]



Observe the pie chart and answer the following questions:

- (i) If one person is selected at random, find the probability that he/she travelled by bus or ship.
(ii) Which is most favourite mode of transport and how many people used it?
(iii) (a) A person is selected at random. If the probability that he did not use train is $\frac{4}{5}$, find the number of people who used train.

OR

- (iii) (b) The probability that randomly selected person used aeroplane is $\frac{7}{60}$. Find the revenue collected by air company at the rate of 5,000 per person. [CBSE 2024]

Long Answer Type Questions

47. Three coins are tossed simultaneously. What is the probability of getting:
(i) at least one head? [CBSE 2024]
(ii) exactly two tails?
(iii) at most one tail?
48. A die is thrown once. What is the probability of getting a number less than 3? [CBSE 2020]

OR

If the probability of winning a game is 0.07, what is the probability of losing it? [CBSE 2020]

49. Two different dice are thrown together, find the probability that the sum of the numbers appeared is less than 5. [CBSE 2020]

OR

Find the probability that 5 Sundays occur in the month of November of a randomly selected year. [CBSE 2020]

50. A bag contains 15 white and some black balls. If the probability of drawing a black ball from the bag is thrice that of drawing a white ball, find the number of black balls in the bag. [CBSE 2017]
51. Two different dice are thrown together. Find the probability that the numbers obtained have
(i) even sum, and
(ii) even product [CBSE 2017]
52. A number x is selected at random from the 1, 2, 3 and 4. Another number y is selected at random from the numbers 1, 4, 9 and 16. Find the probability the product of x and y is less than 16. [CBSE 2016]

53. A card is drawn at random from a well-shuffled deck of playing cards. Find the probability that the card drawn is

[CBSE 2015]

- (i) a card of spade or an ace.
- (ii) a black king.
- (iii) neither a jack nor a king.
- (iv) either a king or a queen.

54. The probability of selecting a red ball at random from a jar that contains only red, blue and orange balls is $\frac{1}{4}$. The probability of selecting a blue ball at random from the same jar is $\frac{1}{3}$. If the jar contains 10 orange balls, find the total number of balls in the jar.

[CBSE 2015]





Explanations

Previous Year Questions

Multiple Choice Questions

- (b) Sample space (S) = $\{6, 7, 8, \dots, 50\}$
 $n(S) = 45$
 perfect squares (E) = $\{9, 16, 25, 36, 49\}$
 $n(E) = 5$
 required probability = $\frac{n(E)}{n(S)} = \frac{5}{45} = \frac{1}{9}$
- (c) 2000 is a leap year
 Total outcomes = 366
 Favourable outcomes = 1
 So, required probability = $\frac{1}{366}$
- (d) We know, $P(E) + P(\bar{E}) = 1$
 $P(E) = 0.79$
 $\Rightarrow P(E) = 1 - 0.79 = 0.21$
 Hence, probability of losing the same game = 0.21
- (a) Total outcomes = $6 \times 6 = 36$
 Let E is event of getting sum of numbers as 2, 3 or 5.
 $E = \{(1, 1), (1, 2), (2, 1), (1, 4), (2, 3), (3, 2), (4, 1)\}$
 Number of favourable outcomes = 7
 $P(E) = \frac{7}{36}$
- (b) Sample space (S) = $\{1, 7, 9, 21, 25\}$
 $n(S) = 5$
 Prime numbers = $\{7\}$
 Number of favourable outcomes = 1.
 Required probability = $\frac{1}{5}$
- (d) We know,
 $0 \leq P(E) \leq 1$
 $\frac{1}{0.89} > 1$ can not be probability of an event.
- (b) Number of total outcomes = 52
 Red ace cards = $\left\{ \begin{array}{ll} \text{Heart} & - \text{Ace} \\ \text{Diamond} & - \text{Ace} \end{array} \right\}$
 Number of red ace card = 2

$$\text{Required probability} = \frac{2}{52} = \frac{1}{26}$$

8. (d) Number of total outcomes = $6 \times 6 = 36$

Let E is an event of getting the sum of two numbers to be more than 10 i.e. 11 or 12.

$$E = \{(5, 6), (6, 5), (6, 6)\}$$

Number of favourable outcomes = 3

$$\text{Required Probability} = \frac{3}{36} = \frac{1}{12}$$

9. (a) Total outcomes (S) = $\{1, 2, 3, 4, 5, 6\}$

$$n(S) = 6$$

Numbers greater than 2 (E) = $\{3, 4, 5, 6\}$

$$n(E) = 4$$

$$P(E) = \frac{n(E)}{n(S)} = \frac{4}{6} = \frac{2}{3}$$

10. (a) Let E is an event of guessing the correct answer

$$P(E) = \frac{x}{6}$$

$$P(\bar{E}) = \frac{2}{3}$$

$$P(E) = 1 - P(\bar{E})$$

$$= 1 - \frac{2}{3} = \frac{1}{3}$$

$$\Rightarrow \frac{x}{6} = \frac{1}{3} \Rightarrow x = 2$$

11. (a) $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$$n(S) = 9$$

Odd prime numbers = $\{3, 5, 7\}$

Number of favourable outcomes = 3

$$P(\text{odd prime}) = \frac{3}{9} = \frac{1}{3}$$

12. (a) Let E is an event of hitting boundary

$$n(E) = 9$$

Number of total outcomes = $n(S) = 45$

$$P(E) = \frac{n(E)}{n(S)} = \frac{9}{45} = \frac{1}{5}$$

Probability of not hitting boundary

$$= P(\text{not } E)$$

$$= P(E) + P(\text{not } E) = 1$$

$$\text{So, } P(\text{not } E) = 1 - \frac{1}{5} = \frac{4}{5}$$

Hence, Assertion (A) is correct and Reason (R) is correct explanation.

13. (c) Total outcomes $(S) = \{(H, H), (H, T), (T, H), (T, T)\}$

$$n(S) = 4$$

$$\text{Favourable outcomes } (E) = \{(H, H), (H, T), (T, H)\}$$

$$n(E) = 3$$

$$P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}$$

14. (a) Total outcomes $= 3 + 5 + 7 = 15$

Favourable outcomes = (neither red nor black balls i.e. white balls = 5 white balls)

Number of favourable outcomes = 5

$$\text{Required probability} = \frac{5}{15} = \frac{1}{3}$$

15. (c) $P(\text{Bad eggs}) = 0.045$

$$\frac{n(\text{Bad eggs})}{\text{Total eggs}} = 0.045$$

$$\Rightarrow \frac{\text{Number of bad eggs}}{400} = 0.045$$

$$\Rightarrow \text{Number of bad eggs} = 0.045 \times 400 = 18$$

$$\Rightarrow \text{Number of good eggs} = 400 - 18 = 382$$

16. (a) We know, $P(E) + P(\bar{E}) = 1$

$$\text{So, } q = 1$$

$$q^2 - 4 = 1^2 - 4 = -3$$

17. (d) $P(E) + P(\bar{E}) = 1$

$$P(E) = 1 - P(\bar{E})$$

$$= 1 - \frac{1}{5} = \frac{4}{5}$$

18. (a) There is only 1 king of hearts in a deck of 52 cards.

$$\text{Required probability} = \frac{1}{52}$$

19. (d) Probability (not Ace) $= 1 - P(\text{Ace})$

$$= 1 - \frac{4}{52} = \frac{12}{13}$$

20. (b) $P(\text{green ball}) = 3P(\text{red ball})$

$$\Rightarrow \frac{n}{5+n} = 3 \times \frac{5}{5+n} \Rightarrow n = 15$$

21. (c) $P(\text{leap year having 53 Sundays}) = \frac{2}{7}$

$$P(\text{getting 53 sundays in a non-leap year}) = \frac{1}{7}$$

22. (b) Total possible outcomes $= (HT, TH, HH, TT)$

$$\therefore \text{Required probability} = \frac{1}{2}$$

23. (d) $P(\bar{E})$ or $P(\text{not } E) = 1 - P(E)$

$$= 1 - 0.65 = 0.35$$

24. (b) Probability

$$= \frac{\text{Number of favourable events in sample space}}{\text{Total number of events in sample space}}$$

$$P(\text{Blue balls}) = \frac{6}{16+8+6} = \frac{6}{30} = \frac{1}{5}$$

25. (c) $P(\text{Not happening of an event})$

$$= 1 - P(\text{Happening of the event})$$

$$= 1 - 0.02 = 0.98$$

26. (a) $x = 1$ $[\because P(E) + P(\bar{E}) = 1]$

$$\Rightarrow x^3 - 3 = -2$$

27. (a) $P(\text{Neither ace nor spade}) = 1 - P(\text{Ace or spade})$

$$= 1 - \frac{16}{52} = \frac{9}{13}$$

28. (d) Probability of any event always $0 \leq P(E) \leq 1$.

29. (d) (E) = Outcomes not possible are $\{(5, 5), (1, 5), (2, 5), (3, 5), (4, 5), (6, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 6)\}$

$$n(E) = 11$$

$$\text{Total outcomes} = 36$$

$$\therefore \text{Number of possible outcomes} = 36 - 11 = 25$$

$$\therefore \text{Probability} = \frac{25}{36}$$

30. (d) Favourable outcomes are 4, 8, 12, i.e. 3 outcomes and total number of outcomes = 15

$$\therefore \text{Required probability} = \frac{3}{15} = \frac{1}{5}$$

Option (d) is correct.

Short Answer Type Questions

31. Total outcomes $= 52 - 1$ black card $= 51$

$$\text{Number of queen(s) of heart} = 1$$

$$\text{Favourable outcomes} = 1$$

$$\therefore \text{Required probability} = \frac{1}{51}$$

32. Number of total cards $= 52 - 3 \times 2 = 46$

(i) Number of cards of club = $13 - 3 = 10$

$$\text{Probability (club)} = \frac{10}{46} = \frac{5}{23}$$

(ii) Number of red cards = 13

$$\text{Probability (red card)} = \frac{13}{46}$$

33. Number of total outcomes = 52

(i) Number of queen(s) of heart = 1

Number of favourable outcomes = 1

$$\text{Required probability} = \frac{1}{52}$$

(ii) Number of jack(s) = 4

$$P(\text{jack}) = \frac{4}{52} = \frac{1}{13}$$

$$P(E) + P(\bar{E}) = 1$$

$$\text{So, } P(\text{not jack}) = 1 - \frac{1}{13} = \frac{12}{13}$$

34. Total possible outcomes are HH, HT, TH, TT

And favourable outcomes are HT, TH, TT

$$\text{Required probability} = \frac{3}{4}$$

35. (i) Probability of getting an even prime number = $\frac{1}{6}$

$$(ii) P(\text{a number greater than 4}) = \frac{2}{6} = \frac{1}{3}$$

$$(iii) P(\text{an odd number}) = \frac{3}{6} = \frac{1}{2}$$

36. Total outcomes = 36

Favourable outcomes $\{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$

Number of favourable outcomes = 5

$$P(\text{sum 8}) = \frac{5}{36}$$

37. $n(s)$ = Total number of alphabets in English = 26.

$n(E)$ = Total number of consonant in English alphabet = 21.

$\therefore$ Probability (Chosen letter is a consonant)

$$= \frac{n(E)}{n(s)} = \frac{21}{26}$$

38. Let E be the event of getting square of a number less than or equal to 4.

S be the sample space. Then,

$$S = \{-3, -2, -1, 0, 1, 2, 3\}$$

$$\Rightarrow n(S) = 7$$

$$\text{and, } E = \{-2, -1, 0, 1, 2\} \Rightarrow n(E) = 5$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{5}{7}$$

39. Total possible outcomes are $(HHH), (HHT), (HTH), (THH), (TTH), (THT), (HTT), (TTT)$ i.e. 8.

The favourable outcomes to the event E 'Same result in all the tosses' are TTT, HHH .

So, the number of favourable outcomes = 2

$$\therefore P(E) = \frac{2}{8} = \frac{1}{4}$$

Hence, probability of losing the game = $1 - P(E)$

$$= 1 - \frac{1}{4} = \frac{3}{4}$$

40. Total outcomes = 1, 2, 3, 4, 5, 6

Prime numbers = 2, 3, 5

Numbers lie between 2 and 6 = 3, 4, 5

41. Sample space = $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$

$$n(S) = 36$$

(i) A = getting a doublet

$$A = \{(1, 1), (2, 2), \dots, (6, 6)\}$$

$$n(A) = 6$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

(ii) B = getting sum of number as 10

$$B = \{(6, 4), (4, 6), \dots, (5, 5)\}$$

$$n(B) = 3$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

42. An integer is chosen at random from 1 to 100

Therefore $n(S) = 100$

(i) Let A be the event that number chosen is divisible by 8

$$\therefore A = \{8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96\}$$

$$\therefore n(A) = 12$$

Now, $P(\text{that number is divisible by 8})$

$$= P(A) = \frac{n(A)}{n(S)} = \frac{12}{100} = \frac{6}{50} = \frac{3}{25}$$

$$P(A) = \frac{3}{25}$$

(ii) Let ' A ' be the event that number is not divisible by 8.

$$\therefore P(A') = 1 - P(A)$$

$$= 1 - \frac{3}{25} = \frac{22}{25}$$

43. Probability of selecting rotten apple

$$= \frac{\text{Number of rotten apples}}{\text{Total number of apples}}$$

$$\therefore 0.18 = \frac{\text{Number of rotten apples}}{900}$$

$$\text{Number of rotten apples} = 900 \times 0.18 = 162$$

44. There are 26 red cards including 2 red queens. Two more queens along with 26 red cards will be $26 + 2 = 28$

$$\therefore P(\text{getting a red card or a queen}) = \frac{28}{52}$$

$$\therefore P(\text{getting neither a red card nor a queen})$$

$$= 1 - \frac{28}{52} = \frac{24}{52} = \frac{6}{13}$$

45. Two dice are tossed

$S = [(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)]$

Total number of outcomes when two dice are tossed
 $6 \times 6 = 36$

Favourable events of getting product as 6 are:

$(1 \times 6 = 6), (6 \times 1 = 6), (2 \times 3 = 6), (3 \times 2 = 6)$

i.e. $(1, 6), (6, 1), (2, 3), (3, 2)$

Favourable events of getting product as 6 = 4

$$\therefore P(\text{getting product as 6}) = \frac{4}{36} = \frac{1}{9}$$

Case Based Type Questions

46. (i) Number of persons who travelled by bus/ship

$$= \frac{36 + 33}{360} \times 120 = 23$$

Number of favourable outcomes = 23

Total outcomes = 120

$$\text{Required probability} = \frac{23}{120}$$

- (ii) Most favourite mode of transport is car.

Number of persons who travelled by car

$$= \frac{177}{360} \times 120 = 59$$

- (iii) (a) $P(\text{used train}) = 1 - P(\text{not used train})$

$$= 1 - \frac{4}{5} = \frac{1}{5}$$

$$\Rightarrow \frac{\text{Number of persons who travel by train}}{\text{Total persons}} = \frac{1}{5}$$

$\Rightarrow$ Number of persons who used train

$$= \frac{1}{5} \times 120 = 24$$

OR

$$(iii) (b) P(\text{aeroplane}) = \frac{7}{60}$$

$$\Rightarrow \frac{\text{Number of persons who used aeroplane}}{\text{Total persons}} = \frac{7}{60}$$

$\Rightarrow$ Number of persons who used aeroplane

$$= \frac{7}{60} \times 120 = 14$$

$$\text{Revenue collected by company} = 14 \times 5000$$

$$= ₹ 70,000$$

Long Answer Type Questions

47. Sample space $(S) = \{(H, H, H), (H, H, T), (H, T, H), (T, H, H), (H, T, T), (T, H, T), (T, T, H), (T, T, T)\}$

$$n(S) = 8$$

- (i) Number of favourable outcomes = 7

$$\text{Probability (at least one head)} = \frac{7}{8}$$

- (ii) Exactly two tails = $\{(H, T, T), (T, H, T), (T, T, H)\}$

Number of favourable outcomes = 3

$$\text{Probability (exactly two tails)} = \frac{3}{8}$$

- (iii) At most one tail = $\{(H, H, H), (H, H, T), (H, T, H), (T, H, H)\}$

Number of favourable outcomes = 4

$$\text{Probability (at most one tail)} = \frac{4}{8} = \frac{1}{2}$$

48. Total number of outcomes = 6

Number of favourable outcomes = 2

$$P(\text{getting a number less than 3}) = \frac{2}{6} = \frac{1}{3}$$

OR

Required probability

$$= 1 - \text{Probability of winning a game}$$

$$= 1 - 0.07 = 0.93$$

49. Total number of outcomes = $6 \times 6 = 36$

Favourable outcomes = $\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1)\}$

Number of favourable outcomes = 6

$$\therefore P(\text{less than 5}) = \frac{6}{36} = \frac{1}{6}$$

OR

In month November 4 sundays are fixed.

But there are two extra days. They may be $\{(Sun, Mon), (Mon, Tues), (Tues, Wed), (Wed, Thurs), (Thurs, Fri), (Fri, Sat), (Sat, Sun)\}$

Number of favourable outcomes = 2

$$\therefore \text{Required probability (5 sundays)} = \frac{2}{7}$$

50. Bag contains 15 white balls.

Let say there be x black balls.

Probability of drawing a black ball

$$P(B) = \frac{x}{15+x}$$

Probability of drawing a white ball

$$P(W) = \frac{15}{15+x}$$

Given that $P(B) = 3P(W)$

$$\therefore \frac{x}{15+x} = \frac{3 \times 15}{15+x}$$

$$x = 45$$

Number of black balls = 45.

51. Two dice are thrown together total possible outcomes = $6 \times 6 = 36$

- (i) Sum of outcomes is even

This can be possible when

$\Rightarrow$ Both outcomes are even

$\Rightarrow$ Both outcomes are odd

For both outcomes to be even number of cases $3 \times 3 = 9$ Similarly,

Both outcomes odd = 9 cases

Total favourable cases = $9 + 9 = 18$

$$\text{Probability} = \frac{18}{36}$$

$$\text{Sum of even outcomes is} = \frac{1}{2}$$

- (ii) Product of outcomes is even

This is possible when

$\Rightarrow$ Both outcomes are even

$\Rightarrow$ First outcome even & the other odd

$\Rightarrow$ First outcome odd & the other even

Number of cases where both outcomes are even = 9

Number of cases for first outcome odd and the other even = 9

Number of cases for first outcome even and the other odd = 9

Total favourable cases = $9 + 9 + 9 = 27$

$$\text{Probability} = \frac{27}{36} = \frac{3}{4}$$

52. x is selected from 1, 2, 3 and 4

y is selected from 1, 4, 9 and 16

Let $A = \{1, 4, 9, 16, 2, 8, 18, 32, 3, 12, 27, 48, 36, 64\}$ which consists of elements that are product of x and y .

$P(\text{product of } x \text{ and } y \text{ is less than } 16)$

$$= \frac{\text{Number of outcomes less than } 16}{\text{Total number of outcomes}}$$

$$= \frac{7}{14} = \frac{1}{2}$$

53. Let S be the sample space of drawing a card from a well shuffled deck.

$$n(S) = 52$$

- (i) There are 13 spade cards and 4 ace's in a deck. As ace of spade is included in 13 spade cards, so there are 13 spade cards and 3 ace's.

A card of spade or an ace can be drawn in = 16 ways

Probability of drawing a card of spade or an

$$\text{ace} = \frac{16}{52} = \frac{4}{13}$$

- (ii) There are 2 black king cards in a deck a card of black king can be drawn in = 2 ways

$$\text{Probability of drawing a black king} = \frac{2}{52} = \frac{1}{26}$$

- (iii) There are 4 Jack and 4 King cards in a deck.

So there are $52 - 8 = 44$ cards which are neither Jacks nor Kings.

A card which is neither a Jack nor a King can be drawn in = 44 ways

Probability of drawing a card which is neither

$$\text{a Jack nor a King} = \frac{44}{52} = \frac{11}{13}$$

(iv) There are 4 King and 4 Queen cards in a deck.
So there are $4 + 4 = 8$ cards which are either King or Queen.

A card which is either a King or a Queen can be drawn in = 8 ways.

So probability of drawing a card which is either

$$\text{a King or a Queen} = \frac{8}{52} = \frac{2}{13}$$

54. Here the jar contains red, blue and orange balls.

Let the number of red balls be x .

Let the number of blue balls be y .

Number of orange balls = 10

Total number of balls = $x + y + 10$

Now, let P be the probability of drawing a ball from the jar

$$P(\text{a red ball}) = \frac{x}{x + y + 10} \Rightarrow \frac{1}{4} = \frac{x}{x + y + 10}$$

$$\Rightarrow 4x = x + y + 10$$

$$\Rightarrow 3x - y = 10 \quad \dots(i)$$

$$\text{Next, } P(\text{a blue ball}) = \frac{y}{x + y + 10}$$

$$\Rightarrow \frac{1}{3} = \frac{y}{x + y + 10}$$

$$\Rightarrow 3y = x + y + 10$$

$$\Rightarrow 2y - x = 10 \quad \dots(ii)$$

Multiplying equation (i) by 2 and adding to equation (ii), we get

$$6x - 2y = 20$$

$$-x + 2y = 10$$

$$5x = 30 \Rightarrow x = 6$$

Substitute $x = 6$ in equation (i), we get $y = 8$

Total number of balls = $x + y + 10$

$$= 6 + 8 + 10 = 24$$

Hence, total number of balls in the jar is 24.

